

Construction and Optimization of Low Energy ($< 240\text{J}$) Miniature Repetitive Plasma Focus Neutron Source

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Abstract

In the recent years, the upcoming diversified applications of neutron sources have compelled thrust towards the research and development of compact pulsed neutron generators. Amongst various options, miniature plasma focus (MPF) device based compact neutron generators have evolved as simple and economical alternative but yet from the application point of view, the neutron yield is far behind the threshold requirement. The potential limitation of MPF devices for practical applications comes from the fact that the neutron output tends to scale roughly as I^4 (where I is the peak discharge current) or as E^2 (where E is the energy stored in the system) and since MPF devices typically operate in sub-kilojoule range, their neutron output is in the range of $10^3 - 10^6$ neutrons/shot (with peak discharge current in the range of 40 – 150kA) depending upon device energy, efficiency and design. In order to make the MPF device useful as compact source of fast neutrons for the widest range of applications, the typical requirement for time averaged neutron output is in the order of $10^7 - 10^{10}$ neutrons/second. To achieve such a time averaged neutron output in MPF devices, it is necessary to operate them in repetitive mode at repetition rate of more than 1Hz.

The work presented in this thesis is mainly based on the design, construction and optimization of indigenously developed three different versions of sub-kilojoule range fast miniature plasma focus devices – FMPF-1, FMPF-2 and FMPF-3 operating at <240J. The designations of the devices have been made on the basis of their pulsed power system configuration and electrical characteristics.

The first version of fast miniature plasma focus device FMPF-1 was primarily made for investigating the performance of the newly designed device at such low energy and to optimize plasma focus tube parameters. After going through several stages of design and experimental iterations of coaxial electrode assembly and other operating parameters, the ‘FMPF-1’ device was successfully demonstrated as a portable neutron source producing maximum average neutron yield of $(1.15 \pm 0.2) \times 10^6$ neutrons/shot at 230J/80kA/5.5mbar Deuterium (D_2) filling gas pressure. The conceived model of ‘FMPF-1’ is extremely compact and aids the feature of portability/table top model. The overall dimensions of the apparatus, which includes capacitor bank, sparkgap switch and the focus chamber is $0.2m \times 0.2m \times 0.5m$ and the total weight of the system is ~25kgs.

The second version of fast miniature plasma focus device FMPF-2 was specifically designed to operate in repetitive mode ranging from 0.2 – 10Hz using newly developed high wattage pulsed power system. Many technological challenges were confronted and resolved for making the successful

operation of the device in repetitive mode up to 10Hz. Using the repetitive miniature plasma focus device FMPF-2, for the first time, the enhancement in time averaged neutron yield has been successfully demonstrated by an order of magnitude, realizing the long stated concept of ‘enhancement by repetition’. While operating at 228J/80kA/6mbar Deuterium (D_2) filling gas pressure, the time averaged neutron yield of $(6.2 \pm 4) \times 10^5$ neutrons/second at 1Hz operation was enhanced to $(6.5 \pm 0.6) \times 10^6$ neutrons/second at 10Hz operation for the burst length of 30 consecutive shots.

In the third version of fast miniature plasma focus device FMPF-3 (a four module upgraded version of FMPF-2), significant improvements were made to enhance the efficiency of high wattage pulsed power system used in FMPF-2 device. As an outcome of rigorous transformations made, the FMPF-3 pulsed power system delivered ~20% higher peak discharge current than FMPF-2 pulsed power system while operating at similar energy level. In FMPF-3 device, at 200J/90kA/5.5mbar Deuterium (D_2) filling gas pressure, the time averaged neutron yield of $(1.4 \pm 0.6) \times 10^6$ neutrons/second at 1Hz operation was enhanced to record yield of $(1.4 \pm 0.2) \times 10^7$ neutrons/second at 10Hz operation for the improvised burst length of 50 consecutive shots.

In the thesis, along with exclusive description on design methodology, construction and optimization exercises of all versions of newly developed miniature plasma focus devices (FMPF-1, FMPF-2 and FMPF-3), the time integrated and time resolved investigations on their neutron and hard X-ray emission characteristics have also been dealt in detail.

List of Publications

Journal Papers:

1. **Rishi Verma**, R. S. Rawat, P. Lee, M. Krishnan, S. V. Springham and T. L. Tan, *Miniature plasma focus device as a compact hard X-ray source for fast radiography applications*, IEEE Transactions on Plasma Science, **38**(4), 652 (2010).
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Chapter 1

Introduction

The research topic of this thesis is the “Construction and Optimization of Low Energy (< 240J) Miniature Repetitive Plasma Focus Neutron Source”. This chapter begins with the introduction of a brief history of plasma focus devices and their universal presence pursued by highlighting the fast growing need of compact neutron generators in the present scenario, illustrating their importance and applications. In the later section, a brief literature survey on the ongoing trend of scaling down the plasma focus devices to sub-kilojoule range is given, followed by research scopes and motivation behind this research project. The chapter then ends by presenting a layout of the thesis.

1.1 Dense Plasma Focus: Brief History and Universal Presence

Dense Plasma Focus (DPF) device is a hydromagnetic coaxial plasma accelerator [1-2] which involves the storage of magnetic energy behind a moving current sheath and the pumping of this energy into pinched plasma column during the rapid collapse phase, producing short duration (~ 10 to 50ns), high temperature ($\sim 1\text{keV}$) and high density plasma ($\sim 10^{19}\text{cm}^{-3}$), which is a rich source of energetic and multiple radiations like fast electrons and soft/hard X-rays ($10^{-1} - 10^3\text{keV}$), fast ions (up to 10MeV) and neutrons (2.45MeV or 14MeV). These intrinsic radiations from the plasma focus device, sets it apart from other devices as a prime candidate for various applications.

The DPF attracted much attention from the scientific community during the 1960's as it was considered as efficient fusion device, producing an intense burst of neutrons when operated with Deuterium/ Deuterium-Tritium as filling gas. This led to intensive research on DPF in laboratories all over the world to attain controlled thermonuclear fusion using this alternative magnetic confinement scheme. Since then, the scope of DPF in fusion research is under active consideration because significant progress in the associated pulsed power technology has enabled the successful operation of

large plasma focus devices, which has led to better understanding of plasma dynamics and neutron production mechanisms. But unfortunately, beyond 500kJ bank energy, the production of neutrons no longer follows established scaling laws and reaches saturation [3-4]. Hence, the essential problem to be resolved in PF research has always been to discover the physics, which dominates the neutron yield limitation/saturation at higher bank energies. Also, it was realized later that the neutrons are not of purely thermonuclear origin rather they are mainly due to beam target mechanism. It is a question related to the neutron production mechanism and plasma dynamics.

The better physical understanding and viability of other fusion technologies such as tokamaks and high-power laser driven fusion experiments downgraded the importance of plasma focus devices as controlled thermonuclear fusion device. But nevertheless, this device was soon recognized as a powerful and compact source of energetic radiations producing fast neutrons [5-6], intense X-rays [7-11], energetic ions [12-14] and electrons [15-16] which have important role in variety of scientific and industrial applications.

In the last few decades, most of the experimental studies performed on plasma focus devices were mainly focused in medium (2-10kJ) [17-24], large 10-500kJ [25-30] and mega-joule [31-32] energy range devices. A brief summary of such worldwide facilities, with their characteristic parameters, optimized for enhanced neutron yield is listed in Table 1.1.

Table 1.1 Characteristics of different Plasma Focus Devices.

Device[ref.] – location	Energy (kJ)	Anode radius (cm)	Peak current (kA)	Pressure (mbar)	Neutron Yield (n/shot)	Operation mode	Energy density parameter $[28E/a^3]$ (J m^{-3})	Drive Parameter S $[I/ap^{1/2}]$ ($\text{kA cm}^{-1} \text{mbar}^{-1/2}$)
PF-1000[31] – Poland	1064	12.2	2300	6.6	2×10^{11}	<i>Single shot</i>	1.6×10^{10}	73.4
FRASCATI[32] – Italy	1000	8	1850	10.6	5×10^{11}	<i>Single shot</i>	5.46×10^{10}	69.10
TAMU[29] – Texas, USA	480	5	1400	13.3	2.6×10^{11}	<i>Single shot</i>	10×10^{10}	76.77
SPEED2 [25-26] – Germany	187	6	4000	2.5	1×10^{11}	<i>Single shot</i>	2.42×10^{10}	421.63
PF-360 [27] – Poland	130	6	1200	5.6	3.8×10^{10}	<i>Single shot</i>	1.7×10^{10}	84.51
DENA [28] – Iran	90	25	2800	1.3	1.2×10^9	<i>Single shot</i>	0.16×10^9	98.23
DPF-40 [30] – China	18	3.2	350	3.3	2.1×10^8	<i>Single shot</i>	1.53×10^{10}	60.20
7 kJ PF [22] – Japan	7	1.75	400	6	5.8×10^8	<i>Single shot</i>	3.65×10^{10}	93.31
GN1 [21] – Argentina	4.7	1.9	350	4	3×10^8	<i>0.02Hz</i>	1.91×10^{10}	92.10
FN II [20] – Mexico	4.6	2.5	350	3.7	3×10^8	<i>Single shot</i>	0.82×10^{10}	72.78
UNU/ICTP [19, 23] – Europe, Africa, Asia^{1,2}	2.9	0.95	170	8.5	1.2×10^8	<i>Single shot</i>	9.5×10^{10}	61.37
NX2 [18] – Singapore[*]	2.9	1.15	410	20	7×10^8	<i>Repetitive, Max 16 Hz</i>	5.3×10^{10}	79.72
BARC [24] – India	2.2	1.05	180	6.2	14.4×10^7	<i>Single Shot</i>	5.32×10^{10}	68.84
PACO [17] – Argentina	2	2.5	250	1.5	5×10^8	<i>Single shot</i>	0.35×10^{10}	81.64

¹Plasma Radiation Sources Laboratory, NIE/ NTU, Singapore[#] At 560kJ Operation² Quaid-i-Azam University, Islamabad, Pakistan

1.2 Miniature Plasma Focus Device: A New Perspective

The first order description of any pulsed plasma device is the energy that is stored in capacitor bank. A comparison between plasma foci of different energies is interesting. Although only a fraction of the initial energy E stored in the capacitor bank is transferred to the plasma, the parameter E/V_p (V_p being the plasma volume) is usually used to characterize the plasma energy density in order to compare different devices. According to the established scaling laws [33] and performed optical investigations [34], the final pinch radius and the maximum pinch length are proportional to anode radius a ($\sim 0.12a$ and $\sim 0.8a$, respectively). Thus, the final plasma volume V_p (prior to the appearance of probable instabilities) is of the order of $\pi(0.12a^2) \times (0.8a) \approx 0.036a^3$, and the plasma energy density at the pinch moment is proportional to $E/V_p \sim 28E/a^3$ [35]. In the wide energy range of plasma focus devices listed in Table 1.1, this value typically lies in the range of $(0.1-10) \times 10^{10} \text{ J/m}^3$.

Other relevant parameter in plasma foci is the called drive parameter $(I/ap^{1/2})$ [33], where I is the peak discharge current, a the anode radius, and p is the Deuterium gas filling pressure for the maximum neutron yield. It has been theoretically and experimentally conjectured that the typical velocity of the current sheath in the axial rundown and radial compression phases (lying in the range of $(0.8-1) \times 10^5$ and $(2-2.5) \times 10^5 \text{ m/s}$ respectively for a wide range of plasma focus devices) is proportional to the drive parameter *i.e.* $I/ap^{1/2}$ [33]. It may be noted from Table 1.1 that for the PF devices ranging from kJ-MJ, the drive parameter is typically in the range of $78 \pm 12 \text{ kA/cm}^{-1} \text{ mbar}^{1/2}$.

Thus, judging from the view point of ‘energy density’ and ‘driver parameter’ (as shown in Table 1.1) of medium [17-24], large [25-30] and mega-joule [31-32] energy range plasma focus devices, it can be interpreted that in all plasma focus machines optimized for Deuterium(D_2) operation, irrespective of their energy and size, have similar magnitude of drive parameter in the axial phase and narrow range of orders of magnitude of compressed energy density in the focus phase that produce pinch plasmas with similar order of temperature and densities. The data shown in Table 1.1 also

indicates that the higher the capacitor bank energy and the peak discharge current are, the bigger is the anode (and hence chamber/device) size. This observation has an important implication, that if the stored energy of the capacitor bank is reduced to sub-kilojoule values then the anode/device size has to be reduced for electrical and dynamical/geometrical parameter matching but the energy density or drive parameter can still be the same as that of medium or large focus facility and hence plasma densities and temperatures of similar magnitudes can be achieved, resulting in similar nuclear and atomic reactions to those occurring in medium/ large energy plasma foci. This means that a radiation spectra similar to that of medium, large or mega-joule focus facility, although at an altogether different scale, can be achieved in a miniature plasma focus device. It translates into the fact that the miniaturization of plasma focus device is possible making this device portable which can then be applied for field applications of commercial importance.

In recent years, many research groups across the globe have shown keen interest in research and development of sub-kilojoule range miniature plasma focus devices.

The experimental study on electron density measurement in sub-kilojoule range, low energy plasma focus device of 400J has been reported by Cristian *et al* [36]. An electron density of $(0.9 \pm 0.25) \times 10^{19} \text{ cm}^{-3}$ was measured at the axis of the plasma column close to the pinch time. This value is at par with the plasma density of a conventional plasma focus devices operating in the energy range of 1kJ to 1MJ.

J. Moreno *et al* [34] made optical observations of the plasma motion (radial and axial) in a very small plasma focus device of 50J, operated in hydrogen at 25kV, to provide the pinching evidence in miniature plasma focus device. A single frame image converter camera of 5ns exposure time was used to obtain plasma images in the visible range. They used six bars around the central anode to serve as cathodes having a hybrid Filippov-Mather configuration of aspect ratio 0.625.

More recently L. Soto *et al* [37] demonstrated the pinch evidence in an ultra-miniature PF device (named as Nanofocus) with submillimetric anode (of radius $\sim 0.8\text{mm}$ and length $\sim 0.5\text{mm}$) operating at 0.1J bank energy.

Hassan *et al* [38] reported optical observations of the current sheath evolution and radial collapse phase in a miniature plasma focus device of $\sim 100\text{J}$, operated in Hydrogen at 15kV and confirmed pinching. An ICCD camera attached to the spectrometer ACTON 750 (with 20ns exposure time) and streak camera, IMACON-500 was used for imaging. In the experimental setup, coaxial electrode assembly constitutes - anode of 9.5mm diameter/ 21.6mm length and vacuum chamber wall of 48mm inner diameter acting as cathode. The system has Mather configuration with the aspect ratio of ~ 2.3 .

Recently many groups have also reported the development of miniature plasma focus devices as portable neutron source [39-44].

M. Milanese *et al* [39] have reported the development of 125J small plasma focus that produces yield in the order of $\sim 10^6$ neutrons/shot. R. K. Rout *et al* [40] have reported development of battery powered tabletop pulsed neutron source based on a 200J sealed miniature plasma focus device that produces average yield of the order of $\sim 10^6$ neutrons/shot in 4π sr. A. V. Dubrovsky *et al* [41] have reported the development of 200J plasma focus device for both neutron and X-ray production.

In the last few years, extensive efforts have been made at the *Chilean Nuclear Energy Commission* to develop MPF as radiation source of X-rays and neutrons in the energy range of few tens to hundreds of joules. Those miniature devices not only demonstrate the evidence of pinching but also produced a good amount of X-rays and neutrons. The reported average neutron yields from such small PF devices are of the order of 10^6 and 10^4 neutrons/shot for 400J [42] and 50J [43] bank energies, respectively.

P. Silva *et al* [43] reported the pinching evidence in a Deuterium-filled plasma focus operating at 50J and 67J by observing the dip in the current derivative signal and the typical peak in the voltage signal which indicates pinch compression, usually observed in several kJ DPF devices. In their setup, the electrode assembly consisted of 29mm long, 6mm diameter copper tube anode and an outer cathode of eight 5mm diameter copper rods, uniformly spaced on a diameter of 27mm. The size of their device was in the order of 25cm × 25cm × 50cm. Very recently neutron emission from this device has been reported by L. Soto *et al* [44]. The average neutron yield for 50J discharges at 6mbar was $(1.2 \pm 0.5) \times 10^4$ neutrons/shot, and $(3.6 \pm 1.6) \times 10^4$ neutrons/shot for 67J discharge at 9mbar.

It is important to note regarding all the Deuterium optimized MPF devices mentioned above [39-44] that they have all been operated in single shot mode and no information on the statistics of their radiation generation reproducibility has been reported.

From the point of view of extreme ultraviolet microlithography application, Cymer Inc. (USA) [45-47] have chosen a low energy plasma focus device of 18J as a potential pulsed plasma EUV source for lithography application at production level. Their MPF device is driven by fast current pulse of magnitude around 50kA. Development of a ~24J miniature hybrid plasma focus as an EUV source, driven by 10kA fast current pulse has been reported by S. R. Mohanty *et al* [48].

The above mentioned ongoing trends conclusively indicate that the research and development of sub-kilojoule range of plasma focus devices is now gaining momentum [36-47], but currently it is in an underdeveloped stage and a lot more effort is needed to develop an efficient and reproducible miniature plasma focus device which would find its way into commercial applications. Experimental research on miniature plasma focus devices of sub-kilojoule range would also provide avenues to test the validity of (i) present theoretical models and (ii) radiation emission scaling laws, in this newly explored low energy regime.

1.3 A Spin-off in Application as Neutron Generator

In the recent years there has been consistently growing demand for compact neutron generators since they can be used as portable non-invasive interrogation tools (*i.e.* without physically touching the object being analyzed) because of the high penetration capability of neutrons. This feature is of significant importance when dealing with old, potentially unstable high explosives, chemical weapons and nuclear materials. At present, the global market potential for compact neutron generators has grown from below \$60 million per annum in the period 1993-2001 (focused on drug trafficking to explosive detection) to over \$750 million in 2006 and is expected to exceed \$1.4 billion by 2010.

An important and growing market for neutron generators is in analyzing bulk materials. The analytical technique for detecting elemental compositions of various materials including nuclear, drugs and explosives is known as pulsed fast thermal neutron analysis (PFTNA) [49]. This technique utilizes the $(n,n'\gamma)$, $(n,p\gamma)$, and (n,γ) reactions to identify and quantify a large number of elements. When a sample is irradiated with a short pulse of neutrons then it becomes activated and emits γ -rays with characteristic and distinct energies. These γ -rays are like the 'fingerprints' of the elements contained in the object. By counting the number of γ -rays emitted with a specific energy, one can deduce the amount of the element contained within the object. In the case of an object that is hidden among other innocuous materials, the identification takes place through the correlation of various chemical elements observed, coupled to the information about the innocuous material itself. A complete PFTNA interrogation tool is composed of compact neutron generator, a gamma radiation detector (like HPGe *i.e.* Hyper-Pure Germanium detector) and a Multi-Channel Analyzer for accumulating the energy spectrum of emitted gamma rays. The stoichiometric relationships are used to convert the elemental information to chemical assays for detecting the elemental content of the major constituents in the bulk material [50]. Other than bulk material analysis, neutrons are also being used in many other

applications like – radiography, tomography, defectoscopy etc. The vast range of applications in which portable neutron sources can be used, are summarized in Table 1.2 [49, 50].

To fulfill these application requirements, there is a large thrust to develop neutron generators that are compact, long-lasting, efficient and inexpensive to construct and yet capable of using safe Deuterium-Deuterium reactions to produce a high neutron yield or flux and can be tailored to meet a variety of specifications.

Table 1.2 Applications of compact and portable neutron generators.

S. No.	Application Field	Specific Applications
1.	Homeland Security	• Explosive detection and identification • Special nuclear materials detection and identification • Land mine detection
2.	Nuclear Geophysics	• Gas and oil well-logging • Ore (uranium) well-logging • Mine mineral mapping and analysis
3.	Neutron Radiography	• Thermal neutron radiography • Fast neutron radiography
4.	Medicine	• Boron capture neutron therapy • Neutron beam therapy
5.	Industrial Processing	• Cement process control • Coal quality analysis • Wall thickness analysis

Neutrons may be produced using a number of techniques including radioactive isotopic sources, electro-physical neutron generators and large accelerators [49-50]. Typical isotopic sources producing continuous fluxes of neutrons are Californium-252 (^{252}Cf), with a half-life of about 2.6 years or Americium-Beryllium (Am-Be, ^{241}Am has a half-life of 458 years). Isotopic neutron sources have the advantage that they have long life and have relatively constant flux of neutrons but the major

disadvantage associated in their use is that they cannot be turned off, therefore they have to be contained within bulky shielding at all times. Also their fluence is limited to $<10^8$ neutrons/second. The main advantage of compact pulsed neutron generators over passive radioactive sources (like ^{252}Cf or Am-Be) is that they simplify safety procedures for deployment and storage because they can be turned off.

Over the last few years accelerator based neutron sources have evolved in the two main streams (i) Large accelerator based neutron source [50], and (ii) Compact and portable neutron sources ; mainly based on either a sealed tube generator [51] or Plasma Focus [39-44].

Even though accelerator based neutron sources (like Spallation Neutron Source (SNS) at Oak Ridge, Tennessee, USA [50]) provide the most intense pulsed neutron beams but they are very expensive, technically complicated and immensely bulky. They have their own set of limitations that keep them out of range for use in many of the commercial and industrial applications.

Sealed tube neutron generators are basically miniature particle accelerators. It consists of a source to generate positively charged ions; one or more structures to accelerate the ions (usually up to ~ 110 kV); a metal hydride target loaded with either Deuterium, Tritium, or a mixture of the two; and a gas-control reservoir, also made of a metal hydride material. The most common ion source used in neutron generators is a cold-cathode, or Penning ion source, which is a derivative of the Penning trap used in Penning ion gauges. The principle of operation is that when Deuterium and/or Tritium gas is introduced into the anode at a pressure of a few mbar, firstly the electric field between the anode and cathodes ionizes the gas and then the Deuterium ions are accelerated and bombarded in a solid target which is loaded with Tritium (or Deuterium) resulting in the production of neutrons via D-D or D-T fusion reaction. Normally, the commercially available compact accelerator neutron tubes are loaded with 1 to 2Ci of Tritium [51]. Although the sealed tube generators are very compact, the major issue in

their utilization for commercial purposes is (i) very high cost of the tubes and (ii) limited operating lifetime (the API 120 sealed tube neutron generator offered from Thermo Electron Corp. gives neutron yield (D-T) of $\sim 2 \times 10^7$ neutrons/second, having operating lifetime of ~ 1200 hours may typically cost $\sim$ US \$ 60k – 80k).

Plasma Focus based neutron generators have evolved from a large, expensive instrument to a compact, affordable product to be commercially viable. Ideally Plasma Focus devices have unlimited lifetime, since all parts of the coaxial electrode assembly are replaceable and Deuterium (or D-T mixture) gas is being used as fuel for neutron production. Plasma focus device is amongst the strongest contenders in the category of coaxial accelerators to be used as portable neutron source. Due to pulsed nature, the plasma focus machine does not pose any activation problem for storage and handling. It produces bursts of neutrons lasting for about 10-100 ns, with energies of 2.5-3, 10–11 or 14 MeV depending on the gas filled in the machine (Deuterium, Deuterium with a Lithium target or a Deuterium–Tritium mixture). Typical threshold requirement of fluence in various applications is 10^6 to 10^8 neutrons/second. To the best of our experience with various plasma focus machines, such neutron fluxes are conventionally obtained from 2-3 kJ PF device [17-24], which are bulky and lack the feature of portability.

According to current scaling estimations, which have been proven to be valid in lower energy regime during this project work, yields of the order of $\sim 10^6$ neutrons/second is practically feasible to obtain in miniaturized versions of tabletop plasma focus device operating at sub-kilojoule range which can be further enhanced by another order of magnitude if the device is made to operate in the repetitive mode. The inherent advantage with ‘Miniature Plasma Focus’ (MPF) device is that they are of low energy, much smaller in size and cost effective in comparison to medium and high plasma foci and also easier to operate in a repetitive regime from few Hz to kHz range since the driving power requirement is low. A possibility of working with DPF on a low energy level and at relatively low cost

makes miniature plasma focus devices commercially viable. The present research and development work aims at demonstrating the potential of miniature plasma focus devices as compact pulsed neutron sources.

1.4 Scope of Research

Due to various potential applications of low energy (sub-kilojoule range) miniature plasma focus (MPF) devices, many scientific and industrial laboratories are taking a keen interest in its research and development. But as mentioned before, exploration of MPF devices has just started only couple of years ago, therefore a lot more research and dedicated effort is required to develop an efficient and reproducible repetitive miniature plasma focus device, with its technology and physics understood and outlined in much more elaborated way, so that it would find its way as commercially viable compact radiation source fulfilling the competing requirements of being portable and cost-effective. The efforts have mainly been concentrated on engineering and optimization of compact sub-kilojoule range plasma focus device so as to make the neutron flux of this fusion-based neutron generator comparable to that of expensive sealed tube neutron sources of similar volume.

Based on above considerations, in this project, we have demonstrated the feasibility of low energy Miniature Plasma Focus (MPF) device along with its applicability as quasi-continuous (repetitive) pulsed neutron source; for being utilized as a portable neutron generator with appreciable amount of neutron flux, while operating at $\sim 0.2\text{kJ}$. We have developed three different versions of low energy repetitive plasma focus devices named as FMPF-1, FMPF-2 and FMPF-3. The preliminary version FMPF-1 device can be operated at maximum repetition rate of 0.5Hz whereas the advanced versions FMPF-2 and FMPF-3 devices can be operated at maximum repetition rate of 10Hz in quasi-continuous mode.

The FMPF-1 and FMPF-2 devices are single module systems (comprising of four $0.6\mu\text{F}/30\text{kV}$ capacitors and eight $0.3\mu\text{F}/30\text{kV}$ capacitors respectively with one trigatron switch) whereas FMPF-3 device is a four module system (with eight $0.3\mu\text{F}/30\text{kV}$ capacitors and four pseudospark switches). The FMPF-1 device, with Deuterium gas operation, produces a maximum average neutron yield of $(1.15 \pm 0.2) \times 10^6$ neutrons/shot at $\sim 230\text{J}/\sim 80\text{kA}$ peak discharge current.

The FMPF-2 device is a first prototype of high repetition rate miniature plasma focus device with newly developed high wattage pulsed power system. This device has been successively fired and tested up to 10Hz repetition rate under different operating conditions. Using pure Deuterium as the filling gas, at $\sim 230\text{J}$ operation (delivering $\sim 80\text{kA}$ peak discharge current) the time averaged neutron yield of $(6.2 \pm 4) \times 10^5$ n/sec at 1Hz operation was enhanced to $(6.5 \pm 0.6) \times 10^6$ n/sec at 10Hz operation for the burst length of 30 consecutive shots.

The FMPF-3 is an advanced version of the FMPF-2 device having superior pulsed power system efficiency along with high repetition rate capability. It delivers $\sim 20\%$ more peak discharge current than the FMPF-2 device and thus produces a higher neutron yield, while operating at a similar charging voltage and energy. Using pure Deuterium as the filling gas, at $\sim 200\text{J}$ operation (delivering $\sim 90\text{kA}$ peak discharge current) the time averaged neutron yield of $(1.4 \pm 0.6) \times 10^6$ neutrons/second at 1Hz operation was enhanced to $(1.4 \pm 0.2) \times 10^7$ neutrons/second at 10Hz operation for the improved burst length of 50 consecutive shots.

The present research and development work includes:

- (a) Conceptualization, designing and development of three different versions of low energy repetitive miniature plasma focus devices with a stored energy of $< 240\text{J}$.
- (b) Design and development of customized, compact pulsed power drivers of minimum practical inductance for the FMPF-1, FMPF-2 and FMPF-3 devices, with available resources.

- (c) Demonstration of the feasibility of production of neutrons, reproducibly, at the lower limit of plasma focus energy at $< 240\text{J}$.
- (d) Performance optimization of the newly developed miniature plasma focus devices by refining/adjusting various operating/design parameters such as bank energy and discharge current, anode shape and its material, anode length, insulator length and its material, gas composition and filling gas pressure to obtain efficient and reproducible plasma pinches for maximum neutron yield.
- (e) The investigation also incorporates numerical experiments that have been performed using the 'Lee Code', to simulate the modifications in the axial and radial phase dynamics along with gross plasma focus properties, with the use of different anode/cathode geometries at different operating parameters.
- (f) Investigations of hard X-ray and neutron emission characteristics for revealing relevant plasma phenomenon and their correlation with neutron production, in sub-kilojoule range miniature plasma focus devices.
- (g) Investigation of the stability in neutron emission in repetitive mode of operation from 1 to 10Hz to reveal the issues/factors responsible for constraining the use of higher repetition frequency in pulsed plasma radiation devices.
- (h) Exploration of unconventional techniques of neutron yield enhancement like usage of composite geometry anode shapes and Deuterium-high Z gas admixture operation.
- (i) Development, customization and calibration of required electrical, X-ray and neutron diagnostics for measuring/monitoring various outputs/parameters.
- (j) Demonstration of the utility of the newly developed devices in other areas of basic and applied research such as in Radiography with nanosecond exposure time and in material modification studies.

In addition to the above mentioned research and development work related to the design, construction and optimization of low energy miniaturized plasma focus device, the characterization study of the FMPPF-1, FMPPF-2 and FMPPF-3 device also helped in understanding and exploring the physics related to plasma focus operation at this limit of low energy leading to formulations of scaling laws of DPF performance in sub-kilojoule range.

1.5 Layout of the Thesis

The organization of this thesis is as follows:

Chapter 1 (Introduction) begins with a general introduction of Plasma Focus with summarized highlights of previous works and brief literature review, followed by importance and applications of, the recently envisioned Miniature Plasma Focus (MPF) device. Research scopes and motivation behind carrying out this project are explained.

Chapter 2 (Plasma Focus Review: Plasma Dynamics and Radiation Emission) starts with basic description of plasma focus device operation and gives an overview of the plasma dynamics of various phases. Following that is a review of some related phenomena like instabilities, micro-instabilities, turbulence *etc.* For investigating design criteria in the low energy region, the neutron yield scaling laws of the PF devices is also discussed in brief. The later part of the chapter illustrates mechanisms responsible for production of X-rays and neutrons. In the last part of the chapter, the relevance of ‘drive parameter’ and ‘energy density parameter’ has been discussed in the context of scalability and energy transfer efficiency.

Chapter 3 (Design and Construction of Fast Miniature Plasma Focus Devices) describes the conceptualization and synthesis of newly developed miniature plasma focus devices (FMPPF-1, FMPPF-2 and FMPPF-3) accompanied by the details on engineering criterion, as well as tradeoffs in design

approach, architecture, pulsed power systems, construction, technical constitution and performance of subsystems.

Chapter 4 (Diagnostics Techniques) provides a detailed description of various diagnostic tools that are employed in the experimental setups. In this chapter, initially, the indigenously designed high bandwidth current derivative probe (used as electrical diagnostic tool for the measuring the electromagnetic parameters of the device) is introduced, followed by the description of other diagnostics that have been used for the detection and time resolved measurements of hard X-rays and neutrons (*i.e.* fast response plastic scintillator coupled with a high gain photomultiplier tube), soft X-rays (*i.e.* Multi-channel channel PIN Diode X-ray Spectrometer) and ions (*i.e.* Faraday Cup for the measurement of typical deuteron energy spectra). The description of the specifically tailored and customized, high sensitivity neutron detection and measurement setup of the ^3He proportional counter has also been detailed along with the briefing on solid state nuclear track detector (SSNTD) based passive technique that has been used for the confirmation of neutron yield by imaging fusion protons.

Chapter 5 (Experimental Results and Discussion) describes the characterization results, analysis and discussion of the experiments performed on all the three versions of indigenously developed miniature plasma focus devices along with performance evaluation of their pulsed power systems. Optimization study of the electrode geometry (such as anode shape, its dimension, insulator sleeve length, cathode structure *etc.*) that affects the neutron yield performance of MPF device is presented. Analysis of experimental results obtained (for single shot and repetitive mode of operation) by using different diagnostic techniques and a discussion of the interpretation of data, for all the three devices, is explained in detail.

Chapter 6 (Conclusions and Future Work) summarizes and concludes the major experimental findings of this research project and outlines possible directions of future work that can be undertaken.

Chapter 2

Plasma Focus Review: Plasma Dynamics and Radiation Emission

The plasma focus is a simple device with complex physics phenomena and multiple dynamic processes. This chapter starts with a basic description of the plasma focus device and its fundamental operation along with a briefing on conventionally followed configurations. A systematic description of plasma dynamics in various plasma focus phases is provided which is followed by a discussion of some pinch related phenomenon like instabilities, micro-instabilities and turbulence. The electrical behavior and the energy transfer process in the plasma focus device are then described to better understand this complex device. The later part of the chapter illustrates the multiple radiation emissions from the dense plasma focus and possible mechanisms of their production. The neutron yield scaling laws of the plasma focus devices are also discussed briefly. At last, the roles of ‘drive parameter’ and ‘energy density parameter’ from the perspective of scalability and energy transfer efficiency are illustrated.

2.1 The Focus Device

One of the most efficient and perhaps the simplest ways of producing high temperature fusion plasma in the laboratory is through the pinch effect. The incarnation of Dense Plasma Focus (DPF) device is mainly related with the intensive efforts laid down during 1960’s towards the development of Z-pinch devices to create and confine dense plasma for attaining controlled thermonuclear fusion. Those classical Z-pinch devices (basically Plasma Guns, first investigated by J. Marshall [52] and H. Alfven [53] between 1958 and 1960), in which the current is passed directly across the length of cylindrical plasma, were observed not to produce very high temperatures since the plasma is disrupted by hydro-magnetic instabilities before sufficient energy is transferred. This shortcoming was

overcome in a modified version of the coaxial plasma gun called ‘Plasma Focus’ [1, 2]. In this device, the pinch formation is delayed, until the discharge current reaches its peak value. The delay is generated by allowing the plasma to travel between a pair of coaxial electrodes before pinching. Hence, DPF is a pulsed plasma device that generates, accelerates and then pinches the plasma due to interaction of current with its self-induced magnetic field, producing compressed, hot and dense plasma. Over forty years of experiments, in a wide energy range of this device, has demonstrated that the short lived pinched plasma attains sufficiently high temperature and pressures necessary for fusion reactions, when filled with Deuterium or Deuterium-Tritium gases. Since the discharge period typically lies in the range of a few hundreds of nanoseconds to microseconds, for this wide range of energies, hence conceptually plasma focus devices can be efficiently designed to be operated in the repetitive regime.

2.2 Structure and Operation

Plasma Focus device was independently discovered by N.V. Filippov [1] and J. W. Mather [2] in the early 60’s. Traditionally plasma focus devices have been characterized as ‘Mather’ and ‘Filippov’ types according to their anode aspect ratio A , with $A = z_0/2a$ in which z_0 is the effective anode length and a is the radius of the anode. The ‘Mather’ configuration is defined by $A > 1$ (typically 5-10) whereas the ‘Filippov’ configuration has $A < 1$ (typically ~ 0.2). Fig. 2.1 illustrates the electrode geometries of the Mather-type and Filippov-type plasma focus devices.

As can be seen from this schematic, fundamentally both configurations of dense plasma focus devices consists of two coaxial electrodes *i.e.* anode and cathode separated by an insulator sleeve. Macroscopically, ‘Mather’ and ‘Filippov’ configurations differ in the direction of motion of the accelerated plasma, which is axial and radial respectively. However both the configurations exhibit similar (1) dynamics of the current sheath, (2) scaling laws for neutron emission and (3) characteristic emission of

energetic ion and electron beams, X-rays, microwaves etc. Eventhough both the devices behave similarly; the 'Mather' type is preferred due to its simpler design, convenient access to various diagnostics, distinguishable phases of current sheath dynamics and high neutron yield for same driver energy [1].

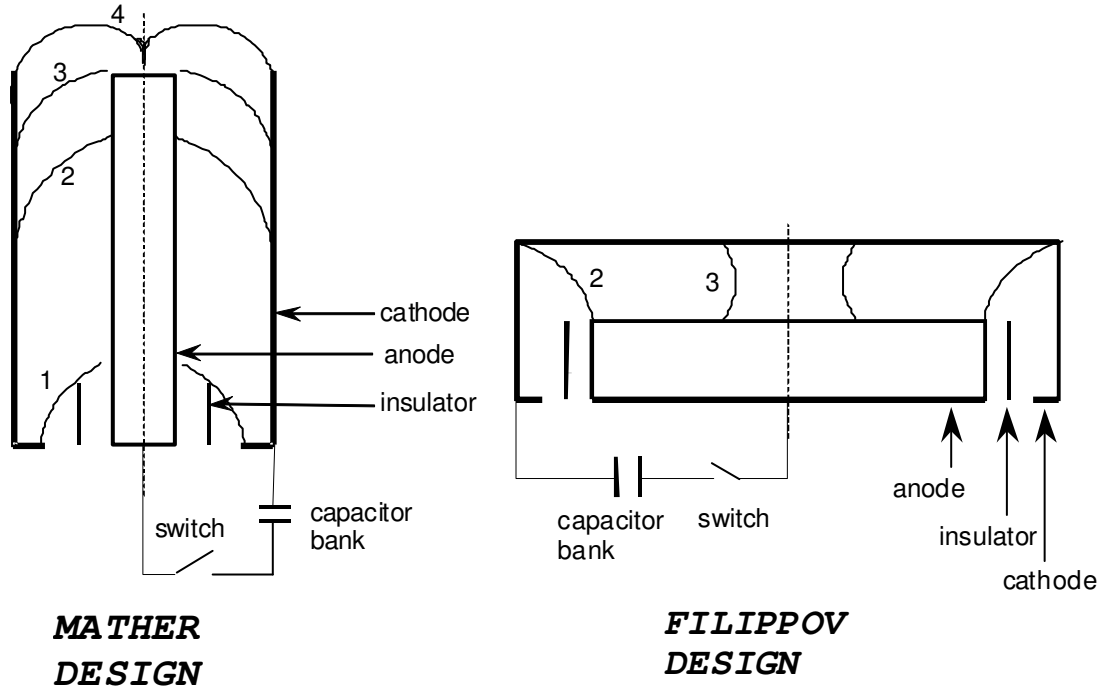


Fig. 2.1 Schematic of Mather and Filippov type plasma focus devices.

Irrespective of configuration, typical plasma parameters over a wide energy range of plasma focus devices are:

- ◆ Electron densities in the range: $5 \times 10^{24} - 10^{26} \text{ m}^{-3}$
- ◆ Electron temperatures in the range: 200eV–2keV
- ◆ Ion temperatures in the range: 300eV–1.5keV
- ◆ Current sheath velocity in the axial phase: $\sim 1 \times 10^5 \text{ ms}^{-1}$
- ◆ Current sheath velocity in the radial compression phase: $\sim 2.5 \times 10^5 \text{ ms}^{-1}$ (typically twice that of the axial speed)

A schematic diagram of the ‘Mather’ version of the DPF, illustrating the operating principle and current sheath dynamics of various discharge phases, is depicted in Fig. 2.2. It is composed of two coaxial cylindrical electrodes, closed and electrically insulated at one end and open at the other end. The insulator provides a voltage standoff between the electrodes at the breach and provides the site for the initiation of the gas breakdown that later results in plasma sheath formation. The electrode assembly is contained in a vacuum chamber filled with the desired gas (*e.g.* H₂, D₂, Ne, Ar or admixture), at a pressure typically ranging from 0.5 to 10mbar, depending on the gas used. The central electrode (*i.e.* anode) of this device is connected to the high voltage terminal of an energy storage capacitor through a fast high current switch and the outer electrode (*i.e.* cathode) is grounded. The electrode assembly is housed inside a metallic vacuum chamber, which is filled with a working gas at an appropriate pressure.

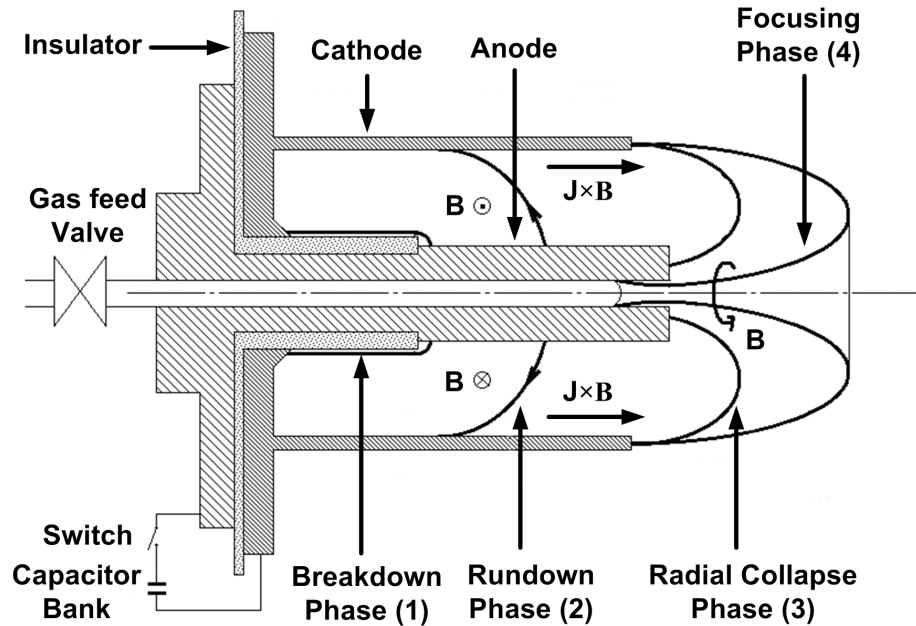


Fig. 2.2 Current sheath dynamics under various phases of plasma focus operation.

When the electrical energy stored in the capacitor bank is rapidly transferred to the electrodes by means of a fast switch; an electrical over-volt discharge rapidly develops into a sheath of plasma,

which evolves axially along the electrode assembly under the effect of $\vec{J} \times \vec{B}$ force. Due to this Lorentz force action, the conducting plasma sheath accelerates towards the open end of electrode assembly *i.e.* from position 1 through position 2 shown in Fig. 2.2. In the final phase (*i.e.* from position 3 to position 4), the sheath collapses on axis with a zippering action forming the pinch plasma column. This zippering action compresses the pinched plasma to a high density and temperature. The pinch has duration of a few tens of nanoseconds, and coincides temporally with a sudden, sharp drop in the total current signal, caused by a decrease in plasma conductivity due to strong confinement. The focus then disrupts due to the growth of the $m = 0$ mode instabilities in the pinch, or from radiative collapse producing intense multiple radiations at the same time.

2.3 The Discharge Phases

Microscopically, the gross plasma dynamics of hot and dense plasma production during the plasma focus device operation is divided into three distinct phases: the breakdown phase, the axial acceleration phase (or axial rundown phase) and the radial phase (or radial collapse phase). Each of these phases is described in detail in following sections.

2.3.1 Breakdown Phase

As mentioned above, when the high voltage pulse is applied between the electrodes of the DPF device filled with the working gas, at an appropriate pressure (usually at few mbar), an azimuthally symmetric electrical discharge is initiated. The seed electrons present between the electrodes in the vacuum chamber are accelerated by the electric field, strongly enhanced by the presence of the insulator sleeve and by the cathode edge. Once the ionization energy of the background gas is reached, multiple ionization avalanches causes the number of charged particles to grow exponentially as predicted by the Townsend law for electrical gas breakdown; the ionized gas develops in a plasma

sheath near the surface of the insulator sleeve. The macroscopic flow of charges connects the electrodes providing a low resistance path for the current to flow from the capacitor bank to the ground of the system [54]. It is remarkable to note that according to Paschen's law [55] the value of the static breakdown voltage in any DPF operating in the range of 1 to 10mbar Deuterium is lower than 1kV (for DC voltage). This value is one order of magnitude smaller than the normally used capacitor bank voltage, which is typically in the range of 10 to 60kV. Thus, the initial breakdown in plasma focus device is always an over-voltage phenomenon, which is further preceded by an electrical breakdown phase that generates the initial plasma through which the discharge current can flow.

The development of the high current discharge in a DPF depends substantially on the initial gas conditions [56, 57], the parameters of the electrodes and insulator [58], the polarity [59] and the initial rate of current rise. Several experimental observations were made to study this phase using different diagnostics like image converter camera, magnetic probe, Rogowski coil etc. Under optimized conditions a sliding discharge develops along the insulator. In addition to the primary sliding discharge, filamentary radial discharges are often observed especially when the pressure is above the optimum [60]. Conditions that control the optimum axis-symmetric breakdown include correct insulator material and electrode-insulator configuration, including the use of a knife-edge cathode [58]. Mather [2] and Kies [61] independently observed that a few initial discharges were always required to form a uniform current sheath. They observed that the insulator surface micro-cracks are filled with evaporated anode metal vapors on the insulator surface during these conditioning discharges. The metal vapor increases the surface conductivity of the insulator and helps in the formation of a uniform current sheath. Once the breakdown has completed, current flows axis-symmetrically from the anode to the cathode across the insulator surface, and within a time of 100-300ns, due to the Lorentz force, the current sheath lifts off from the surface of the insulator sleeve, setting the stage for the axial rundown phase.

In the recent years different computational models have been developed for simulating the complex phenomena occurring in the ignition/ breakdown phase of the plasma focus device operation [62, 63].

2.3.2 Axial Acceleration Phase

After the arrival of the current sheath at the inner surface of the outer electrode, it is accelerated axially along the Z-axis towards the open end of the electrodes by its own $\vec{J} \times \vec{B}$ force. The radial component of $\vec{J} \times \vec{B}$ pushes the current sheath towards the outer electrode. Since the magnetic pressure P has a radial dependence, as shown below:

$$P = \frac{B_\theta^2}{2\mu_0} = \frac{\mu_0 I^2}{8\pi r^2} \quad (2.1)$$

The axial component of the force is stronger near the anode due to the $1/r$ dependence of both the current density J_r and the magnetic field B_θ . This leads to a higher velocity of the current sheath near the surface of the central electrode than near the outer electrode and hence the current sheath becomes more canted as it accelerates axially. The current sheath thus has a parabolic shape during the axial phase while maintaining its axis-symmetric character [64].

The current sheath in axial acceleration phase could be thought of as a strong ionizing shock wave that is driven by a magnetic piston and ionizes the swept up gas as it propagates. This moving shock and the magnetic piston carry a finite volume of compressed plasma between them [65]. The axial acceleration phase ends when the current sheath reaches the open end of the electrodes and enters into the next phase *i.e.* the radial collapse phase.

In the axial phase an important condition is that the current sheath should arrive at the end of the anode at an instant close to the first maximum of the discharge current, preferably immediately after this moment. This condition is a common requirement for all pinch devices; it represents the

optimum energy transfer from the capacitor bank to the pinched plasma. For each device, optimization studies provided the time difference between the current maximum and the end of the axial acceleration phase for an optimum energy transfer under a given set of operating parameters. The plasma temperature during the rundown phase ranges from 30 to 60eV.

The snowplow model [66] has been proposed to describe the dynamics of the axial rundown phase. This model has been proven to predict reasonably well the current sheath velocity and duration of the phase [67, 68], and also the sheath profile by its 2D version [69]. The fitting of experimental data to the computational model [65, 70] requires two important loss factors to be taken into account in the axial acceleration phase, namely the *current shedding* and the *mass loss*. The *current shedding* (i.e. loss or the leakage of the discharge current) occurs due to the slow moving diffused current layer behind the main current sheath and also due to the current retained at the insulator surface whereas the *mass loss* (i.e. escape of swept mass, carried by the current sheath) results from the pressure gradient in the slanted current sheath [64].

For the observation of axial rundown phase different diagnostic techniques, like magnetic probe measurements, Schlieren imaging, shadowgraphy, interferometric investigations, image converting cameras and other optical detection systems have been employed in the past. The measured axial speed and thickness of the plasma sheath typically lies between $1.5 - 15\text{cm}/\mu\text{s}$ and $1 - 4\text{cm}$ respectively, for the wide energy range of plasma focus devices [71]. Various experimental observations made during the axial acceleration phase show that only a fraction of total discharge current (typically 60 to 70%) takes part in the final focus phase [64, 72, 73].

2.3.3 Radial Phase

At the end of the axial rundown phase, the current sheath sweeps around the end of the inner electrode (usually the anode) and finally collapses due to the radially inward

$\vec{J} \times \vec{B}$ force. This collapse occurs within 50 to 200ns, for medium to high energy DPF devices, depending on the device characteristics. The radial collapse velocities range from 7 to 60cm/ μ s, depending on the geometry of the electrodes, initial gas pressure, current sheath structure and electrical characteristics of the device. During the radial phase, the rapid change of inductance results in an induced electric field in the plasma column. Since the discharge current for this phase is almost constant, the voltage induced V_d (across plasma diode) is given by:

$$V_d = I \frac{dL}{dt} \quad (2.2)$$

Here, I is the peak discharge current and dL/dt is the rate of change of inductance.

This phase is important due to the attainment of extremely high energy density, transient character, and the intense multiple radiation emission including nuclear fusion products when operated with Deuterium or Deuterium-Tritium mixtures. Based on the reported experimental data, the radial phase can be subdivided into four sub-phases, namely, compression phase, quiescent phase, unstable phase and decay phase. In the following sub-sections these sub-phases are discussed in detail.

2.3.3.1. Compression Phase

In the final phase of the compression, the pinch plays a crucial role in the plasma focus. The radial phase starts with the rapid collapse of the azimuthally symmetrical but non-cylindrical, funnel-shaped plasma sheath towards the axis under the influence of the inward $\vec{J} \times \vec{B}$ force. This radial implosion ends when the plasma column reaches the minimum radius ($r = r_{min}$) with the plasma density at its maximum value ($\sim 10^{19} \text{ cm}^{-3}$). This instant is usually taken to represent $t = 0$ for the PF discharge.

Shock heating is the main heating mechanism inside the plasma before the front of the current sheath meets at the Z-axis; for this heating process, the $\sim 300\text{eV}$ ions play a much more important role than the $\sim 100\text{eV}$ electrons [74]. After the plasma structure is transformed into a plasma column and the

shock wave (reflected on itself at the axis) meets the magnetic piston, Joule heating becomes the main heating mechanism [75] for the central part of the plasma column, which does not yet feel the effect of the piston. For the formation of the final focus, the plasma column is compressed adiabatically. The magnetic field penetrates the plasma column rapidly at the end of this phase, associated with a sharp increase in the plasma anomalous resistivity. This results in an increase in the total resistance of the system [76]. The sharp voltage spike and the dip in the current (related to the large increase in the plasma column impedance) which are observed for a typical plasma focus discharge, as presented in Fig. 2.3, are enhanced due to the anomalous resistive effects [75].

Spectroscopy [77], interferometry [78], laser scattering [75] and some other diagnostic tools can be employed to determine the maximum value of the electron density and the electron and ion temperatures. The final electron temperature can reach 1 – 2keV [79].

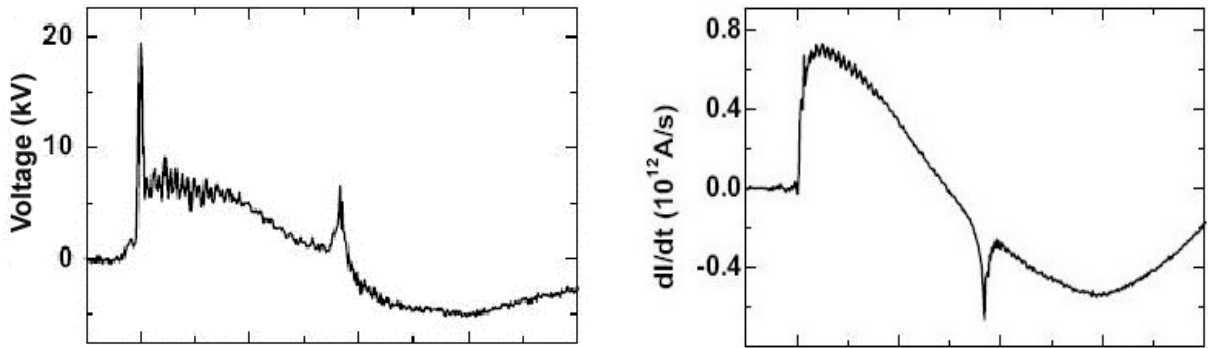


Fig. 2.3 Typical oscillogram of voltage and current derivative signals in PF discharge.

According to the Bennett equilibrium that exist during the pinch phase, $I^2 \approx 3.2 \times 10^{12} NT$; the final temperature depends on the current I and the linear density N , following the relation $T \sim I^2/N$. This relation indicates that: the lower the linear density, the higher the temperature. Also the electron temperature is independent of the minimum radius r_{min} of the pinched plasma column.

At the end of the compression, the pinch column is formed, which then stagnates for a brief period of time. The diameter and length of the compressed plasma column are typically about 1/10 of

the anode radius, respectively, and thus depends on the dimensions of the anode [33]. The plasma density at this stage is $\sim 10^{19} \text{cm}^{-3}$.

Based on the snowplow equation, computational models can be developed for the radial phase. A zero radius column is obtained when these models are applied to the radial compression (which is clearly un-physical). To overcome this, either the introduction of a retarding kinetic pressure term, or the use of criterion for the minimum radius as the Larmor radius were used for different models.

The slug model, originating in shock wave theory, is a major improvement in the plasma focus modeling. The plasma sheath is considered as having a finite thickness, and the result is a non-zero final radius for the pinch column; unfortunately, this model gives non-realistic results towards the end of the compression phase. Lee [80] showed that all these methods are not energy-consistent and should be replaced by an energy-balance model. This energy-balance criterion provides the correct end-point for the implosion trajectory, and gives the correct quasi-equilibrium radius. A complete energy-consistent trajectory is obtained when the energy-balance criterion is combined with the slug model [81]. In addition, the computed pinch length and the minimum radius agree well with experimental measurements [33].

2.3.3.2. Quiescent Phase

After stagnation, the quiescent phase indicates the beginning of the expansion of the plasma column in the axial as well as in the radial directions. The rate of expansion in the radial direction is hindered by the confining magnetic pressure but the plasma expands unhindered in the axial direction resulting in the so-called “fountain”-like geometry of the current sheath. Thus, an axial shock front is formed. The sharp change in the plasma inductance (dL/dt), which started during the compression phase, induces an electric field in the plasma column. This electric field accelerates the ions and the electrons in opposite directions. The relative drift velocity between the electrons and ions increases and

approaches the electron thermal velocity, leading to formation of a ‘Plasma Diode’. At this stage, micro instabilities like electron-cyclotron and various forms of beam-plasma instabilities start to grow. The plasma column is magnetically confined, and the temperature of the plasmas further increases by ohmic heating. In addition, the magnetically confined column is hydro-magnetically unstable to sausage ($m=0$) and kink ($m=1$) instability modes.

The pinch lifetime t_p can be defined, based on these observations, as the duration between the first plasma compression (as seen on the current derivative signal) and the instant of the onset of the $m=0$ instability. This pinch lifetime is many times larger than the radial Alfvén transient time of the pinch column [82]. Among other reasons behind this, the strong axial flow, the two-dimensional curvature of the current flow, the $\vec{B} \times \nabla \vec{B}$ drift, and the influence of the self-generated axial magnetic field are also mentioned in the reported literature [82].

2.3.3.3. Unstable Phase

During the plasma focus evolution, the unstable phase is the richest in associated phenomena like $D-D$ reaction products (neutrons, protons), hard and soft X-ray emission, fast deuterons/ions and electrons. Fig. 2.4 shows the schematic of the emission of radiation (photons and particles) from the plasma focus device after the focusing event.

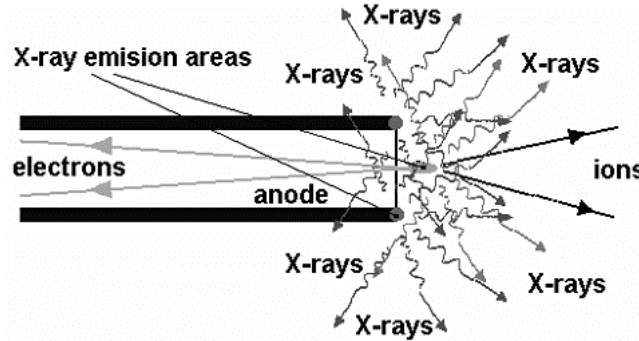


Fig. 2.4 Particle and photon emission from a plasma focus device; the X-ray photons are emitted from the plasma region and from the anode rim.

Due to the growth of $m=0$ instabilities, the induced electric field is enhanced, and this accelerates the electrons towards the inner electrode (anode) and the ions in the opposite direction. At the same time, an axial ionization wave-front has been reported [75]. Using time-resolved interferometry, the measured peak velocity of this front in Deuterium is about $120\text{cm}/\mu\text{s}$, confirming that the ionization is caused by the fast deuterons [75, 83]. This ionizing front overtakes the axial shock front (formed due to the continued axial expansion of the plasma column during the radial phase). The beginning of this ionizing front coincides with the beginning of the hard X-ray and neutron pulses. The ionization front develops later into a bubble-like structure with several density gradients [75, 84]. The first gradient is attributed to the ionization front that separates the ambient gas from the ionized structure.

Due to the bombardment of the anode by runaway electrons, especially for solid anodes, a fairly large amount of impurities are injected into the plasma column. This decreases the Pease-Braginskii current, causing the plasma column to implode near the central electrode [85]. The implosion proceeds sequentially along the plasma column. This breaking up corresponds to the second density gradient observed in interferometric holograms [75]. This density gradient moves away from the anode at a typical speed of $20\text{cm}/\mu\text{s}$, which is much slower than the ionization front. The disruption of the plasma column continues until it is broken completely. The electron drift velocity can be estimated from the current density and the plasma density. For a typical set of device and plasma pinch parameters, the value obtained is $\sim 10^7\text{m/s}$. This is higher than the electron thermal velocity of $\sim 10^6\text{m/s}$ estimated from the electron temperature, the result being an important energy injection into the plasma, which leads to a strong plasma heating. The electron temperature increases up to $4\text{--}5\text{keV}$ as indicated by the large amount of measured Bremsstrahlung radiation [82].

2.3.3.4. Decay Phase

The decay phase is the last phase of the radial collapse, as well as the last stage of interest for the plasma focus dynamics. During the decay phase, a very large, hot and thin plasma cloud is formed due to the complete breaking of the plasma column. A large amount of Bremsstrahlung radiation is emitted during this phase. The soft X-ray emission rises abruptly during the decay. At this stage, the neutron pulse, which started at the beginning of the unstable phase, reaches its peak value.

Thus the plasma column breaks up and decays. The high electric field induced during this period gives rise to the emission of beams of energetic ions and electrons. For a Mather-type device with an anode radius of 1cm, the implosion time is ~100ns while the lifetime of the pinched plasma column is ~20ns [33]. Strong emission of X-rays and neutrons also occur during this phase.

2.4 Plasma Instabilities

The current produces a magnetic field around the plasma column which thermally insulates the hot plasma from the wall and exerts an inward radial force that confines the plasma. The plasma instabilities are major obstacle in the way of progress towards the confining of the plasma for controlled thermonuclear fusion. These instabilities impose limitations on the amount of current and pressure that can be confined by the magnetic field.

Plasma described by MHD (Magnetohydrodynamic) equations admits a gross number of instabilities. Especially, a cylindrical plasma confined by a magnetic field produced by a current at the surface of the pinch plasma will develop a radial perturbation δr which is of the form [86, 87]:

$$\delta r = \xi_0 e^{i(m\theta + Kz - \omega t)} \quad (2.3)$$

where ξ_0 is the amplitude of displacement and K is the wave number. The coefficient m is the poloidal mode number and it determines the azimuthal periodicity of δr . When $m=0$, the perturbations

are independent of azimuthal coordinate (*i.e.* sausage instability) where as, when $m=1$, the pinch is perturbed helically (*i.e.* kink instability).

In the following subsections, a brief introduction is given for major types of instabilities observed in the plasma focus device.

2.4.1 Rayleigh-Taylor Instability

Rayleigh-Taylor instability occurs whenever a dense and incompressible fluid is supported by another fluid of a lower density in an accelerating field. In plasma, Rayleigh-Taylor instability can occur because the magnetic field acts as a light fluid supporting the plasma – a heavy fluid. The main contribution to the accelerating field is the one due to the magnetic field lines curvature, $g_{eff} \sim v_{Ti}^2/R$, where v_{Ti} is the ion thermal velocity, and R is the boundary radius of curvature [88].

In a magnetically-confined plasma, the interface between the plasma and the magnetic field is found to be unstable during the radial compression [88]. Therefore, in the plasma focus device, at the end of the compression phase the boundary of the plasma column is fluted due to this instability.

Peacock *et al* [89] with the help of the interferometric studies confirmed the presence of the Rayleigh-Taylor instability in the compression phase of the plasma focus. The growth time t_i of the Rayleigh-Taylor (RT) instability in accelerated dynamic pinches is given by:

$$t_i = \sqrt{\frac{2\pi\lambda_p}{g}} \quad (2.4)$$

Here, g is the sheath acceleration and λ_p is the dominant wavelength of perturbation [90]. In this relation, it may be noted that instability growth time varies inversely with sheath acceleration. Hence, this type of instability dampens when the radial velocity of current sheath is decreased by any means.

2.4.2 Sausage ($m=0$) Instability

A major problem encountered in the early fusion experiments was that the plasma column would spontaneously pinch itself off due to magneto-hydro-dynamic instabilities. The growth of $m=0$ instability (*sausage instability*) is due to a poloidally symmetric radial perturbation which constricts the plasma column slightly in the place where magnetic pressure exceeds the kinetic pressure (∇P) of the plasma [91], as shown in Fig. 2.5.

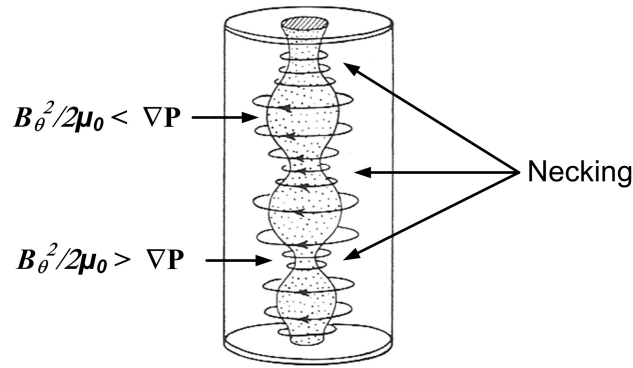


Fig. 2.5 Schematic presentation of $m=0$ instability.

The longitudinal current density (J_z) flowing through a smaller cross-sectional area of the constricted part of the pinch produces a stronger magnetic field (B_θ), whereas the field at other parts remains unchanged. The stronger field around each constricted region exerts a greater inward force on the plasma column so that the column constricts further. The subsequent rapidly changing magnetic fields at each constriction induces a large longitudinal electric field which accelerates the ions/deuterons within the plasma up to the energies of hundreds of keV, leading to a burst of multiple radiation emission. The sausage instabilities have been reported in plasma focus devices by several researchers [91-93].

The sausage instability can be stabilized by permeating the plasma column with a longitudinal magnetic field comparable in strength to the poloidal field around the column. One way to consider

this stabilization mechanism is by noting that any constriction of the plasma column requires energy to compress the longitudinal magnetic field within the plasma. A magnetic field in a conducting fluid may be thought of as exerting pressure perpendicular to the field lines and tension along the field lines. Compressing the longitudinal field within the plasma then results in a restoring force due to the increasing magnetic pressure.

2.4.3 Kink ($m=1$) Instability

Another type of MHD instability that has been observed in the plasma focus is the $m=1$ (*kink*) instability, in which the plasma column twists into a helical shape like a corkscrew, as shown in Fig. 2.6.

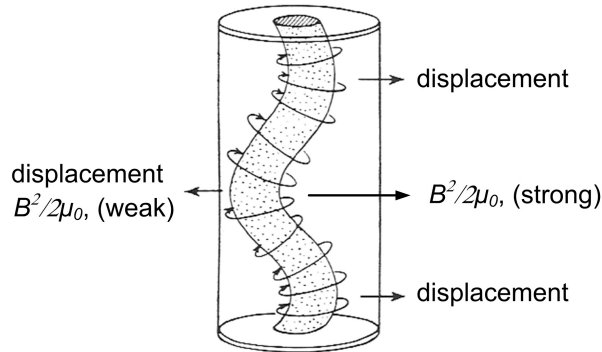


Fig. 2.6 Schematic presentation of $m=1$ instability.

The mechanism for this kink instability becomes clear by noting that when any part of the plasma column is bent, the poloidal field on the inner edge of the bend becomes stronger than the field at the outer edge. The resulting magnetic pressure then pushes the column to bend it further. Rawat *et al* [92] reported the observation of kink instability using soft X-ray pinhole imaging.

In order to stabilize this $m=1$ kink mode, the longitudinal magnetic field must be made strong enough and the plasma column wide enough so that no part of the magnetic field between the plasma and the wall closes upon itself once along the length of the plasma column. This is known as the

Kruskal–Shafranov criterion [88]. Roughly speaking, the tension associated with the longitudinal magnetic field prevents the plasma column from bending into a kink.

2.4.4 Microinstabilities and Turbulence

The existence of microinstabilities and turbulence in the plasma focus was reported by Bernard *et al* [75, 93]. They used the light scattering diagnostic techniques to study the magnitude of density fluctuations as a function of direction and observed that the excitation of waves along the current direction was more intense (1.5 times) than along the direction perpendicular to it. This variation of the scattered light indicates not only the existence of the microinstabilities, but also the correlation of the microinstabilities with the drift velocity between the electrons and ions due to the current flow along the electrode axis.

The observation of the anomalous resistivity of the pinched plasma, non-thermal radiation in the microwave range and bursts of high energy ions and electrons also indicates the presence of microinstabilities and turbulence in the dense plasma focus [93]. Schonbach *et al* [94] observed reproducible microwave signals in the frequency range of 2.6 to 18GHz. Gerdin *et al* [95] have also observed broadband microwave emission with an upper limit correlated inversely with the pinch radius.

During the radial phase, the change in the plasma inductance results in an induced electric field along the pinch. At the end of the quiescent phase, the growth of the $m=0$ instability enhances the induced electric field locally. The emission of electrons starts as soon as the induced electric field is higher than 300kV/cm, the estimated value for the critical field strength for run-away electrons [82]. These runaway electrons move towards the anode and produce hard X-rays through electron bombardment of anode material. The electron energy estimated from hard X-ray energies to be in the range 50 to 500keV [96]. Hence, the ordered velocity of the beams, estimated from the hard

X-ray energy, exceeds the thermal velocity of the plasma extracted from the electron temperature. Under this condition, it has been shown theoretically that various types of beam-plasma interactions occur, giving rise to a number of other instabilities [97].

2.5 Electrical Behavior and Energy Transfer Process

As a first approximation, the electrical behavior of the device can be easily understood by considering an equivalent circuit for the plasma focus device, as depicted in Fig. 2.7.

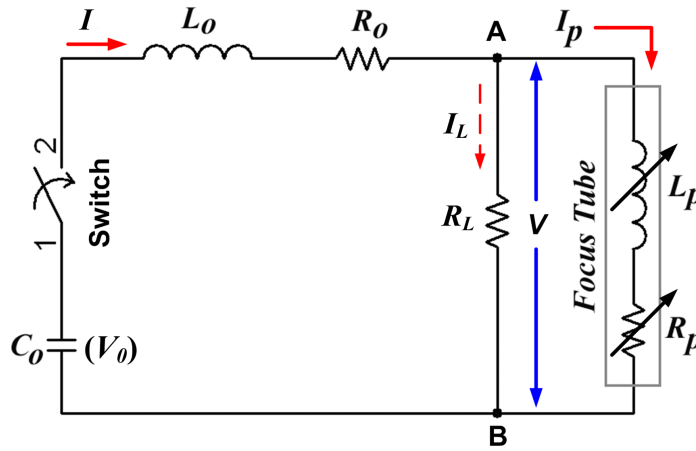


Fig. 2.7 Equivalent circuit for the plasma focus.

This electrical circuit consists of a capacitor bank C_0 , charged at a voltage V_0 , an equivalent series inductance L_0 and an equivalent series resistance R_0 (correspondingly contributed by the capacitors, transmission line, connections and switches). The plasma is considered as having an inductance and resistance, L_p and R_p , with temporal dependence. Here, I is the discharge current, I_p is the current flowing through the plasma ($\approx f_c \times I$, here f_c is current shedding factor) and I_L is the leakage current in the plasma tube, representing the current that remains along the insulator surface and does not flow through the accelerated plasma sheath.

According to Krichoffs laws, the focus tube potential difference (i.e. between point A and B in the electrical schematic shown in Fig. 2.7) can be written as [98]:

$$V = V_0 - \frac{1}{C_0} \int_0^t Idt - R_0 I - \frac{d}{dt} (L_0 I) = R_p I_p + \frac{d}{dt} (L_p I_p) \quad (2.5)$$

The focus tube is considered as a standard cylindrical coaxial conduction tube with anode height z_0 , anode radius a and cathode radius b . The focus tube inductance L_p associated with the plasma in the axial and radial phase is given as:

$$\text{Axial Phase :} \quad L_{p1} = \frac{\mu_0}{2\pi} \left(z_0 \ln \left(\frac{b}{a} \right) \right) \quad (2.6)$$

$$\text{Radial Phase:} \quad L_{p2} = \frac{\mu_0}{2\pi} \left(z_p \ln \left(\frac{b}{r_p} \right) \right) \quad (2.7)$$

Here, z_p and r_p are the length and the radius of the pinch plasma column.

Hence, the total focus tube inductance is $L_p \approx L_{p1} + L_{p2}$ and the voltage that is induced across PF due to inductive component alone may be given as:

$$V = \frac{\mu_0}{2\pi} \left(z_0 \ln \left(\frac{b}{a} \right) + z_p \ln \left(\frac{b}{r_p} \right) \right) \frac{dI_p}{dt} + \frac{\mu_0}{2\pi} \left(\ln \left(\frac{b}{r_p} \right) \frac{dz_p}{dt} - \left(\frac{z_p}{r_p} \right) \frac{dr_p}{dt} \right) I_p \quad (2.8)$$

The resistance associated with the plasma column in the radial phase can be written as:

$$R_p = \frac{z_p}{\sigma \pi r_p^2} \quad (2.9)$$

Upon rearranging the equation 2.5 and performing the substitutions, the rate of change of current for the radial phase may be given as [98]:

$$\frac{dI}{dt} = \frac{V_0 - \frac{1}{C_0} \int_0^t Idt - (R_0 + f_c R_p) I + \frac{\mu_0}{2\pi} f_c I \left(\frac{z_p}{r_p} \frac{dr_p}{dt} - \ln \left(\frac{b}{r_p} \right) \frac{dz_p}{dt} \right)}{L_0 + \frac{\mu_0}{2\pi} f_c \left(z_0 \ln \left(\frac{b}{a} \right) + z_p \ln \left(\frac{b}{r_p} \right) \right)} \quad (2.10)$$

As can be observed from above expression (2.10), the large change in the plasma impedance in the final stage of the radial compression (principally in the inductance) produces the ‘dip’ observed in the dI/dt signal and the peak observed in the voltage signal (as shown in Fig. 2.3). These features in the electrical signals are considered to be evidence of an efficient pinch (or focus formation) in the plasma focus device. The energy that is transferred into the plasma focus basically comes from the energy stored in the capacitor bank, which is given as:

$$E = \frac{1}{2} C_0 V_0^2 \quad (2.11)$$

This energy goes into plasma focus tube in three ways [98] –

$$(i) \text{ Inductive energy} \quad W = \int_t I_p L_p \frac{dI_p}{dt} dt \quad (2.12)$$

$$(ii) \text{ Piston work:} \quad W = \int_t I_p^2 \frac{dL_p}{dt} dt \quad (2.13)$$

$$(iii) \text{ Joule heating:} \quad Q_{joule} = \int_t I_p^2 R_p dt \quad (2.14)$$

The inductive energy represents the change of the energy stored in L_p by the change of the discharge current. Here, it is important to note that a part of this input energy is stored in the focus tube inductance as $\approx 1/2 L_p I_p^2$. Hence, the energy stored in the focus tube inductance is determined by both, the total inductance L_p of focus tube and the net current I_p flowing through the plasma. It is remarkable to note that since this energy is not dissipative; therefore when the current drops, it also acts as an energy source for pinch [98]. The piston work is the main path to transfer energy into the plasma focus tube. Half of the piston work is used to increase L_p by pushing the conductive plasma sheath, consequently it increases the energy stored in L_p . The other half is converted into plasma energy. The joule heating is induced by the plasma resistance R_p and the flow of current through it. It directly heats

the plasma. The mentioned energy transfer process as depicted by flow chart shown in Fig. 2.8, and may be analyzed and derived using circuit equations illustrated below.

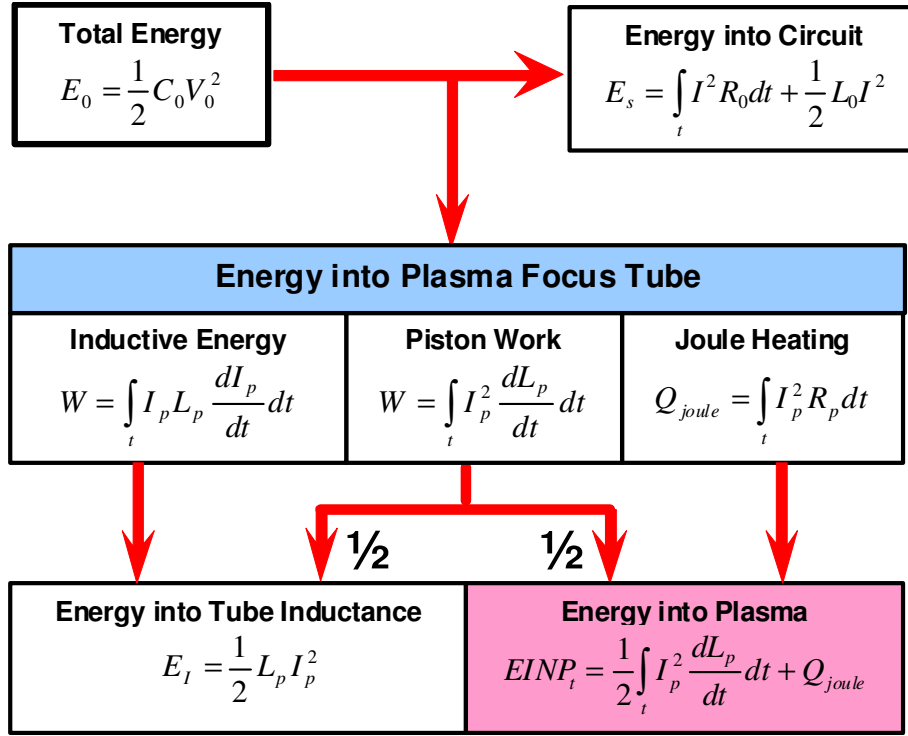


Fig. 2.8 Energy transfer process in plasma focus

Since, V is the voltage appearing across the focus tube and I_p is current flowing through the plasma the power input to the tube is given by [98]:

$$P_{total} = I_p V = I_p L_p \frac{dI_p}{dt} + I_p^2 \left(R_p + \frac{dL_p}{dt} \right) \quad (2.15)$$

The power transferred into plasma is:

$$P_{plasma} = P_{total} - \frac{d}{dt} \left(\frac{1}{2} L_p I_p^2 \right) = I_p^2 \left(R_p + \frac{1}{2} \frac{dL_p}{dt} \right) \quad (2.16)$$

The first term in this equation is induced apparently by the joule heating which heats the plasma directly. The second term is the power contributing to the dynamics of the system (kinetic energy, and also the thermal energy, ionization energy, etc.). The term $\frac{1}{2} \frac{dL_p}{dt}$ can be seen as a "dynamic resistance" of the plasma [75, 93]. Therefore, the total energy transferred into plasma is [98]:

$$EINP = \int_0^t I_p^2 \left(R_p + \frac{1}{2} \frac{dL_p}{dt} \right) dt \quad (2.17)$$

Hence, the net energy transferred into plasma ($EINP$) is mainly contributed by joule heating and half of the piston work. Further, this transferred energy goes into the final plasma column in the forms of: kinetic energy (E_k) internal energy (U) ionization and excitation energies (E_i) and radiation energy (Q_{rad}) [98].

Based on the analysis of the ascribed energy transfer processes, it is conclusively deduced that for improvising the energy transfer efficiency: (i) the circuit resistance R_0 and inductance L_0 must be minimum for pumping larger discharge current I and hence the current I_p flowing through the plasma (ii) the focus tube must be designed with minimal inductance L_p (by preferably using a short anode) to allow more energy to go into the plasma [98].

2.6 Radiation Products from Dense Plasma Focus

As mentioned before, the plasma focus is a powerful and compact source of energetic radiations producing fast neutrons [2-6], intense X-rays [7-11], energetic ions [12-14] and electrons [15,16]. General properties of these radiation products such as, emission times, energy spectra, and their correlation with plasma focus dynamics were first studied by groups lead by Mather [2,99,100] and Bernstein [101,102]. In the following sections, a brief description of the multiple radiation

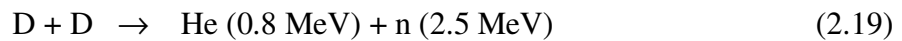
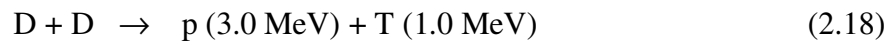
products and their emission characteristics is provided. Particular attention is given to the emission characteristics of neutrons and X-rays.

2.6.1 Neutrons

Neutron emission has been observed from sub-kilojoule to mega-joule energy range plasma focus devices with Deuterium or a Deuterium-Tritium mixture as filling gas. The general nature of emission seems to remain nearly same at all energies. A small fraction, approximately 4% of the total amount of Deuterium present in the focus zone, contributes 100 to 300keV deuterons [103, 104]. Tiseanu *et al* [105] concluded that in a small energy plasma focus device, the medium energy deuterons (50-100keV) with a relatively broad angular distribution are responsible for most of the neutron yield. Space resolved neutron spectroscopy shows that $\approx 80\%$ of the neutrons are produced by $\approx 100\text{keV}$ deuterons [106]. The neutron emission in high energy banks is observed distinctly in two phases [107] of duration 100-200ns each, one peaking during compression phase and the other after focus break-up. But in small and medium energy devices, the emission generally starts from the end of collapse phase and maximizes during the turbulent phase. It is likely that due to short lifetime of focus in low energy devices, the two neutron pulses merge to have a single pulse.

2.6.1.1 Neutron Production Mechanism

Using Deuterium gas, PF devices produce fusion D–D reactions, generating a fast-neutron pulse ($\sim 2.5\text{MeV}$, of tens to hundreds of nanoseconds duration) and protons (leaving behind ^3He and T). The basic Deuterium-Deuterium (D-D) fusion reactions are:



with approximately fifty percent probability for each reaction.

While the physics of the plasma focus devices has been extensively studied over several decades, the neutron production mechanism is still an important question that is being investigated on different plasma focus devices operating over wide ranges of energies with different operating conditions. The principal mechanisms for neutron production in plasma focus devices have been characterized as thermonuclear and non-thermonuclear. In the earliest investigations, the neutrons produced were assumed to be of thermonuclear origin [1, 2]. In the thermonuclear mechanism, D–D reactions are produced by the thermal collisions between deuterons in the bulk of the pinch plasma column of focus device and therefore an isotropic emission of 2.45MeV neutrons is expected. However, ever since the advent of plasma focus devices, the neutron flux has been observed to be anisotropic, with more emission in the axial direction as compared to radial directions [102, 108]. Moreover, the average neutron energy in the axial direction is found to be greater than the typical average energy of isotropic thermonuclear neutrons (2.45MeV), which strongly suggests the presence of other neutron emission mechanisms in addition to the purely thermal one.

In 1974, Forrest and Peacock [109] performed ion and electron temperature measurements through ruby laser scattering and found that the temperature of Deuterium ions (T_i) $\leq 1\text{keV}$ was too low for a purely thermonuclear interpretation of the neutron production. These findings pointed to the existence of non-thermal processes in which the D–D reactions are produced by accelerated (suprathermal) deuterons, colliding with the thermal deuterons in the plasma bulk and the neutral gas atoms outside the pinched plasma column (beam-target effect) [75] attributing anisotropy in the neutron yield. Experimentally, anisotropy measurements are made by using two detectors simultaneously, positioned in the axial (0°) and radial (90°) (or rearward (180°)) directions. The ratio of yields measured (Y_{0°/Y_{90°) in the respective directions is defined as the fluence anisotropy. This factor has typically been reported to be ~ 1.2 to 3, for a wide energy range of plasma focus devices operating in the kJ [18, 110, 111] to MJ [31] range.

The conventional explanation of the ion acceleration involves production of strong localized electric fields which result from the destruction of the plasma column due to the development of magneto-hydrodynamic instabilities, especially, the $m=0$ sausage instability [91,112,113]. Bernstein and Hai [114] postulated that the high energy deuterons are generated by an acceleration mechanism, in which the ion velocities are randomly oriented because of curved motion in the magnetic field of the pinch. In the last few decades significant efforts have been led by various research laboratories [12, 115-118] to investigate the velocity distribution of deuterons producing neutrons and different models have been deduced e.g. the particle trap model [79], converging ion model [108], crossed-field acceleration model [119] and gyrating particle model [120] but till date, the entire mechanism of the ion acceleration and neutron production has still not been fully understood and investigations at various fronts are ongoing [20,121-123]. However, experimentally it has been widely evidenced that the major contribution to the neutron yield comes from medium (50-100keV) [105,106] and low energy deuterons (<50keV) [124].

2.6.1.2 Neutron Yield Scaling

Based on more than forty years of experimental results of dimensionally optimized Deuterium operated plasma focus devices (with a wide range of energies and currents from 1kJ to 1MJ and 100kA to 1MA [17-32]), the total neutron yield Y follows empirical scaling relationships with capacitor bank energy E , peak discharge current delivered by capacitor bank I , and the current flowing through the plasma during focus called the pinch current I_p .

The observed scaling of neutron yield with energy E stored in the driver, typically follows the relation [54, 125]:

$$Y = 10^6 \times E^{(1.5-2.5)} \quad (E \text{ in kJ}) \quad (2.20)$$

The variation in the empirical energy scaling proportionality constant (in the range of 1.5 to 2.5) depends mainly on the operating parameters of the PF device, which should be optimized in terms of external inductances, charging voltage, electrode geometry, filling pressure, gas purity *etc.*

According to the Bennett relation [126], plasma temperature depends on the current I and the density n , hence from the pressure balance pinch relation $B^2/8\pi = nKT$ (with B the magnetic field, n the plasma density, K the Boltzmann constant and T the plasma temperature), the peak current appears to be the most important parameter that affects the total neutron yield. The evidenced neutron yield scaling with peak discharge current in various experiments [54, 125,127,128] may be correlated as:

$$Y = 10 \times I^{(3-4)} \quad (I \text{ in kA}) \quad (2.21)$$

One of the major causes of the observed variation/ uncertainty in the peak discharge current scaling proportionality constant (in the range of 3 to 4) is due to the fact that the actual current that goes into the pinched plasma, differs distinctly from device to device, even in the PF devices of same energy, delivering similar level of peak discharge current (this is due to variation in the snowplow efficiency and electro-dynamical processes of the machines) [129].

Hence, the above mentioned observations indicate that the study of neutron yield empirical scaling with pinch current should be of more relevance for attaining a consistent scaling law. But there are serious constraints in the experimental measurement of pinch current. Lee and Saw [128] have deduced an effective and simple method for the estimation of pinch current by realistically simulating the experiment using the Lee Model [70]. Recently, neutron yield scaling has been thoroughly reviewed by S Lee *et al* [128,129], using the five-phase Lee model (RADPFV5.13) [70], and it has been found that the neutron yield scaling with pinch current follows the relation:

$$Y = 2 \times 10^{11} \times I_p^{4.7} \quad (I_p \text{ in MA}) \quad (2.22)$$

It has been shown by S. Lee *et al* [128,129] that neutron yields obtained for a wide energy range of plasma focus devices, fit more closely with the pinch current scaling relationship, as mentioned above. Hence, it is hereby emphasized that the most relevant parameter for neutron production is the pinch current I_p and not the peak current I delivered by capacitor bank, which is commonly used as a scaling parameter.

It may be noted that neutron yield is enhanced by approximately two orders of magnitude ($\sim 10^2$), when operated with a D-T gas mixture instead of pure Deuterium gas filling because of the higher fusion cross section for the D-T reaction.

Neutron yield scaling from the perspective/ hypothesis of neutron production mechanisms may also be analyzed as mentioned below [130,131].

As it has been commented in the previous section, the two widely accepted main processes in the production of the total neutron yield (Y) in plasma focus device are: thermonuclear fusion and ion beam–target fusion *i.e.* $Y = Y_{th} + Y_{b-t}$, where Y_{th} is the thermonuclear component and Y_{b-t} is the beam–target component, hence scaling laws for the neutron yield is also investigated considering contributions of thermonuclear and beam target component.

The rate equation for thermonuclear production follows [35]:

$$Y_{th} \approx n^2 \langle \sigma v \rangle V_p t_p \quad (2.23)$$

Here, V_p is the volume of the pinched plasma column, t_p its duration (i.e. lifetime of the pinch) and $\langle \sigma v \rangle$ is the fusion cross section which is a function of the mean ion temperature T . In plasma focus device operation these factors have dependence of $t_p \sim 2a$ and $V_p \sim 0.38a^3$ (a is the anode radius) thus the thermonuclear component of the neutron yield will be proportional to a^4 (a^3 from the volume dependence and another factor a from the pinch lifetime dependence). The rate equation can now be interpreted as:

$$Y_{th} \approx n^2 a^4 \langle \sigma v \rangle \quad (2.24)$$

As described earlier, the drive parameter S (i.e. $I/ap^{1/2}$) is constant over the range of devices, (I/a) may be considered to have an almost fixed value (the square root of the pressure, $p^{1/2}$, varies very little i.e. less than 40% in comparison to the variation of I i.e. ten times or more). This gives $Y_{th} \propto I^4$ [35].

But as proposed by Lee and Serban [130], if the value of I/a is increased, the speed will increase causing the temperature (T) to increase with $(I/a)^2$. Increasing T will lead to an increase in a fusion cross section $\langle \sigma v \rangle$ proportional to T^x , with $x \sim 4$, and thus $\langle \sigma v \rangle \sim (I/a)^8$. Though, $Y_{th} \propto \langle \sigma v \rangle$ (volume) (pinch lifetime) therefore,

$$Y_{th} \approx (I/a)^8 (a^3) (a) \approx (I/a)^4 I^4 \quad (2.25)$$

Thus, if (I/a) is a constant $Y_{th} \sim I^4$, else if a is kept constant then $Y_{th} \sim I^8$ as noted from above [82].

The rate equation for phenomenological beam-target yield is written in the form [132, 133]:

$$Y_{b-t} \approx n_b n_i (r_p^2 z_p) (\sigma v_b) t_p \quad (2.26)$$

where, n_b is the number of beam ions per unit plasma volume, n_i is the ambient ion density, r_p is the radius of the plasma pinch with length z_p , σ the cross-section of the D-D fusion reaction, n_b the beam ion speed and t_p is the beam-target interaction time assumed proportional to the confinement time of the plasma column.

Total beam energy is estimated to be proportional to $L_p I_p^2$ [132,133], a measure of the pinch inductance energy, L_p being the focus pinch inductance. Thus the number of beam ions is $N_b \sim L_p I_p^2 / v_b^2$ and n_b is N_b divided by the focus pinch volume. Note that, $L_p \sim z_p \ln(b/r_p)$, that $t_p \sim r_p / v_b$, and that $v_b \sim V_d^{1/2}$ where V_d is the disruption-caused diode voltage [91,133]. Here, b is the cathode radius. It is

reasonable to assume that V_d is proportional to V_{max} , the maximum voltage induced by the current sheath collapsing radially towards the axis. Hence, beam target component should scale as [132,133]:

$$Y_{b-t} \approx C_n n_i I_p^2 z_p^2 \ln(b/r_p) \sigma / V_{max}^{1/2} \quad (2.27)$$

where I_p is the current flowing through the pinch at the start of the slow compression phase; r_p and z_p are the pinch dimensions at end of that phase. C_n is a constant, which in practice is determined experimentally. The D-D cross-section is highly sensitive to the beam energy. It is known from experiments that the ion energy responsible for the beam-target neutrons is in the range 50-150keV [105,106], and for smaller lower-voltage machines the relevant energy could be at 30-60keV [134].

2.6.1.3 Neutron Yield Enhancement and Impurity Effect

Even though, till date, neither the mechanism of neutron production is fully understood nor the predictive models have been that successful but enhancement of neutron yield has been attained experimentally in various ways. Koh *et al* [18] reported a very wide and high pressure operation regime for NX2 plasma focus with a five fold increase in neutron yield with the use of tapered anode geometry. Shyam and Rout [24] have observed that neutron emission from a plasma focus is significantly affected by the electrode material. Electrodes of *Al*, *Ti*, *Cu*, stainless steel and *Cu-W* alloy were used. The electrode's resistance towards erosion was found to be a major factor for high neutron yield. The *Cu-W* alloy electrode provides the least erosion of all the materials that were used. Due to its low erosion, the neutron yield with a *Cu-W* alloy anode was twice that for *Al*.

Bernstein and Hai [114] have shown yield enhancement correlation with anomalous resistivity that causes ohmic heating. Thein *et al* [135] reported the increase in the D-D neutron yield with increase in ion acoustic wave instability amplitude. Neutron emission obtained by injecting a pulsed relativistic electron beam into a dense plasma focus device was studied by Freiwald *et al* [136]. It was observed that neutron yield increased to five times the average value obtained without beam injection.

The influence of a mesh type β -source for pre-ionization on neutron emission from two low energy Mather-type plasma focus devices was reported by Zakaullah *et al* [137]. The neutron yield was found to increase up to 25%, provided the strength of the β -source did not cross a certain critical limit.

S. Ahmad *et al* [138] have reported the effect of pre-ionization induced by depleted Uranium (${}_{92}\text{U}^{238}$) around the insulator sleeve on the neutron emission of (1.8–3.3kJ) plasma focus device. The pre-ionization enhances neutron yield, about $(50 \pm 5) \%$, broadens the neutron emission pressure range and improves shot-to-shot reproducibility of plasma focus operation for neutron emission.

Yap *et al* [139] have reported enhancement in the neutron yield using Deuterium-Argon admixtures. Babazadeh *et al* [140] reported enhancement in the yield by a factor of 1.5 to 3.5 times with different anode geometries using Deuterium-Krypton admixture. Maximum neutron yield was obtained by adding Krypton about 1% and 1.5% with the Deuterium gas.

Zoita *et al* [141] studied the influence of the radiation processes on the plasma evolution during the pinch and post-pinch phases in experiments with Neon-seeded Deuterium discharges carried out on a 28kJ/ 60kV device. They reported dramatic changes in the radiation characteristics of the discharge as well as in the pinch configuration. High aspect ratio (length/diameter >10) pinches emitting large amounts of soft X-rays accompanied by efficient emission of neutrons (and hard X-rays) in the absence of macroscopic instabilities were produced. They also concluded that the line radiation as well as the main continuum radiation in the spectral range of 3-4keV are emitted mainly from the intense bright spots having dimensions of about $50\mu\text{m}$.

2.6.2 X-rays

Plasma focus devices have been demonstrated as potential X-ray sources (when operated with the high-Z gases like Ne, Ar, Kr and Xe, or with H_2 , D_2 and N_2 as well) for various applications such

as lithography ($\sim 0.9\text{-}1.5\text{keV}$) [7-10], microscopy ($\sim 0.25\text{-}2.5\text{keV}$) [142] and micromachining ($\sim 4\text{keV}$) [143].

For a focused plasma at $T_e = 1\text{keV}$, its X-ray continuum is expected to peak at $\lambda_o = 6.2\text{\AA}$, which is in the soft X-ray region. On the other hand, for high-Z focus plasma, the characteristic line emission also falls in the soft X-ray region. For example, for nearly fully ionized Argon or Neon plasma at a suitable high temperature, the K_α line radiation has wavelengths of $4.2\text{\AA}$ and $12.132\text{\AA}$ respectively.

The electromagnetic radiative processes of the plasma focus are: quasi-equilibrium thermal radiation from macroscopic plasma structures and radiation due to electron beams interacting with non-plasma targets (*i.e.* electrodes) and/or with periodic electron density structures [144]. The radiation spectrum of the plasma focus in the X-ray region covers a wide range from below 0.3keV up to over 100keV in a time span ranging from few nanoseconds to few hundred nanoseconds. Hence, on the basis of energy, the X-rays emanating from PF devices can be broadly categorized as soft and hard X-rays.

2.6.2.1 X-ray Emission Processes

A brief qualitative description of various processes of X-ray emission in the plasma focus device is presented below [145]:

Bremsstrahlung (Free-Free) Radiation

Bremsstrahlung radiation, in the classical picture, occurs whenever a charged particle is accelerated or retarded. It is mostly caused by the acceleration of electrons in the Coulomb field of ions. Since the initial and final states are continuous, the spectrum of the emitted radiation is also continuous. In highly ionized high-Z plasma Bremsstrahlung can be a dominant radiation process, with a spectral emission coefficient per unit wavelength peaking at about:

$$\lambda_{\max} = \frac{6200}{KT_e} A^\circ \quad (2.28)$$

The emission decreases rapidly with decreasing wavelength.

Recombination (Free-Bound) Radiation

A free electron can be captured into a bound state of an ion, thereby reducing the ionic charge by unity. The excess energy of the electron may be emitted as a photon, giving free-bound or recombination radiation. The emission spectrum is a continuum for each bound state with a low-frequency cut-off (the recombination limit) corresponding to the minimum energy needed to remove the electron from the bound state. The shape of the spectrum depends on the free-electron energy distribution and on the energy-dependent capture cross section into the bound state.

A process that competes with this radiative recombination is three-body recombination. This occurs when two free electrons simultaneously collide with an ion, one being captured and the other carrying away the excess energy. No radiation is emitted, and so this process, which becomes more probable at higher densities, should in principle be minimized in plasma devices to be used as a continuum X-ray sources. However, because high electron densities are necessary in X-ray sources, this is not possible in practice. Another process that does not result in radiation is dielectronic recombination.

Line (Bound- Bound transition) Radiation

Line radiation occurs due to electronic transition between the discrete or bound energy levels in atoms, molecules or ions. The transition of an atom or ion A of charge Z from a level p to a lower level q can be written as:

$$A_p^Z \rightarrow A_q^Z + \gamma \quad (2.29)$$

When an ion, an atom or a molecule is in the excited state, it will undergo a transition to a lower state or the ground state through spontaneous or stimulated emission. The energy of the emitted photon γ is given by the difference of energies of the initial E_i and the final levels E_f . Thus, the energy of the emitted photon can be written as:

$$h\nu = E_f - E_i \quad (2.30)$$

Since the energy levels of the atom are quantized, these emissions appear as discrete packets of energy, or “lines”. This radiation is also known as characteristic radiation because it represents the characteristic properties of the emitting ion, atom or molecule. The characteristic X-ray spectra are associated with certain series of lines, termed K , L , M ... etc. series. The K series of the spectra arises from the transition from higher orbits ($n = 2, 3, \dots$) to the K -shell and similarly for the other series. In order of importance and line intensity, these lines are designated as α , β , γ ... etc. The α line is associated with $\Delta n = 1$ transition and is usually the most intense. The intensities of β , γ ... lines sequentially decrease and are associated with $\Delta n = 2, 3, \dots$ respectively.

Hard X-ray Radiation

The cause of the dominant hard X-ray emission is attributed to the non-thermal high energy electron beams striking the anode surface [144]. The intense transient high voltage V_p induced across pinched plasma due to the rapid change in inductance occurring during the pinch compression, causes the acceleration of electron beam that, in turn determines the X-ray energies. Neglecting any ohmic contributions, the pinch voltage (V_p) can be calculated by [146]:

$$V_p = \frac{d(L_p I_p)}{dt} \quad (2.31)$$

where I_p is the pinch current and L_p , is the focus tube inductance associated with the plasma motion in axial and radial phase.

Since the total current remains essentially constant during the compression, the pinch voltage can be estimated, considering a constant pinch length, as [146]:

$$V_p \cong -\frac{\mu_0 I z_p}{2\pi} \frac{1}{r_p} \frac{dr}{dt} \quad (2.32)$$

Here μ_0 is the magnetic permeability of vacuum, dr/dt is the convergent radial velocity and z_p and r_p are the pinch length and radius, respectively.

Time-resolved measurements show that the power law spectrum ($dNH_X/dE \sim E_{be}^{-x}$, with $x = 2 \sim 4$, E_{be} energy of the electron beam in keV and NH_X - the hard X-ray flux - in photons per cm^2 over the detector surface) found for the plasma focus in the range 50 ~ 500keV is the result of the time integration over spectra produced by quasi-monochromatic electron beams with average energies changing over a ten nanosecond time scale [147].

The hard X-ray emission was correlated with the neutron production when Deuterium was used as the working gas: the neutron yield is found to be large when the hard X-ray emission is large [2].

2.6.2.2 X-ray Emission Studies in Plasma Focus Devices

The emission of soft X-rays from the plasma focus has been studied for many years and various diagnostic techniques have been employed. These diagnostics include: (i) pinhole cameras (with one or multiple pinholes and appropriate filters) to record the time-integrated spatial/spectral distribution of the X-ray emitting region, (ii) microchannel plates used in conjunction with pinhole cameras for the time-resolved spatial distribution of soft X-rays, (iii) semiconductor detectors with appropriate filters for space-integrated temporal/spectral distribution of the soft X-ray emission, (iv) crystal X-ray spectrographs for space-resolved spectral analysis, (v) X-ray streak photography for the temporal distribution of X-rays in one space dimension (radial or axial), and (vi) X-ray films, X-ray resists and X-ray pulse calorimeters for absolute soft X-ray yield measurements.

From various experiments reported in the past, it has been confirmed that the soft X-ray emitting source is situated on the electrode axis at the open end of the electrode system in the vicinity of the anode face. The time-integrated images show that it has a roughly cylindrical shape. Its dimensions vary from below 1mm to over 10mm in diameter. These gross dimensions have been shown to be related to the anode radius [33], with an axial length from a few mm to a few cm. Spectral filtering of the X-ray image reveals a fine structure presenting intensely radiating small macroscopic entities (called by various authors "hot spots" or "micropinches" [144]) as well as filaments inside and along the axis of the pinch. These observations are more evident when the PF device is operated with high-Z gas admixtures.

Choi *et al* [148] reported X-ray "hot spots" with typical dimension of about 100 μ m which occur within 5ns of the first compression (the focus formation) during the period of the stable phase of the pinch. They concluded that the "hot spots" are not the manifestation of the $m=0$ instabilities which disrupt the pinch column.

For a low energy Deuterium-filled plasma focus devices the soft X-ray emission extends over a time interval of about 20-70ns and features three peaks: the first peak is attributed to X-ray emission by the dense pinched plasma column at the end of the compression phase; the second one is associated with the unstable plasma column; and the third peak (corresponding to the disruption phase) is caused by emission from the face of the inner electrode [149].

The X-ray emission characteristics depend strongly on the plasma focus device operating regimes and parameters (e.g. gas filling composition and pressure, stored energy, peak discharge current, driver impedance, electrode material, shape, profile and configuration, polarity of the inner electrode, etc.). Among them, the composition of the gas and the filling pressure has the strongest influence. When Deuterium is used, the working pressure for the highest X-ray yield does not coincide with the neutron-optimized output pressure [82].

Rawat *et al* [92] investigated the effect of insulator sleeve length on Neon X-ray emission from UNU-ICTP plasma focus device and reported that the insulator sleeve length is one of the major parameters that can severely affect the average SXR yield from a plasma focus device. They found that average SXR yield increase with increasing sleeve length up to certain limit but increased beyond the optimum length, results in a decrease in the SXR yield. Zhang *et al* [150] have reported a strong dependence of the Neon soft X-ray yield on the current sheath curvature angle, which can be tailored by insulator sleeve length and filling gas pressure.

For an X-ray optimized plasma focus device, the scaling law of the X-ray output Y_X as a function of the peak discharge current I and the pinch radius, r_p , is empirically given as [151]:

$$Y_X = \frac{I^4}{r_p^2} \quad (2.33)$$

This dependence, similar to the scaling law found for Z-pinchs, indicates a better operation at higher voltages for a given stored energy. However, the exact mechanisms by which high intensity X-rays are emitted in the plasma focus are still controversial and more experimental work is being done to optimize the X-ray production for practical applications.

2.6.3 Ions

Energetic ion beams have become a subject of current interest for its applications in ion implantation [152], surface modification [153], thin film deposition [154], semiconductor doping [155] and short-lived isotope production [156].

The disruption of plasma column during the radial phase induces an electric field, which accelerates the ions and electrons in opposite direction with very high velocities. The ions are accelerated during the evolution of macroscopic $m = 0$ instabilities [91] towards the top of the chamber in a conical fashion with energies ranging from tens of keV to a few MeV.

Various mechanisms have been proposed for ion acceleration [120,157]. The conditions, in which these accelerated ions are emitted from DPF, have vital importance both from the viewpoint of underlying physics and their applications in various fields. In the past, several theoretical and computational models related to ion production and acceleration have been developed [79, 108, 114, 118-120]. However, none of the models is in satisfactory agreement with the experimental measurements carried out over the wide range of DPF bank energies.

Most of the earlier experiments for diagnosing the ion beams were carried out in Deuterium medium to investigate the correlation between neutron production and deuteron acceleration. Sadowski *et al* [158] compared the characteristics of ion beams obtained from the different small energy (5 to 50kJ) DPF devices and found that the intensity of ion beams depends upon electrode geometry, bank energy, working gas type and working gas pressure.

Time integrated ion pinhole images recorded at different DPF laboratories have revealed a complex microstructure of the registered Deuterium ion beams. The pinhole image consists of a small central image surrounded by another ring-shaped image. It is believed that the central image is formed due to the emission of numerous high-energy ion micro-beams along the Z-axis whereas the ring-shaped image is formed due to low energy ion bunches that are deflected by the azimuthal magnetic field of the plasma column [158]. It has been reported that the energy spectrum of ion beam obey the power law $\frac{dN_i}{dE} \propto E_{bi}^{-x}$, where E_{bi} is the ion energy, N is the ion number and x ranges between 2 to 5 [159, 160].

Gullickson *et al* [5] reported that the maximum ion emission occurs along the Z-axis (at 0° angle) and the ion fluence decreases gradually with increasing angle inside a conic geometry. On the contrary, Sadowski *et al* [161] observed a distinct drop of ion fluence at 0°. However, Kelly *et al* [162] measured angular distribution of fast deuterons in a 5kJ DPF and observed that the number of ions detected in the 90° direction is approximately 50% of the corresponding number in the 0° direction.

2.6.4 Electrons

Energetic electrons beams have long been observed in the plasma focus, with energy much in excess of the discharge voltage. Electron beam interaction with the metallic anode is proposed as the source of the hard X-ray emission [163].

The production of accelerated electron in DPF is linked with $m=0$ instabilities [91,108]. The growth of this instability in plasma column during the unstable phase enhances an induced electric field. This enhanced electric field together with magnetic field accelerates the electrons towards the anode. These electrons are accelerated to such a high extent that they attain the relativistic kinetic energy. Another explanation is that the plasma resistance in pinch column increases rapidly [93], while the plasma current remains almost constant. Therefore, the voltage along the pinch column increases, so the electric field along the axis also increases. Under this mechanism, the electron beams can be measured through a hollow anode by Rogowski coil and by some electron spectrometer and analyzer [164]. Electron emission was studied in different PF with stored energy from below 1kJ to 1MJ, with a charging voltage from 10 to 200kV and a maximum current from 100kA to 3MA. Several processes were proposed to explain the electron beam acceleration up to energies more than 100 times higher than the applied voltage [164].

The conditions to extract the electron beam for measurements are not very simple. The strong magnetic field from the focus area traps the very low energy electrons, so these electrons are not “seen” by the detectors. For slightly higher energies, the electrons interact strongly with the filling gas. Only the very high energy electrons emitted along the anode axis can be considered not to have suffered important interactions with the plasma, the gas or the electrode system; however, they are prone to interact among themselves.

The high voltages involved make the direct coupling to the oscilloscope very difficult. The existence of the filling gas makes the direct electron beam measurements (separated from neutralizing

background current) very laborious. Most experiments employ the X-ray emission from a target (which could be the anode itself) for electron beam measurement.

The energy distribution measured by various authors seems to be drastically influenced by the device used as the electron detector [165]. Therefore, it is very difficult to compare different experiments reporting results with one order of magnitude difference for the peak of the energy distribution. Most measurements on both electrons and ion beams indicate that the peak energy is neither related to the capacitor charging voltage, nor to the energy stored in the capacitor bank. Small machines reported high energy (more than 1MeV) electrons [166]. Different components of the emitted electron beam have different temporal characteristics. The low energy component (energies below 150keV) starts at the moment when the shockwave reaches the electrode axis and reflects on itself. The emission lasts the whole dense plasma lifetime, long after the pinch disruption [93].

Typical parameters of the electron beam measured by Schmidt *et al* [167] in a 1.6kJ plasma focus device are: maximum electron energy of 1MeV and particle number $\sim 10^{14}$. The energy and current in electron beams were estimated by Harries *et al* [168] by observing the X-ray emission from plasma focus. Their result shows that the beam energy for electrons over 50keV is of the order of 1J and current of the order of 200A. Again the observation of X-ray energies >50keV from anode surface on axis implies that electrons are traveling in a beam towards the anode. Therefore, a strong electric field exists between the dense focus and the anode that is sufficient to accelerate the electrons to energies of the order 100keV over a distance of the order of 1cm. A total electron beam current of 17kA was estimated by Stygar *et al* [159] for a 12.5kJ plasma focus device.

For use of plasma focus in material synthesis applications, recently Zhang *et al* [15] have reported that hydrogen should be the first choice for thin film deposition as it produces the highest electron beam charge and higher energy (from 50 to 200keV) electrons. Neon is the next best choice as it gives the next highest electron beam current with mid-energy (from 30 to 70keV) electrons.

2.7 Energy Density and Drive Parameter

An interesting feature of the optimally designed plasma focus devices is that the energy density of the focused plasma and the drive parameter is practically constant over the whole range of devices, independently of their bank energy. However, the more energetic the machine, the higher is the pinch plasma volume, the corresponding lifetime, and the outcoming radiation yield. The scalability of plasma phenomena along with constancy in energy density and drive parameter assure that the plasma dynamics, and the consequent plasma and radiation characteristics, are mainly the same for the smallest and the largest plasma focus devices. A brief explanation of the constancy of these two determining parameters is mentioned as under.

The ‘Energy density parameter’ termed as E/V_p , where E is the initial energy stored in the capacitor bank and V_p is the final pinch volume. The mechanisms for radiation of energetic particles depend mainly on plasma energy density. According to the study performed by Lee and Serban [33] and optical diagnostics by Soto *et al* [35,36,169] the final pinch radius (before appearance of instabilities) is typically of the order of $0.12a$ and the maximum pinch length is of the order $0.8a$. Thus, the final plasma volume V_p is of the order of $0.036a^3$, and the plasma energy density E/V_p in the pinch phase is proportional to $\approx 28E/a^3$ [170]. It is noticeable from performance parameters of the devices listed in Table 1.1 that macroscopically ‘Energy density’ parameter remains consistently within the range of $(0.1-10) \times 10^{10} \text{ J/m}^3$ for a wide bank energy range of plasma focus devices.

Drive parameter/ speed factor mainly represents a measure of the speed of the plasma in both axial and radial phases and it is defined as $S = (I_p/a)/\sqrt{\rho}$, in which I_p is the peak discharge current flowing through the pinch, a is the anode radius and ρ is the ambient gas density. Its consistency (average value of $89 \pm 8 \text{ kA/cm/Torr}^{1/2}$) in medium to large energy range of Mather-type plasma focus devices has been established previously by Lee and Serban [33]. Having a drive parameter within this

range implies that all surveyed machines operate with same axial and radial speeds and also the plasma temperature does not vary much.

Role of drive parameter in energy transferred to the plasma column is realized from the expression of energy per unit mass in the final plasma column i.e.:

$$\frac{EINP_{slug}}{mass} \sim const \times \frac{\frac{\mu_0}{4\pi} I_p^2 z_p \ln\left(\frac{b}{r_{min}}\right)}{\rho \pi a^2 z_p} \sim const 2 \times \frac{I^2}{\rho a^2} = const 2 \times S^2 \quad (2.34)$$

Here, r_{min} is the final column radius and its mass is measured by $\rho \pi a^2 z_p$. Here $\ln(b/r_{min})$ varies at a very slow rate compared to a^2 , and can be considered constant. Drive parameter is also considered as a measure of the drive magnetic pressure per unit density, or the drive magnetic energy per unit mass. From equation (2.34), it is clear that the drive parameter is directly related with the energy density in the final plasma column and consequently the plasma temperature. For enhancing the speed factor, either current should be increased or the anode radius/ gas pressure must be reduced.

After providing detailed description of plasma dynamics through various phases and emission of various particles/ radiation in plasma focus device we will now discuss the design and construction of three different versions of newly developed miniature plasma focus devices in the next chapter.

Chapter 3

Design and Construction of Fast Miniature Plasma Focus Devices

This chapter describes the complete pathway of development of the three different versions of Fast Miniature Plasma Focus devices, designated as FMPF-1, FMPF-2 and FMPF-3. In the naming of these devices suffix 'Fast' has been added to signify their inherent sub-microsecond duration quarter time period (i.e. in the range of a few hundreds of nanoseconds). In this chapter along with the description of the basic rules of thumb generally applied in pulsed power system design and plasma focus tube design; the relevant information about the conceptualization, design, construction and technical constitution of all the three versions of FMPF devices are thoroughly described and elaborated along with the characterization results of their sub-systems.

3.1 Introduction

Three different versions of the fast miniature plasma focus device, designated as FMPF-1, FMPF-2 and FMPF-3 were developed under this project, as repetitive compact pulsed neutron sources.

FMPF-1 is developed as a test bed for investigating the operation and optimization of the performance of the miniature plasma focus device operating at such low energies. It is a single module system that delivers peak current of $\sim 81\text{kA}$ into the focus tube (under short circuit condition), when the capacitor bank of $2.4\mu\text{F}$ is discharged at 13kV . This system can be operated at a maximum repetition rate of 0.5Hz .

FMPF-2 is developed as a first prototype of high repetition rate miniature plasma focus device based on optimized electrical specifications for the FMPF-1 device. In the FMPF-2 device, although the capacitance of the energy storage bank remains the same as FMPF-1 device, the number of

capacitors were doubled for reducing the inductance and resistance of the circuit and enhancing their shot-life under repetitive mode operation. It delivers peak current of $\sim 83\text{kA}$ into the focus tube through a single sparkgap switch, when the capacitor bank of $2.4\mu\text{F}$ is discharged at 13kV (under short circuit condition). For the FMPF-2 device, a totally new upgraded pulsed power system was devised with high performance components (like switch, capacitors and triggering system) to customize the device operation in repetitive mode up to 10Hz .

FMPF-3 is an optimized version of the repetitive miniature plasma focus device as a pulsed neutron source that can be operated in quasi-continuous mode up to 10Hz . It is a four module system with each module being equipped with a pseudospark switch and two parallelly connected $0.3\mu\text{F}$ capacitors. The triggering system is entirely different from that used in FMPF-1 device. A high performance 4-channel triggering system has been indigenously designed and implemented in the FMPF-3 device. This device delivers a peak current of $\sim 95\text{kA}$ into the focus tube under short circuit discharge conditions (when four $0.6\mu\text{F}$ modules are synchronously discharged in parallel at 13kV) at repetition rates up to 10Hz .

More details of FMPF-1, FMPF-2 and FMPF-3 devices are described under the following sections along with a brief description of the design approach and considerations of pulsed power system design.

3.2 Design Approach

The designing of plasma focus devices is a complex procedure and it is interlinked with many aspects such as the choice of the electrical circuit parameters, the careful choice of major and auxiliary pulsed power components (depending on matching requirements and their commercial availability) and the coaxial electrode design for optimum device performance. In this section, firstly the design procedure for pulsed power systems is reviewed and then attention is focused on the design of the

coaxial electrodes of the focus tube assembly.

The first step in designing a plasma focus device is the estimation of the stored energy and peak current delivering capacity of its pulsed power system, on the basis of the expected operational capabilities. For doing so, primarily the charging voltage is chosen and then accordingly, all the other electrical parameters are deduced.

In our project, since the objective was to develop a ~200J system, we first needed to decide the typical operating voltage range for estimating the required capacitance. As commercially available energy storage class capacitors are rated typically up to 30kV with shot-life of $>10^6$ charge/ discharge cycles withstanding 80% voltage reversal and as our device were to be operated in the repetitive regime (in burst mode), therefore to obtain a higher shot-life we decide to operate the system in the range 12 to 15kV. By operating the capacitors at 40 to 50% of their maximum charging voltage, life expectancy improves significantly (discussed later in detail in the followed section 3.3.2) [171]. Moreover, the another advantage of operating the device in the range of 12 to 15kV is that the system can be made compact enough (as large clearances do not have to be provided to prevent high voltage leaks); which is an important consideration in achieving low inductance.

For the selected range of operating voltage, the capacitance for ~200J system lies in the range of 2.77 μ F to 1.77 μ F. With approximated maximum system inductance of 30nH, the peak discharge current was expected to be ~100kA.

Again from a consideration of the maximum ratings of commercially available energy storage capacitors (of moderate energy and size), as the peak discharge current delivering capacity (limited by their equivalent series inductance (ESL) and equivalent series resistance (ESR)) of capacitor typically lies in the range of 25 to 50kA (maximum); therefore to meet the total peak discharge current delivery requirement, an appropriate number of capacitors are connected in parallel forming a capacitor bank. It may be noted that the parallel configuration of capacitors contributes effectively in reducing the

inductance and resistance of the system.

After finalization of the capacitor bank parameters, the selection of the switch is made for fast energy transfer and then the pulsed power system is constructed in a minimum possible inductance configuration and its performance is investigated by short circuit tests. Short circuit test results yield the residual parameters of the system (like resistance and inductance) and quarter time period ($T/4$) of the current signal.

The final stage of design trade off for the optimized performance of the plasma focus device involves designing the coaxial electrodes of the focus tube assembly such that their physical dimensions match the plasma sheath run down time to coincide with the peak value of discharge current (roughly about a quarter time period of discharge current), in order to produce the highest magnetic pinch pressure and plasma temperatures. The axial plasma sheath velocity v_a scales with the corresponding magnetic field $B_\theta (\propto \text{peak discharge current } I)$, gas density ρ , and anode radius a , as [8]:

$$v_a = \sqrt{\frac{B_\theta^2}{2\mu_0\rho}} \quad (3.1)$$

where μ_0 is the vacuum magnetic permeability and the driving magnetic field B_θ is:

$$B_\theta = \frac{\mu_0 I}{2\pi a} \quad (3.2)$$

As indicated from the relations above, the anode radius and gas pressure are the key parameters to be adjusted to obtain an optimum plasma sheath velocity and hence a high pinch temperature. Typical velocities of the current sheath are of the order of $1 \times 10^5 \text{ ms}^{-1}$ in the axial phase and of the order of $2.5 \times 10^5 \text{ ms}^{-1}$ in the radial compression phase [35]; and so a preliminary approximation for the anode length can be obtained from:

$$z_0 = v_a [(T/4) - (a/v_r)] \quad (3.3)$$

Here, a is the anode radius, v_a and v_r is the sheath velocity in axial and radial phases and $T/4$

is the quarter time period of the discharge signal obtained from the short circuit test.

The anode radius is approximated on the basis that the ‘drive parameter’ has been experimentally observed to be in narrow range for Deuterium operated plasma focus devices ranging from kilojoules to hundreds of kilojoules [33]. As investigated by Lee and Serban [33], it is about $89(\text{kA/cm})/(\text{Torr})^{1/2}$ with a remarkably small deviation of $\sim 10\%$ over a wide energy range of plasma focus devices [33]. Since the peak discharge current of the system is known from the short circuit test, therefore for a typical Deuterium filling operation at few Torr, the anode radius is approximated using the relation $89 \approx I/ap^{1/2}$ (I – peak discharge current in kA, a – anode radius in cm and p – filling gas pressure in Torr) [33]. It may be worth mentioning here that though the validity of the speed parameter for sub-kilojoule range plasma focus was not established by any research group, we believed that there was no specific reason for it to be invalid at the lower energy range, and hence we still used it as the guiding principle in roughly estimating the anode radius of our FMPF series devices.

The cathode radius is decided by consideration of the focus tube inductance. For maximum ideal energy transfer, the inductance of the focus tube L_p is designed to closely match the equivalent circuit inductance L_o of the pulsed power system. Mismatch in inductance is actually the major cause that induces ‘current shedding effect’ *i.e.* a portion of the discharged current stays near the back wall insulator sleeve, without contributing to the plasma sheath. The expressions for estimating the maximum focus tube inductance (associated with the plasma motion in the axial and radial phases) have been mentioned in the previous chapter under section 2.5 (Eq. 2.6 and 2.7). It is important to note that pinch current is also limited by the focus tube inductance and therefore due weightage must be given to this parameter during the designing. The height of the cathode rods are conventionally kept similar to the anode height.

It is important to mention that, since depending on the plasma sheath position, the inductance of the focus tube changes dynamically (from a minimum, corresponding to the breakdown phase and to

a maximum at the pinch stage) and it modifies the electrical parameters of the equivalent circuit (like peak discharge current I and quarter time period $T/4$ of the signal), therefore there is a probability that the estimated physical dimensions of the coaxial electrode assembly by the above mentioned design procedure may not perfectly match with the electrical characteristics of the system. Experimental iterations are required for fine tuning the design of the electrodes to achieve optimal device performance and radiation yields. The major role of the above mentioned focus tube design procedure is mainly to provide startup values for initializing the experiment.

3.3 Considerations of Pulsed Power System Design

To operate the plasma focus, high voltage (~tens to a few tens of kV) and high current (>10kA) electrical pulses of a few hundred ns to a few μ s duration are required. This necessity is fulfilled by the pulsed power system which mainly comprises: (i) capacitor bank for energy storage, (ii) triggerable switch for rapidly transferring the stored energy from the capacitor bank to the load (which is the dynamic plasma across the electrode assembly inside the focus chamber), (iii) connections used for paralleling the capacitors and connecting switch to the plasma load, and (iv) HV charger for charging the capacitor bank to a suitable voltage. The mentioned constituting components - capacitor bank, triggerable switch and connections are supposed to have the following common characteristics: (a) low inductance, (b) ability to withstand the desired high voltage, (c) high current carrying capacity, and (d) ability to withstand heavy mechanical stresses due to the high magnetic fields.

In a typical pulsed power system, the high power electrical pulses of a few μ s duration are produced by charging the capacitor bank to a suitable voltage (using a high voltage charger) and then discharging it through a fast switch to the load *viz.* the plasma focus. However, to obtain high current pulses, the capacitors and the connections from the capacitors to the load and the load itself must be carefully designed.

In the following subsections, the electrical response of a typical pulsed power system is illustrated along with a briefing on the important technical issues to be considered for the selection of major components of the pulsed power system.

3.3.1 Electrical Response

The governing LCR circuit equations that characterize the electrical response of the pulsed power system are illustrated as under. The simplified electrical equivalent of a typical pulsed power system is shown in Fig. 3.1. This electrical circuit consists of a capacitor bank of total capacitance C_0 , charged to a voltage V_0 , an equivalent series inductance L_0 and an equivalent series resistance R_0 (correspondingly contributed by the capacitors, transmission line, connections and switches).

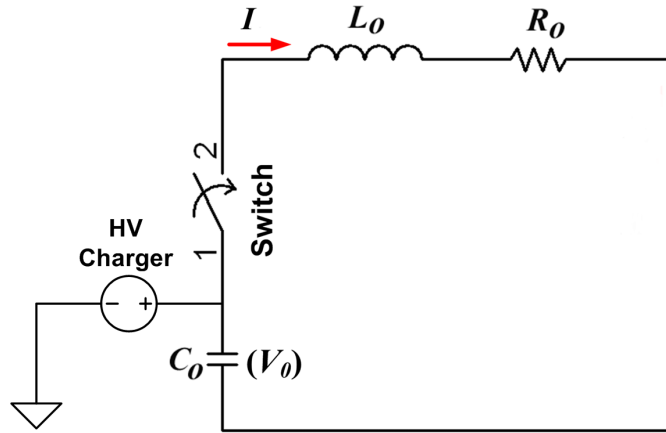


Fig. 3.1 Equivalent circuit of the pulsed power system.

When the energy stored in the capacitor bank is discharged through fast switch then the current flowing in the circuit is described by following equations:

$$I(t) = (V_0 / \omega L_0) e^{-\alpha t} \sin \omega t \quad (3.4)$$

$$\text{Here, } \alpha = R_0 / 2L_0; \quad \omega = \sqrt{\frac{1-r^2}{L_0 C_0}} \quad \text{with} \quad r = \sqrt{\frac{R_0 C_0}{4L_0}}$$

In typical plasma focus experiments, the external resistance (R_o) is usually low ($\sim m\Omega$) and the capacitors usually have very low equivalent series resistance, therefore, the peak discharge current (I) and its period (T) of sinusoidal oscillation in the circuit can be evaluated as:

$$I = V_0 \sqrt{\frac{kC_0}{L_0}} \quad (3.5)$$

$$\text{with } k = e^{-\pi r / \sqrt{1-r^2}} \quad (3.6)$$

$$\left(\frac{dI}{dt} \right) = \frac{V_0}{L_0} \quad (3.7)$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{L_0 C_0} \quad (3.8)$$

Here, k is defined as the voltage reversal factor (*i.e.* the ratio of the voltages of any two consecutive extremes). As can be seen from equations (3.5) and (3.6), to get high peak discharge current (I): (a) sufficiently high charging voltages must be used, (b) capacitors must have low equivalent series inductance (ESL) and low equivalent series resistance (ESR), and (c) the overall inductance L_o and resistance R_o in the circuit (contributed by switches, connections, collector plates *etc.*) must be as low as possible. The important considerations for the selection of major constituents of pulsed power system are briefly discussed under following sub-sections.

3.3.2 Capacitors

A capacitor bank constitutes of a multiple number of energy storage capacitors connected in parallel. Energy storage capacitors are the costliest and most important component of the pulsed power system and their choice is generally a compromise between technical requirement, cost and performance. The important parameters relevant to the choice of capacitors are (i) ESL, (ii) ESR, (iii) voltage reversal (k), and (iv) shot-life (Λ) (*i.e.* charge-discharge cycles) [171,172]. As has been

discussed and illustrated through mathematical expressions (Eq. 3.5 – 3.7) in the previous section; for obtaining higher peak currents the capacitors must have the ESL and ESR values as low as possible.

The ESL and ESR value of any capacitor depends on manufacturing technology and it is usually provided by the manufacturer in the specifications. However, it is always recommended for the users that ESL and ESR for individual capacitors must be investigated before constituting the bank.

Pulse discharge operation, particularly with under-damped (ringing) discharges (which is expected in low impedance drivers, with reversal as high as 80%), places large internal stresses on components. It is important to note that shot-life (Λ) mainly depends on two operating parameters:

- (i) Charging voltage (V_0) and
- (ii) Voltage reversal (k).

An empirical expression that have been developed to predict the shot-life of a pulse discharge capacitor is given by [172]:

$$\Lambda_{op} = \Lambda_r \times \left[\frac{\omega_r}{\omega_{op}} \right]^{1.6} \times \left[\frac{V_r}{V_o} \right]^{7.5} \quad (3.9)$$

Here, notations Λ , V and ω denote shot-life, charging voltage and ringing factor respectively for rated (r) and operating (op) conditions. The relationship between ringing factor and voltage reversal is given by [172]:

$$\omega = \sqrt{1 + \frac{1}{\left[\frac{2}{\pi} \ln(k) \right]^2}} \quad (3.10)$$

As shown in equation (3.9), capacitor charging voltage is the most important life determinant, with 7.5 as exponent. A small reduction in voltage leads to a huge increase in capacitor shot-life. Typically, shot-life enhancement by a factor of 1000 can be achieved by operating the capacitor at ~50% of its rated value [171]. Voltage reversal also has a significant impact with 1.6 as exponent. It is

normally expressed in the range of $0 < k < 1$. When the voltage reverses, the mechanical stresses on the capacitor are quite high. As regards budgetary considerations, it is important to note that, approximately, the cost of the capacitor doubles with every ten-fold increase in the shot-life.

3.3.3 Triggerable Switches

The triggerable switch in a pulsed power system is incorporated to transfer energy from the capacitor bank to the load at a preset voltage and time. There is a wide range of available pulsed power switches that operate on various technologies like – Sparkgap, Railgaps, Thyatron, Ignitron, Pseudospark, etc. The selection of a particular type of switch is done on the basis of desired switching performance. There is always a compromise between the cost, complexity in switch operation and the obtained switching performance. The general technical specifications to be considered for the selection of an appropriate switch are - (a) voltage hold-off range, (b) peak current transferring capacity, (c) inductance, and (d) lifetime of the switch (depending on cumulative charge transfer).

In repetitive pulsed power systems, due to the requirement of successive high power discharges (of high voltage and high current), the switch prerequisites are very demanding. To lower the maximum current (detrimental even because of its high time-derivative) and the total inductance per switch, their parallelization is mandatory. The drawback of parallelization lies in the increased complexity of the system from both the electrical and operational viewpoints. For obtaining synchronized operation of the parallel switches, the specifications of major concern are (i) breakdown time delay t_{bd} (*i.e.* the time lag between arrival of the voltage pulse at the trigger electrode and the start of main gap conduction) and (ii) jitter σ_{bd} (*i.e.* the measure of standard deviation in the breakdown time delay). The main operating/ working parameters that prominently affect the t_{bd} and σ_{bd} are the voltage across the main gaps, operating gas pressure and the type of gas. The role of triggering pulse characteristics and scheme is also very dominant in achieving minimum breakdown time delay along

with low jitter. The nominal ‘breakdown time delay’ and ‘jitter’ of the switch under parallel operation must be typically <50ns and <10ns respectively for synchronized discharge.

The issue of critical concern that limits the repetitive operation of the switch is the ‘recovery time’. This parameter is cross-linked with many factors like switching technique (type of switch), operating voltage, insulating gas, charge transferred and electrode material.

3.3.4 Connections

There are two alternatives for making the connections (between the capacitors to switch and switch to load); either to use a number of coaxial cables in parallel or to use a flat plate transmission line. Both of these schemes have their own pros and cons and have to be chosen in consideration with application requirement. For large capacitor banks with multiple discharge modules operating in parallel, use of coaxial cables is beneficial, since it is easy to lower the inductance by parallelization and to adjust the ‘transit time isolation’ between the modules (for compensating jitter) by adjusting the cable length. Furthermore, the outer conductor of the cable also helps in partially screening the electromagnetic noise induced by the pulsed current transition. Although the use of coaxial cables has various benefits along with higher flexibility, it must be noted that the lumped inductance of such grades of coaxial cables (*e.g.* RG213, URM 67, RG218 are commonly used) is typically ~230nH/m. Hence, unless heavy parallelization of cables is used (like in large banks of multiple discharge modules), it is difficult to obtain connection inductance in the range of ~10-20nH.

The flat plate (sandwich) transmission lines are advantageous in comparison to coaxial cables in following respects – (a) they can have much lower inductance and higher current carrying capacity as compared to coaxial cables, (b) their maintenance is easier, (c) interfacing/ end connections to capacitors and discharge switch is convenient along with high reliability, (d) reduction in overall size and complexity of the system, and (e) it is more economical and long lasting. However, there are two

major limitations (i) if multiple discharge modules are operated in parallel; higher end switching performance (t_{bd} and $\sigma_{bd} < 10\text{ns}$) is an important pre-requisite for the synchronized discharge, since practically the ‘transit time isolation’ between modules is negligible due to limitation on size of collector plates, and (ii) the flexibility of keeping the load physically away in any position with respect to the capacitors is eliminated.

The inductance (L_T) contributed by per unit length of parallel plate transmission line for condition ($d \ll W$; where d is clearance between the plates and W is the width of the plate) is given by expression:

$$L_T = \mu_0 \frac{d}{W} \text{ Henry / meter} \quad (3.11)$$

As evident from equation (3.11), the clearance between the plates d ; decided by the dielectric thickness should be kept as small as possible along with high voltage insulation standoff for minimal inductance.

When the flat plate transmission lines are used (especially in large devices with currents in MA level) owing to much smaller clearance, the electro-dynamical force acting on the plates is quite high because of the interaction between opposing magnetic fields. Moreover since the current is oscillatory, the buckling is also “forced oscillatory”. For the purpose of absorbing the transient electro-dynamical force; constructionally, the flat plat transmission lines are made sufficiently heavy and rigid to prevent their separation from altering to any significant extent during the discharge.

3.3.5 High Voltage Charger

A key element of the compact and repetitive pulsed power systems is the high voltage charger that charges the capacitor bank up to a preset level within certain duration. It is important to note that the maximum attainable pulse repetition frequency (PRF) of any pulsed power system is primarily limited by the peak power rating (P_{pk})/ charging rate capacity (J/s) of the HV charger [173,174].

Hence, considerable attention is required in the selection of HV charger ratings to match the desired frequency of operation. Along with the increase in rep-rate requirement, the power requirement of the HV charger also increases correspondingly. The most important technical specifications to be considered for the selection of a compatible HV charger is maximum charging rate (J/s) and charging voltage range.

In the following sub-sections procedure for the estimation of charging time and protection of HV charger against voltage reversal is discussed.

3.3.5.1 Determining Maximum Charging Cycle Rate

The peak charging rate (J/s) of the HV charger rating determines the capacitor charging time and average charging rate determines the total power deliverable by the power supply. The sketch shown in Fig. 3.2 illustrates a typical charging cycle of a constant current power supply [173,174].

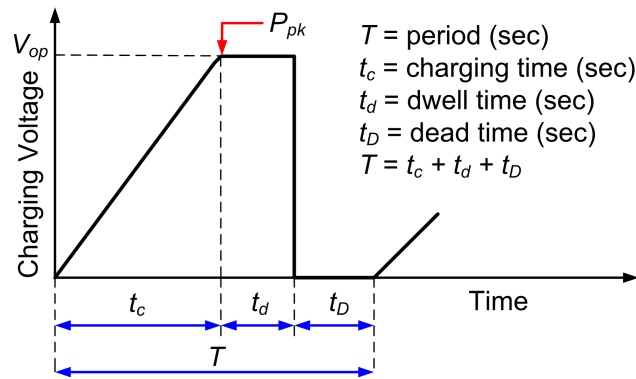


Fig. 3.2 Typical charging cycle waveform.

The profile depicts the charging voltage as a function of time. The peak power P_{pk} occurs at maximum charging voltage V_o at end of charging time t_c . After t_c , the power supply has a dwell time t_d , during which it may need to supply a small amount of charge to maintain the capacitor voltage, after which the capacitor is discharged. As shown in typical charging profile, T is the time period for one

charging cycle, and t_D is the dead time during which the power supply output is inhibited (to allow faster recovery of energy transfer switches). In repetitive charging applications, for the estimation of the maximum charging rate (*i.e.* number of charge cycles per second in Hz), it is important to consider dwell time t_d and dead time t_D along with charging time t_c [173,174] as their durations may be considerable, depending upon the application.

Using the peak power rating of the power supply, charge time t_c is estimated using the equation below [174] –

$$t_c = \frac{0.5 \times C_0 \times V_0 \times V_{rated}}{P_{pk}} \text{ seconds} \quad (3.12)$$

Here, P_{pk} is the peak power rating of the charger in Watts, C_o is bank capacitance in Farads, V_{rated} is power supply maximum voltage rating in Volts and V_o is the voltage up to which capacitor bank is charged in Volts. The durations for dwell time t_d and dead time t_D are decided by the user as per application requirement.

3.3.5.2 Protection against Voltage Reversal

An illustrative circuit diagram of a pulsed power system connected to a switching power supply (HV Charger) is shown in Fig. 3.3. If this circuit is under-damped (which is the case with plasma focus as load as the impedance varies from open to short circuit impedance with time), the voltage on the capacitor will reverse. This reversal can damage the diodes in the output section of the power supply if the magnitude of the reversal is high enough to cause excessive current flow in the diodes [175]. Even in cases where the load does not cause voltage reversal, a fault in the load may cause voltage reversal leading to power supply failure. The output diodes would fail rapidly under these conditions and potentially damage the output section of the charger.

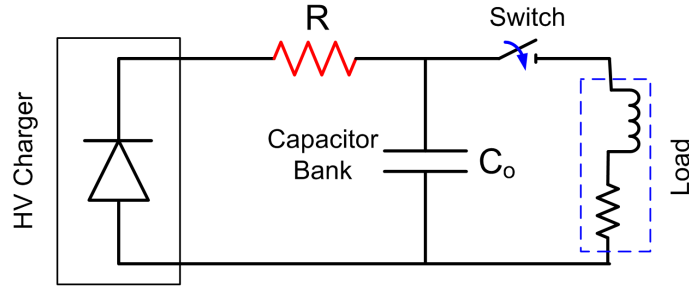


Fig. 3.3 Basic pulsed power system connected to switching power supply.

The power supply must be protected from voltage reversal by electrically isolating the supply during discharge. This may be done either by a passive protection or with an isolation switch. An isolation switch at first offers an attractive option because the power supply is completely removed from the circuit. But there can be problems associated with this scheme. Faults can occur when a switch pre-fires during charging, which can lead to failure of the power supply and the switch. This problem may be avoided if the switches used have a very low probability of pre-firing, such as solid-state switches or if a solid-state isolation switch is used that can be opened rapidly under fault conditions. But solid-state switches are costly and very sensitive to electromagnetic interference. A simple and robust method to limit/reduce the effect of reflected energy (from the system to power supply) is to place a series resistance R of typically 50 to 500 Ω (of power rating $\sim$ 200-300W), between the power supply and the capacitor bank [176] as shown in Fig. 3.3.

3.4 The FMPF-1 Device

The main objective at the first stage was to construct an efficient pulsed power system for exploring the performance of the newly designed first version of fast miniature plasma focus FMPF-1 device operating at about 200J.

In this section, we illustrate the description and characteristics of various subsystems of the FMPF-1 device and give the relevant information about their performances.

3.4.1 Capacitor Bank

The design and construction of pulsed power system for the fast miniature plasma focus FMPF-1 device was conceived in consideration with readily available resources for fast track development.

As conceptualized under section 3.2, for the selected operating voltage range of 12-15kV, the capacitance in the range of $2.77\mu\text{F}$ to $1.77\mu\text{F}$ was required for building the pulsed power system with a stored energy of $\sim 200\text{J}$; hence, with the availability of $0.6\mu\text{F}/30\text{kV}$ energy storage capacitors in our laboratory, we decided to construct a capacitor bank of $2.4\mu\text{F}$ capacitance by connecting 4Nos. of $0.6\mu\text{F}$ capacitors in parallel.

For measuring the ESL (equivalent series inductance) and ESR (equivalent series resistance) values of these capacitors; a short circuit test was performed on each capacitor by connecting low inductance sparkgap switch close to the discharge end. A recorded oscillogram of the discharge current trace at 9kV operation is shown in Fig. 3.4.

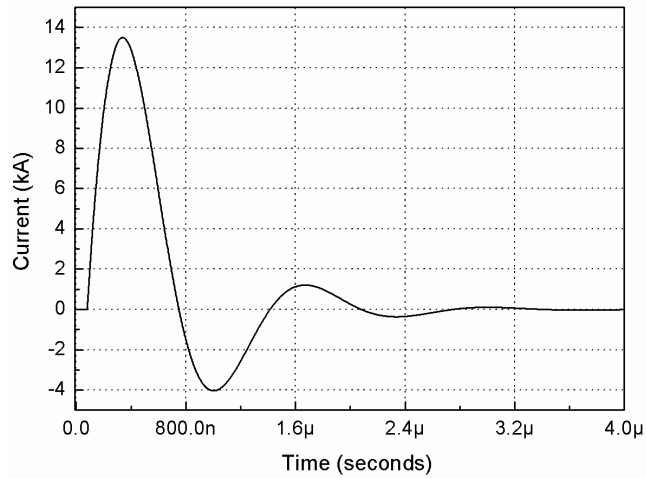


Fig. 3.4 Oscillogram of short circuit discharge current trace (9kV operation).

It may be observed from the current trace that the voltage reversal is reasonably low ($\sim 35\%$), indicating the presence of large inductance and resistance in the circuit. By using the fundamental LCR circuit equations for damped sinusoidal oscillation, mentioned in section 3.3.1; the equivalent series

inductance (ESL) and the resistance (ESR) are estimated to be $\sim 70\text{nH}$ and $\sim 250\text{m}\Omega$ respectively. Obtained values of high ESL and ESR for these capacitors are the main factors for limiting the peak discharge current to $\sim 22\text{kA}$ at $\sim 13\text{kV}$ charging voltage. Based on measurements performed, the ratings of capacitors used in the construction of capacitor bank for FMPF-1 device are summarized below:

Table. 3.1 Specifications of Capacitors used in FMPF-1.	
Capacitance	$0.6\mu\text{F}$
Maximum Charging Voltage	30kV
Operating Voltage	$8 - 15\text{ kV}$
Maximum Stored Energy @ 30kV	270J
Series Inductance (ESL)	$\sim 70\text{nH}$
Series Resistance (ESR)	$\sim 250\text{m}\Omega$
Voltage Reversal	35%
Maximum Current	$22\text{kA} @ 13\text{kV}$
Shot-life	$>10^6$ charge discharge cycles
Physical Dimensions (H×B×W)	$30\text{cm} \times 16\text{cm} \times 6\text{cm}$
Weight	$\sim 5\text{kgs}$

3.4.2 Switch Development and Characterization

For rapidly transferring the energy stored in the capacitor bank to the load, pressurized triggered sparkgap switches are technically the most viable and economical option as they are simple in construction, low inductance, have simple trigger requirements and are easy to operate. The advantages of a pressurized sparkgaps in comparison with atmospheric-pressure sparkgaps are: (i) smaller conductive resistance, (ii) smaller inductance, and (iii) a faster switching time due to the shorter gap distance and higher field strength. For a uniform or nearly uniform gap filled with air, the breakdown voltage V_{bd} is a function of the air pressure p and the gap distance d , and can be expressed as [177]:

$$V_{bd} = 24.4pd + 6.53\sqrt{pd} \quad (3.13)$$

where p and d are the air pressure in bar and the gap distance in cm respectively.

The triggerable sparkgap switches are available in various configurations such as trigatron, direct over-volted gaps, field distortion gaps, railgaps etc. Amongst these, trigatron type sparkgap switches are the most compact and offer the lowest cost with minimal maintenance. In FMPF-1 device an indigenously designed inexpensive and compact 100kA, 25kV hold-off trigatron sparkgap switch has been used for transferring the energy from capacitor bank to load. It has been custom designed to be embedded with in the transmission plate assembly thereby eliminating the use of cables in system.

The major advantage of trigatron sparkgap geometry is that, with trigger pin concentric and within one of the main electrodes, only a relatively low trigger voltage is required to initiate the discharge. Under optimum triggering conditions, the switch developed for FMPF-1 was reliably triggerable in the range of 8kV to 18kV. In trigatron switches the hold off voltage can be easily changed over a wide range by adjusting the internal gas pressure [178]. The designed switch is pressurized with Nitrogen (N_2). For reliable switching behavior, the switch is used with a Thyristor driven floating trigger module [179].

The construction, operation and characterization results of the developed switch are detailed in the following sub-sections.

3.4.2.1 Construction and Operation of Trigatron Switch

Major factors that have been considered during the customization of switch development are: (a) low inductance, (b) minimum breakdown time delay, (c) reliable operation, and (d) low erosion at high current levels. For attaining these characteristics due weightage in design has been given to issues like - uniformity of the electrode erosion, sensitivity to the main and trigger gap distances (defined later), electrode shape, electrode material, gas pressure, flush gas flow pattern, triggering method and switch recovery. A cross sectional diagram of the developed sparkgap switch is shown in Fig 3.5.

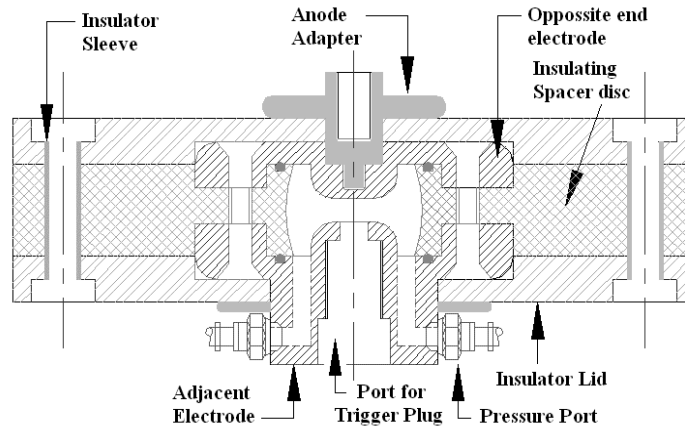


Fig. 3.5 Cross-sectional view of the trigatron switch.

Trigatron switches employ two main discharge electrodes, called the opposite (O) and adjacent (A) electrodes and an insulated trigger pin (T). These electrodes are made up of Brass. Brass has been chosen as electrode material since its erosion rate is low as compared to other materials and is also relatively cheap. A commercially available motorbike spark-plug is used as the trigger electrode. The main insulator, between, is Delrin, chosen for its resistance to surface tracking and high impact strength. Throughout, Nitrile rubber o-rings have been used for sealing the assembly. To further strengthen the high voltage insulation and impact withstand capacity Delrin enclosures have been used on both the sides, covering exposed portions of top and bottom electrodes. The main electrodes carry current once the breakdown initiates. The effective diameter of brass electrodes is 9.5mm and the main gap d_g in between the opposite and adjacent electrodes is 6mm. The trigger gap d_t between the trigger pin and adjacent electrode is 1mm. The electrode gaps are flushed continuously using pressurized nitrogen. As shown in the schematic, the flush gas flows in and out through the radial pressure ports in the trigger end electrode of the sparkgap switch. Physical dimensions of the switch excluding trigger plug is 100mm (ϕ) $\times$ 30mm (w). The estimated inductance of the switch is $\sim 10\text{nH}$. Photographs of the transmission plate embedded switch assembly are shown in Fig. 3.6.

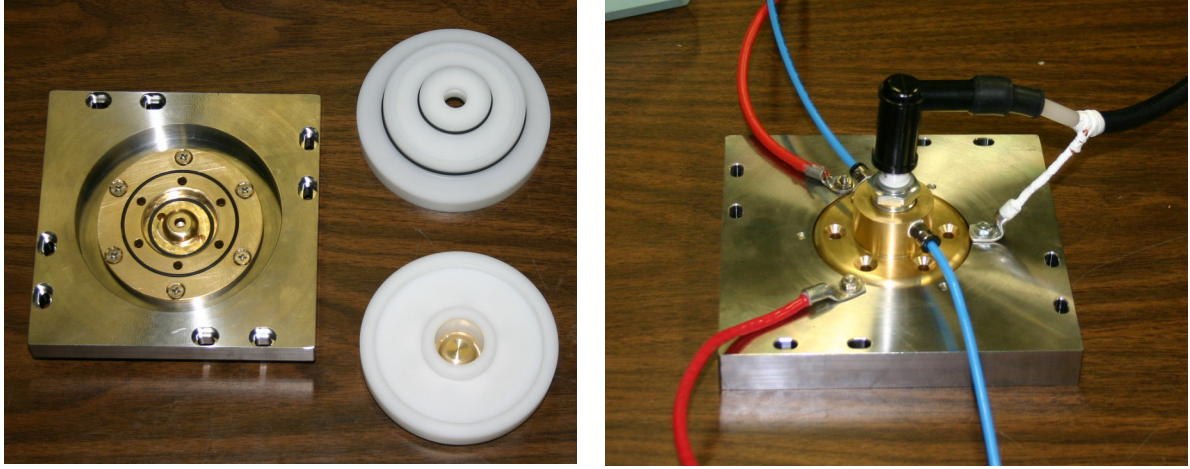


Fig. 3.6 Photographs of transmission plate embedded switch assembly.

The basic physical mechanism responsible for triggering a high voltage breakdown in the trigatron has been investigated in details by many researchers [178, 180-182]. There are basically two phases in closing of sparkgap when a trigger pulse with peak voltage V_t is applied in the presence of main gap voltage V_g , the initial phase of streamer formation takes place at the high electric field region surrounding the trigger pin. This is followed immediately by the second phase of arc formation which then fully closes the switch. Breakdown time lags (i.e. time delay t_{bd}) associated with the first and second phases are known as the statistical (t_s) and formative time lags (t_f), respectively. Most of the jitter is known to be associated with the statistical delay [183].

Performance of the sparkgap switch has been evaluated in the heteropolar mode with load series configuration for capacitive discharge application [178]. In the heteropolar mode, applied trigger voltage V_t and main gap voltage V_g are of opposite polarity. The mean electric field between the trigger pin and adjacent electrode is [178]:

$$E_t = \frac{V_t}{d_t} \quad (3.14)$$

Similarly, the mean electric field between high potential opposite electrode and the trigger is given as [178]:

$$E_g = \frac{V_g - V_t}{d_g} \quad (3.15)$$

The possible discharge initiation mechanisms are illustrated as [184] – (i) BAE (Breakdown at Adjacent Electrode) Mode: if $E_t > E_g$, then the trigger pulse first causes breakdown to the adjacent electrode (BAE) and the resulting UV radiation and plasma provide a source of ionization leading to discharge across the main gap. (ii) BOE (Breakdown at Opposite Electrode) Mode: if $E_t < E_g$, then the trigger pulse first forms breakdown streamers directly to the opposite electrode (BOE) due to applied field, the ionization density avalanches until arc channel is formed and resistance suddenly decreases across the main gap. In general, it is mentioned that operation in pure BAE mode results in longer breakdown time and excessive jitter [184]. Break down time and associated jitter is best attained in heterolpolar operation with simultaneous BAE and BOE initiation using the critical trigger voltage V_{tc} given by [181,184]:

$$\frac{V_{tc}}{V_g} = \frac{-d_t}{d_g - d_t} \quad (3.16)$$

Hence, the magnitude of mean electric field across the two gaps will be equal when $V_t = V_{tc}$. For $V_t > V_{tc}$, BAE operation will occur first while for $V_t < V_{tc}$ BOE happens first. For the developed trigatron switch the value of V_{tc}/V_g is 20%. The characteristic performance of the switch in series of shots have been taken above the critical value using a short circuit as load are discussed in the following section.

3.4.2.2 Switch Characterization

Switch characterization experiments were performed in the voltage range of 4kV to 25kV in ‘isolated load series configuration’ as shown in the schematic in Fig. 3.7. For adjusting the hold off voltage in the range of 4 to 25kV, the Nitrogen filling gas pressure inside the switch was accordingly adjusted in the range of 3 to 16psi. A negatively going trigger pulse of ~25kV was supplied by the

indigenously developed trigger module. The residual short circuit inductance works as load (ZL).

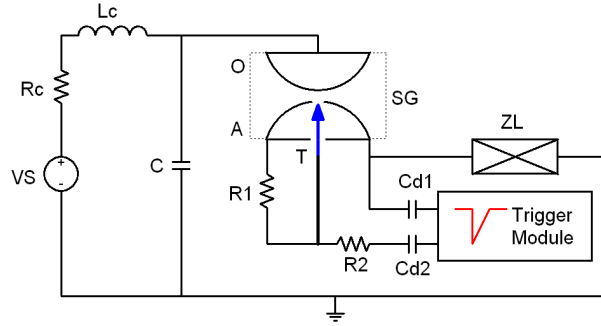


Fig. 3.7 Schematic of configuration used in sparkgap characterization.

For diagnostics, custom designed Rogowski coil and high voltage probe (Tektronix P6015) have been used along with Yokogawa DL9140, 5GS/s, and 1GHz oscilloscope for data capture.

3.4.2.2.1 Operating Regime for Stable Trigger Performance

For keeping the inductance of the switch as small as possible, the distance between the electrodes has been kept small *i.e.* 6mm. But at smaller distance the chances of the switch to pre-fire increases. Hence to prevent pre-firing the switch is pressurized with Nitrogen gas. Measurements concerning the gap voltage and N₂ fill pressure (close to atmospheric pressure, and that is why pressure is mentioned in unit psi) are shown in the graph in Fig. 3.8.

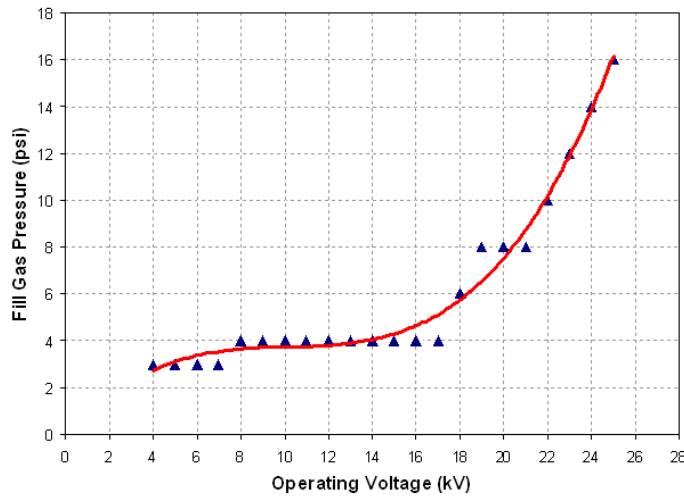


Fig. 3.8 Operating voltage as function of gas pressure.

It is observed that up to 18kV, only a marginal increase in pressure (from 3psi to 6psi) is required for increasing the hold-off voltage but in the operating range of 19kV to 25kV, there is a steep rise in the filling gas pressure (from 7psi to 16psi) indicating the limitation of switch hold-off capacity. The reliable operation regime of the sparkgap switch has been shown in the graph in Fig. 3.9. As shown in the graph, for reliable triggering, main gap voltage ranges from 7kV to 22kV.

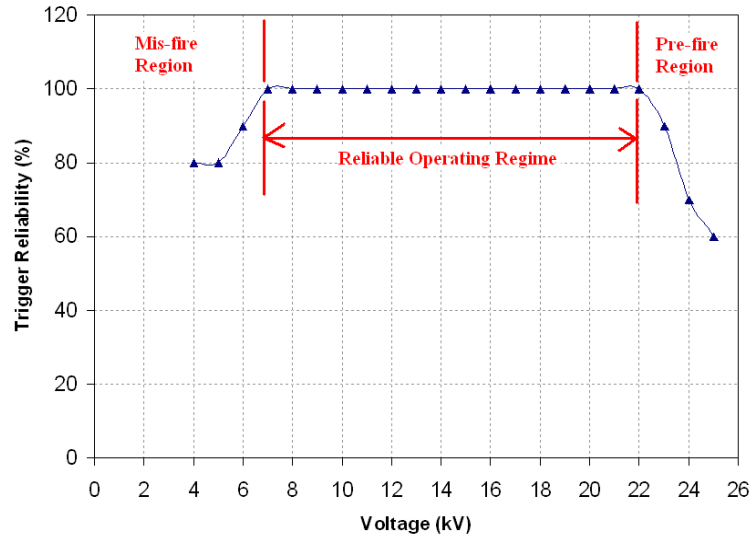


Fig. 3.9 Operating Voltage as function of trigger reliability (%).

3.4.2.2.2 Breakdown Time Delay Measurement

Time delay measurements mainly indicate the influence of the electric field across the main gap and filling gas pressure. The breakdown time delay of the switch is characterized as the difference in timing of two signals *i.e.* $t_2 - t_1$. Here, t_1 is the time of arrival of trigger voltage pulse at the trigger pin and t_2 is the time at which the sparkgap switch becomes fully conducting. A simple empirical description of the behavior of breakdown in high pressure gases has been given as [184]:

$$\rho t_{bd} = 97800 \left(\frac{E}{\rho} \right)^{-3.44} \quad (3.17)$$

Here, ρ is the gas density in g/cm^3 , t_{bd} is the breakdown time in seconds and E is the magnitude of mean electric field across the gap in kV/cm. It can be seen from the above relation that the

breakdown time is a strong function of electric field across the gap $\propto E^{-3.44}$ and thus depends on the voltage applied to the switch as well. Experimentally this has been observed as indicated in the graph in Fig 3.10 and Fig. 3.11. It can be observed that at lower voltages as the mode of operation is mainly BAE therefore time delay is considerably longer and the behavior of the switch is erratic. But at higher voltages the delay reduces significantly in the range of 8kV to 18kV.

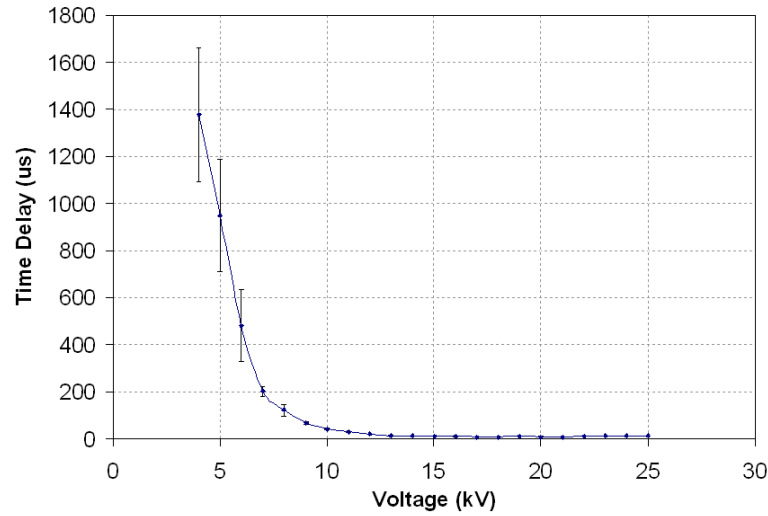


Fig. 3.10 Time delay (t_{bd}) as function of main gap voltage.

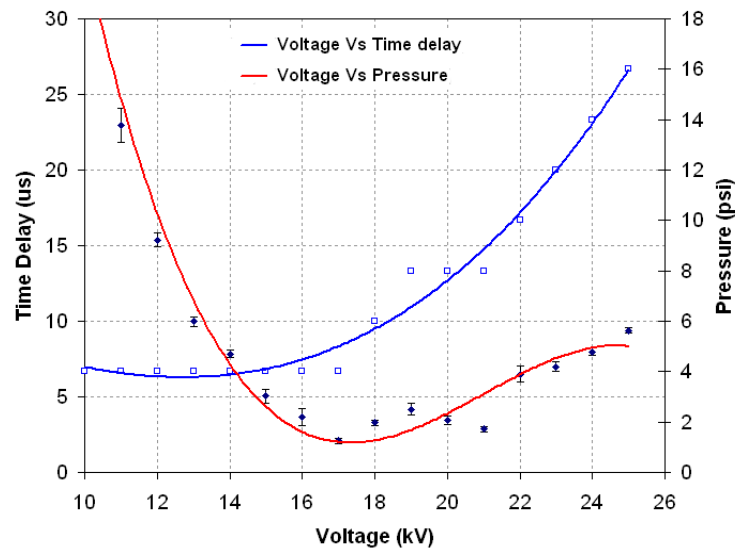


Fig. 3.11 Breakdown time delay as function of pressure.

Upon analyzing the characterization data in the operation range of 10 to 25kV, it is observed that when the pressure starts to play a dominating role in voltage hold-off in the operation range of 18 to 25kV then delay also starts increasing as depicted in the graph in Fig. 3.11. This upward trend in time delay for >18kV operation, indicates the limit of hold off capacity of the sparkgap.

3.4.2.2.3 Lifetime Estimation

The lifetime of the sparkgap depends on many factors such as electrode material, electrode geometry, electrode erosion rate per shot and pulse repetition rate [185]. Depending upon the maximum permissible electrode volume-loss for maintaining stable switching, the lifetime of the switch can be estimated. The electrode material loss in each shot depends on coulomb transfer loading (i.e. net charge transferred during the each shot) of the sparkgap switch. In this sparkgap switch the electrodes have been designed to maintain stable switching up to erosion depths of 2mm (cathode together with anode) [181]. Low erosion is the most important factor to be considered in the selection of electrode material. Spark erosion rates of various materials that are normally used as sparkgap electrodes are shown in Table 3.2.

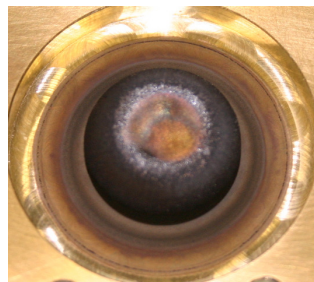
Table 3.2 Spark erosion rate and density of various electrode materials [185].

Material	Density (gm/ cm ³)	Erosion rate (μcm ³ C ⁻¹)
Elkonite (Cu-W)	12.5 – 17.2	5.44
Brass	8.4 – 8.75	7.59
Stainless Steel (SS)	7.85	9.14
Aluminum	2.75	17.37

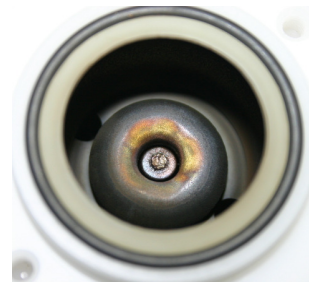
The erosion results are presented in terms of the volume of eroded material per coulomb of charge transferred (cm³ C⁻¹). As seen from the above data Elkonite has the lowest erosion rate, but it is too costly and not easy to machine. Aluminum is the cheapest option but it has highest erosion rate of all the materials along with poor mechanical strength. Brass has been chosen as compromise between various factors like low spark erosion characteristics, mechanical and thermal properties, availability

and low cost.

According to Kousoubis and MacGregor [185], the erosion rate for brass is $\sim 8 \times 10^{-6} \text{ cm}^3/\text{C}$. Thus for 200J with a maximum coulomb transfer of $\sim 80 \text{ mC}$ per shot, the switch has an estimated useful life of $\sim 2 \times 10^6$ shots amounting to total coulomb transfer of $> 140 \text{ kC}$ over its working life. In the above estimation it has been assumed that electrode erosion rates for anode and cathode is uniform. Erosion of electrodes after ~ 5000 shots of operation is shown below in Fig. 3.12.



Opposite end electrode



Adjacent end electrode

Fig. 3.12 Sparkgap electrodes showing erosion after ~ 5000 shots of operation.

Based on experimental characterization results, specifications of the newly developed sparkgap switch are summarized in Table 3.3.

Table 3.3 Specifications of indigenously developed trigatron switch.	
Maximum Operating Voltage	25kV
Gas Pressure Range	2 – 20psi (pressurized nitrogen gas)
Operating Voltage Range	8kV – 18kV
Maximum Current	100kA
Series Inductance	$< 5 \text{ nH}$
Breakdown time delay (t_{bd})	$< 50 \mu\text{s}$
Trigger Voltage	$\geq -20 \text{ kV}$ (typically)
Main Electrode Material	Brass (9.5 mm diameter)
Trigger Electrode	Commercial sparkplug
Insulating Material	Delrin
Discharge Gap	6mm
Physical Dimensions	100mm (dia.) $\times$ 30mm (height)
Weight	$\sim 1 \text{ kg}$

It may be noted that observed breakdown time delay of $<50\mu\text{s}$ is due to the rise time limitation of applied trigger pulse characteristics, performance of which is illustrated in the following section.

3.4.3 Triggering System

Sparkgap switches endure high voltage and high current operation while achieving precise timing, but its switching performance is mainly decided by the applied trigger pulse characteristics (produced by the trigger module). A major factor that affects the performance of the sparkgap is the trigger pulse rise time. The schematic of the circuit is shown in Fig. 3.13.

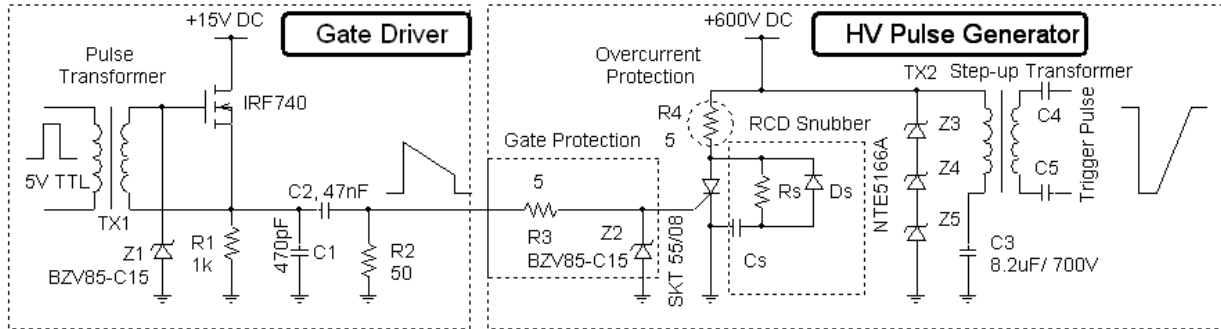


Fig. 3.13 Simplified circuit diagram of the trigger module.

As shown in the schematic SKT 55/08 thyristor from Semikron Inc. has been used as discharge switch. It has a breakdown voltage rating of 800V, with an average current rating of 55A and peak current carrying capability of about 1300A. To improve the reverse recovery and to reset the transformer in repetitive mode an anti-parallel diode DSE112-12A is connected across it. It is rated at 1200V, 11A and has reverse recovery time of 50ns. To avoid complications in switching, the thyristor is used for discharging capacitor through transformer primary in the source grounded configuration.

The operation of the circuit is as follows: when a low voltage pulse ($\approx 5\text{V}/\mu\text{s}$) is fed as an input to trigger module (through the pulse transformer), the gate drive unit (based on MOSFET, #IRF740 operation) generates a $+15\text{V}/0.6\mu\text{s}$ pulse that is applied to the gate of the thyristor. The thyristor

conducts immediately and discharges the capacitor (of 8.2 μ F charged up to 700V) through the primary winding of the step-up transformer. A commercially available flyback transformer has been used for stepping of voltage pulse as it has an epoxy encapsulated high voltage secondary winding. A high voltage pulse of more than ~24kV is induced across the secondary winding of the step up transformer as shown in the graph in Fig. 3.14.

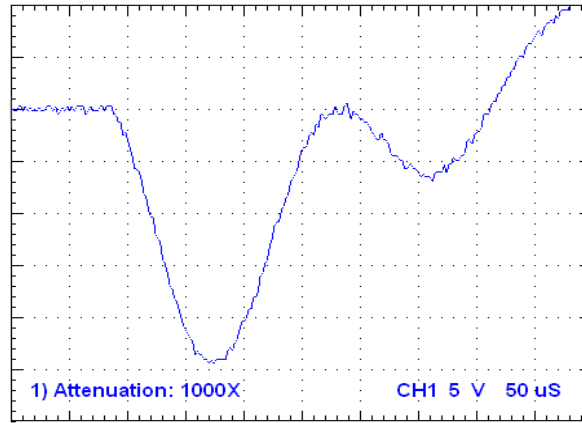


Fig. 3.14 High Voltage output waveform of trigger pulse.

The rise time and half width of the pulse are about ~50 μ s and 100 μ s, respectively. For bringing the connections out from primary and secondary, short length RG58 coaxial cables have been used to avoid any increase in inductance.

The critical issue in solid state drivers is the survivability of switches. As in transformer-coupled topologies, the maximum voltage developed across the device is normally at turn-off because at that point the rate of change in current (di/dt) is highest [186]. The switching element in such applications must have a voltage rating high enough to handle the maximum input voltage, reflected secondary voltage and voltage spike which may occur due to the transformer leakage inductance (*i.e.* $V_L = L_l(di/dt)$). Typically, the voltage spike due to the leakage inductance (L_l) of the transformer is estimated to be thirty percent of the applied voltage. Thus, operation at 600V ensures a safe margin of operation for the thyristor.

To prevent damage due to over voltage transients RCD charge-discharge snubber protection has also been implemented. To ensure over voltage and di/dt protection, a 15V zener diode, # BZV85-C15 and series resistance of 5Ω has been included in the circuit. Another resistance of 5Ω has been used in the discharge circuit to avoid excessive loading on the switching device. The purpose of using decoupling capacitors (C4 & C5 of rating 4.7nF/40kV) at the final output is to isolate the trigger circuit from the DC charging circuit of the pulsed power system. Specifications of the indigenously developed trigger module are summarized in Table 3.4.

Table 3.4 Specifications of trigger module.	
Operating mode	Single pulse (manual), External
Trigger input	Electrical (5V TTL/ 5V – 15V)
Maximum output voltage	- 25kV
Pulse rise time	~50 μ s
Pulse duration	~100 μ s
Energy stored	~1.5 Joule
Load impedance	Direct short to infinite
Pulse repetition rate	5 Hz (maximum)
Input power requirements	200 – 265VAC, 1 ϕ , 50Hz

3.4.4 Plasma Focus Tube

The focus tube assembly comprises a coaxial electrode system, insulator and vacuum chamber. Following the design procedure discussed under section 3.2 (and on the basis of obtained short circuit results) four different anode geometries were tested for the optimization of the FMPF-1 device, and are shown in Fig. 3.15. (Performance evaluation for each geometry has been detailed in section 5.2.2 of chapter 5).

The typical dimensions of tested anodes are: length $\sim 16 \pm 1$ mm and diameter ~ 12 mm. A circular sharp edge at the lower end of the anode promotes symmetrical ignition of the current sheath around anode periphery. The selection of appropriate materials for anode and insulator sleeve was done in accordance with results obtained during optimization experiments and comparative studies reported earlier for enhanced neutron yield.

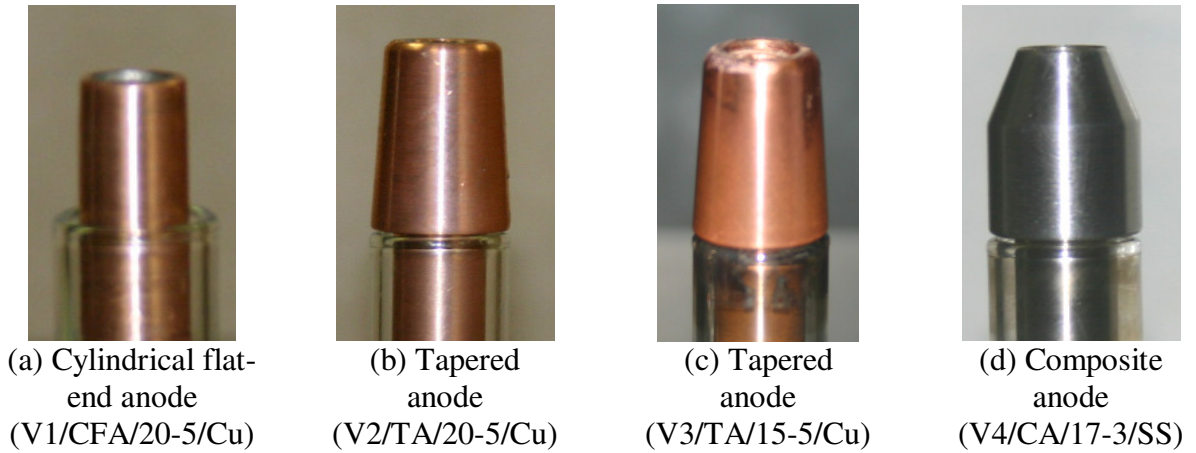


Fig. 3.15 Various anode geometries/ designs that have been tested.

Although in the preliminary phase, experiments have been performed with copper anodes but it was experienced that shot-life of the insulator degrades faster with the use of copper anodes due to its higher erosion rate and low melting point. The deposition of vaporized copper condensates on the insulator surface over the period of time, due to which the surface resistivity of insulator reduces significantly and results in current leakage or shunting, thereby effectively reducing the pinch current I_p and hence the focusing [187]. The microscopic (SEM) image of the insulator surface pattern of such damage (after ~5000 shots) is shown in Fig. 3.16.

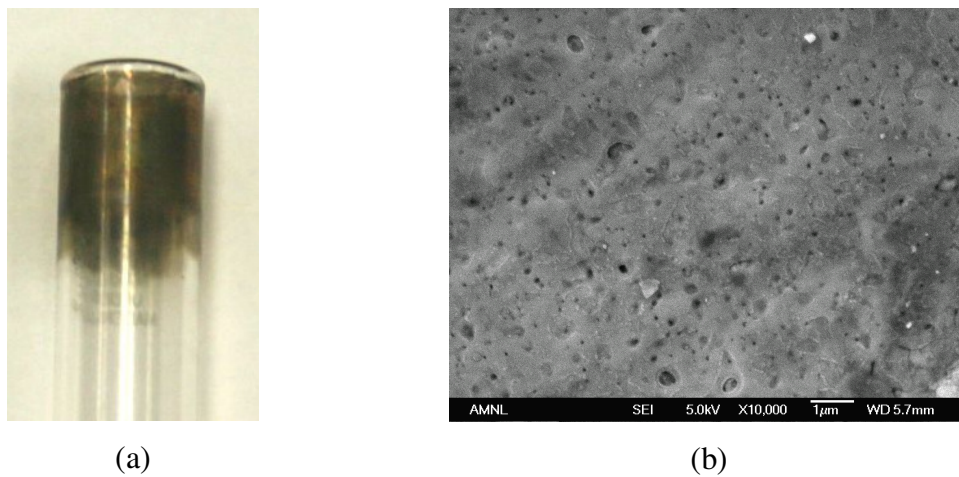


Fig. 3.16 (a) Damaged Pyrex insulator sleeve and (b) SEM image of insulator surface pattern after ~5000 shots of operation.

For the reason of experimental evidence mentioned above, in major phase of the experiments stainless steel (SS) anode were used. Stainless steel anodes have also been reported effective in producing neutron yields higher than copper anodes [188].

In the selection of insulator sleeve material, one of the prime considerations is the dielectric constant. A study done by Beg *et al* [189] indicated significant enhancement in neutron yield with the use of high dielectric constant sleeve material. Dielectric constants of various sleeve materials have been compared in the Table 3.5 [189,190].

Table 3.5 Dielectric constants and densities of various insulators.

Material	Dielectric Constant	Density (gm/ cm ³)
Alumina/ Ceramic	4.5 – 8.4	4.14
Pyrex	4.5 – 6.0	2.05
Nylon	3.55	1.09
Perspex	2.76 – 3.12	1.08
Teflon	2.04	2.04
Quartz	3.7 – 4.2	2.17

Ceramic has the highest dielectric constant, but as a compromise between electric characteristics, easy availability and low cost; in our device, ‘Pyrex’ has been used. An insulator sleeve of Pyrex glass with a breakdown length and thickness of 5mm and 2mm, respectively, was placed between the anode and cathode. A photograph of the anode assembly is shown in Fig. 3.17.

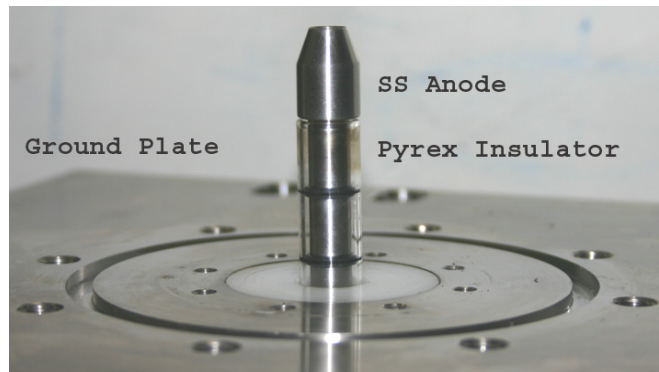
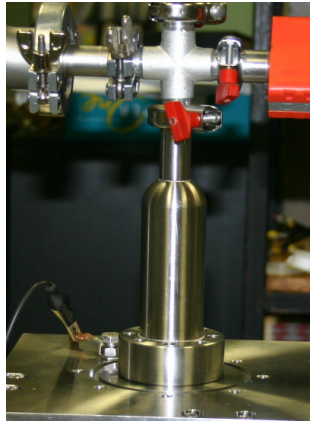
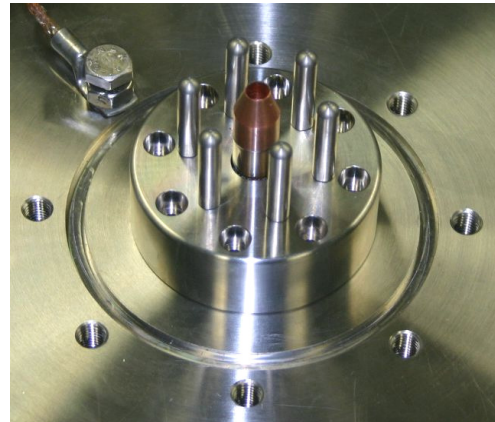


Fig. 3.17 Coaxial electrode assembly arrangement in FMPF-1.

As cathodes, tubular and squirrel cage cathode geometries have been experimented with and their performance has been thoroughly compared and analyzed in section 5.2 of chapter 5. A photograph of the coaxial electrode assemblies with the use of (a) tubular and (b) squirrel cage cathode geometries is shown in Fig. 3.18.



(a) Tubular cathode



(a) Squirrel cage cathode

Fig. 3.18 Cathode geometries/ designs that have been experimented.

The advantage of the ‘tubular cathode’ is that while serving the purpose of cathode, its chamber walls also acts as vacuum envelope hence the compactness of focus tube assembly is enhanced. However, during optimization experiments it was evident that the snowplow efficiency is severely affected by use of tubular cathode geometry (reported later in detail in section 5.2.5 of chapter 5). The chamber wall of the tubular cathode used is 30mm inner diameter and it has been fabricated from stainless material. The wall thickness is 5mm. The squirrel cage cathode assembly consists of six 6mm diameter, 30mm long SS rods, uniformly spaced on a coaxial circumference of 30mm diameter, concentric with the anode axis. While using squirrel cage cathode, a separate chamber with customized ports for attaching various diagnostics has been used as a vacuum housing. A photograph of the vacuum chamber used, for the squirrel cage geometry cathode, with various diagnostic attached to it is shown in Fig. 3.19.

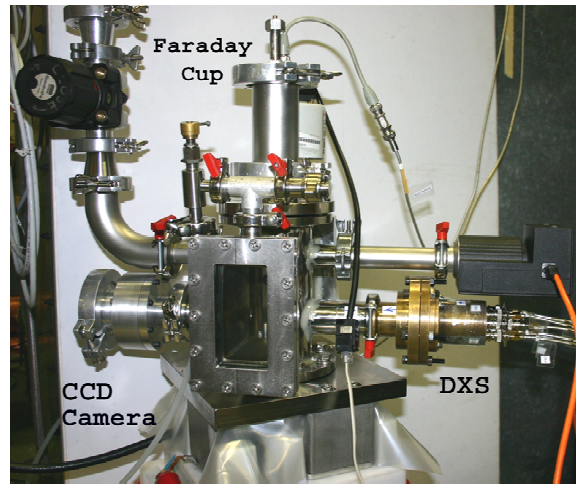


Fig. 3.19 Photograph of the vacuum chamber with various diagnostic attachments.

3.4.5 Connections and Assembly

The capacitor bank of FMPF-1 was made of four $0.6\mu\text{F}$, 30kV low inductance capacitors (total weight $\sim 20\text{kg}$), connected in parallel in a compact layout through a common transmission plate assembly (made up of SS304) of size $0.2\text{m} \times 0.2\text{m}$ as shown in Fig. 3.20.

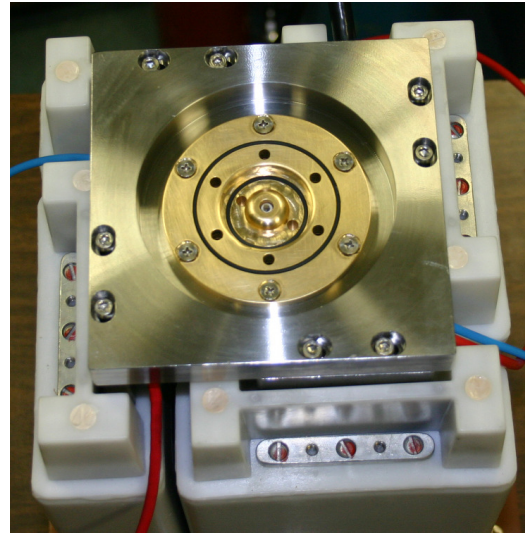
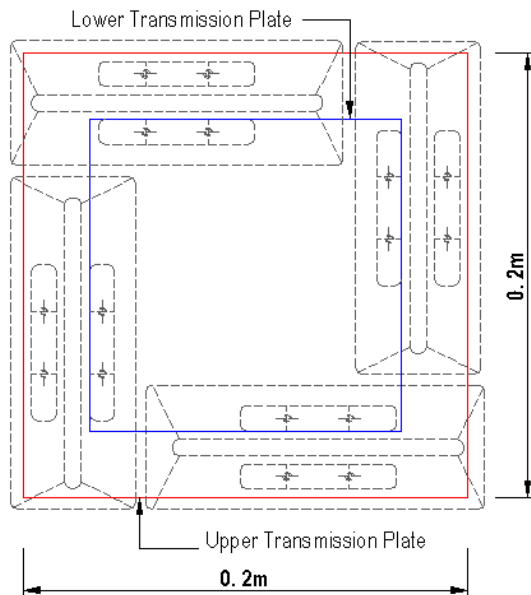


Fig. 3.20 Schematic and actual layout of the capacitors in staggered configuration.

As insulation between the flat plate transmission lines, four layers of 125 μm thick Mylar were used. Mylar has a dielectric strength of typically 17kV/25 μm *i.e.* 17kV/mil. Since, the shot-life of the dielectric degrades as $V_0^{-7.5}$ and the DC breakdown strength of the dielectric scales as square root of the dielectric thickness (t) given as:

$$V_B \propto \sqrt{t} \quad (3.18)$$

The overall shot-life of the 500 μm thick Mylar insulation (having breakdown strength of 76kV DC) at 13kV working voltage corresponds to $\sim 10^6$ charge-discharge cycles which typically matches with the lifetime of the capacitor. Thus, the provided insulation thickness ensures reliable operation without insulation failure. Using the expression mentioned under section 3.3.4, the total inductance contributed by the transmission plate assembly along with its connections is $\sim 8\text{nH}$. A schematic of the integrated assembly is shown in Fig. 3.21.

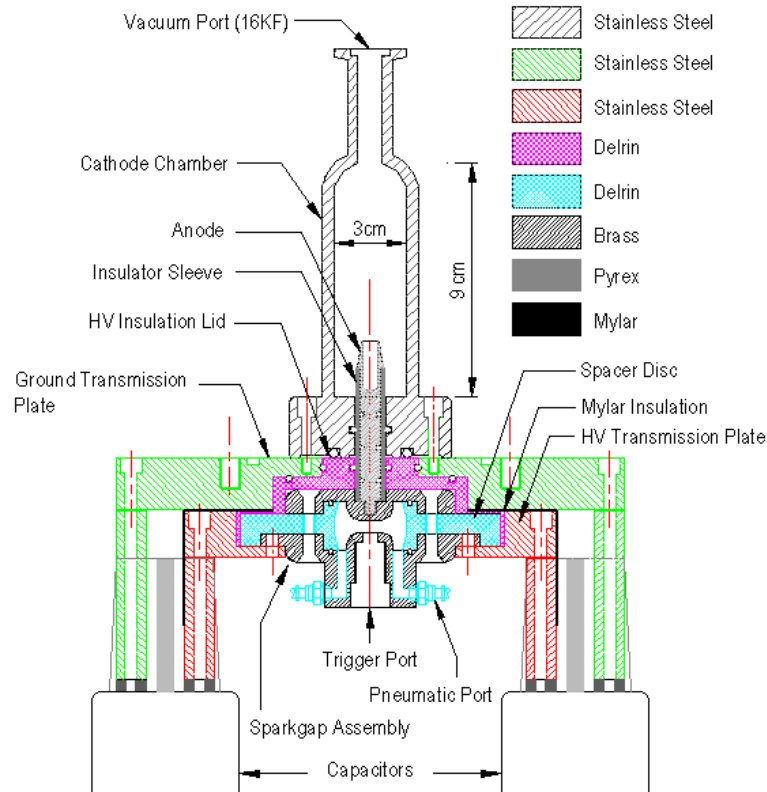
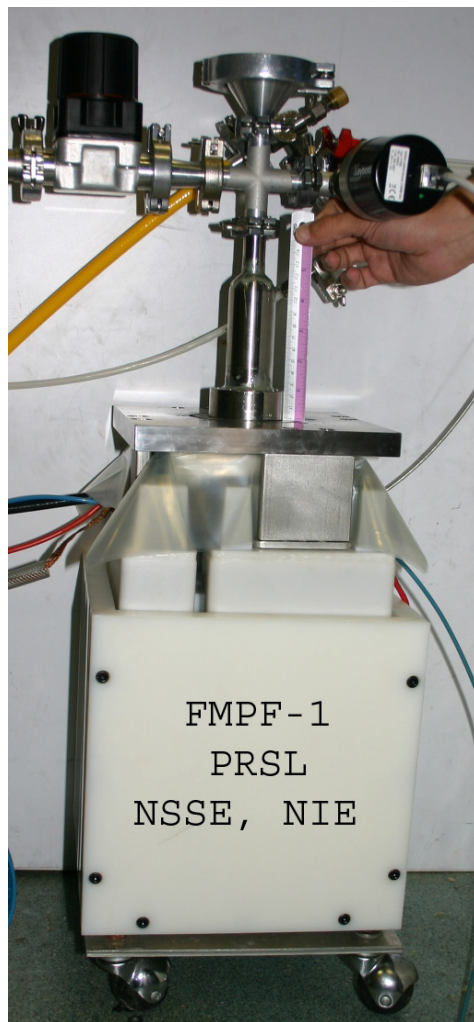


Fig. 3.21 Schematic of integrated assembly of FMPF-1 with tubular cathode.

As is can be seen from this schematic, the connections between capacitor bank, sparkgap and plasma focus electrode assembly have been minimized by embedding the sparkgap within the transmission line assembly of the capacitor bank and by directly interfacing the plasma focus head to the discharge end of sparkgap. The measured total driver bank inductance ($\approx$ capacitor bank + transmission line + sparkgap) is about $27 \pm 2 \text{ nH}$. A photograph of the integrated assembly of FMPF-1 device is shown in Fig. 3.22 (a) along with subsystems (b-d) used in the experimental setup.



(a) The FMPF-1 device



(b) Vacuum System



(c) Trigger Module



(d) Charging Power Supply

Fig. 3.22 Integrated assembly of FMPF-1 and sub-systems.

It is important to note that in the FMPF-1 device, the transmission plate assemblies have been made up of 10mm thick SS304 plates (of size $0.2\text{m} \times 0.2\text{m}$) so that the inertia of the plate assembly overcomes the stress (*i.e.* repulsive force) generated due to the flow of large currents. If the parallel plate assembly had been made light weight, using thin sheets, then it may have had a tendency to fly apart during flow of discharge current and the significant amount of delivered energy may be converted into mechanical deformation rather than going in to plasma load. Therefore, plate assemblies were intentionally designed to be heavy.

3.4.6 HV Charging Power Supply

A constant voltage power supply ‘HV07-XR2-15K’ (procured from M/s Anashan Leadsun Electronics Co. Ltd., China) has been used for charging the modular capacitor bank of FMPF-1 device. It is a 15kV/100mA power supply especially designed for charging capacitors in pulsed power applications. The average power rating is 1.5kW. It incorporates a high-frequency series-resonant inverter, operating at 53 kHz for efficient generation of the output power. A high-performance control module precisely regulates the output voltage according to manual presets along with inbuilt overload and short circuit protection. The output voltage is adjustable over the range of 2 to 15kV. A $1\text{k}\Omega$ high power wire wound resistor has been used for the charging of capacitor bank. This resistor also serves the purpose of protecting the power supply from short circuit conditions.

Several key experiments that were performed on the FMPF-1 device are reported and discussed in detail in chapter 5 in section 5.2.

3.5 The FMPF-2 Device

After gaining experience with the FMPF-1 device and obtaining reliability in the performance of single shot discharges, as an next step for obtaining repetitive performance; the second version of the fast miniature plasma focus device ‘FMPF-2’ was designed and constructed.

From the obtained results of characterization experiments on the optimized FMPF-1 device, it was realized that even though the design and integration of the device was excellent resulting in a very compact system, the efficiency of pulsed power system suffered considerably due to the high equivalent series inductance (*ESL*) and resistance (*ESR*) of the capacitors used (*i.e.* *ESL* $\sim 70\text{nH}$ and *ESR* $\sim 250\text{m}\Omega$). The peak discharge current under short circuit conditions was limited to $\approx 80\text{kA}$ at 200J . Also the ratings of capacitors, sparkgap switch and high voltage charging power supply used in FMPF-1 device restricted operation to effectively single shot mode (as the maximum repetition rate was limited to 0.5Hz).

Hence, the main objective that has been taken into consideration in the design of first version of repetitive fast miniature plasma focus FMPF-2 device, for the enhancement in time averaged neutron yield; was to customize and operate the pulsed power system for repetitive mode, while considering important measures necessary for the upgrading of the pulsed power system.

In the following sub-sections, after describing the upgrading approach and major design considerations for the FMPF-2 device; details of the upgraded components, their performance and the technical constitution is thoroughly described and elaborated.

3.5.1 Upgradation Approach and Design Considerations for FMPF-2

As mentioned earlier, the low efficiency (*i.e.* low peak discharge current delivering capacity) of the FMPF-1 pulsed power system was mainly due to capacitors; therefore the first target was to design an efficient capacitor bank of $\sim 200\text{J}$ energy; using commercially available energy storage class capacitors, which have low equivalent series inductance (*ESL*) and low equivalent series resistance (*ESR*) and are rated up to 30kV , with shot-life of $>10^6$ charge/ discharge cycles, withstanding 80% voltage reversal in repetitive mode.

Another important consideration for a low inductance capacitor bank is the shot-life/ life

expectancy of the energy transfer switches, since the system will be made to operate in repetitive mode. As an implication of low driver inductance, voltage reversal in the circuit rises considerably and the discharge oscillates for a longer duration, even after termination of the pinch. This leads to significant electrode damage in deployed switches (*e.g.* sparkgap), because when the voltage reverses polarity, it drives ion/electrons into either end of electrodes and this causes ablation and sputtering of the electrodes. Ablated electrode material gets coated over dielectric spacers used for separation of the electrodes, leading to erratic triggering/ pre-firing of switches as more shots are fired. Hence, to overcome this problem either of two schemes may be implemented (i) using a single switch with sufficiently large coulomb transfer capacity (ii) using multiple switches in parallel to synchronously transfer the current from different modules in to the load. Both of these schemes have their pros and cons; in the first scheme although switching and triggering requirements are much simplified and higher end performance (low rise time and high dV/dt) of the trigger pulse characteristics is not required the drawback is that (a) inductance contribution by the single switch is significant, and (b) shot-life of the switch considerably degrades due to loading of higher coulomb transfer. In the second scheme of deploying multiple switches in parallel, brings advantages of reduced overall switch inductance and enhanced shot-life (since the coulomb loading is shared), but for “synchronized discharge”, the switching and triggering requirements are highly demanding *i.e.* switches with inherently low breakdown time delay t_{bd} and simultaneously generated multiple trigger pulses having very low jitter, low rise time and high dV/dt are required. Also sufficient measures are needed to avoid the cross flow of energy, between the switching modules along with the arrangement for individual charging of modules. Thus, a judicious choice of the scheme is required to achieve optimum performance of the system.

For running the pulsed power system in the repetitive mode, the requirement for high capacity constant power charger (CPC) is a must. As these special chargers are very costly, depending upon the

requirement of repetition rate and the energy to be stored in the system, an appropriate power rating of the charger must be chosen. Also a recharge command and control circuit is needed to be interfaced between the CPC and capacitor bank for operation in repetitive mode [191]. Since we planned to operate the system in repetitive mode up to a maximum of 10Hz, the CPC rating was worked out accordingly and is illustrated in the upcoming section.

3.5.2 Capacitor Bank

Energy storage capacitors play a key role in deciding the performance and efficiency of pulsed power systems and their choice is generally a compromise between technical requirement, cost and performance. The 2.4 μ F capacitor bank of the FMPF-2 device comprises a single module of eight 0.3 μ F, 30 kV low inductance plastic case capacitors (#37323) procured from General Atomics, USA. For comparing the performance of these newly procured GA capacitors with locally produced capacitors as used in FMPF-1, a short circuit test was performed in the common setup by connecting a low inductance sparkgap close to the discharge end. The recorded oscillogram of this test on both the capacitors at 10kV discharge is shown for comparison in Fig. 3.23.

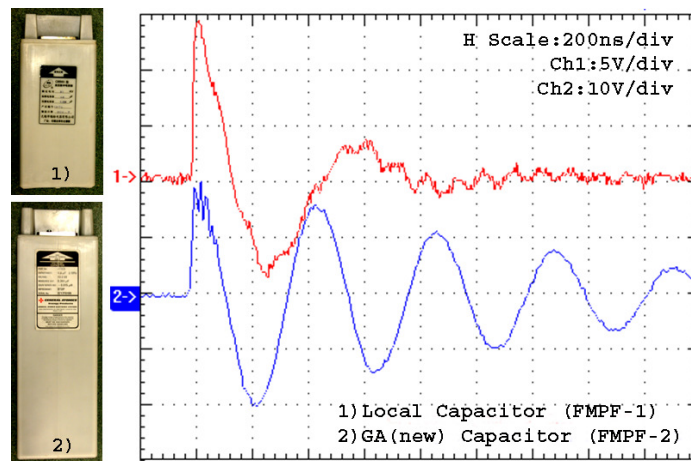


Fig. 3.23 Oscillogram of current derivative signal under short circuit tests of capacitors.

A very significant difference in reversals/ ringing in the discharge traces may be observed. In

the discharge trace for the local capacitors utilized in FMPF-1, the voltage reversal is considerably lower ($\sim 35\%$) indicating high equivalent series inductance (ESL) and resistance (ESR)) for the capacitors, conversely in the discharge trace of newly imported (GA) capacitors, the voltage reversal has been considerably high ($>80\%$) indicating low equivalent series inductance (ESL) and resistance (ESR)) of these capacitors. The detailed specifications of capacitors (#37323, GA) used in the construction of FMPF-2 capacitor bank are summarized as under:

Table. 3.6 Specifications of Capacitors used in 'FMPF-2'.	
Capacitance	$0.3\mu\text{F} (\pm 10\%)$
Maximum Charging Voltage	30kV
Maximum Stored Energy @30kV	135J
Series Inductance (ESL)	$\sim 15\text{nH}$
Series Resistance (ESR)	$\sim 60\text{m}\Omega$
Voltage Reversal (rated)	20%
Maximum Current	50kA
Shot-life	$>10^8$ charge discharge cycles
Physical Dimensions (H×B×W)	$470\text{mm} \times 150\text{mm} \times 58\text{mm}$
Weight	5.5kgs
Model No./ Make	#37323/ General Atomics, USA

The layout of arrangement of capacitors in FMPF-2 device is shown in Fig. 3.24.

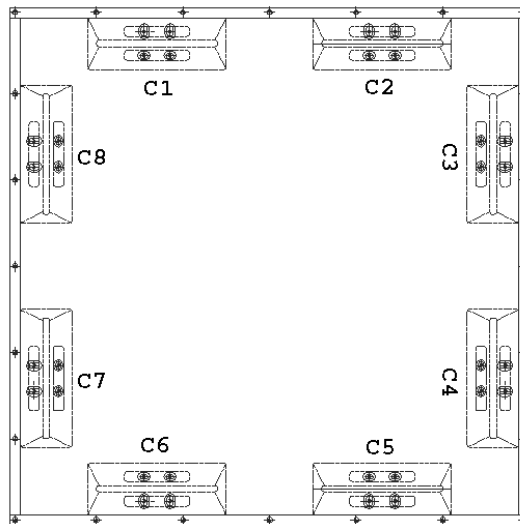


Fig. 3.24 Layout of the capacitor bank in FMPF-2 device.

As shown in the schematic, the FMPF-2 device has enclosed structure of size $0.6\text{m} \times 0.6\text{m} \times 0.75\text{m}$ and the mainframe is made up of high mechanical strength, extruded Aluminum alloy 6061 plates. The eight capacitors were arranged symmetrically on the four sides of the box in the mainframe. Each capacitance is rated as $0.3\mu\text{F}/30\text{kV}$ and the total capacitance has been measured to be $2.4\mu\text{F}$. This enclosed structure has specifically been chosen to prevent interference of discharge generated electromagnetic noise with outside diagnostics.

As mentioned under section 3.3.2, the shot-life (life expectancy) of capacitors is crucial for repetitive pulsed power systems and is severely affected by charging voltage and reversal in the circuit hence, using the empirical expressions illustrated in Eq. 3.9 and 3.10, the cumulative effect of voltage reversal in the circuit and actual operating voltage on shot-life of capacitors can be observed from the plots shown in Fig. 3.25.

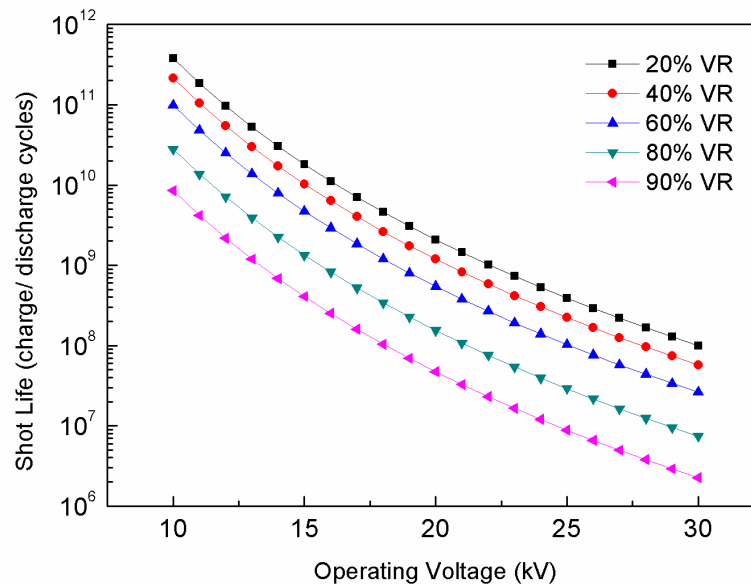


Fig. 3.25 Shot-life of pulse discharge capacitor (#37323, GA, USA)

as a function of operating voltage and voltage reversal (VR or k) in the circuit.

It may be realized from above plots that the capacitor voltage and the voltage reversal in circuit are the two most important life determining parameters. A minor reduction in operating voltage and voltage reversal results in a considerable increase in shot-life and vice-versa. In our case the capacitors

used are rated at 30kV maximum charging voltage with 20% reversal, but though typically with plasma focus as load, ~80% of voltage reversal was expected in the circuit therefore we have chosen the maximum operating voltage as 15kV which is 50% of the maximum rating. By operating the capacitor in the range of 12–15kV *i.e.* at ~40–50% of its rated value with 80% reversal, the life expectancy enhances by more than two orders of magnitude for similar circuit condition (for *e.g.* at 80% voltage reversal in the circuit, the shot-life enhances from 7.41×10^6 at 30kV operation to 1.34×10^9 charge/discharge cycles at 15kV operation), which corresponds to a cumulative shot-life of 3 years of continuous operation at a repetition rate of 10Hz.

3.5.3 Triggerable Sparkgap Switch

In FMPF-2 device owing to familiarization with operational characteristics and simplicity in the trigger requirements, a gas pressurized 100kA/ 40kV hold-off trigatron sparkgap switch model SG-101M-75C procured from R. E. Beverly III and Associates, USA has been used [192]. The photograph of this switch is shown in Fig. 3.26 and the detailed specifications are summarized in Table 3.7.

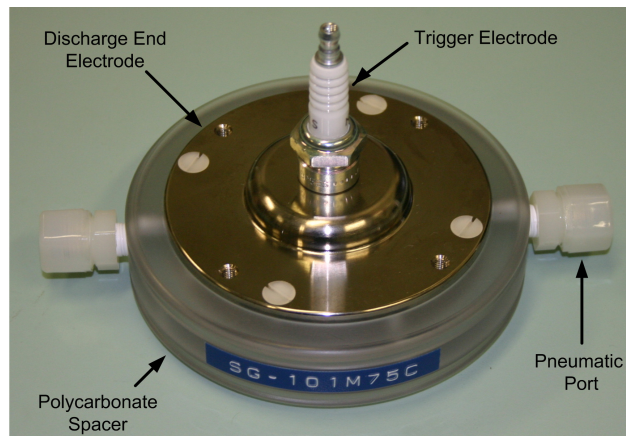


Fig. 3.26 Photograph of SG-101M-75C trigatron switch.

The inductance of the switch is $<20\text{nH}$ and it is capable of operating in repetitive mode up to 10Hz. A commercially available spark-plug is used as the trigger electrode. The electrodes are of

sintered low erosion material *i.e.* Elkonite/ W-Cu alloy to ensure long life and a low probability of misfire or prefire [185].

Table 3.7 Specifications of Trigatron Sparkgap Switch Model SG-101M-75C.	
Operating Gas/ Pressure Range	N ₂ / 2 – 20psi
Maximum Operating Voltage	40kV
Operating Voltage Range	10kV – 40kV
Maximum Current	100kA
Switch Inductance	< 20nH
Breakdown time delay (t_{bd})	< 50ns
Trigger Voltage	≥ 20 kV (typically)
Main Electrode Material	Sintered W-Cu Alloy
Trigger Electrode	Commercial sparkplug
Insulating Material	Polycarbonate
Discharge Gap	6.35 mm
Physical Dimensions	127mm (dia.) × 40.6mm (height)
Weight	~1 kg

The operating voltage curve as a function of gauge pressure for Nitrogen fill is shown in Fig. 3.27.

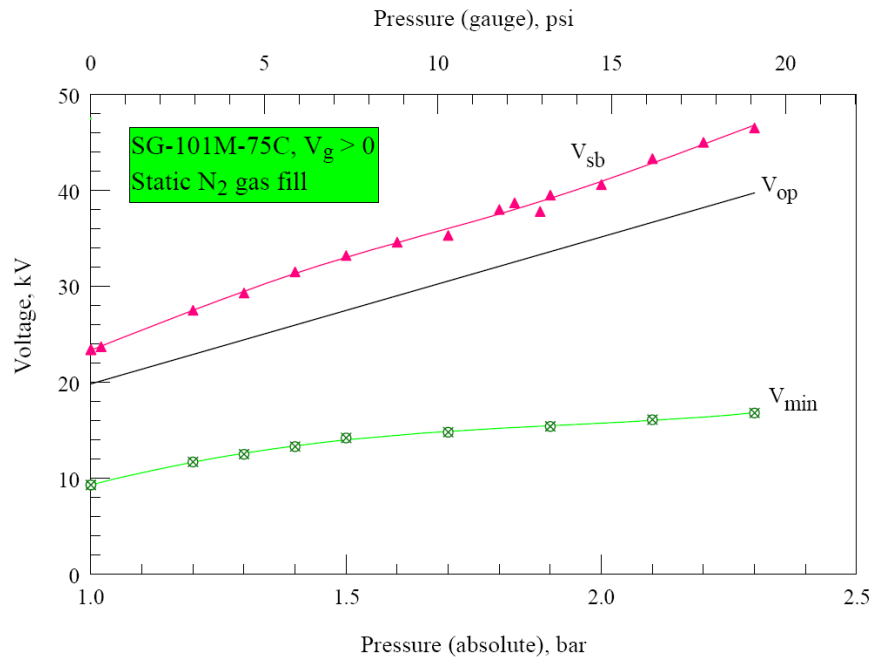


Fig. 3.27 Operation curve as function of gas pressure for trigatron switch.

Under optimum triggering conditions, the switch is triggerable in the range 10kV to 40kV by (i) choice of the filling gas and (ii) adjustment of the internal gas pressure [178]. For lower operating ranges ($V_{op} \sim 6\text{-}19\text{kV}$) Hydrogen (H_2) is recommended, where as for moderate ($V_{op} \sim 18\text{-}33\text{kV}$) and higher ($V_{op} \sim 20\text{-}40\text{kV}$) operating ranges, $\text{SF}_6(10\%):\text{Ar}(90\%)$ mixture and Nitrogen (N_2) gases are recommended respectively [192]. For reliable switching, operating voltages are typically chosen at 85% of the self breakdown voltages (SBV) at respective pressures.

If the filling gas is not continuously purged during repetitive operation mode, the stale gas fill may either reduce the self-breakdown voltage dramatically or may result in runaway (self triggering). Hence for maintaining speedy recovery and stable break down characteristics, and the flushing of hot spark residues after every discharge is necessarily required. The minimum gas flow rate/ purge velocity is function of internal gas volume γ (cm^3), discharge repetition frequency f (pulses per second) and exchange fraction ε . An exchange fraction of $\varepsilon = 1$ corresponds to one complete internal gas volume replacement followed by each discharge. The empirical relation for estimation of minimum gas flow rate (dV_{gas}/dt) is [193]:

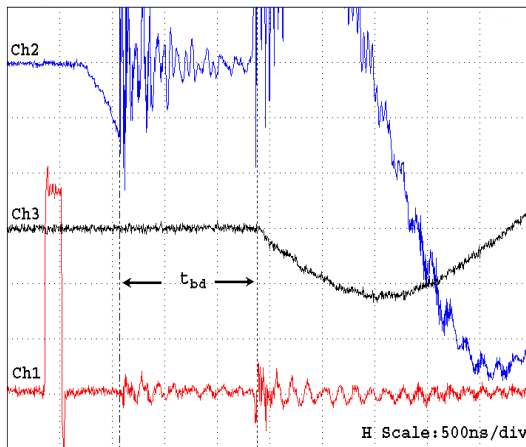
$$\frac{dV_{gas}}{dt} = 0.06 f \varepsilon \gamma \text{ (LPM)} \quad (3.19)$$

Since the effective volume of the sparkgap switch model SG-101M-75C is $\sim 70\text{cm}^3$ therefore for the operation of the switch in the range of 1-10Hz, the minimum gas flow rate requirement (*i.e.* $\varepsilon = 1$) is 4-40LPM (liters per minute). It may be noted that even though switches with Hydrogen as filling gas may be operated at the highest repetition rates along with fastest turn-on and smallest electrode erosion [193], in the present device, mainly due to stringent safety considerations (as a consequence of high flushing rate requirement) Nitrogen has been used as the free-flow gas instead of Hydrogen. Since with use of Nitrogen, pressurization is only required for $>20\text{kV}$ operation, therefore in our case at $\sim 15\text{kV}$ operating voltage, only free running gas condition was maintained at the desired

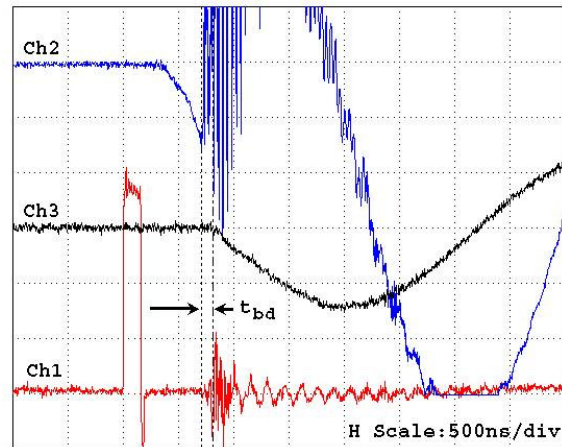
flow rate. However during the experiments, a marginal increase in the residual pressure of <1psi was observed due to continuous flushing.

As has been discussed and illustrated under section 3.4.2.2.2, the breakdown time delay in pressurized sparkgap switches is mainly influenced by electric field across the main gap and the filling gas pressure. In our setup, the switch has been used under free running gas condition, therefore the breakdown time delay is mainly affected by voltage applied to the switch.

For investigating the breakdown time delay characteristics of the SG-101M-75C trigatron switches at various voltages, characterization experiments were performed in the voltage range of 12kV to 18kV in ‘isolated load series configuration’ [193] as shown in the schematic in Fig. 3.7. All data have been collected under dynamic N_2 gas flow conditions while maintaining the flow rate of 5 LPM. A trigger pulse of -28kV was supplied by one of the channels of the trigger system. The residual short circuit inductance represents the load (ZL). For the diagnostics, custom designed Rogowski coil and high voltage probe (Tektronix P6015) were used along with Yokogawa DL9140, 5GS/s, and 1GHz oscilloscope for data capture. The sample oscilloscopic traces depicting the breakdown time delay measurement of sparkgap switch at charging voltages - 12kV, 14kV, 16kV and 18kV have been shown in Fig. 3.28(a-d).



(a) Operating at 12kV



(b) Operating at 14kV

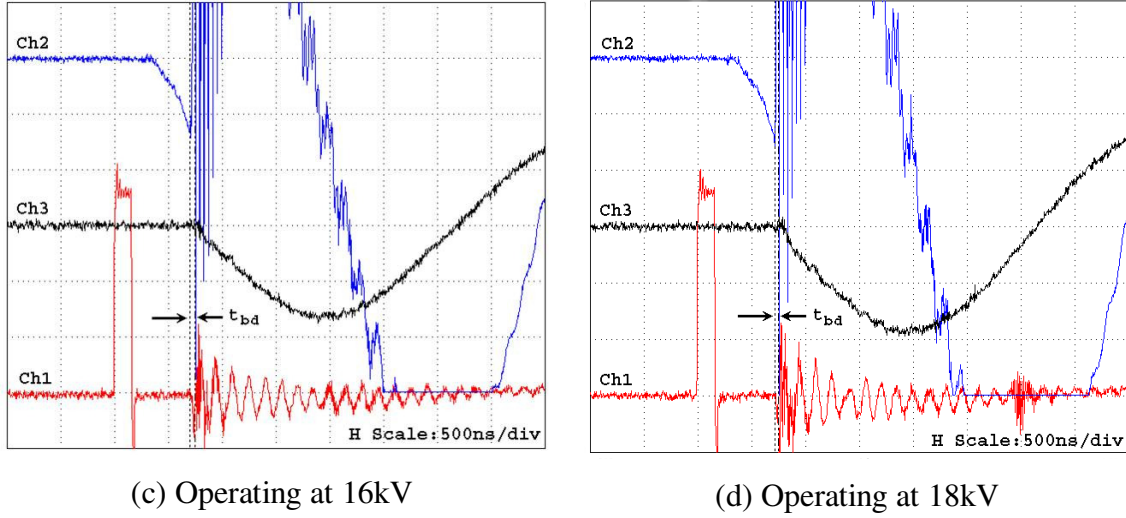


Fig. 3.28 Oscillograms showing variation in the breakdown time delay at different operating voltages.

In the oscillograms shown, Ch1 is the master trigger input signal, Ch2 is the HV trigger signal and Ch3 is the discharge current signal. The breakdown time delay t_{bd} of the switch is characterized as difference in timing of arrival of trigger voltage pulse at the trigger pin and the time taken by sparkgap switch to become fully conducting.

The experimental observation of variation in breakdown time delay of the SG-101M-75C trigatron switches, as function of main gap voltage (*i.e.* operating/ charging voltage) under freely running Nitrogen gas condition is shown in the graph in Fig. 3.29. It may be noted that each point shown in the graph is an average of 50 shots at the respective voltages.

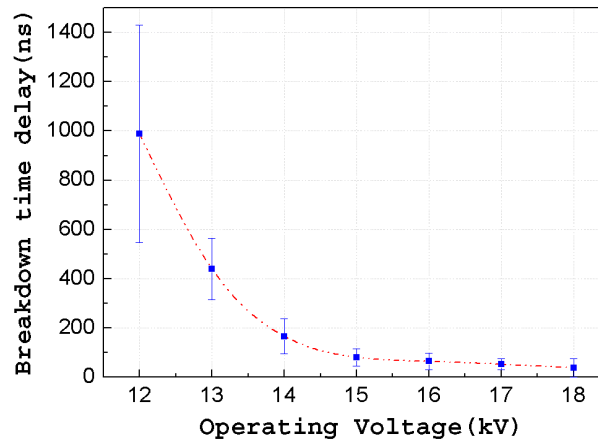


Fig. 3.29 Time delay (t_{bd}) as function of operating voltage for SG-101M-75C trigatron switches.

It can be observed from the trend seen in the graph that at lower voltages, the time delay is considerably longer, and there is a large deviation (at 12kV, t_{bd} is $988\pm441\text{ns}$), indicating erratic behavior of the switch. With successively higher operating voltages, the delay reduces significantly (at 18kV, t_{bd} is $38\pm22\text{ns}$).

This investigation revealed the constraint that for using multiple SG-101M-75C trigatron switches in parallel, to obtain synchronized breakdown, the operating voltage must be $\geq 18\text{kV}$. But as discussed in the previous section, since we decided to operate the capacitor bank in the range of 12-15kV for enhancing shot-life of the capacitors at higher voltage reversal ($\sim 80\%$), therefore single switch operation has been implemented in the present device configuration instead of multiple switches in parallel.

The advantage with use of single switch operation is that the breakdown time delay t_{bd} is of no practical concern for efficient device performance, however the life-time of the switch is sacrificed at the cost of this benefit. Utilization of multiple switches in parallel has the benefits of enhanced shot-life (since the coulomb transfer is shared) and reduced inductance contribution but it increases the complexity of the system hardware and its overall size (and crucially it is not possible to attain synchronization due to the longer breakdown time delay in the 12-15kV range).

3.5.4 Triggering System

Triggering requirements for the trigatron sparkgap switch model SG-101M-75C demand a fast pulse having a peak voltage $> 20\text{kV}$ (typically $> V_{op}/3$, where V_{op} is the switch operating voltage), with a rise time of $\sim 0.1\text{kV/ns}$, and an energy of $> 5\text{mJ}$ [193]. The complete triggering system comprises (i) a customized control module PG-103D4-02 pulser, and (ii) an optically isolated trigger head– THD-01 along with its epoxy encapsulated high voltage trigger transformer, procured from M/s R. E. Beverly III and Associates, USA. Brief descriptions of these sub-systems are mentioned in next sub-sections.

3.5.4.1 Control Module

The PG-103D4-02 pulser is a compact control module that facilitates optical trigger pulses (in infrared range) up to four channels for various customized applications [194]. Operating modes include single pulse (manual fire) and repetitive (internally clocked with range of selection up to 10Hz) operations. The external trigger input is optically isolated; surge protected and compatible with TTL–12 V signals. A view of the front panel controls with input and rear panel outputs are shown in Fig. 3.30 and 3.31 respectively.



Fig. 3.30 View of front panel controls and input of the PG-103D4-02 pulser.



Fig. 3.31 View of rear panel outputs of the PG-103D4-02 pulser.

In rear panel view (refer Fig. 3.31), it may be noticed that PG-103D4-02 facilitates two additional output pulses along with the four optical trigger pulses, these are ‘Sync’ and ‘Recharge’. The ‘Sync’ pulse is a fast output pulse, facilitated for the synchronization of trigger pulses with external diagnostics and instrumentation. The ‘Recharge’ command signal is provided for synchronizing and controlling an external constant-power charging supply. The pulse shape for these two output signals are shown in Fig. 3.32.

It may be noted from the oscilloscope traces in Fig. 3.32 that ‘sync’ and ‘recharge’ pulses have pulse durations (full width at half maximum *i.e.* FWHM) of 150ns and 25ms respectively. The ‘recharge’ pulse has a key role to play in repetitive operation of pulsed power systems, which is to inhibit the high voltage constant-power charger [194]. The momentary inhibition of charging, followed by each discharge, allows time for the switch to recover before high voltage is reapplied to the energy-

storage capacitors during the subsequent recharge cycle. This feature is particularly important in repetitive mode operation.

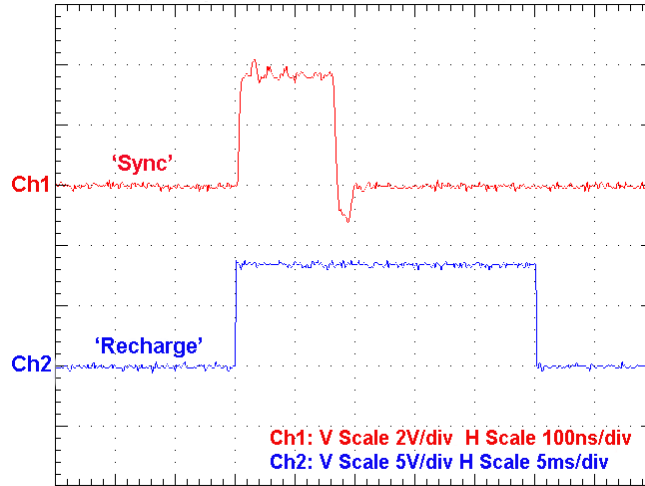


Fig. 3.32 The 'Sync' and 'Recharge' signal output waveforms.

The interfacing of the high voltage constant power charger with the 'recharge' command signal (gate) is described in detail in the following section.

3.5.4.2 Trigger Head and Pulse transformer

Two other important constituents of the trigger system are the trigger head and the pulse transformer – THD-01 (procured from M/s R. E. Beverly III and Associates, USA). The integrated view of a single channel trigger system with interconnected sub-components is shown in Fig. 3.33.

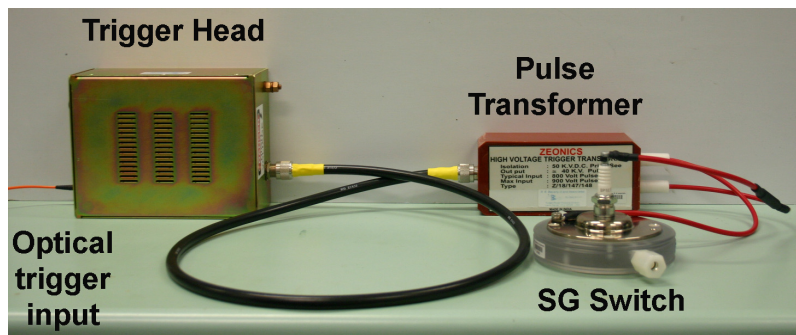


Fig. 3.33 Photograph showing interconnection between subsystems of single channel trigger system.

As may be seen from the photograph shown in Fig. 3.33, the trigger head receives its fire command from the PG-103D4-02 pulser through a 10m long multimode optical fiber link (Amphenol, ST 62/125) that operates at 820nm wavelength. These utilize HFBR-1412/2412 transmitter-receiver links. In the trigger head the optical input is converted to an analog signal and the corresponding output produced is a fast 800V pulse, which is then fed as input to the pulse transformer via a 1.2m long double-shielded coaxial cable RG 214/U. The pulse transformer amplifies 800V input pulse to a high voltage $\sim 30\text{kV}$ output trigger pulse for switching the trigatron sparkgap.

The delays between respective outputs - 'sync', 'recharge' and 'HV trigger output' were measured and are shown in Fig. 3.34.

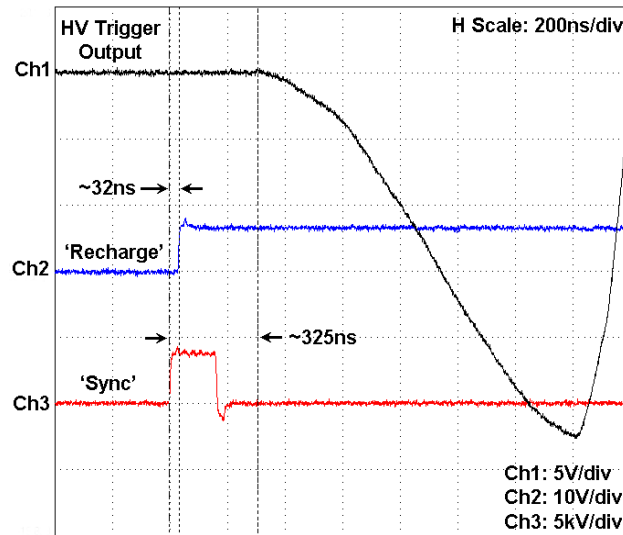


Fig. 3.34 Time resolved information of single channel trigger outputs.

The measured inherent delays between 'sync'- 'recharge' and 'sync' – 'HV trigger output' pulse are $\sim 32\text{ns}$ and $\sim 325\text{ns}$ respectively. The obtained rise time and peak amplitude of 'HV trigger output' are 700ns and $\sim 28\text{kV}$, respectively. The Tektronix HV probe - P6015A was used for measuring high voltage trigger output. A negative polarity for the trigger pulse is necessary for operating the trigatron switches in heteropolar mode. Sparkgaps triggered with fast rising voltage fronts exceeding

30kV/ μ s also help in the ignition of spatially uniform discharge streamers along the insulator, which essentially leads to sheath formation [178,193].

The electrical specifications of the trigger head and pulse transformer assembly are summarized in the Table 3.8.

Table 3.8 Specifications of Trigger Head and Pulse Transformer.	
Operating mode	Single pulse (manual), External
Trigger input	Electrical (5V TTL/ 5V – 15V)
Maximum output voltage	- 36kV
Pulse rise time	$\sim 0.7\mu$ s
Pulse energy	~ 160 mJ
Transformer isolation	50kV DC
Pulse reproducibility	< 5% variance shot to shot
Load impedance	Direct short to infinite
Pulse repetition rate	10 Hz (maximum)
Input power requirements	200 – 265VAC, 1 ϕ , 50Hz

3.5.5 Plasma Focus Tube

The view of coaxial electrode arrangement along with vacuum chamber is shown in Fig. 3.35.

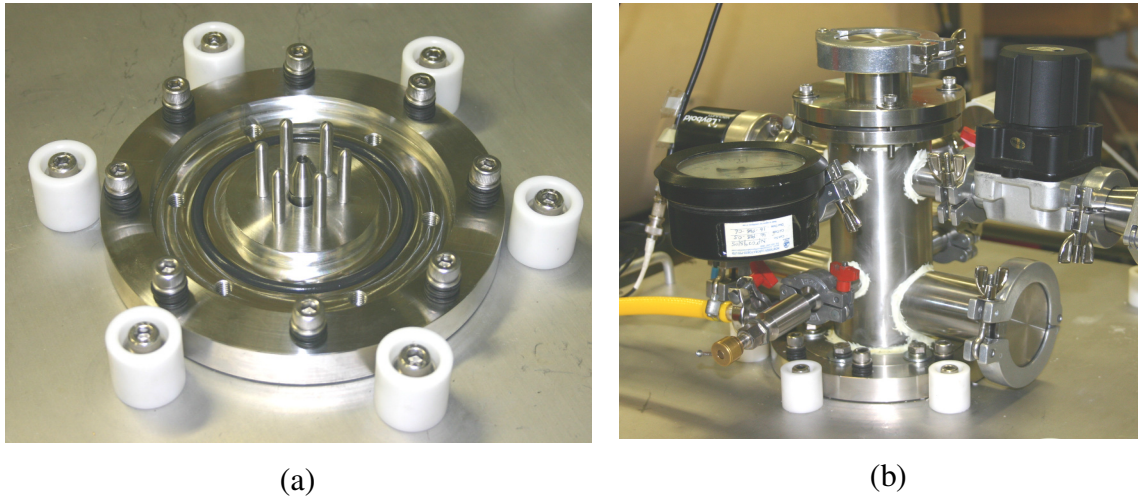


Fig. 3.35 Coaxial electrode arrangement and vacuum chamber used in FMPF-2 device.

In FMPF-2 device, since the amplitude and quarter time period of peak discharge current remained approximately the same as for the FMPF-1 device, the effective dimensions of the anode and

cathodes have been kept same, as mentioned under section 3.4.4. However, the coaxial electrode assembly arrangement was modified in compatibility with FMPF-2 collector plate design.

3.5.6 Connections and Assembly

The arrangement for the capacitor, the connecting plates, the sparkgap switches, and the triggering system were made systematically so as to minimize the circuit inductance and produce a symmetric discharge current. As can be seen from the constructional drawing (refer Fig. 3.36), the inter-connections between modules, sparkgap and plasma focus head have been minimized by using a flat plate transmission line assembly.

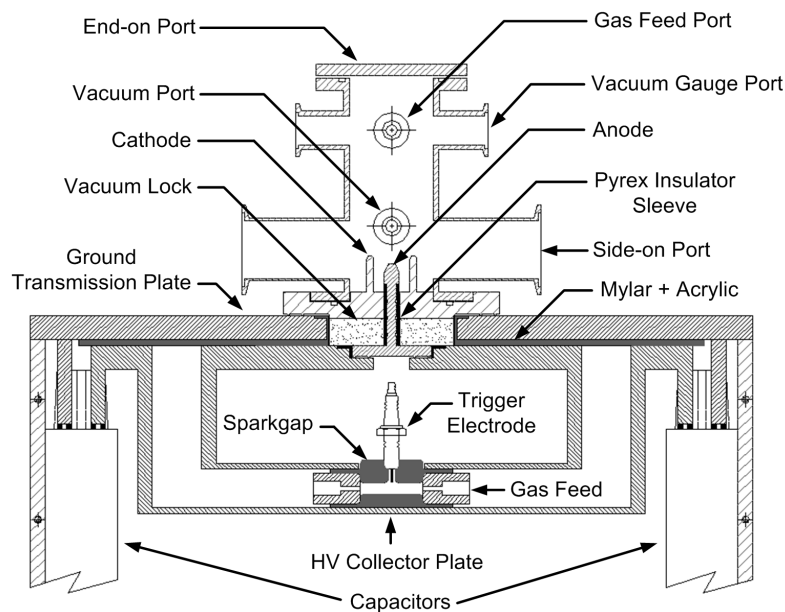


Fig. 3.36 Construction details of FMPF-2 device.

The earth plates of the capacitors were directly connected to the mainframe and the discharge end (HV) connections of all modules, and were collectively terminated on a common central plate that is connected to the input plate (charging end) of the sparkgap switch. The output plate (discharging end) of the sparkgap is connected to the central transmission plate that directly interfaces to the plasma

focus head. A 5mm thick acrylic sheet along with four layers of 125 μ m thick Mylar sheets have been used as insulation between the transmission plates.

The collector plate assembly was successfully tested to withstand 25kV DC without any insulation breakdown/ flashover. The use of coaxial cables was purposely eliminated to reduce inductance, resistance and frequent maintenance of the device. The measured total driver bank inductance contributed by – capacitor bank inductance + transmission lines + sparkgap + focus tube is about $\approx 56 \pm 3$ nH. A photograph of the integrated assembly of the FMPF-2 device is shown in Fig. 3.37. The assembly has a footprint of 0.6m $\times$ 0.6m, with height of 0.7m. The vacuum chamber extends above the top surface level.

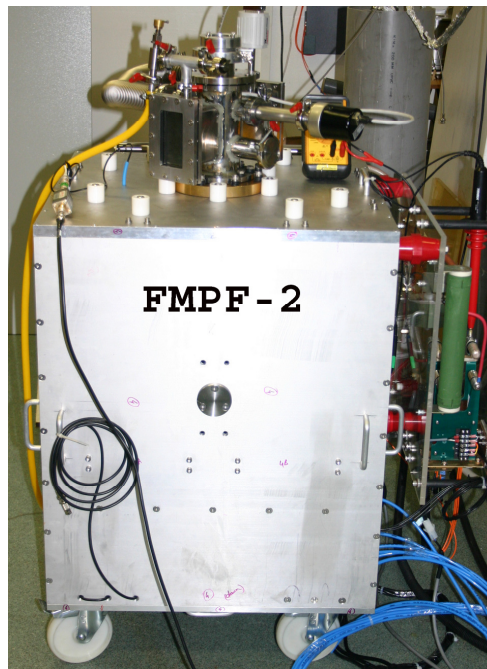


Fig. 3.37 Integrated assembly of FMPF-2 device.

3.5.7 Constant Power Charger

A constant power charger ‘EMI HVPS 802L’ (procured from M/s *a.l.e.* Systems, USA) has been used for fast and repetitive charging of the modular capacitor bank. The HVPS 802L is especially designed for rapid charging of energy storage capacitors, with an average power exceeding 8kW [195].

The output voltage of the HVPS 802L is fully adjustable over the whole range 1 to 40kV with 9kJ/s and 8kJ/s being the peak and average charging rates, respectively. The major specifications of the charger are summarized in Table 3.9.

Table 3.9 Specifications of constant power charger ‘EMI HVPS 802L’	
Input voltage/ current	180 – 264V/ 40A AC, 3 ϕ , 50Hz
Peak charging rate	9kJ/s
Input efficiency	85% @ full power
Output voltage	1 – 40kV
Output current	450mA
Average output power	8kW

The full output capacity is able to supply a peak charge rate of 9kJ/sec. The output feature of HVPS 802 is shown in Fig. 3.38(a-b) [195].

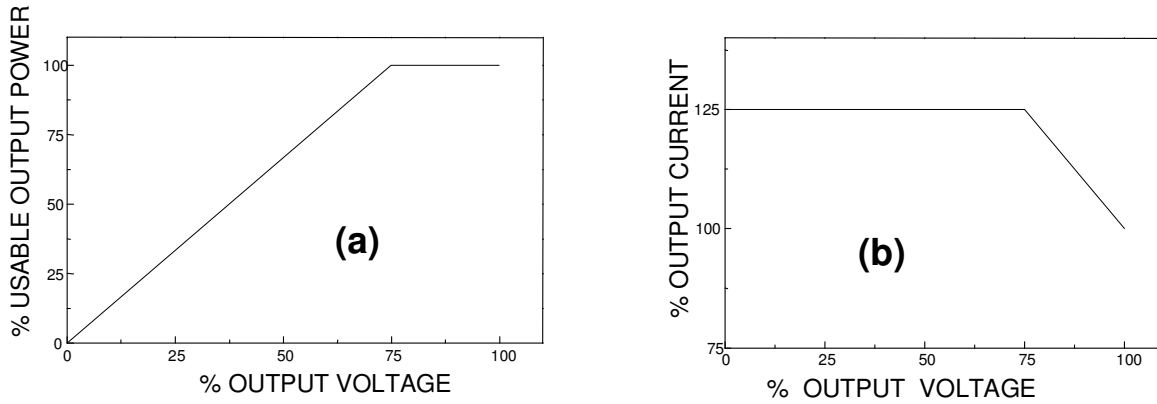


Fig. 3.38 The electrical output feature of ‘EMI HVPS 802L’ power supply.

From 100% down to 75% of the rated voltage, the HVPS802L delivers constant peak charge rate P_{pk} of 9kJ/sec. Below 75% rated voltage, the current is constant and the charging rate depends on voltage setting [174, 195].

For charging in the range of 75% to 100% of the rated voltage, the charging time is [195]:

$$t_c = \frac{0.5 \times C_0 \times V_0 \times V_{rated}}{P_{pk}} \quad \text{seconds} \quad (3.20)$$

For charging below 75% of the rated voltage, the charging time is [195]:

$$t_c = \frac{0.5 \times C_0 \times V_0 \times (0.75 \times V_{rated})}{P_{pk}} \quad \text{seconds} \quad (3.21)$$

Following above relations, for FMPF-2 device, we estimated that for charging the 2.4 μ F capacitor bank up to 15kV maximum, using a 40kV rated model 802 power supply, having the peak power rating of 9kJ/sec; the charging time will be ~60ms (refer Eq. 3.21). Thus by allowing durations of <15ms as dwell time and ~25ms as dead time for the sparkgap switch recovery; the maximum repetition rate attainable is 10Hz.

An imperative feature of HVPS 802L is that it is facilitated with an INHIBIT connection. Enabling of the INHIBIT connection, intermittently disables the charging for the specified duration and allows time for the energy transfer switches to recover before the voltage is re-applied for the subsequent recharge cycle. Hence, dead time is necessarily required in repetitive mode operation.

3.5.8 Control Scheme for Operation in Repetitive Mode

The block diagram representation of operation and control scheme for repetitive FMPF-2 device has been shown in Fig. 3.39.

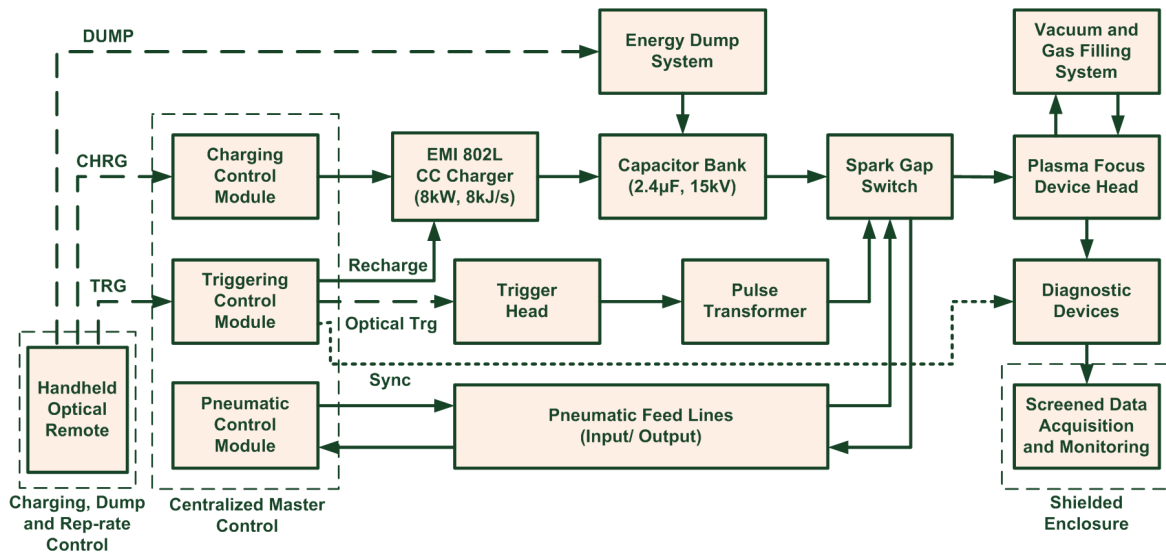


Fig. 3.39 Block diagram representation of operation and control scheme for FMPF-2 device.

As indicated in the diagram; the control, operation, monitoring and protection of various subsystems ascribed in the previous section is done via the centralized master control. The main objective of having a centralized main frame console for FMPE-2 operation is to protect sensitive apparatus from malfunction by sequential interlocking between various modules during experimental runs. For the enhanced safety of the operating personnel from dangerous events (in view of the presence of tens of kV DC), the major controls for operation of the device *i.e.* charging and triggering in pulsed/ repetitive modes at desired frequency and emergency stop/ dump are provided on a stand alone handheld optical remote. The isolation is provided by a galvanic separation barrier consisting of 3-Channel optical fiber cable assemblies. These utilize HFBR-1523/2523 transmitter-receiver links that can easily work up to 50m. The module also facilitates running the system at pre-adjusted repetition rates of 0.2Hz, 0.5Hz and 1Hz to 10Hz. The analog rep-rate frequencies are generated using an LM555 timer chip in astable oscillator mode. The photograph of the battery operated handheld optical remote is shown in Fig. 3.40.



Fig. 3.40 Optically isolated handheld remote control.

The two important subsystems, CPC and trigger console have necessarily to be interfaced for repetitive mode of operation. In the repetitive mode of operation, sequentially the charge command is

sent followed later by a trigger. Upon receipt of charge command in the charging control module, the grounding system interlocked with the capacitor bank decouples, and the constant current charger (EMI 802L) charges the bank up to its preset level. After the preset time interval, when the trigger command is sent to the trigger module PG-103D4-02 as an external trigger, it simultaneously generates one optical and two analog pulses. (i) An optical trigger signal is produced at 820nm wavelength and it is sent to the trigger head THD-01 via a 10m long multimode optical fiber link (Amphenol, ST 62/125). These utilize HFBR-1412/2412 transmitter-receiver links. In the trigger head the optical input is converted to an analogue signal and the corresponding output produced is a fast 800V pulse, which is then fed as input to a pulse transformer via 1.2m long double shielded coaxial cable RG 214/U. The pulse transformer amplifies 800V input pulse to a high voltage ~30kV output trigger pulse for switching the trigatron sparkgap. Since there is no direct electrical connection between the control module and the trigger head, noise immunity is greatly improved and the chances of spurious triggering is largely eliminated. (ii) The concurrently produced analogue 'RECHARGE' pulse of +12V for 25ms duration is fed to a constant current charger to temporarily inhibit the power supply for the specified interval.

In repetitive mode of operation, it is necessary to allow time for the sparkgap switch to recover before the voltage is reapplied to the energy-storage capacitors during the subsequent recharge cycle [191] following each trigger pulse. The sample charging waveform acquired during the test run with 0.3 μ F capacitor in repetitive mode of operation at 10Hz is shown in Fig. 3.41.

It may be noticed that after the trigger pulse is applied, charging is inhibited for 25ms duration (allowing the sparkgap switch to recover) and then after this delay the capacitor recharges to the preset voltage. (iii) Another concurrently produced analogue 'SYNC' pulse of 5V/ 150ns is a fast output pulse that facilitates the synchronization of the trigger pulse with external diagnostics and instrumentation.

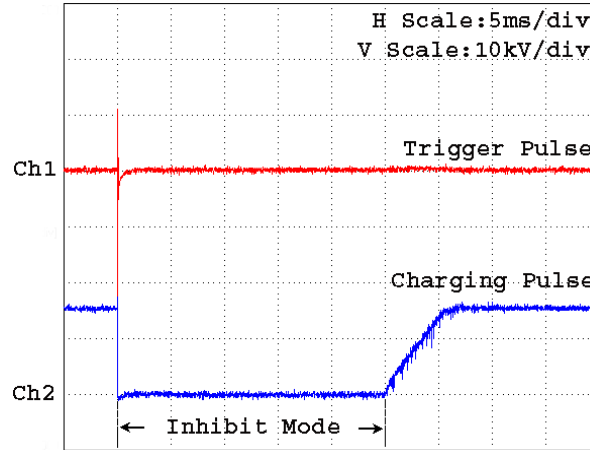


Fig. 3.41 Sample charging waveform during INHIBIT mode of operation.

The pneumatic control console performs the role of maintaining consistent purging of Nitrogen (N_2) gas at the required flow rate (depending upon discharge repetition frequency) through the sparkgap switch. Flexible polyurethane tubing's were used as the pneumatic feed lines. The experiment demonstrating the enhancement in time averaged neutron yield by the repetitive operation of FMPF-2 device is reported and discussed in detail in chapter 5, in section 5.3.

3.6 The FMPF-3 Device

After successfully demonstrating the repetitive operation of miniature plasma focus up to 10Hz, using the prototype single module FMPF-2 device and results of which are discussed in detail in chapter 5 under section 5.3; the final four-module version FMPF-3 was devised and optimized. The FMPF-3 is basically an upgraded version of FMPF-2 device with a pseudospark switch based modified synchronous switching system.

The shortcomings evident in the operation of FMPF-2 device were: (i) although the design and integration of all components of pulsed power system was done in the lowest practical inductance configuration but since a single switch was used (due to the limitation of large variation in breakdown time delay and jitter of the trigatron type sparkgap switches); the inductance contribution due to

connection of all the capacitors to a common switch and the switch itself was significant, and (ii) erosion rate of sparkgap electrodes was found to be undesirably high since the single switch was operated close to its maximum current handling limit. Moreover, the voltage reversal in the system was quite high causing ablation and sputtering of the electrodes. The coating of dielectric spacer with ablated electrode material further resulted in high voltage tracking leading to erratic triggering as well as pre-firing of the switch. In addition to the above mentioned causes, also from the applications point of view, use of a pressurized sparkgap switches is not convenient due to the large flushing gas flow rate requirement (4-40LPM) for operation of the device in repetitive mode in the range of 1-10Hz.

In order to eliminate the above mentioned deficiencies and to further enhance the electrical performance of the FMPF-3 device, energy transfer through a single trigatron sparkgap switch is replaced with four parallel operating pseudospark switches (each switch being attached to 0.6 μ F module of two capacitors). As mentioned before, the added advantage of deploying multiple switches in parallel is that along with the reduced inductance, the lifetime of each of the switches is increased very considerably, since the total coulomb transfer is shared amongst the switches connected in parallel.

The modular layout of the FMPF-3 device along with details of the modified switching system is detailed as under.

3.6.1 Layout of Modular Assembly

The modular layout of the FMPF-3 device assembly is shown in Fig. 3.42. The constructional drawing shown there, illustrates the physical inter-connections between the modules, the pseudospark switches and collector plate assembly for the FMPF-3 device. The incorporation of the four pseudospark switches has been made systematically so as to minimize the circuit inductance and to ensure symmetric flow of the discharge current. The measured total inductance, contributed by:

capacitor bank + transmission lines + sparkgap + focus tube is about $\approx 34 \pm 2 \text{ nH}$.

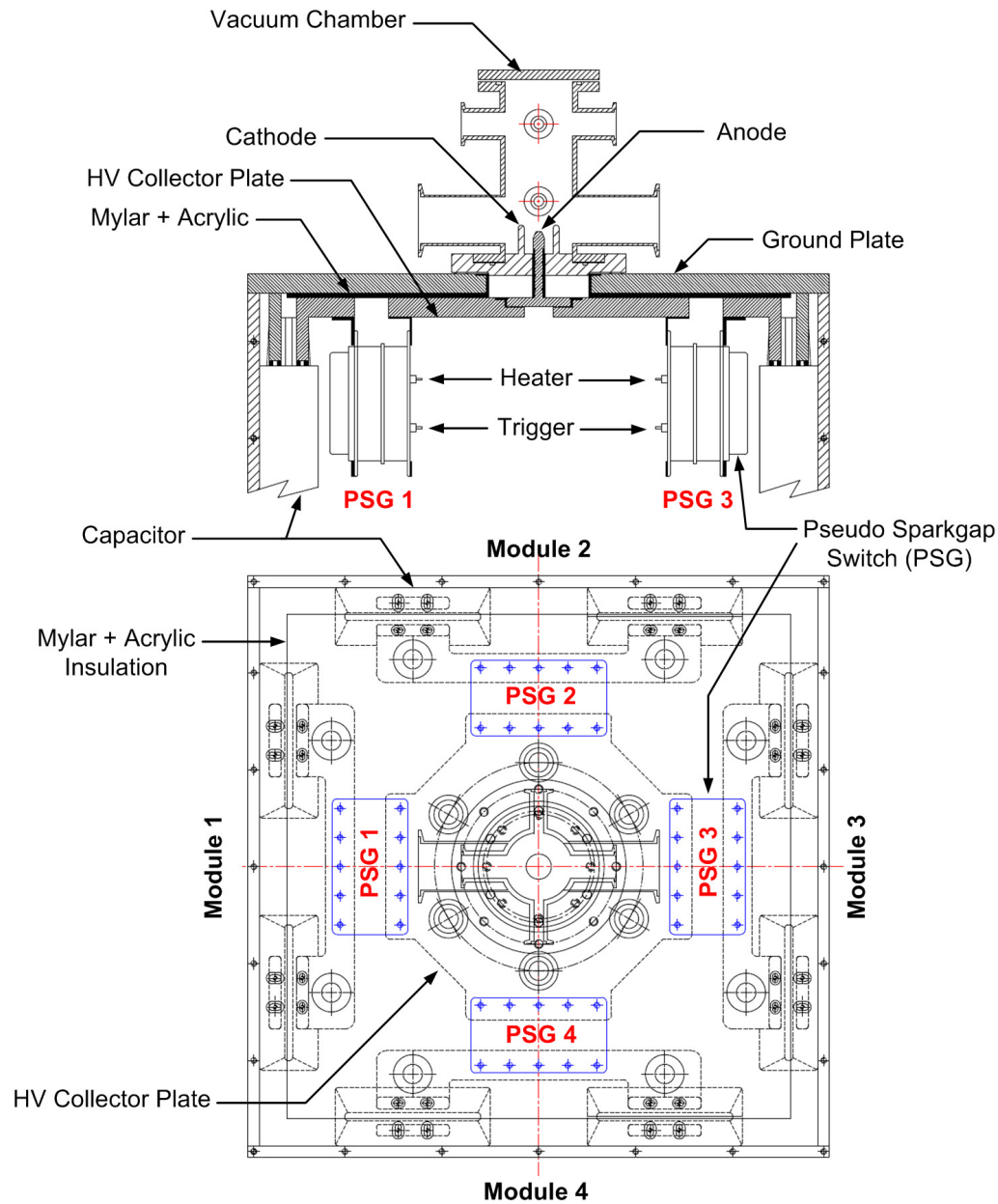


Fig. 3.42 Constructional layout of FMPF-3 device assembly.

3.6.2 Configuration of Main Electrical Circuit

The configuration of main electrical circuit for the FMPF-3 device is depicted in Fig. 3.43. In the schematic shown, the four modules of capacitor bank (M1~M4; each connected with two $0.3 \mu\text{F}$

capacitors in parallel) are individually attached with four in-line single pseudosparkgap (PSG) switches. On the high voltage side, the four capacitor modules are charged up to 15kV (maximum) in parallel by the constant power charger. Then the PSG switches (PSG1~PSG4) are fired simultaneously by the 4-channel high performance triggering system to discharge the stored energy of modules (M1~M4) into the plasma focus head.

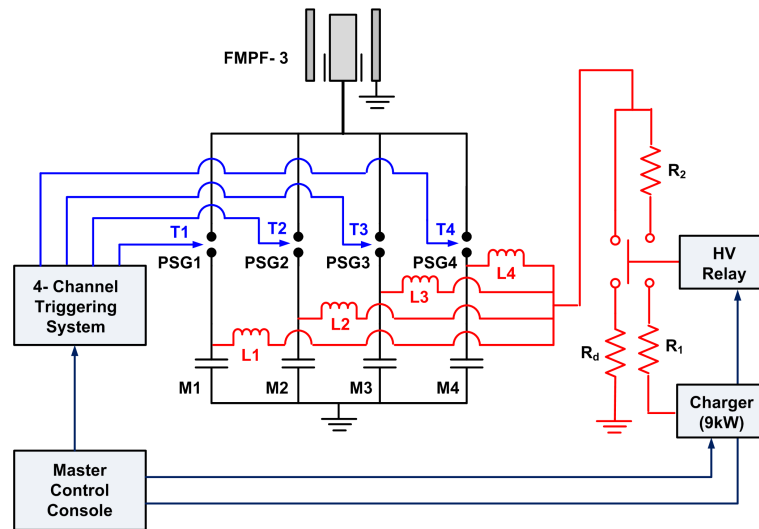


Fig. 3.43 Configuration of main electrical circuit.

To protect the charger from high reverse voltages exceeding maximum ratings, two resistors R_1 and R_2 ($1k\Omega$, 300W) are used in the charging circuit [191,195]. The four inductors $L1\sim L4$ (of $60\mu H$ each) have the specific function of isolating the modules during discharge. In the situation where one of the switches fails to fire, the attached module discharges through inductors $L1\sim L4$ and then through the plasma focus chamber. Hence, the fault current under such misfire conditions is lowered because of the large inductance in the discharge path and thus the likelihood of the cross flow of energy between the modules is significantly reduced.

The operating sequence of subsystems – 4 Channel triggering system, charger and HV relay is mutually synchronized via the ‘master control console’. The main function of the high voltage relay

E30-DT-40-1-31-BD (procured from M/s Ross Engg. Corporation, CA, USA) is to connect the charging terminals with the capacitor bank, only when the charging command is enabled, and otherwise to keep all the capacitors in dump mode by default. The control coil of HV relay is rated as 230V/0.5A and has DC isolation of 30kV. Two bulk ceramic composition disc resistors (R_d) rated as $1k\Omega \pm 5\%$ (AB399 procured from M/s HVR International, USA) are used in series for energy dumping. Each of the resistors features a peak energy absorption capacity of 24kJ. The mounting arrangement of charging and discharging sub-systems/ accessories assembled on an acrylic sheet of size 0.5m $\times$ 0.5m is shown in Fig. 3.44.

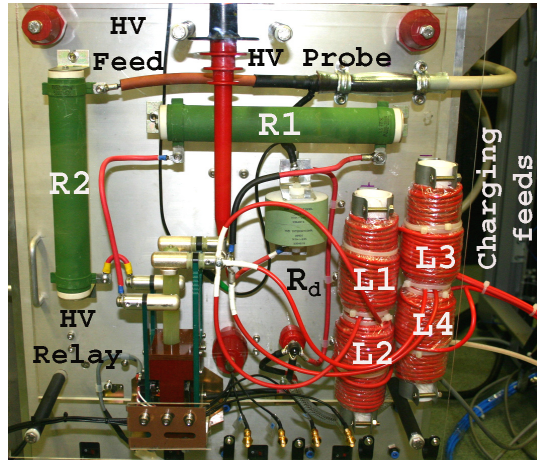


Fig. 3.44 Mountings of charging/ discharging subsystems and accessories.

3.6.3 Pseudospark Switch

In the FMPPF-3 device, a low pressure switching device with a cold cathode, called pseudosparkgap (PSG) switch (model TDI1-150k/25 procured from Pulsed Technologies Ltd., Russia) has been used [196]. Alternatively this type of switch is also known as a Copper Arc Thyatron. Similar to the classical Hydrogen thyatron, the range of operating pressure for the PSG switch corresponds to the conditions of the left side of the Paschen curve, where the electron free path for ionization is much in excess of the main electrode separation. PSG switches have stacked ceramic-

metal construction, with Hydrogen as a buffer gas in the pressure range 0.2 to 0.6mbar (in operational mode). The cathode (trigger) assembly is fitted with a special semiconductor discharge initiator (igniter). The inductance of the switch is $<20\text{nH}$ and it has been tested for operation in repetitive mode up to 10Hz. The appearance and constructional details of the TDI1-150k/25 pseudospark switch is shown in Fig. 3.45 [197].

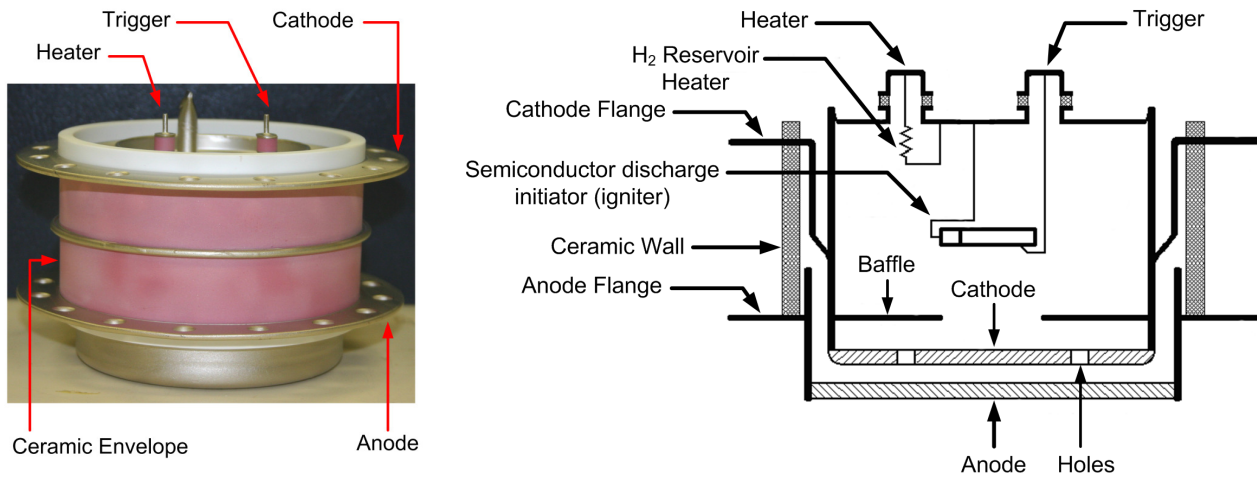


Fig. 3.45 Appearance and constructional schematic of TDI1-150k/25 Pseudospark Switch.

As shown in the schematic, the PSG consists of two parallel plate electrodes as anode and cathode. The cathode has axially bored holes in it enabling Hydrogen gas to enter the anode-cathode gap region (from the heated reservoir). The hole diameter is comparable to the spacing between the electrodes, being $\sim 6\text{mm}$. For the external triggering of the switch, a considerable pre-breakdown electron current is required to ignite a high current discharge in the main gap. The trigger unit is used to provide this electron current. When the trigger voltage of suitable amplitude is applied to the igniter, the electric field formed between the cathode and igniter causes electrons from the resulting discharge plasma to be injected through the cathode openings into the space between the cathode and anode of the low pressure sparkgap, igniting the main discharge [198]. A heater is also used in this device for heating the Hydrogen reservoir that releases the working gas inside the PSG chamber. The final

pressure inside the switch depends on the current flowing through the heater. The electrical specifications of the PSG switch model TDI1-150k/25 are summarized in Table 3.10 [196,197]:

Table 3.10 Specifications of Pseudospark Switch Model TDI1-150k/25.	
Peak Anode Voltage	25kV
Operating Voltage Range	3kV – 20kV
Maximum Current	150kA
Switch Inductance	< 20nH
Jitter	< 4ns
Trigger Voltage/ Rate of rise	$\geq -3\text{kV}/ \geq 5\text{kV}/\mu\text{s}$
Heater Voltage	3 – 7V AC
Heater Current	$\leq 2.5\text{A AC}$
Frequency of operation	10Hz (tested)
Readiness time	~ 5 minutes
Physical Dimensions	150mm (dia.) $\times$ 110mm (height)
Weight	~3.5kg

The two most advantageous characteristics of TDI1-150k/25 type PSG switches are; (i) they have stable breakdown characteristics and remain unaffected by ambient conditions (unlike pressurized sparkgaps) as these are sealed switches. This feature leads to higher reliability in trigger along with low jitter (<4ns) in operation, and (ii) the lifetime of this switch (defined in terms of ‘accumulated charge transferred’ *i.e.* $5 \times 10^5 \text{ C}$) provides $>10^6$ shots in case of a 200J plasma focus device. This means that by using TDI1-150k/25 type PSG switches, the system can be operated in repetitive mode at 10Hz for >2 weeks of continuous operation without requiring maintenance/ repairs. The lifetime of the switches is an important consideration for obtaining consistency in system performance and meeting demands of commercial/ industrial applications.

3.6.4 The 4-Channel Triggering System

For obtaining the synchronized operation of pseudospark switches for all four modules, a high performance 4-channel triggering system of high repetition rate (10Hz), low rise time (<10ns) and low

jitter ($\pm 1\text{ns}$) has been indigenously designed and implemented for the FMPF-3 device. The electrical schematic of this 4-channel triggering system is shown in Fig. 3.46.

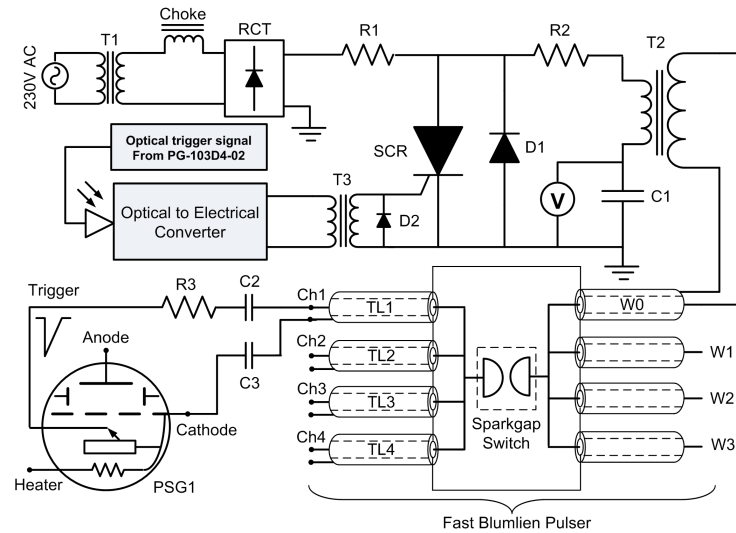


Fig. 3.46 Electrical schematic of repetitive 4-Channel triggering system.

(Note: $R1=100\Omega$, 50W; $R2=0.25\Omega$, 40W; $R3=25\Omega$, 40W; $C1=220\mu\text{F}$, 400V; $C2-C3=2.5\text{nF}$, 40kV DC; $T1$ =Isolation transformer, 240V, 500VA; $T2$ =Motorbike ignition coil, 350V/20kV; $T3$ =Pulse transformer, 1:1, PT24E; $D1=1\text{N}5408$; $D2=1\text{N}4007$; SCR=Thyristor, 30TPS16; $\text{PSG1}\sim\text{PSG4}$ =Pseudospark Switches, TDI1-150k/25; $\text{TL1}\sim\text{TL4}$ =transmission lines of 6m length (RG59, 75 Ω); $W1\sim W3$ =6m length of RG59, 75 Ω cables with one end open; $W0$ =Charging feed to Blumlien of 6m length of RG59, 75 Ω cable; V =Voltmeter, 400V DC; RCT=Bridge rectifier, KBPC3506; Choke=lamp ballast, 400mH, 100W).

The triggering system mainly comprises two major sections (a) Impulse charger and (b) Fast Blumlien pulser. The charging of the Blumlien is done by the impulse charger that produces $\sim 100\mu\text{s}$ long pulse of $\sim 20\text{kV}$ amplitude under no load condition (shown in Fig. 3.47). This high voltage charging pulse is produced by discharging an electrolytic capacitor $C1$ ($220\mu\text{F}/400\text{V}$) through the customized primary of the commercially available motorbike transformer $T2$, using a SCR as a control switch. The resistances $R1$ and $R2$ in the charging and discharging circuit have been correspondingly used to enable faster recovery of the SCR switch during repetitive mode operation. The resistance $R1$ moderates the charging rate of capacitor $C2$, whereas resistance $R2$ inserted in the discharge circuit lowers the reversal (and thus the current and conduction time), allowing rapid recovery of SCR switch.

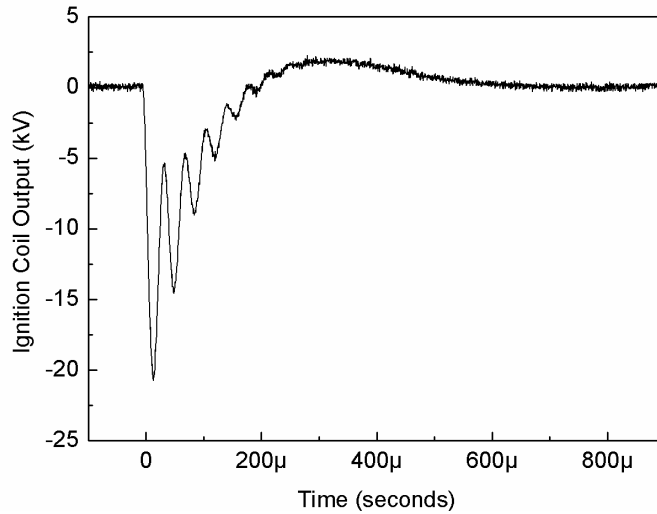


Fig. 3.47 Impulse charging waveform of Blumlien pulser.

The fast Blumlien pulser comprises a two electrode sparkgap and four charge (W0~W3) and discharge cables (TL1~TL4), each of length 6m. The construction of this pulser is shown in Fig. 3.48. The -20kV pulse produced by impulse charger, charges the transmission lines (W0 ~ W3) until the sparkgap breaks down. The four transmission lines (TL1 ~ TL4) are discharged by the sparkgap operation, thereby producing fast rising nanosecond duration impulses (Ch1 ~ Ch4) to further trigger the four PSG switches (PSG1 ~ PSG4). The triggers are provided to respective channels through decoupling capacitors (*e.g.* C2, C3 shown in Fig. 3.46) to protect the triggering system from high voltage DC, and from the backflow of energy during the discharge.

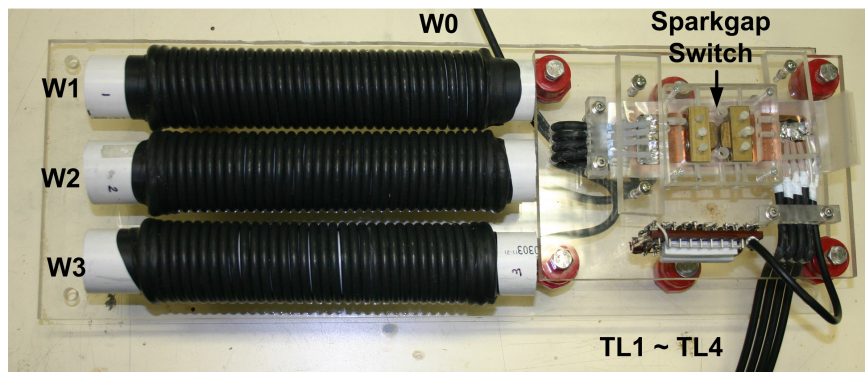


Fig. 3.48 Indigenously designed and constructed 4-Channel Blumlien pulser.

The Blumlien produces a high voltage pulse of duration $\tau = 2L/V_F$, where V_F is the velocity of propagation of electromagnetic wave in the cable and L is the physical length of cable. The rise time of the leading edge of the pulse is mainly limited by the inductance of the (a) sparkgap switch and (b) connections. As shown in Fig. 3.49, in our system the rise time of leading edge has been $<10\text{ns}$ and the duration of the pulse $\sim 60\text{ns}$ (FWHM).

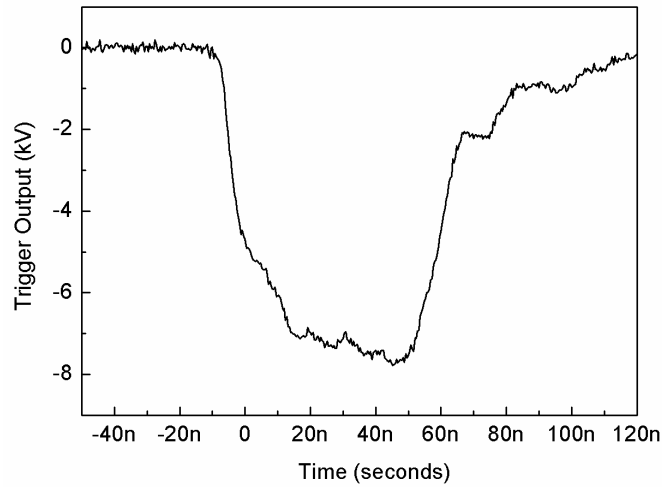


Fig. 3.49 Sample oscillogram of Blumlien pulser output.

The major electrical specifications of the indigenously designed and developed repetitive 4-Channel triggering system are summarized in Table 3.11.

Table 3.11 Specifications of the 4-Channel triggering system.	
Trigger pulse peak voltage	-8kV
Trigger pulse peak current	<300A
Trigger pulse rise time	<10ns
Trigger pulse duration	$\approx 60\text{ns}$
Jitter	$\pm 1\text{ns}$
Rate of rise (dV/dt)	$\geq 5\text{kV}/\mu\text{s}$
Repetition rate	0.2 – 10Hz
No. of channels	4
Triggering	Optical/ Electrical/ Manual
Isolation	25kV DC

3.6.5 The 4-Channel H₂ Generator Heating System

As mentioned before, a heater is used inside this device for heating H₂-reservoir which releases the working gas inside the PSG chamber [197,198]. The electrical schematic of 4-Channel heater circuit is shown Fig. 3.50.

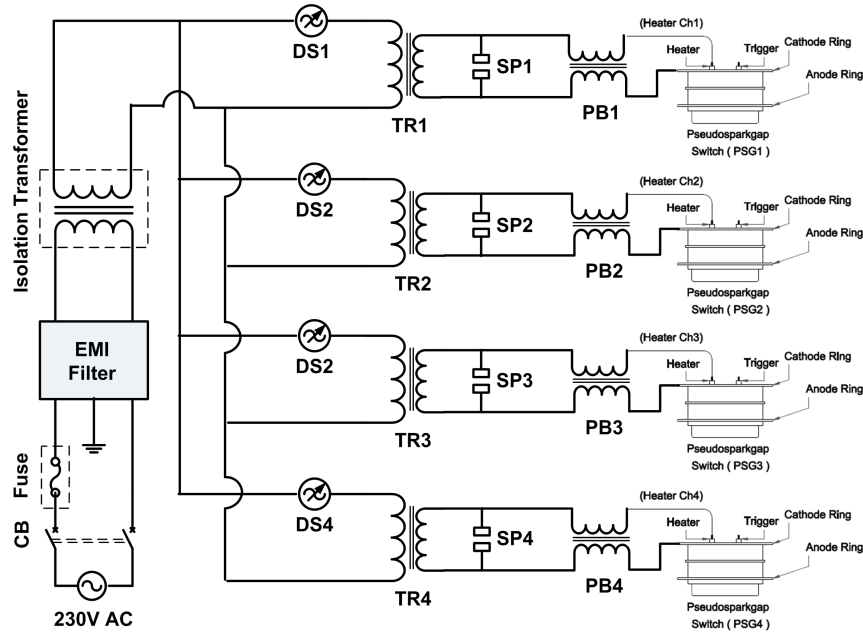


Fig. 3.50 The 4-Channel heater circuit.

(Note: EMI power line filter=240V, 10A; Isolation transformer=240V, 500VA; DS1~DS4=Dimmer switches, 0-240V, 500W; TR1~TR4=230V/6V, 30VA transformer with >5kV isolation between primary and secondary windings; SP1~SP4=Surge arrestor, EPCOS-90; PB1~PB4=Pulse blocking inductors, ~200μH).

For controlling the pressure inside the pseudospark switches four individual voltage sources (TR1~TR4) are used across each channel. An appropriate voltage/ current in the range of 3 -7V/ 1.5 – 2.5A (AC) is carefully adjusted across each PSG switch by the setting of the Dimmers (DS1~DS4) of the corresponding channels. Current flowing through the heaters of respective channels plays crucial role since it determines the operating pressure of the gas in the inter-electrode spacing and consequently the voltage hold-off capacity along with breakdown time delay [197,198]. It is important to note that since one of the terminals of the transformer is connected to the discharge end of the PSG switch, and hence high voltage isolation of >5kV DC is required between the primary and secondary

windings of the transformers used. For additional protection, surge arrestors (SP1~SP4) have also been used across secondary windings of transformers of each channel along with pulse blocking inductors (PB1~PB4) in series.

3.6.6 Scheme of Operation for FMPF-3

The block diagram representation of operation and control scheme for repetitive FMPF-3 device is shown in Fig. 3.51.

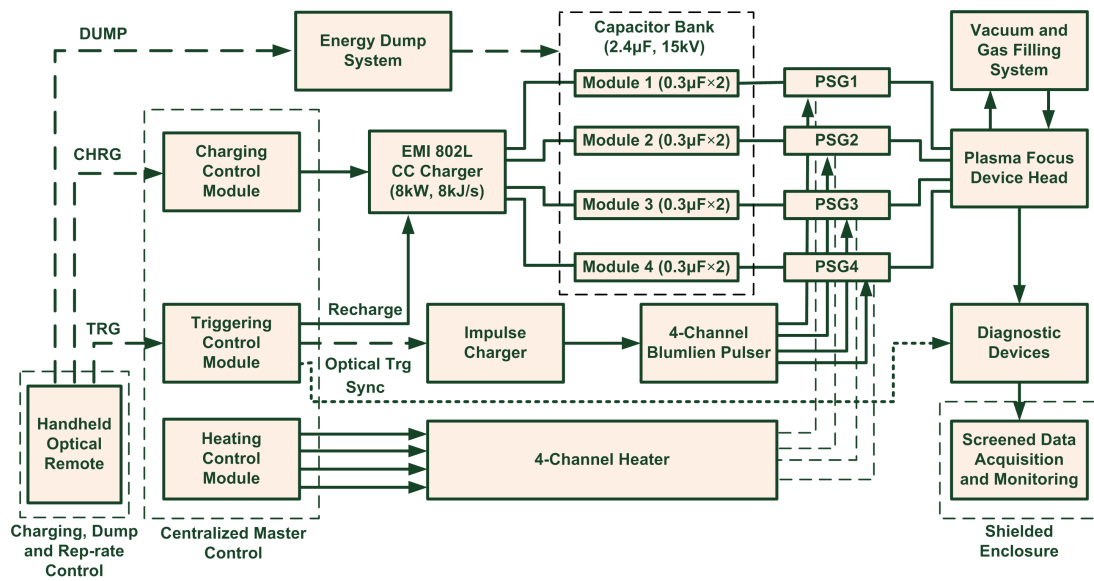


Fig. 3.51 Block diagram representation scheme of operation in FMPF-3 device.

As indicated in the diagram; the fundamental scheme of operation in the FMPF-3 device is similar to the FMPF-2 device (as mentioned in section 3.5.8), except for the changes that (i) charging of the four modules of the capacitor bank is done individually while using isolation inductors in series, (ii) for discharging stored energy of modules into plasma focus load, four pseudospark switches (PSG1~PSG4) are synchronously triggered by a high performance 4-Channel Blumlien pulser, and (iii) a 4-Channel heater is also incorporated for controlling the Hydrogen gas pressure inside each of the pseudospark switches.

The control sequence of various modules during single/ repetitive mode of operation remains the same as described in section 3.5.8. The plasma focus tube parameters in FMPF-3 are also similar to the one used in FMPF-1 (and FMPF-2) device since the amplitude and duration of quarter time period of peak discharge current remains in the close range of tolerance for optimum device performance.

The experiments that were performed on the most optimized version of repetitive miniature plasma focus 'FMPF-3' device are reported and discussed in detail in chapter 5 under section 5.4.

3.7 Vacuum Pumping System

Vacuum pumping for the FMPF devices employed a tabletop model 'PT-70 Dry' (procured from M/s Oerlikon Leybold, Germany) which is a turbo molecular pump system with oil free scroll back pump (model - SC5D). The pumping speed of system is 60liters/second. As shown in Fig. 3.22(b), this system is compactly assembled as a standalone unit facilitating installation of pumps and their controllers for high vacuum pumping and measurement.

For measuring the vacuum over a wide range, a PENNINGVAC transmitter PTR90 has been used. It combines cold cathode ionization with the Pirani principle. This allows for complete coverage of the pressure range from 5×10^{-9} mbar to atmospheric pressure by a single transducer. The cold cathode system is ignited directly through switching-on the internal high-voltage at the optimum ignition pressure. For the absolute measurement of filling gas pressure, CERAVAC transducer CTR90 (from M/s Oerlikon Leybold, Germany) has been used. It has accuracy of 0.15% in the range of 0 to 10mbar. A two channel compact gauge controller with digital readout unit, CENTER TWO (from M/s Oerlikon Leybold, Germany), was used to display the pressure measurements from these active sensors. It facilitates simultaneous display of all channels. The entire vacuum system was supported on anti-vibration mount for noiseless operation.

3.8 Data Acquisition System

Most of the plasma focus devices operated with medium energy range capacitor bank produces electrical discharges with quarter periods ($T/4$) in the range of $1.5\mu\text{s}$ - $4\mu\text{s}$; hence the pre-requisites of bandwidth are not that high for the implemented diagnostics. In miniature plasma focus device, since the capacitor bank is of low energy ($\sim 200\text{J}$), the associated plasma dynamics is much faster in comparison with conventional devices. In our FMPF devices, the quarter discharge time is $\sim 400\text{ns}$ and the expected pinch lifetime is of the order of a few nanoseconds. Hence, a fast high-bandwidth data acquisition system is required to record the signals from the applied diagnostics. In the present experimental setup, the data acquisition system consists of two Yokogawa digital storage oscilloscopes (model - DL9140) and one Tektronix TDS 340 real-time digital oscilloscope and a computer. The digital storage oscilloscopes have sampling speed of 5GS/s and bandwidth of 1GHz . Four channels of each oscilloscope (so altogether 8 channels) are used simultaneously in various experimental studies.

For evaluating and optimizing the device performance and investigation of the radiations and charged particles emitted from these three different miniature plasma focus devices *viz.* FMPF-1, FMPF-2 and FMPF-3, different time resolved and time intergraded diagnostics were employed. The working principles and design descriptions of these diagnostics are discussed in detail in the next chapter.

Chapter 4

Diagnostic Techniques

Diagnostics in plasma focus devices are mainly required to fulfil two major objectives: (i) to investigate the performance of the pulsed power system that drives plasma focus device, and (ii) to obtain information about the hot, dense and short-lived plasma and the multiple radiation components such as X-rays, electrons, ions and neutrons which typically last from a few ns to several 100ns, depending upon device size, energy and other operating parameters. One of the major tools to investigate the phenomena occurring inside the pinched plasma without perturbing it is to obtain physical information from emitted radiations (i.e. from high energy photons and energetic particles). In consideration of the above mentioned requirements, electrical diagnostics (for monitoring the electromagnetic parameters of the device) and radiation diagnostics (to perform time resolved and time integrated measurement of X-rays and neutrons) that have been designed and used for investigating the performance of newly developed fast miniature plasma focus devices (FMPF-1, FMPF-2 and FMPF-3) are detailed in the following sections, along with a pre-discussion of the associated complications in their diagnostics.

4.1 Complications in Diagnostics of FMPF devices

Diagnostics for fast sub-kilojoule range fast miniature plasma focus (FMPF) devices are more complicated than kilojoule range plasma focus devices due to the following reasons:

(i) In medium and high energy range plasma focus devices [17-30], the quarter time period ($T/4$) of electrical discharges is typically in the range of few microseconds ($1 - 5\mu\text{s}$) but since in the sub-kilojoule range of miniature plasma focus device [39-44] it is at the sub-microsecond level, typically ranging from 150ns to 500ns, the associated plasma dynamics are much faster in comparison

with conventional kilojoule devices. In such fast discharges, the bandwidth/response time of the electrical diagnostics become very critical for observing fast dynamic events like focusing (pinch) phase which may have a typical life time of only a few nanoseconds.

(ii) Another major issue associated with the diagnostics of ‘fast miniature plasma focus devices’ is the presence of large amount of electromagnetic noise, since the pulsed power drivers are inherently fast with low inductance. In FMPF devices, electromagnetic noise is mainly generated from the Mylar/ Acrylic separated low inductance parallel plate assembly due to transmission line effects and secondly from the low inductance fast switches (mainly sparkgaps). In our experiments with the unshielded FMPF-1 device, the electromagnetic noise produced during the discharge was found to be so strong that it resulted either in the complete malfunctioning of the employed diagnostics or significant decrease in the signal quality. For eliminating noise interference, all of the implemented diagnostics along with the cables were effectively shielded and radiation diagnostics were installed inside the shielded enclosure (i.e. Faraday Cage).

(iii) Commonly used neutron detectors such as activation counters (Indium/Beryllium based) and Bubble detectors, that have been used for kJ range plasma focus [18, 199], are not effective for measuring yields below the order of $\sim 10^6$, which is the expected yield from miniature plasma focus devices operating with a typical energy of $< 400\text{J}$.

(iv) The compatibility of neutron detector response time is also a critical issue for the measurement of neutron yield produced in discharge during the repetitive mode of operation in the range of 1 – 10Hz. For example, Indium or Beryllium or Silver activation based neutron counter depend on counting the activated material over several half-lives of their decay, which are typically of durations of the order seconds or minutes. Such diagnostic methods therefore cannot be employed for shot-to-shot neutron yield measurements at 10Hz operation, which is the ultimate aim of our project.

To overcome the above listed complications, a high bandwidth electrical diagnostic and high sensitivity neutron diagnostics with fast response time (for application in repetitive operation) was implemented in our experimental setup [200] for investigating the performance of the newly developed FMPF devices as pulsed neutron source.

4.2 Electrical Diagnostics

It is well known that the current derivative signal, obtained by Rogowski coil, is a key diagnostic that provides important information regarding the plasma dynamics in the axial acceleration and radial collapse (pinch) phases of the discharge. In the FMPF devices, indigenously designed, high bandwidth (350MHz) Rogowski coil (having response time of $<3\text{ns}$) was used. The conceptual schematic (a) and electrical equivalent (b) of the Rogowski coil is shown in Fig. 4.1.

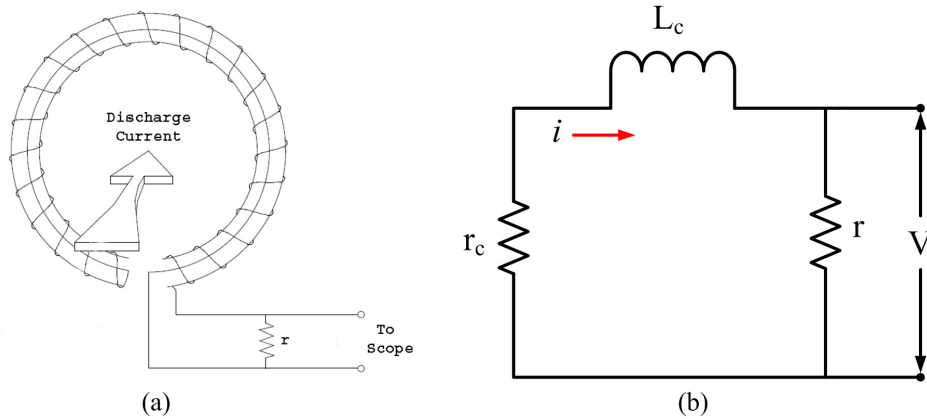


Fig. 4.1 Conceptual schematic of the (a) Rogowski coil and (b) electrical equivalent.

The Rogowski coil is a multi-turn solenoid formed into the shape of a torus. The discharge current passes through the coil, as shown in Fig. 4.1(a). The time varying discharge current dI/dt induces an emf, $k_I(dI/dt)$ in the coil. As shown in Fig. 4.1 (b) above, L_c and r_c are the inductance and resistance of the coil respectively, and i is the induced circuit current in the coil. The coil is terminated with a low-inductance current viewing resistor (CVR) r . The equation for the equivalent circuit is:

$$L_c \frac{di}{dt} + (r + r_c)i = k_1 \frac{dI}{dt} \quad (4.1)$$

Here, k_1 is a constant.

A Rogowski coil can be designed to operate in either differentiating or integrating mode by selection of the proper circuit parameters, as illustrated below:

(i) Integrating mode ($L_c \gg (r + r_c)$): Upon solving the circuit equation for condition $L_c \gg (r + r_c)$, the output voltage measured across the coil is –

$$V_o = ri = \left(r \frac{k_1}{L_c} \right) \times I = k_2 I \quad (4.2)$$

Thus, the output measured is proportional to the transient discharge current I going through it.

(ii) Differentiating mode ($(r + r_c) \gg L_c$): Upon solving the circuit equation for condition $(r + r_c) \gg L_c$, the output voltage measured across the coil is –

$$V_o = ri = \left(\frac{rk_1}{r_c + r} \right) \frac{dI}{dt} = k_3 \frac{dI}{dt} \quad (4.3)$$

Here, the output voltage measured across coil is proportional to the derivative of the discharge current (dI/dt). In our FMPF devices high bandwidth Rogowski coils have specifically been designed to operate in differentiating mode for observing fast dynamic changes in the device impedance.

Since the rise time of the Rogowski coil is limited by the length of winding, in our implemented designs the winding length has been kept $<15\text{cm}$, and they have been made from 5mm wide copper strip. This strip winding technique limits the stray inductances and enhances flux coupling [186] thereby attaining high-bandwidth response. A current viewing resistor of 50Ω has been used for

termination. The arrangement of short length strip wound Rogowski coils used for the FMPF devices are shown in Fig. 4.2.



Fig. 4.2 Arrangement of strip wound Rogowski Coils in (a) FMPF-1 and (b) FMPF-2 device.

For measuring the absolute value of the peak discharge current, calibration of the Rogowski coil is required. To obtain the calibration factor, firstly the current derivative output of the Rogowski coil is passively integrated using an RC integrator of $2\mu\text{s}$ times constant ($2\text{k}\Omega$, 1nF). As the plasma focus is a low impedance device, the obtained trace of the discharge current flowing through the circuit is a damped sinusoid. A sample oscilloscope trace of the passively integrated Rogowski coil output for a short circuit discharge is shown in Fig. 4.3.

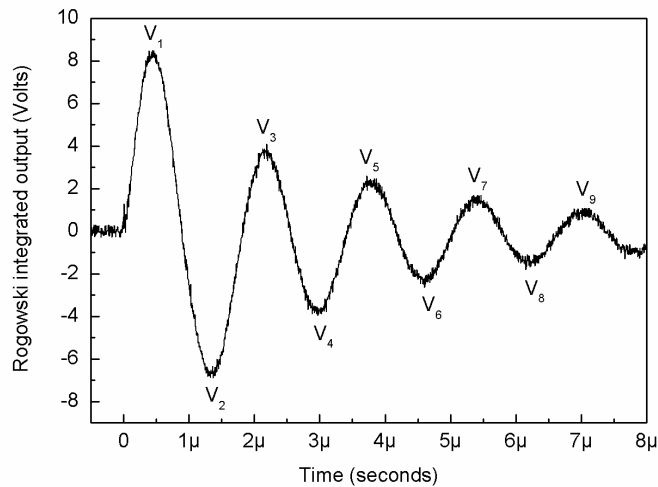


Fig. 4.3 A passively integrated Rogowski output signal.

In an LCR discharge system for calibrating the integrated output of a Rogowski coil, the relation given in [201] is used:

$$I_1 \text{ (kA)} = \frac{\pi C_0 V_0 (1+k)}{T} \quad (4.4)$$

In the expression shown above, the charging voltage V_0 and capacitance C_0 are known values whereas the information about time period T and reversal ratio k are obtained from the experimental current trace, as shown in Fig. 4.3.

The reversal ratio k is an average of the decreasing amplitudes of the consecutive half-cycles V_n ($n = 1, 2, 3, \dots$) of the Rogowski integrated output signal, is estimated by:

$$k = \frac{1}{n-1} \left(\frac{V_2}{V_1} + \frac{V_3}{V_2} + \dots + \frac{V_n}{V_{n-1}} \right) \quad (4.5)$$

Also for the determination of T , the average of several cycles is taken.

Thus, by measuring k and T from the oscillogram, I_1 may be computed in kA using Eq. 4.4 and compared with the value obtained from oscilloscope trace V_1 in volts. The calibration factor f for the Rogowski coil can then be written as:

$$f = \frac{I_1}{V_1} \text{ kA/Volt} \quad (4.6)$$

The calibration factors estimated for the passively integrated outputs of the Rogowski coils used for the FMPF-1, FMPF-2 and FMPF-3 devices were 10kA/Volt, 10.5kA/Volt and 10.5kA/Volt respectively.

4.3 Time Resolved Neutron and Hard X-ray Diagnostics

The time-resolved hard X-ray and neutron measurements, correlated with other diagnostics, improve the overall understanding of the temporal evolution of the transient plasma. These

measurements are helpful in determining the mechanisms of the emission of the hard X-rays and the neutrons. In the following paragraphs, a brief description of the ‘scintillator photomultiplier detector’ and principle of the ‘neutron time of flight’ technique is elaborated.

4.3.1 Scintillator Photomultiplier (SPMT) Detector

Neutrons and high-energy photons impinging upon a suitable scintillation material produce light pulses, which can be detected with a photomultiplier (PM). The choice of the photomultiplier is dependent on the experimental requirements. The photo-tubes vary widely as to their spectral response, quantum efficiency, gain, noise level, maximum output current, physical size and overall structure. When a suitable PM tube has been chosen, there remains a selection of a tube base circuit appropriate to the signal duty cycle for the experiment.

The output voltage pulse for a typical scintillator-photomultiplier combination can be calculated from the following equation:

$$V = \frac{dN}{dt} g \varepsilon e G R \quad (4.7)$$

Here, N - number of scintillation photons; g - light-collection efficiency; ε - quantum efficiency of photo-cathode; G - gain of photomultiplier tube; R - terminating resistance.

It is possible to obtain a few nanoseconds resolution with the combination of a fast multiplier and a selected scintillation material. The detection efficiency for fast neutrons in a plastic scintillator depends upon the neutron energy, the geometry and the type of plastic scintillator.

In our experiments, for acquiring time resolved signals of neutrons and hard X-rays with quantum energies of tens to hundreds of electron volts a 14-stage high-gain photomultiplier tube (EMI 9813BK) coupled with NE-102A plastic scintillator (of thickness-40mm and diameter-52mm). NE-102A has a decay time constant of ~2.4ns [200]. A model PM28B high voltage power supply from

Thorn EMI electron tubes provided -1800V bias to the photomultiplier tube. The various components of SPMT detector assembly have been shown in Fig. 4.4.

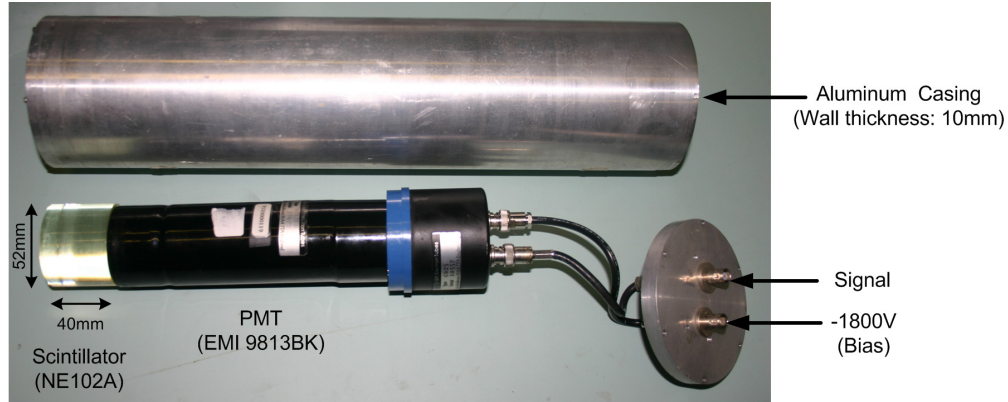


Fig. 4.4 Components of SPMT detector assembly.

The plastic scintillator is coupled by an optical joint (with high viscosity silicon grease similar to Dow Corning DC-200) to the photomultiplier front face. For high light collection efficiency, aluminized Mylar is wrapped around the scintillator. The entire scintillator photomultiplier tube assembly is jacketed inside 400mm long aluminum casing of 10mm wall thickness to effectively shield it from electromagnetic noise and visible light.

4.3.2 Neutron Time of Flight Technique

Particle velocities can be measured by their time of flight over a known distance through the use of fast time-resolved detectors [202]. The use of the time-resolved secondary emission particle detectors allows a comparison of the transit time spread over the observed signal structure from the source at various distances. A measurement of the time interval for similar signal structures then determines the velocity of the particle. The expression for determining the energy of the particle of mass m having velocity v with total kinetic energy W , is $W=1/2 mv^2$.

In principle, it is possible to obtain the entire energy spectrum if the particles are emitted in a

burst by recording the time dependent rate of particles that are incident on the detector placed at distance x from the source.

The energy resolution is given by $\Delta E/E = 2\Delta T/\tau$, where ΔT is the emission duration, $\tau = x/v$ is the average time-of-flight of the particle from the source to the detector, and v is the average velocity [202]. The best accuracy for such a measurement is limited by the value of the emission duration or by the intensity. This theoretical examination of the energy resolution shows that the time-of-flight method can be used for a source emitting short particle pulses with a broad energy distribution and not too large average energy. The energy resolution can be improved by increasing the flight path x , but the maximum value of the flight path is limited by the intensity of the emission of the source [202]. It is noteworthy to mention here that for the emission duration ΔT ($\approx 50\text{ns}$) being much higher than the plastic scintillator time resolution ($\approx 2\text{ns}$), the detector timing resolution can be neglected. Thus, for a particle source characterized by long emission duration, or high average velocities and narrow spectra, this method is difficult to use. However, the average energy of the bulk of neutrons can be determined using a peak-to-peak method of analysis.

4.4 Total Neutron Yield Measurements

Time integrated neutron diagnostic techniques for the measurement of average total yield, involves implementation of both active and passive detection techniques like ^3He proportional counter and CR39 Solid State Nuclear Track Detector (SSNTD). Details regarding the customized setups used in our experiments along with a pre-discussion of the principles of detection by these methods are presented below.

4.4.1 ^3He Proportional Counter

The typical neutron yield per shot expected for a miniature plasma focus device (using existing

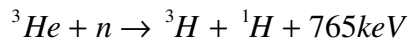
scaling laws) is of the order of $\sim 10^3 - 10^6$ neutrons/shot (with peak discharge current in the range of 40 – 150kA) depending upon device energy, efficiency and design [39-44]. This compelled us to tailor high sensitivity, gas filled thermal neutron detector *i.e.* ^3He Proportional Counter setup for measuring low yields [200, 203]. To achieve high neutron to gamma discrimination ratio it is necessary that the employed detector have minimal gamma-ray sensitivity.

Table 4.1 lists the neutron detection efficiency (i.e. interaction probability for neutrons of the specified energy entering the detector face at normal incidence) and approximate gamma-ray sensitivity for various neutron detection techniques [204]. The detection efficiency mentioned is for a single pass through the detector at the specified energy.

Table 4.1 Neutron detection efficiency and gamma ray sensitivity for some detectors [204].

Detector Type	Neutron Active Material	Incident Neutron Energy	Neutron Detection Efficiency (%)	Gamma Ray Sensitivity (R/h)
Plastic Scintillator	^1H	1MeV	78	0.01
Loaded Scintillator	^6Li	thermal	50	1
Methane (7 atm)	^1H	1MeV	1	1
^3He (4 atm), Ar (2 atm)	^3He	thermal	77	1
^3He (4 atm), CO_2 (5 %)	^3He	thermal	77	10
BF_3 (0.66atm)	^{10}B	thermal	29	10

It can be concluded from the above chart that, ^3He gas-filled detectors are the most suitable because of their high neutron (thermal) detection efficiency (typically $\sim 77\%$) and extremely low gamma ray sensitivity. The detection principle is based on the nuclear reaction shown below and illustrated in Fig. 4.5.



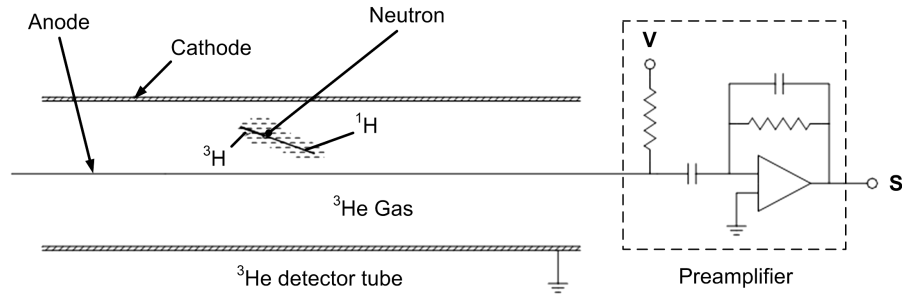


Fig. 4.5 Detection of neutrons using ^3He proportional counter.

As may be seen from the given nuclear reaction, the neutron absorption causes the break-up of the nucleus into a Tritium nucleus, ^3H and a proton, ^1H . The triton and proton share the 765keV reaction energy. Thus, the amount of energy deposited by this nuclear reaction leads to ionization of the gas. When operated in proportional mode, the ionization produced by the primary electrons initiates the multiplication process near the anode wire leading to neutron detection. The ionization potential of Helium is about 25eV. So about $\sim 25,000$ ions and electrons ($\sim 4 \times 10^{-15}\text{C}$) are produced per neutron absorption.

The cross section for the $^3\text{He}(n,p)$ reaction is 5330b for thermal neutrons. It is important to note that cross section has strong dependence on the incident neutron energy E , having roughly $1/\sqrt{E}$ dependence. Typically ^3He has 77% efficiency for thermal neutrons ($\sim 0.025\text{eV}$), 2% at 100eV, 0.2% at 10keV and roughly 0.002% at 1MeV [204]. Because of this strong energy dependence ^3He gas filled neutron detector tubes are customarily surrounded by a moderating medium like paraffin wax with thickness of $\sim 10\text{cms}$ to maximize the counting efficiency. The optimum thickness of the moderator was estimated using MCNPX simulation for D-D neutrons. In our setup, a high sensitivity ^3He neutron detector tube – RP-P4-1636-203 from GE Reuter-Stokes (having nominal sensitive length of 36” and 2” diameter) was used in proportional counter mode along with low noise fast rise time charge sensitive Preamplifier – CAEN A424A and Amplifier-ORTEC 485 [200].

The principle function of the preamplifier is to integrate the (very small) charge pulses from the detector and provide impedance matching to drive the signal cables between the counter and the amplifier. The preamplifier derives its power from the main amplifier. It also provides connection from the HV power supply to the counter (i.e. ^3He detector tube) and contains a high voltage capacitor which insulates the high voltage from the preamplifier input. TENNELEC TC 948 (0 to +1500V) has been used as the NIM high voltage module. For providing power to various transistorized modules viz. amplifiers, high voltage supplies etc. standard NIM bin module 4001A from ORTEC was used. Layout of the overall detection and instrumentation set-up is shown in Fig. 4.6 [200].

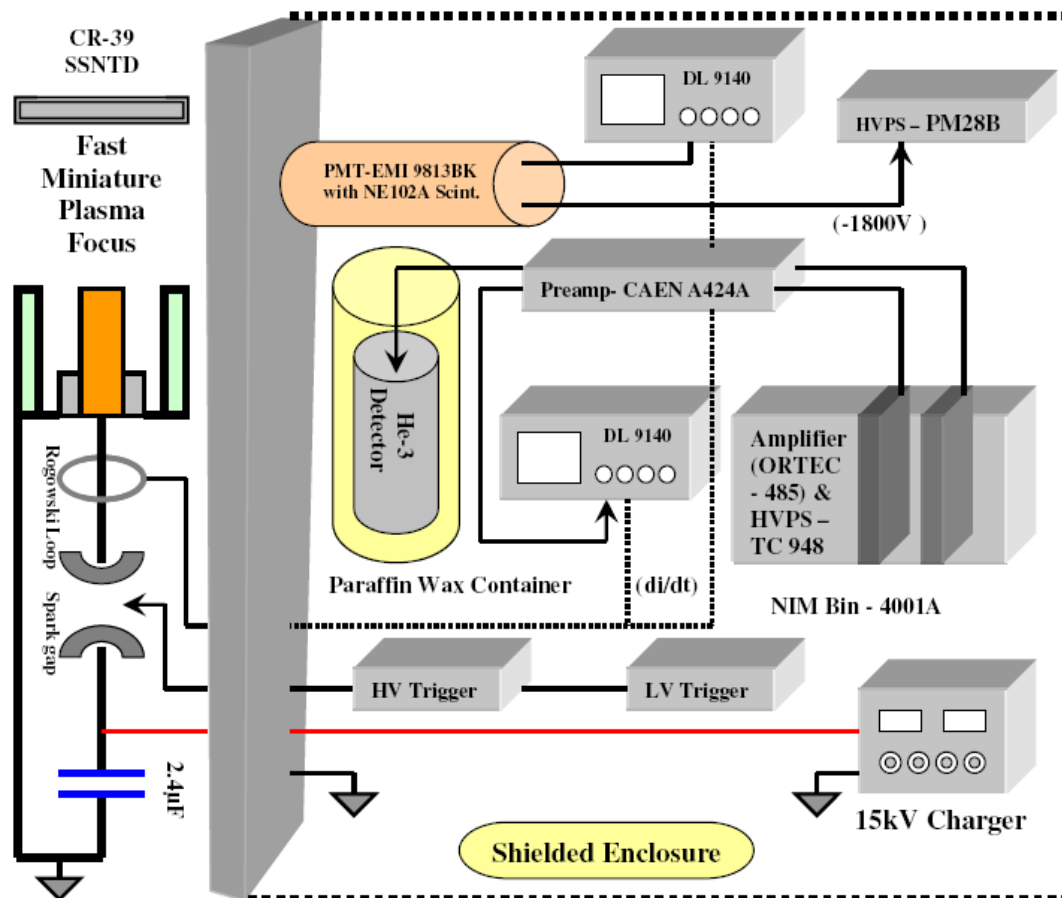
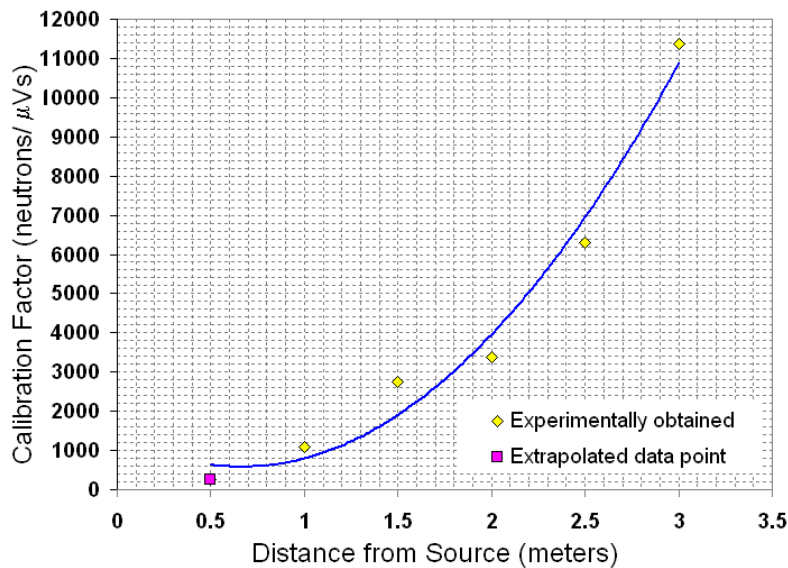


Fig. 4.6 Overall layout of neutron detection and instrumentation set-up.

To calibrate the ^3He detection system (with moderator included) a Beryllium activation counter (previously cross calibrated with an Indium detector [199]) and NX2 plasma focus device was used. Both of these detectors, newly adapted high sensitivity ^3He and the Beryllium activation counter were used simultaneously in the NX2 plasma focus device, detecting the neutron yields of the order of 10^6 to 10^7 neutrons per shot (it is noteworthy to mention that NX2 was deliberately operated away from its neutron optimized regime (giving $\approx 10^8$ neutrons/shot) to facilitate the cross calibration of the ^3He detector). We estimated the calibration factor for the ^3He proportional counter placed at distances of 1m to 3m (at 0.5m intervals) from the NX2 pinch. The variation in the calibration factor with distance is plotted and tabulated in Fig. 4.7. For each of the distances shown in Fig. 4.7, the calibration factor was estimated using an average of 20 NX2 shots. It may be noted that the calibration factor for 0.5m distance of ^3He proportional counter from the neutron source point has been extrapolated, because the NX2 has an access limitation of $>1\text{m}$ for the large ^3He detector tube. The bias voltage was kept fixed at +650V during the entire calibration process [200].



Distance (meter)	Calibration factor ($\mu\text{Vs/ neutrons}$)
0.5	3.70×10^{-3}
1.0	9.27×10^{-4}
1.5	3.65×10^{-4}
2.0	2.97×10^{-4}
2.5	1.59×10^{-4}
3.0	8.80×10^{-5}

Fig. 4.7 Calibration curve for ^3He proportional counter (at bias voltage of +650V).

It is noteworthy to mention here that +650V is a relatively low operating/bias voltage. For single neutron detection, a bias voltage of ~1400 to 1600V would be used. Due to the burst nature of PF neutron yield, many neutrons are absorbed in ^3He detector within several tens of μs and the preamplifier integrates the resulting charge for all absorptions. Therefore to avoid saturation in the preamplifier, the ^3He detector is operated with a lower bias voltage and therefore, a much lower gas gain than would be employed for single neutron detection.

4.4.2 Solid State Nuclear Track Detector (SSNTD)

Other than the use of ^3He proportional counter; the neutron yield from FMPF devices were also confirmed with passive detection technique using Solid State Nuclear Track Detector (SSNTD). The major advantage of this technique is that it is insensitive to photons. The plastic under commercial names CR-39/ PM355 are most widely used SSNTD's for the detection of charged particles and neutrons [122, 123]. Basically this plastic material is transparent thermo-set polymer of 'Allyl Diglycol Carbonate' with the chemical formula $\text{C}_{12}\text{H}_{18}\text{O}_7$.

The detection principle of nuclear track detectors has been summarized by Fleischer *et al* [205]. When a highly ionizing charged particle is incident on a plate of plastic, then due to inelastic interaction its energy deposition along the trajectory causes radiation damage within a cylindrical region a few nm in radius. As a result, it leaves a latent track of broken molecular bonds along the trajectory of the particle. The latent track is then revealed and enlarged by etching the plastic with a suitable chemical such as sodium hydroxide (NaOH) or potassium hydroxide (KOH). The typical etching conditions are 6M aqueous solution at 70°C. The etchant removes material preferentially along the particle's trajectory. Specifically, two simultaneous etch rates controls the development of conical etch pits: the general etch rate, V_G which removes the bulk of material isotropically, and the track etch rate, V_T which etches along the particle path. The sensitivity of the material to particles of a

given type and energy is usually defined as the etch rate ratio $s \approx V_T/V_G$, it is related to the linear energy transfer (LET) that is responsible for etchable radiation damage. Since the etch rate is larger within the track of broken bonds, a conical pit is produced, whose diameter depends, for a given etching time, on the charge and energy of the particle that produced it. Furthermore, the ellipticity of the pit gives information about the angle of incidence of the charged particle. These pits can be easily observed with an optical microscope.

For filtering out the charged particles with energies below a certain threshold, filters of appropriate thickness are used in front of the CR-39/ PM355 detector. We have used ‘Kapton’ film as the filter in our experiments. Table 4.2 summarizes the results of SRIM simulation that have been done for estimating the threshold energies of charged particles (for mainly protons and deuterons) passing through the Kapton filter layer with 50 μm , 75 μm , 100 μm thickness.

Table 4.2 Threshold energies of charged particles for passing through Kapton filter.

Kapton filter Thickness	Threshold energy of Proton	Threshold energy of Deuteron
50 μm	1.8MeV	2.3MeV
75 μm	2.3MeV	3.0MeV
100 μm	2.7MeV	3.6MeV

Results obtained from the SRIM code [122, 134] show that a 50 μm thick Kapton window would allow to pass 2.45MeV neutrons and the 3.03MeV protons from the fusion reactions but it will stop the other charged products of D-D fusion reactions such as 1.01MeV tritons and the 0.82MeV ^3He nuclei. It will also stop the lower energy deuterons as well as impurity ions resulting from electrode erosion. The detection efficiency of the CR-39/ PM355 for protons at near normal incidence is expected to be 100%.

4.5 Time Resolved Soft X-ray Measurements

For the time-resolved soft X-ray measurements from the plasma focus, the choice of the detector depends on the information required. The purpose of our experiment was to get information on the temporal evolution of the soft X-ray emission over specific spectral regions of interest.

For the detection of the soft X-rays there are an increasing number of semiconductor detectors e.g. surface barrier, ion-implanted and avalanche junction diodes, etc. These semiconductor detectors are also sensitive to charged particles, provided the particle deposits part of its energy in the sensitive layer of the detector. When a semiconductor detector absorbs an X-ray photon of energy $h\nu$, a number of $h\nu/\omega$ electron-hole pairs are produced, where ω is the average energy needed to create an electron-hole pair. The collected charge is therefore proportional to $h\nu$, the photon energy. For silicon at room temperature, $\omega = 3.62\text{eV}$. The quantum detection efficiency is given by [206]:

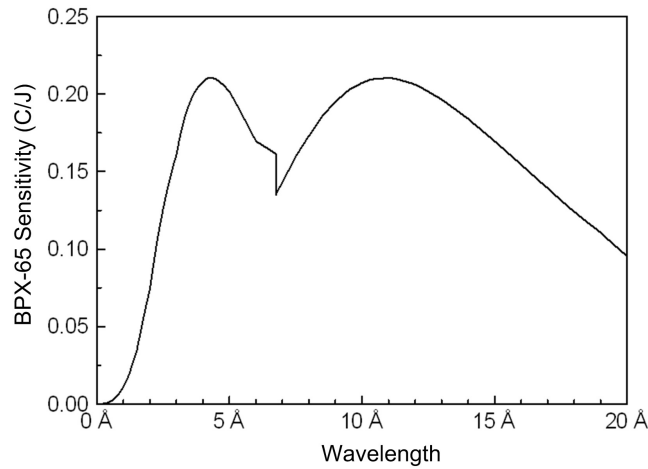
$$Q = \frac{h\nu}{\omega} e^{-\mu_d t_d} (1 - e^{-\mu_s t_s}) \quad (4.8)$$

where the first exponential term describes the absorption in the dead layer (on the front surface of the detection area) of thickness t_d , and the second one gives the efficiency for stopping X-rays in the depletion (sensitive) region of thickness t_s . μ_d and μ_s are the linear attenuation coefficients in the dead layer and sensitive layer, respectively. Hence, it can be seen that there is an energy range over which semiconductor detectors can be used. The lower limit is determined by the absorption in the dead layer and the upper limit by incomplete absorption (as thickness of sensitive region may not be enough to absorb the higher energy photons). Lower energy X-rays can be detected using semiconductor detectors if the dead layer is thinner. The pair creation energy ω is in general a function of the photon energy. It is assumed that ω is a constant for photon energies well above the band gap of Si ($\sim 10 \times E_g$). Since both the electrons and the holes have large mobilities, and since the collection distances are

short, it is possible to achieve relatively short collection times. This is an important advantage for measurements of pulsed X-ray sources.

BPX-65 PIN diodes (a detector normally used in the optical range) were chosen for our system. The glass window was removed for the application of X-ray detection. The BPX-65 PIN diode sensitivity $Q(\lambda)$ below 20 Å along with its major technical specifications is presented in Fig. 4.8.

A five-channel X-ray spectrometer with filtered BPX-65 PIN diodes [207] was used in our experiment. Appropriately selected filter pairs were used to maximize the detection within a particular band of the expected emission spectrum. Details of the filter combinations used in our experiments are provided in Chapter 5. The diode in each channel was reverse biased at -45V, and a capacitor signal decoupling circuit was used, as illustrated in Fig. 4.9.



Specification of BPX-65 PIN diode	
Radiant sensitive area	: 1mm ²
Intrinsic Si wafer thickness	: 10μm
Dead layer thickness	: 0.5μm
Rise time (typical) (@ 900 nm)	: 0.5ns
Dark current	: < 5nA

Fig. 4.8 BPX-65 sensitivity in wavelength range below 20Å and its major specifications.

The formula used to estimate the X-ray yield at source into 4π steradians per shot is given as [208]:

$$E_{4\pi} = \frac{4\pi R^2 A}{R_0 A_D S} \quad (4.9)$$

Here, R is the distance between the X-ray source and the detector, A is the area under the X-ray signal obtained on oscilloscope, R_0 is the termination resistance (50Ω), A_D is the radiant sensitive area of

BPX65 detector, and S is the overall sensitivity of the detector in a particular wavelength range with the inclusion of attenuation of the X-rays by filter and gas.

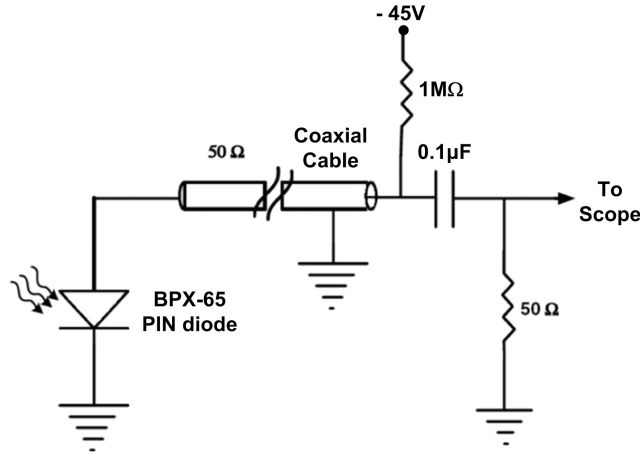


Fig. 4.9 Electrical schematic of the biasing and decoupling circuit.

4.6 Ion Measurements using Faraday Cup

Faraday cup is an important diagnostic tool for obtaining information about the velocity/energy distribution of deuterons in Deuterium operated plasma focus devices. The schematic of Faraday cup used in our setup along with biasing and decoupling circuit has been shown in Fig. 4.10.

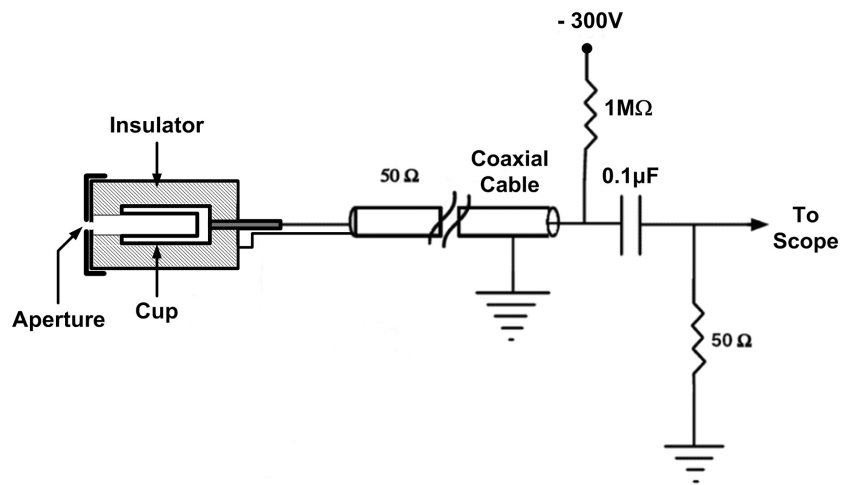


Fig. 4.10 Biased Faraday cup collector for ion energy measurement.

It consists of a defining aperture of 0.5mm diameter and a 10mm deep collector cup (of copper) biased at a negative potential of -300V and placed ~5mm behind the aperture. The physical dimensions were chosen to match with the impedance of coaxial cable used for signal transmission (i.e. 50Ω). The characteristic impedance of the Faraday Cup was adjusted to 50Ω using the simple relation [209]:

$$Z = \frac{138.2}{\sqrt{\epsilon_r}} \log_{10} \frac{D}{d} \text{ ohms} \quad (4.10)$$

where D = inside diameter of the outer electrode, 35mm; d = outside diameter of the inner electrode, 8mm and ϵ_r = dielectric constant of spacer material, 3.2.

The dielectric spacer used in between the cylinder is of cast Nylon. The accompanying electrons associated with the ion beam must be prevented from entering the Faraday cup. This is accomplished either by negative electrical bias or by applying a transverse magnetic field. We used electrical biasing in our experiments. The output signal from Faraday cup may be denoted as:

$$V = IR \quad (4.11)$$

where I is the ion beam current, R is a resistor (50 ohms) across which V is measured. A negative voltage V_{bias} is applied to the Faraday cup, such that [209]:

$$|V_{bias}| > V_c + V + V_e = \frac{1}{C} \int Idt + IR + V_e \quad (4.12)$$

where V_c is the voltage change on the capacitor caused by the ion current source charging the capacitor and V_e is the potential to repel the electron beam.

Secondary electrons are generated at the collector surface by the high energy ions bombarding it. This results in an overestimation of the ion current. In general, only a fraction of secondary electrons contribute to the ion current if their only path to ground is through the incoming beam. If the applied bias potential is enough to repel all the primary electrons near the aperture, the neutralization of the beam in the gap (between collecting surface and aperture) is done by the secondary electrons [209].

Thus, by applying negative bias to the collector, the stray electrons are decelerated and deflected away from the collector; limiting their contribution to the measured ion current. As a rule of thumb, while working with different gases in plasma focus device, the minimum negative bias voltage to be applied to the Faraday cup during ion current measurement is estimated using the relation shown below [210]:

$$|V_{bias}| > \left(\frac{m_e}{m_i} \right) E_b + V \quad (4.13)$$

where m_e is the mass of electron, m_i is the mass of ion, E_b is the maximum expected beam energy and V is voltage that is observed without biasing the collector. The voltage observed in our collector without biasing was $\sim 3V$. In reference with the above considerations, our ion collector was biased at $-300V$ for Deuterium experiments; since ion energies up to $1MeV$ are expected.

Since the biased Faraday cup is placed at known distances from the pinch in the end-on direction, it is possible to find the ion velocity distribution from the time of flight of the ions. If the ions were created and accelerated in a short time interval compared to the flight time, the speeds of the ions, v_i , can thus be measured at a distance d from the target using the relation:

$$v_i = \sqrt{\frac{2E_i}{m_i}} = \frac{d}{t} \quad (4.14)$$

Where E_i is the kinetic energy of the ion and m_i is the mass of the ion.

The analysis of ion signal obtained using the biased Faraday cup is presented in Chapter 5.

After discussing the various diagnostics tools and techniques, experimental results obtained with various anode geometries, including the time resolved and time integrated radiation measurements that have been performed for evaluating the performance of FMPF devices, will be presented in the following chapter.

Chapter 5

Experimental Results and Discussion

In this chapter, the neutron emission results, analysis and discussion of the optimization experiments performed on three different versions of Deuterium filled miniature plasma focus devices – FMPF-1, FMPF-2 and FMPF-3 are presented. As elaborated in Chapter 3, the main difference between three devices of equivalent energy is the variation in their pulsed power system efficiencies and capabilities that affects the neutron yields.

5.1 Introduction

To optimize the performance of the newly developed FMPF devices as efficient pulsed neutron sources; the behavior of machines have been systematically investigated during each stage of development of the different FMPF devices such as: changing anode length and shape [200], cathode structure [211], working gas pressure [211,212], high-Z gas admixture operation [213,214] hard X-ray and neutron yield anisotropy measurements [215] and repetition rate [216]. Along with the evaluation of circuit parameters, the correlation of different radiation products (like X-rays and neutrons) with different operating parameters (gas pressure, operating voltage, repetition rate etc.) and anode dimensions is presented. The investigation incorporates numerical experiments that have been performed using the ‘Lee Code’ [70], to simulate the axial and radial phase dynamics along with gross plasma focus properties in terms of expected radiation product, with the use of modified electrode assemblies. For characterizing the FMPF devices as compact neutron sources, experimental studies on time resolved measurements of neutrons (and hard X-rays) along with time integrated measurements of fusion neutrons/protons, in the axial and radial directions are also performed and reported [215]. In the final section, experimental results on enhancement in time averaged neutron yield by operating the

FMPF-2 and FMPF-3 device in repetitive mode up to 10Hz, has been analysed in detail while revealing the consequences and limitations of high repetition rate operation [216,217].

5.2 Results of FMPF-1 Device

The ‘FMPF-1’ being the first version of low energy miniature plasma focus device; has mainly been used as test bed in our experiments for investigating the effect of various modifications and the corresponding changes in gross dynamics of miniature plasma focus device operation at ~0.2kJ [211-214]. The results and experience obtained from successive modifications in design and operating parameters of the FMPF-1 device in relation to its electrical and radiation yield performance have been used as a guide in conceiving the design and operation of upgraded versions of FMPF devices *viz* FMPF-2 & -3.

In the following sections, along with the preliminary investigations of short circuit discharge characteristics (for the estimation of equivalent circuit parameters and pulsed power system efficiency); anode/ cathode optimization experiments have been discussed in detail with respect to neutron and X-ray emission results [200, 211-215]. The main objective of performing optimization experiments on FMPF-1 device was to (i) demonstrate the production of neutrons from a device operating at such low energy, (ii) to optimize the device efficiency for obtaining maximum neutron yield/shot, (iii) to map out a pathway of developing repetitive version(s) of FMPF devices with upgraded pulsed power system efficiencies, and (iv) use this device as test bed for gauging the adequacy of new diagnostics such as ^3He proportional counter based neutron yield measurements which must not only have higher sensitivity to low neutron yields (expected from lower energy MPF) but also have fast response time so that it can be used on future versions of repetitive FMPF devices.

5.2.1 Short Circuit Discharge Characteristics

After construction of FMPF-1 device, the foremost investigation performed on this device was the study of discharge characteristics to electrically characterize and investigate the efficiency of its pulsed power system while driving the MPF as load. The study of short circuit discharge characteristics is also necessary for the approximation of the anode dimensions (radius and length), since those are primarily decided by the time duration ($T/4$) for the discharge current I to reach its peak value and to determine the peak discharge current attainable at various charging voltages [33].

As the evaluation of circuit parameters and the efficiency of the pulsed power system (*i.e.* the capacitor bank) is mainly determined by the peak discharge current it can deliver, short circuit tests were performed for the measurement of same. A schematic of the electrical equivalent for the short circuit test is shown in Fig. 5.1.

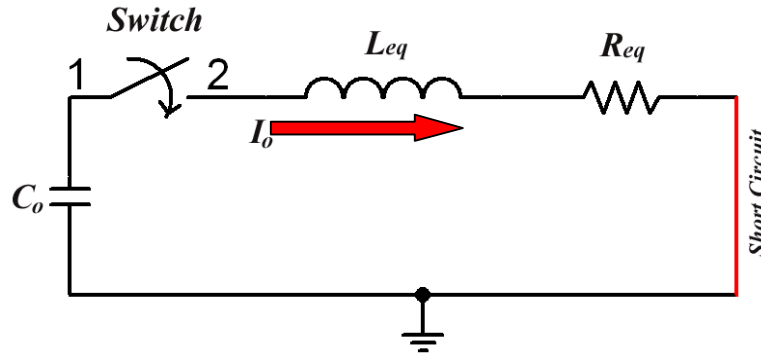


Fig. 5.1 Equivalent circuit for short circuited discharge.

Here, L_{eq} and R_{eq} represent the net inductance and resistance in the discharge circuit contributed by – capacitor bank + sparkgap + transmission plates + connections.

The short circuit tests at various working voltages were experimentally performed by discharging the capacitor bank at the highest safe pressure inside the vacuum chamber filled with Argon. Since, in very high pressure discharges, the breakdown essentially occurs at the closed end of

coaxial electrode assembly and subsequently very high filling gas pressure does not allow the current sheath to move down the tube avoiding any inductive or dynamic loading of the discharge current by the current sheath motion, hence it is equivalent to short circuiting the plasma focus at the closed end of its electrode assembly near the insulator sleeve [218]. In our experiments we have used up to 40mbar Argon during the short circuit discharges, while operating in the range of 9-14kV. Sample oscillograms of short circuit discharge current traces at 9, 11 and 14kV operation are shown in Fig. 5.2.

The sinusoidal damped oscillation waveforms of short circuit discharge current traces, shown in Fig. 5.2 were obtained using a Rogowski coil along with a passive integrator (described in section 4.2). The discharge currents reach to their peak value ($T/4$) at about 400ns. It is important to note that the peak discharge current I is mainly limited by the effective inductance and resistance contributed by various FMPF-1 system components, shown as series equivalent circuit inductance L_{eq} and resistance R_{eq} in the equivalent circuit of short circuit discharge set-up shown in Fig. 5.1.

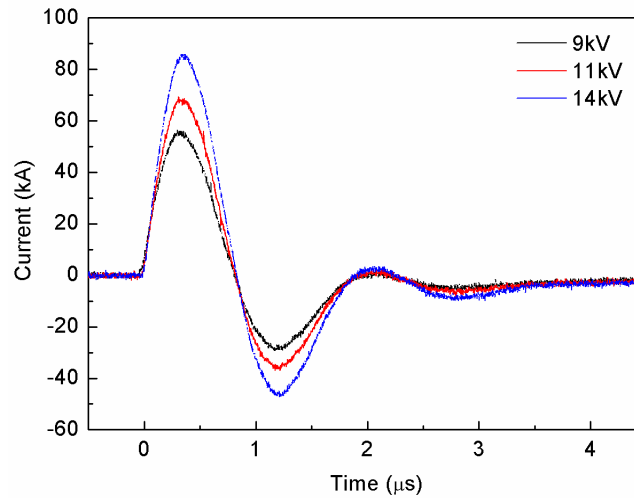


Fig. 5.2 Sample oscillograms of short circuit discharge current traces at various voltages.

The methodology that was used for the measurement of the discharge current I and performance limiting parameters L_{eq} and R_{eq} in the circuit, from the experimentally obtained discharge current waveforms at various discharge voltages, is as follows. Firstly from the obtained oscilloscope

traces of short circuit discharge current; the reversal ratio k and time period T of the waveform are determined (as explained in section 4.2) then the peak current I_l in the first half of the cycle is approximated by using the relation shown in Eq. 4.4 [201]. The expressions used for measurement of L_{eq} and R_{eq} in the circuit are given as [201]:

$$L_{eq} = \frac{T^2}{4\pi^2 C} \quad (5.1)$$

$$R_{eq} = -\frac{2}{\pi} \sqrt{\frac{L_{eq}}{C}} \times \ln(k) \quad (5.2)$$

From the above mentioned relations, for the operating voltage range of 9 – 14kV, the estimated average values of series equivalent inductance L_{eq} and resistance R_{eq} are 27 ± 2 nH and 66 ± 3 mΩ. The estimated higher value of equivalent circuit resistance R_{eq} is consistent with the experimental observation of highly damped discharge current traces shown in Fig. 5.2. The investigated peak discharge currents (under short circuited condition by gas filling at high pressure) at different operating voltages/ energies, for FMPF-1 device, has been collectively summarized and plotted in Fig. 5.3.

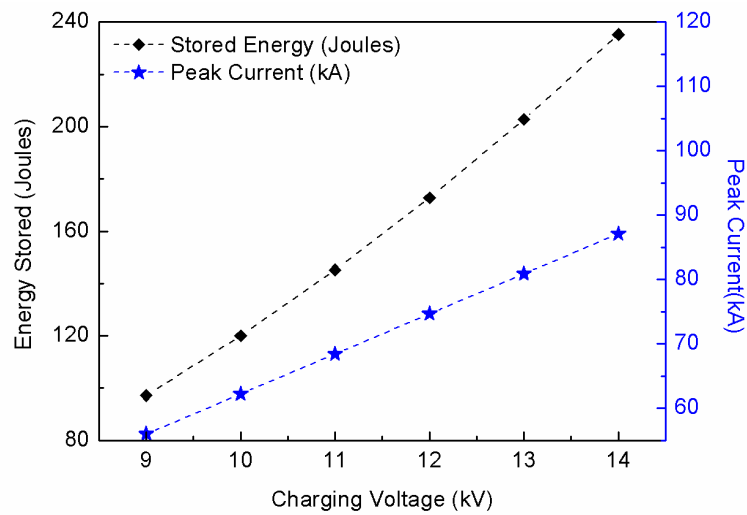


Fig. 5.3 Maximum peak currents in FMPF-1 device at different charging voltages.

The main electrical parameters for the FMPF-1 device, as obtained through short circuit tests, are summarized in Table 5.1.

Table. 5.1 Main electrical characteristics of FMPF-1 device.	
Energy bank capacitance (C_o)	2.4 μ F
Maximum charging voltage (V_o)	14kV
Maximum stored energy @14kV (E_o)	235J
Maximum current (under short circuit) (I_{sc})	87kA @14kV
Typical operating voltage range	12 – 14 kV
Eq. circuit inductance (L_{eq})	27 $\pm$ 2nH
Eq. circuit resistance (R_{eq})	66 $\pm$ 3m Ω
Voltage Reversal (k)	33%
Quarter time of discharge ($T/4$)	~400ns
Maximum discharge repetition rate (DRR)	0.5Hz

It is important here to note that even though the pulsed power system of the FMPF-1 device has a low system inductance due to its compact design and integration, its efficiency (*i.e.* higher peak discharge current at same energy level of capacitor bank) is limited by high equivalent series resistance that is mainly contributed by the capacitors (refer section 3.4.1). Thus, we can conclude that in the selection of capacitors, low equivalent series resistance (ESR) is equally important as low equivalent series inductance (ESL), to obtain high discharge current.

5.2.2 Optimization of Anode Parameters

The plasma dynamics involved with DPF discharges is complicated and does not lend itself to accurate predictive modeling. So far various simulation codes like the SNOWFLOW model and its derivatives, and MHD codes have been developed and used with varying degrees of success to analyze the dynamics but experimental exploration is still very necessary in order to optimize the DPF design, and find the best compromise between various design trade-offs. In this section, we report the experimental study of different anode shapes and their effect on plasma focus dynamics, especially when the shape is tapered towards the open end.

As illustrated in the section 3.2, after obtaining important information from the investigation of short circuit discharge characteristics on (i) the time duration after which the discharge current reaches to its peak value ($T/4$), and (ii) the amplitude of the discharge current I at the corresponding charging voltage, along with the preliminary approximations of average axial transit speed of the current sheath of $\approx 5\text{cm}/\mu\text{s}$; the anode dimension were estimated as: length $\approx 20\text{mm}$ and radius $\approx 4.6\text{mm}$ (anode length was estimated by matching plasma run down time to coincide with peak discharge current and the anode diameter was estimated assuming speed factor of $89\text{kA}/\text{cm}/(\text{Torr})^{1/2}$ at $13\text{kV}/81\text{kA}$, 5mbar D_2 operation). Using the above mentioned anode dimensions as guiding numbers, the optimization experiments were started. The wall of tubular vacuum envelope, of 30mm inner diameter was used as the cathode. An insulator sleeve of Pyrex glass with a breakdown length of 5mm was placed between the anode and cathode. The four anode designs that have been studied are shown in Fig. 5.4(a) - (d).

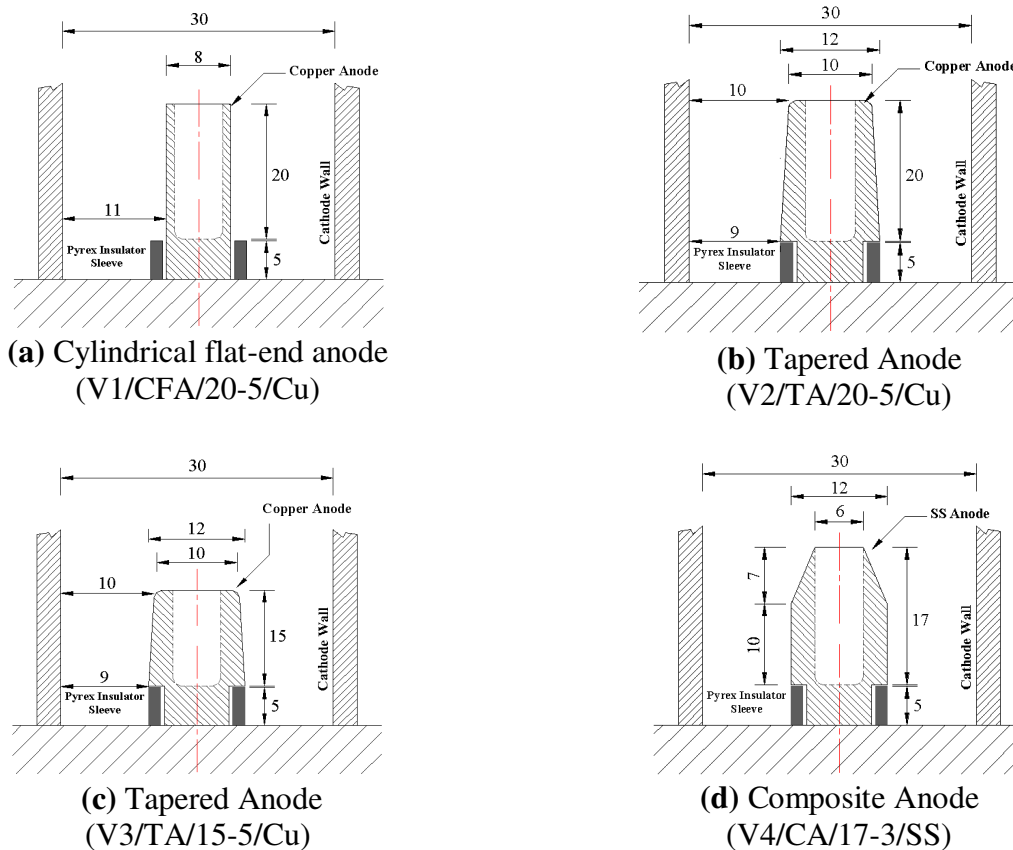


Fig. 5.4 Schematic of various anode geometries that have been experimentally studied.

All the implemented anode designs are hollow because this feature has been reported to produce higher neutron yields with better stability of the pinched plasma, by comparison with solid anodes [32,219,220]. For all the four configurations shown in Fig. 5.4(a-d), we scanned the Deuterium filling pressure in a wide range to investigate the optimum focusing region (recording 20 shots at each selected pressure value). After every set of five shots, the filling gas was purged and fresh gas was filled to minimize the effects of impurities accumulated in the working gas. The negative peak (dip) of the current derivative signal obtained from the Rogowski coil is taken as the fiducial time reference for all time-resolved measurements. The charging voltage was kept fixed at 12kV throughout the investigation.

The current derivative signal, is a key parameter that provides relevant information regarding the plasma dynamics in the axial acceleration phase and radial collapse (pinch) phase of the plasma focus device. The successful pinch compression is verified by a strong dip in the current derivative signal which is due to the rapid change in plasma impedance [93,121]. According to Bernard *et al* [93] the increase in plasma impedance is a necessary condition for good operation of plasma focus devices for efficient production of neutrons and X-rays. They attributed the large increase in the plasma impedance to (i) fast changing plasma inductance $L_p(t)$, and (ii) growth in anomalous resistance $R_p(t)$ caused by the collision of electrons with plasma waves, when they reach a high intensity level. Therefore in our experiments, the current derivative signal of discharge current was used to investigate the effect of anode design/gas parameters on the pinching characteristics of the miniature plasma focus device. The magnitude of this distinct, sharp dip in the current derivative signal is considered to be a measure of the pinching efficiency. Higher magnitude current derivative dip implies stronger pinching [82, 221] resulting in efficient radiation emission from the pinched plasma column. Representative di/dt traces for the selected anode designs, at their optimized D₂ filling gas pressures, are shown in Fig. 5.5(a) – (d).

Pinching efficiency (measured in terms of peak amplitude of the dip in the current derivative signal) for different anode geometries is shown collectively in Fig. 5.6 for comparative analysis. The Fig. 5.6 also includes the ‘Speed factor’ curves for the investigated anode designs.

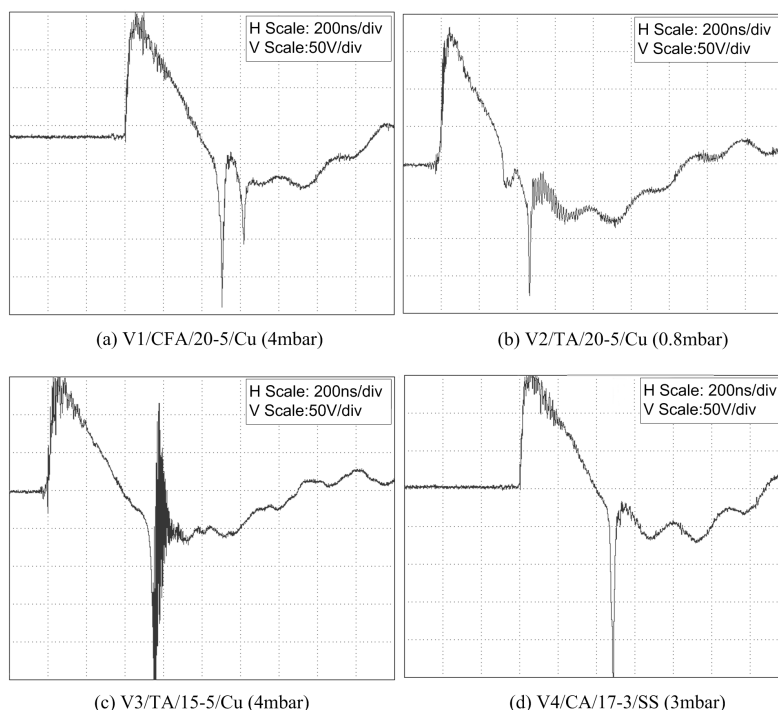


Fig. 5.5 Representative di/dt patterns for the experimented anode designs.

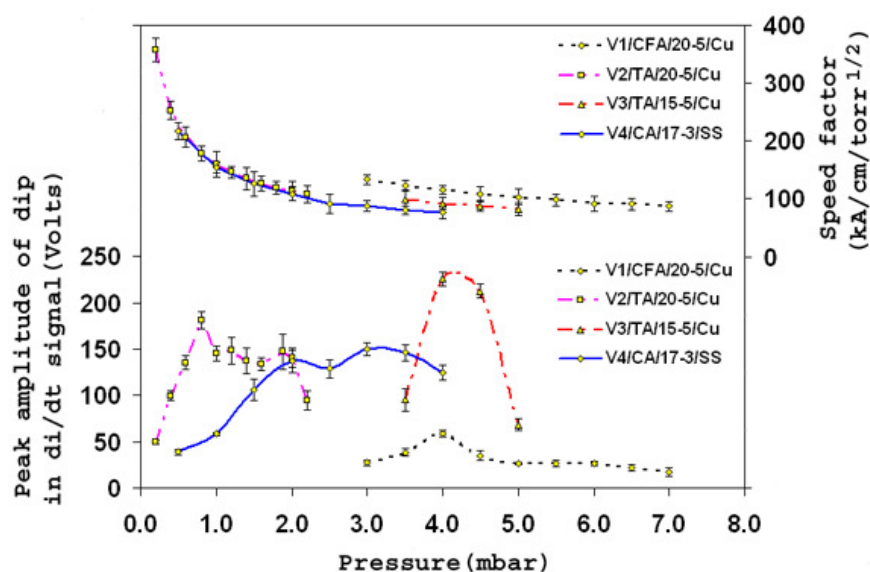


Fig. 5.6 Focusing peak amplitude and Speed factor for various anode geometries.

The ‘Speed factor’ (also referred to as the Drive parameter) i.e. $(I/a)/\sqrt{p}$ (here, I is the peak discharge current in kA, a is the anode radius in cm and p is the filling gas pressure in Torr), is a measure of the driving magnetic energy per unit mass associated with peak discharge current and is considered for analyzing the performance of different anode designs. Across the wide energy range of Deuterium optimized, Mather type plasma focus devices, the drive parameter has a typical value of $89 \pm 8 \text{ kA/cm/Torr}^{1/2}$ [33]. The characteristic curves shown in Fig. 5.6 illustrate that operating regimes and pinching characteristics of plasma focus device are significantly influenced by anode design/ geometry. For the cylindrical flat anode geometry (V1/CFA/20-5/Cu), pinching efficiency was found to be low, along with poor reproducibility. This was probably due to either failure of capacitor bank to discharge or delay in the ignition of the breakdown phase across the insulator which in turn causes further delay in the current sheath buildup time [38, 150]. As the miniature plasma focus has a relatively short quarter time period, its pinching efficiency is significantly affected by the delay in breakdown phase. As uniform initiation of plasma sheath depends crucially on the electric field between the anode and cathode [63], to enhance electric field and support rapid formation of surface discharge streamers over insulator; in the rest of the anode designs/ geometries, the lower end of the anode diameter was extended up to the insulator outer surface with a sharp edged periphery, as seen in Fig. 5.4(b) – (d) to improve the initiation of the breakdown phase. This resulted in significant improvements in pinching efficiency and higher reproducibility. For the second anode design V2/TA/20-5/Cu (see Fig. 5.4(b)), even though good pinching efficiency was achieved at lower operating pressures, we failed to detect neutrons or hard X-rays. It can be seen from the graph shown in Fig. 5.6 that the speed factors at various pressures for this anode are relatively high ranging from 250 to $113 \text{ kA/cm/(Torr)}^{1/2}$ in the pressure range of 0.4mbar to 2mbar. Such high-speed factors may cause effective separation of the magnetic piston from the plasma layer, thereby resulting in an inefficient collapse of the plasma layer in the radial phase as the piston continues to drive axially [222].

However, this anode design was found favorable for producing high soft X-ray yields [214].

In order to reduce the speed factor (to bring it closer to the typical range of $89 \pm 7 \text{ kA/cm} / (\text{Torr})^{1/2}$) a third anode geometry – V3/TA/15-5/Cu (see Fig. 5.4(c)) was used with the anode length reduced to 15mm (from 20mm of the second anode design). Reduction of the anode length increases the operating pressure to match the arrival time of current sheath at the anode top with the quarter time period of the capacitor bank. As expected, this anode design performed best in a relatively high pressure regime, with the optimum pinching pressure shifted to 4mbar (as compared to 0.8mbar for the second configuration (refer Fig. 5.6)). However, due to significantly reduced length it focused efficiently but in a very narrow pressure range of 4 to 4.5mbar.

To broaden the operating pressure range for optimum pinching, a fourth composite configuration – V4/CA/17-3/SS (see Fig. 5.4(d)) was tested. This anode is made of stainless steel as it has low spark erosion characteristics as compared to copper [188]. In this design, the length of the anode was increased from 15mm to 17mm and a taper was incorporated over the last 7mm of the anode length with the radius gradually tapering from 6mm to 3mm. This decrease in anode radius from 5mm (of third design) to 3mm (for fourth design) enhanced the linear current density (*i.e.* I/a) and hence increased the compression efficiency of the pinch, since dimensions and life time of pinches scale according to anode radius [33].

As indicated from characteristic curves shown in Fig. 5.6, in experiments with the fourth composite anode geometry, efficient and consistent pinching was observed over a wide range of operating pressures, from 1.5 to 4mbar.

5.2.3 Deciding Factors for Efficient Focusing

After performing the experimental study on various anode geometries, it was realized that there are two major issues – speed factor and ignition/initiation process (in the breakdown phase), which are

critical for efficient performance of miniature plasma focus device. Their role in deciding the performance is discussed below.

5.2.3.1 Speed Factor

It is remarkable to note that for the neutron optimized regime in miniature plasma focus device, speed factor/ drive parameter has higher and lower end cut-offs. Especially when the speed factor is increased much over $90\text{kA/cm}/(\text{Torr})^{1/2}$, the device no longer operates in the neutron optimized regime. The lower speed limit is easily understandable in terms of poor coupling of the magnetic piston to the driven plasma layer, indicated by a low magnetic Reynolds number R_m , or in terms of low specific energy [222]. For the higher speed limit (as witnessed in our experimental study, anode design - V2/TA/20-5/Cu gave good focusing efficiency, but neither neutron or hard X-ray radiation were detected), it appears that high R_m leads to an effective separation of the magnetic piston from the plasma layer. The plasma layer then collapses radially, whilst the magnetic piston is still driving axially [201]. Such a large separation is not conducive to an intense focus. Other possible mechanisms responsible for the upper speed limit are due to (i) plasma–anode interactions or, more specifically, the ion (and sometimes, electron) current component, (ii) development of Rayleigh–Taylor instabilities, and (iii) at high velocities the mean free path increases and subsequently, the shock becomes diffusive, hence the drive is ineffective.

Practically drive parameter/ speed factor is enhanced by using an anode of tapered design (as shown in Fig 5.4(b-d)) and evidenced by the results shown in Fig. 5.6. As the magnetic flux density ($B = \mu_0 I / 2\pi a$) around the conductor through which current I flows varies with the conductor radius $a(z)$ and the magnetic pressure is given as $B^2/2\mu_0$ i.e. $\approx I^2/a^2$, therefore the magnetic energy per unit volume associated with the current scales up as the current sheath accelerates along the tapered anode, because of steadily reducing anode radius and increasing magnetic flux density. Thus mass swept by

the current sheath i.e. $\rho_o(\pi(b^2 - a^2)z_o)$ along the anode length at any position z_o , increases rapidly as current sheath moves up [223]. For these reasons tapered geometries have significantly higher focusing efficiency by comparison with cylindrical flat anode geometries.

Another advantageous implication of using tapered geometry is that axial transit time of the current sheath (given by $t_a \approx \sqrt{4\pi^2 \rho_o (b^2 - a^2) z_o^2 / (\mu_o I^2 \ln(b/a))}$) reduces because the ratio $(b^2 - a^2) / \ln(b/a)$ decreases, as a reduces. Therefore, to synchronize t_a with the current rise time of the capacitor bank, the ambient gas pressure should be raised. Consequently a plasma focus device with a tapered anode operates at higher gas pressure [33, 130, 223]. And as it is well known that thermonuclear fusion yield follows the equation - $Y \approx n^2 \langle \sigma v \rangle V \tau$, where n is the number density of ions, V is the volume of the focus, τ is the confinement duration and $\langle \sigma v \rangle$ is fusion cross-section (function of mean ion temperature T) in the plasma focus pinch column. Hence, for a tapered anode with the ability to achieve efficient focus formation at higher pressure, there is a relative increase in n (i.e. number density) of deuterons in the pinched plasma column which increases the thermonuclear neutron yield [33].

5.2.3.2. Ignition Process in Breakdown Phase

In this experimental study, it has been realized that the anode geometry strongly influences the ionization growth rate in the breakdown phase of plasma focus discharge. Anode shapes extending over the insulator (Fig. 5.4(b-d)) sleeve were found to have much better focusing efficiency and reproducibility by comparison with non-extended designs (Fig. 5.4(a)). The following issues play a critical role in the ignition process –

1. The availability of a small number of free electrons in the almost neutral gas, during the first few nanoseconds of the discharge are of vital importance for the ignition process as they are a

pre-requisite for ionization growth [138]. This is the reason for the improvement in reproducibility after a few conditioning shots.

2. The ignition/breakdown process depends crucially on the applied electric field [63], therefore the anode geometry, plays an important role in enhancing the local fields at initiating points. The sleeve length and magnitude of the voltage pulse are also important consideration.
3. Ionization growth rate is also influenced by the rise time of the high voltage pulse (i.e. dV/dt) applied across the insulator because, depending upon the rise time of pulse, free electrons are accelerated to make inelastic collisions with neutral molecules resulting in rapid ionization and a resulting electron avalanche. This avalanche ultimately leads to the formation of surface discharge streamers over the insulator surface. Hence, streamer formation time is a function of rise time of the high voltage pulse appearing across the insulator (which is limited by the switch inductance, used for applying the high voltage pulse).

It is important to note that the quarter time period is inherently short for sub-kilojoule plasma focus devices and hence the ignition process at the start of the breakdown phase plays a very important role in determining the overall performance of the device. Yordanov *et al* [63] have reported in their study that delay in ignition/ non-uniform discharge initiation of plasma sheath formation results in an unbalanced plasma front traveling down the electrode, and consequently in poor focusing.

5.2.4 Preliminary Results on Radiation Emission from FMPF-1

After performing a detailed study on focusing efficiencies of various anode designs, systematic experiments were performed using tapered and composite anode designs (V2/TA/20-5/Cu and V4/CA/17-3/SS respectively) with the objective of optimizing the radiation emission. These two anode designs were chosen on the basis of consistently good focusing efficiency.

5.2.4.1 Soft and Medium Energy X-ray Measurements

As reported in section 5.2.2; when the miniature plasma focus device was operated with a tapered anode geometry V2/TA/20-5/Cu, then even after getting good focusing efficiency, we failed to detect neutrons and hard X-rays as is to be expected with Deuterium operation. Hence, as a part of a further investigation on this experimental finding, time-resolved measurements were carried out to obtain information about the soft X-ray emission spectrum when the second anode geometry V2/TA/20-5/Cu is used.

It is known that the soft X-ray emission characteristics depend strongly on the operating regimes and parameters e.g. gas filling composition and pressure, stored energy, discharge current, driver impedance, electrode material, shape, profile and configuration, polarity of the inner electrode. But amongst all these parameters, composition of the gas and the filling pressure have the strongest influence. Therefore, in our study, we also explored the effect of Deuterium-Krypton (D_2 -Kr) admixtures on plasma dynamics and thermal X-ray production.

The end-on soft X-ray emissions were measured using three channels of DXS (diode X-ray spectrometer, described in section 4.5) containing silicon *PIN* diodes (windowless BPX-65), placed at distance of 420mm from the anode tip. Appropriate filter pairs were used to maximize detection within particular bands of the emission spectrum. The sensitivity of the *PIN* diodes was cross calibrated using common filter. In the present investigation the three channels of the DXS were covered with (i) 20 μ m Al, (ii) 125 μ m aluminized Mylar and (iii) 16 μ m Co along with 125 μ m Mylar.

Emission of soft thermal X-rays from the hot and dense plasma is due to the electron-ion interaction processes such as Bremsstrahlung, recombination emission and line emission as described in section 2.6.2. It is to be noted that in the case of Kr, the most prominent soft X-ray line emission is from M-shell radiation, with a wavelength of 4nm i.e. 300eV [224]. Since we also intended to study the role of admixtures in enhancing the thermal X-rays from Deuterium plasmas therefore the selection

of filters was made in such a way that none of them had significant transmission below 900eV, so as to exclude the contribution from the 4nm (300eV) line emission from Kr. Filter pair combinations of 20 μ m Al and 125 μ m aluminized Mylar gave soft X-ray yield in the range of 900eV – 1600eV. Filter combination of 16 μ m Co along with 125 μ m aluminized Mylar covers the spectral range of medium energy X-rays from 3.2keV to 7.7keV. The above filters were chosen to observe the influence of admixture on low (soft) to medium energy X-ray emission characteristics. A typical current derivative signal (di/dt) of plasma focus discharge and X-ray signals, through different filter channels are shown in Fig. 5.7.

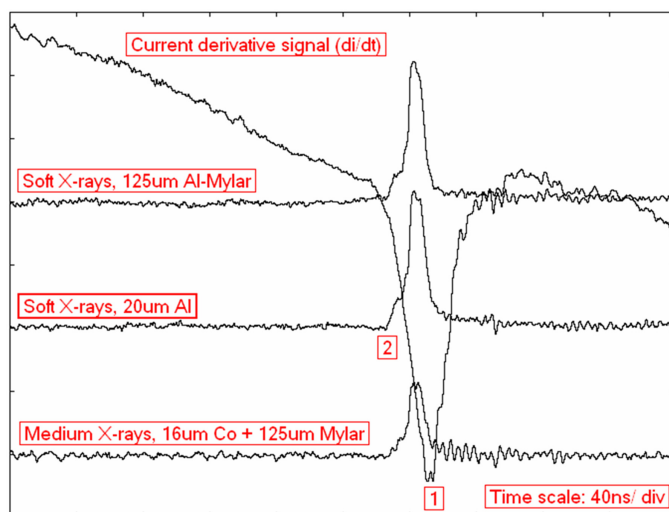


Fig. 5.7 Sample oscilloscope traces of current derivative and DXS signals.

The methodology discussed earlier under section 4.5 has been used to calculate the average X-ray yield at source into 4π steradians per shot. It is important to note that, throughout the experiment, the stored energy of the system was kept at 200J and the volumetric ratio of admixture was kept constant at D₂: Kr = 9:1 for all filling gas pressures. The results mentioned and discussed below are the averages for 15 shots at every filling gas pressure. The variation in average X-ray yield in the spectral

range of 0.9keV to 1.6keV and 3.2keV to 7.7keV as a function of gas pressure for Deuterium and D₂-Kr admixture are shown in Fig. 5.8 [213].

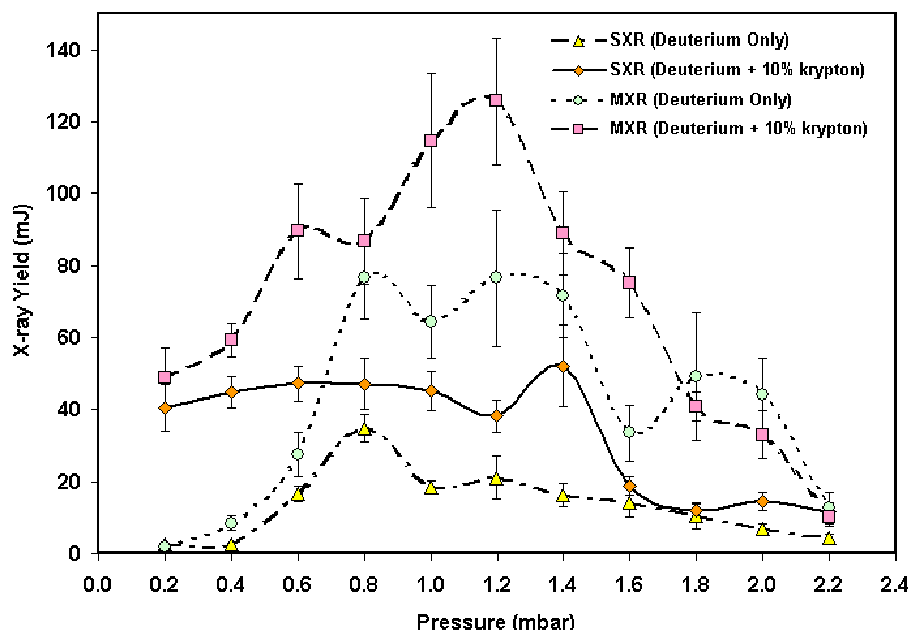


Fig. 5.8 X-ray yields in the spectral range of 0.9keV to 1.6keV (SXR) and 3.2keV to 7.7keV (MXR) for D₂ and D₂-Kr admixture at different filling pressures.

When Deuterium alone is used, the highest soft X-ray yield of $\sim 35.7 \pm 3.7$ mJ is observed at 0.8 mbar and average soft X-ray yield of $\sim 20.7 \pm 3.4$ mJ was produced in the optimum focusing range of 0.6 mbar to 1.4 mbar. When D₂-Kr admixture was used, the highest soft X-ray yield of $\sim 57.1 \pm 4.6$ mJ was observed at 1.4 mbar and average soft X-ray yield of $\sim 46.4 \pm 6.5$ mJ was obtained in the ‘broadened’ optimum pressure range of 0.2 mbar to 1.4 mbar. This >2 fold increase in soft X-ray yield and broadening of the optimum pressure range for D₂-Kr admixture operation was also observed in the spectral range of 3.2 to 7.7 keV as shown in Fig. 5.8. With Deuterium alone, highest medium energy X-ray yield of 78.5 ± 8.43 mJ was observed at 0.8 mbar and average yield of 72.38 ± 9.6 mJ was obtained in the optimum pressure range of 0.8 to 1.4 mbar for 3.2 to 7.7 keV X-rays. Upon use of D₂-Kr admixture peak X-ray yield of 125.62 ± 11.8 mJ was obtained at 1.2 mbar and average X-ray yield of $\sim 107.6 \pm 12.6$ mJ was obtained in the ‘broadened’ optimum pressure range of 0.6 mbar to 1.4 mbar.

One of the most important observations is that the average X-ray yield in operation with D₂-Kr admixture at lower pressures specifically in the range of 0.2 to 0.4mbar resulted in a >17 fold enhancement (from 2.52±0.41mJ to 43.75±5.59mJ) in 0.9 to 1.6keV range, where as a >10 fold enhancement (from 5.12±1.4mJ to 54.1±6.6mJ) was obtained in the spectral range of 3.2 to 7.7keV. It may be noted that the X-ray yields are calculated at the source i.e. the pinched plasma column. Hence, an order of magnitude increase in X-ray yields in both the spectral range at lower operating pressures for D₂-Kr admixture has major advantage of significantly reduced absorption of X-rays as it travels through the gaseous medium to reach resist coated substrate surface for applications like lithography and micro-machining. The higher transmission of X-rays for low pressure D₂-Kr admixture operation will result in higher X-ray flux per shot at the resist and hence can provide sufficient flux in fewer discharges for optimum exposure.

It is evident from the results discussed above that an admixture not only plays an important role in enhancing X-ray yield but it also broadens and stabilizes the operating regime. Experimental results shown in Fig. 5.9 indicate that by using a volumetric 9:1 D₂-Kr admixture the plasma dynamics are significantly influenced causing enhancement in focusing peak amplitude and focusing duration. The magnitude of dip of the negative spike in the current derivative signal corresponding to the focus phase, which is considered as a measure of focusing efficiency, for the case of 9:1 D₂-Kr admixture is on an average ~1.5 times more in comparison to Deuterium only gas discharges.

The average life-time of the pinches (i.e. focusing duration) in the entire operating range, measured from FWHM of current derivative signals, averages to 38±2ns for Deuterium only and to 46ns±4ns for 9:1 D₂-Kr admixture. This indicates the stabilization of pinch column for admixture operations. Peacock *et al* [225] reported that a few percent of admixture gas aids in the transfer of directed kinetic energy in the implosion into thermal energy in the dense plasma and helps to stabilize the dense pinch, which is in agreement with our findings. Relative increase in the focusing duration

coupled with enhanced focusing efficiency for D₂-Kr admixture explains the enhancement in the X-ray yields along with broadening of the operating pressure regime.

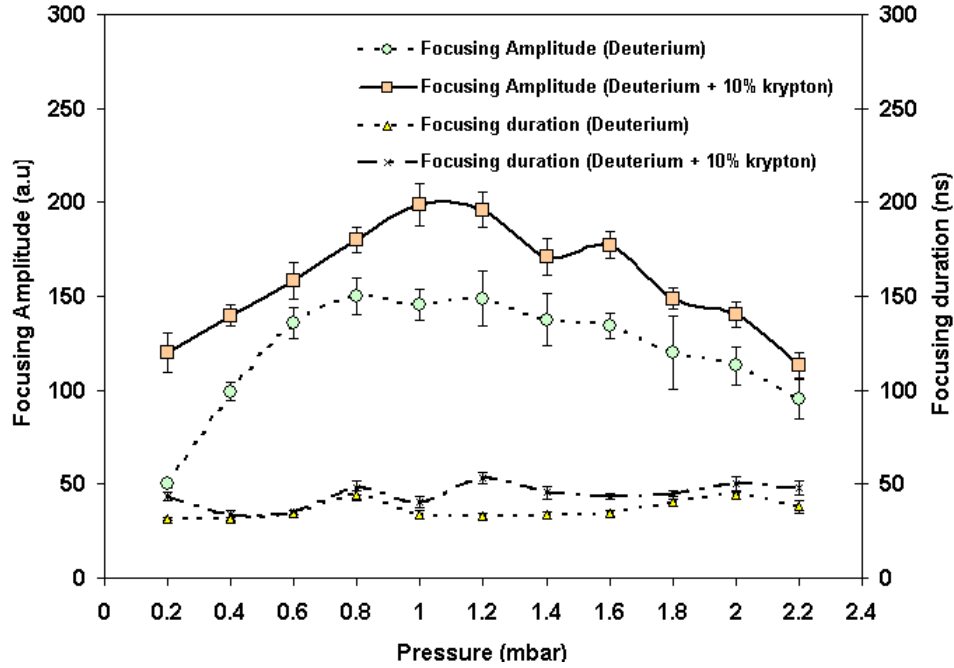


Fig. 5.9 Focusing peak amplitude and focusing duration for D₂ and D₂-Kr admixture at different filling pressures.

The time resolved investigation of X-ray and current-derivative signals for Deuterium and D₂-Kr admixtures also indicate that there is an early onset of X-ray emission for the admixture. Fig. 5.10 shows the variation in time difference between the peak of the radial collapse phase in the current derivative signal (point-1 in Fig. 5.7) and the beginning of soft X-ray emission instant (point-2 in Fig. 5.7) at different gas pressures. It shows that for D₂-Kr admixture, soft X-ray emission generally begins about 26 ± 4 ns (averaged over entire pressure range) before peak compression is reached. Whereas for Deuterium only discharges, soft X-ray emission begins about 18 ± 3 ns (averaged over the entire pressure range) before peak compression is reached.

The early onset of X-ray emission for admixture points to the fact that plasma is heated more effectively which can be attributed to the fact that the admixture increases the plasma density through

higher compression (i.e. smaller pinch radius) [226]. Though the average characteristic time durations (FWHM) of X-ray emissions in both cases (with and without Kr) remain about $17\text{ns} \pm 3\text{ns}$, but higher X-ray yield with admixture is essentially due to improved pinch stabilization and higher plasma density through improved compression.

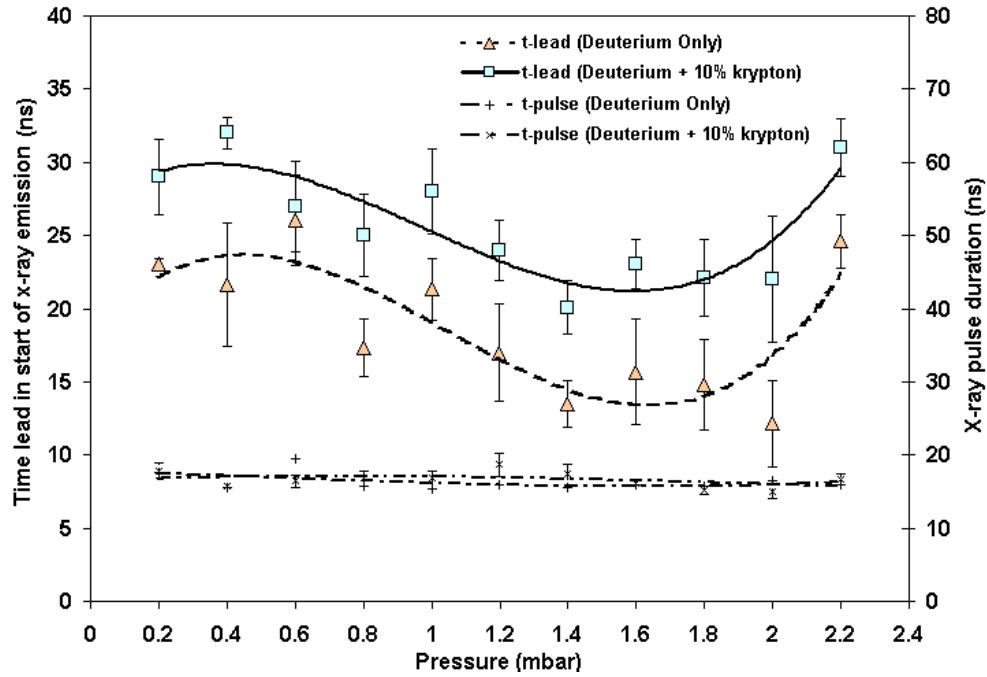


Fig. 5.10 Time lead in start of X-ray emission (t-lead) and time duration of X-ray pulse (t-pulse) for D_2 and D_2 -Kr admixture at different filling pressures.

Thus, in this investigation we found that with Krypton seeding in Deuterium there is significant enhancement in the X-ray yields and the broadening of operating pressure regime, which has been explained on the basis of relative increase in the focusing duration coupled with enhanced focusing efficiency. An order of magnitude enhancement in X-ray yields at low pressures for admixture operation has been obtained [213]. Hence, from an applications point of view, admixture operation is useful in achieving high performance device efficiency for lithography and micro-machining applications.

5.2.4.2 First Results of Neutrons and Hard X-ray Emission from FMPF-1

As indicated from characteristic curves shown in Fig. 5.5, in experiments with the fourth composite anode geometry i.e. V4/CA/17-3/SS, efficient and consistent pinching was observed, over a wide range of operating pressures from 1.5 to 4mbar. Implementation of the composite geometry gave consistent neutron production over a wide range of pressures. In this section, a study of the first measurements of neutrons and hard X-rays, using composite anode design, is reported in detail. Production of neutrons and hard X-rays were systematically scanned over the entire pressure range to investigate the optimum regime and obtain time-resolved information. A typical oscilloscope trace of di/dt and current trace for this anode geometry is shown in Fig. 5.11.

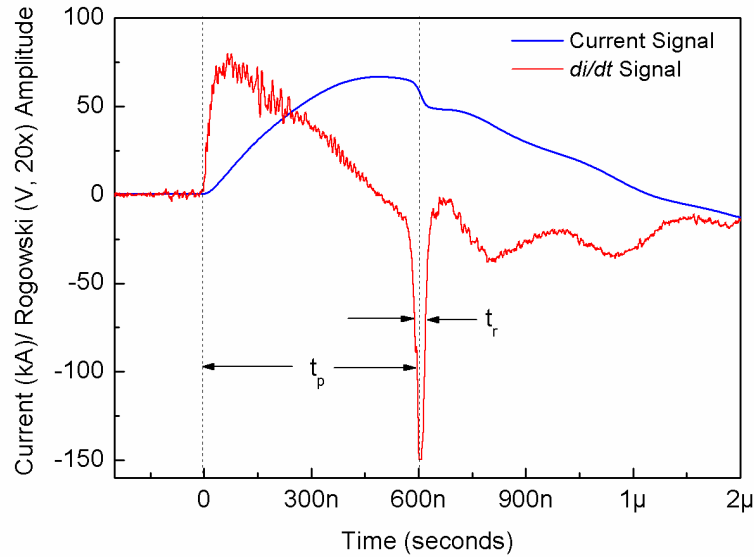


Fig. 5.11 Typical current derivative and current trace for shots in D₂ using composite anode design V4/CA/17-3/SS at 3mbar pressure.

The variation in time-to-pinch from the breakdown phase (which includes axial acceleration phase and compression phase), defined as t_p in Fig. 5.11, and the radial phase duration measured using FWHM of the dip in current derivative signal), defined as t_r in Fig. 5.11, at different filling gas pressures is shown in Fig. 5.12. The characteristic time t_p increases with the increase in operating

pressure due to slowing down of the current sheath in the axial phase. However, the radial phase duration is almost constant over a wide range of Deuterium pressure from 0.5 to 4.0 mbar indicating the consistent focusing in FMPPF-1 device.

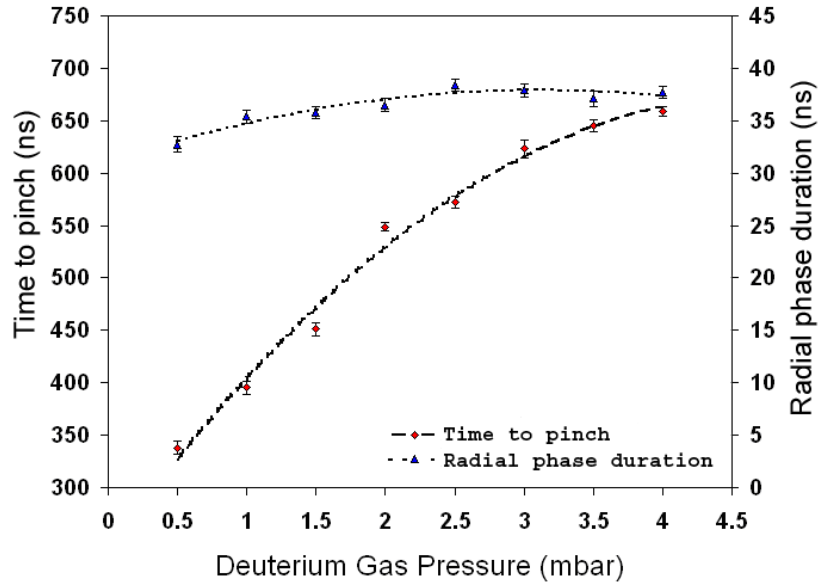


Fig. 5.12 Time-to-pinch and radial phase duration (FWHM) versus Deuterium gas pressure.

Layout of the overall detection and instrumentation setup has been illustrated under section 4.4.1 and shown in Fig. 4.6. The ^3He detector was placed in the side-on direction at a distance of 0.5m from focus. A typical oscilloscope obtained from the ^3He detector signal is shown in Fig. 5.13.

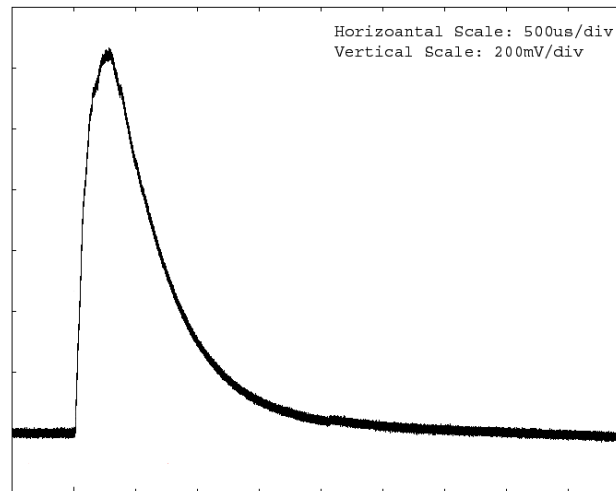


Fig. 5.13 Time integrated oscilloscope trace of ^3He detector output.

The variation of neutron yield with Deuterium filling gas pressure for the composite anode design V4/CA/17-3/SS is shown in Fig. 5.14.

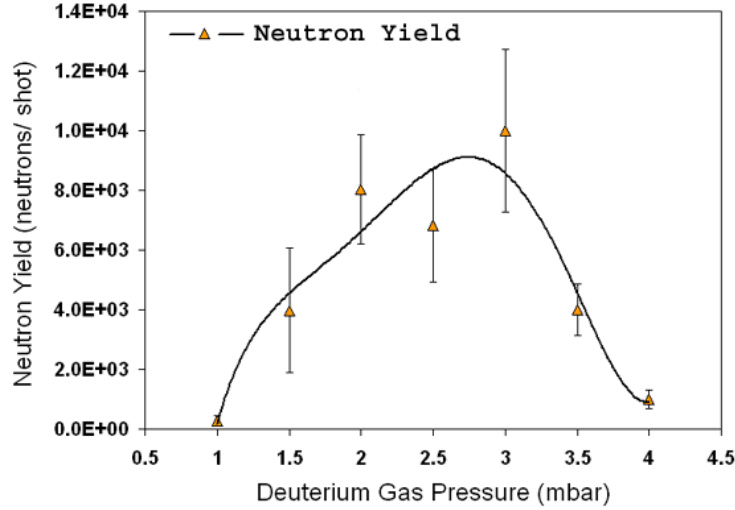


Fig. 5.14 Total neutron yield versus Deuterium filling pressure.

Comparing the graph shown in Fig. 5.14 with the pinching efficiency graph shown in Fig. 5.6 confirms that the neutron yield maximizes at the highest amplitude of dip in the di/dt signal. The measured average neutron yield (Y_{90°) was $1 \pm 0.27 \times 10^4$ neutrons/shot at 3mbar gas pressure.

It is observed in Fig. 5.14 that there is an optimum pressure which produces a maximum neutron yield. This can be explained using the effect of ambient gas pressure on thermonuclear and beam target mechanisms. From a thermonuclear point of view, we can follow the explanation provided by Moreno *et al* [227], according to which the optimum neutron yield can be achieved provided the peak current occurs simultaneously with the pinch (with the neutron yield proportional to I^4). This condition, when expressed mathematically was shown to involve an interdependence among anode length, charging voltage and filling gas pressure [227]. If two of the parameters are kept fixed, then the third can be fine tuned to satisfy the condition for obtaining the optimum yield. Moreover, logically, the initial increase in filling gas pressure increases the plasma density in the pinch, increasing thereby

the reaction rate and therefore neutron yield. But above a certain critical pressure, which depends on the other operating parameters of the focus machine, increasing the pressure does not increase the neutron yield as the time-to-pinch increases further and the pinch does not occur simultaneously with peak current resulting in lower heating of the pinch plasma and thus a lowering of neutron yield. From the beam target point of view, one needs strong Rayleigh Taylor (RT) instability formation for efficient production of accelerated deuterons to produce higher neutron yield [228]. It may be noted that the growth time for RT instability is given as $t_i = \sqrt{2\pi\lambda_p/g}$ where g is acceleration of the current sheath and λ_p is the wavelength of perturbation [229]. At lower pressures, the growth rate of RT instabilities will be too high (because of higher acceleration of the current sheath) to allow the formation of well defined pinch plasma column, resulting in poor neutron yield. Near the optimum operating pressure, the current sheath acceleration is adequate for efficient instability formation. This results in strong instability generated deuteron beams resulting in higher neutron yield by efficient beam-target mechanism. At pressures significantly above the optimum pressure, the acceleration of the current sheath may be too slow to allow for efficient RT instability formation resulting in less efficient beam-target interaction leading to low neutron yield. It may be noted that anisotropy investigation is reported in detail in a later section of this chapter, revealing the dominance of either of the neutron production mechanism in this sub-kilojoule range miniature plasma focus device [216].

The scintillator-photomultiplier detector is located in the side-on position, 0.5m from the focus in order to separate in time, the hard X-ray (HXR) pulse from the neutron pulse. The current derivative and scintillator-photomultiplier signals were recorded simultaneously by a fast digital oscilloscope DL9140; both channels were triggered at the same time and similar lengths of cables were used for both diagnostics. The oscilloscope traces of typical current derivative and scintillator-photomultiplier signals for a shot at 3mbar Deuterium gas pressure are shown in Fig. 5.15.

The first peak is of non-thermal hard X-rays produced by the interaction of energetic electrons with the copper anode. It is important to mention here that there is an inherent delay of ~ 30 ns in the photomultiplier tube signal, causing the HXR peak to shift significantly from the peak of the current derivative signal as seen from graphs shown in Fig. 5.15. The second peak is confirmed to be the neutron peak as 2.45 MeV D-D neutrons are expected to arrive about 24 ns after the first HXR peak that matches well with the observed time difference of 23.8 ns between the two peaks.

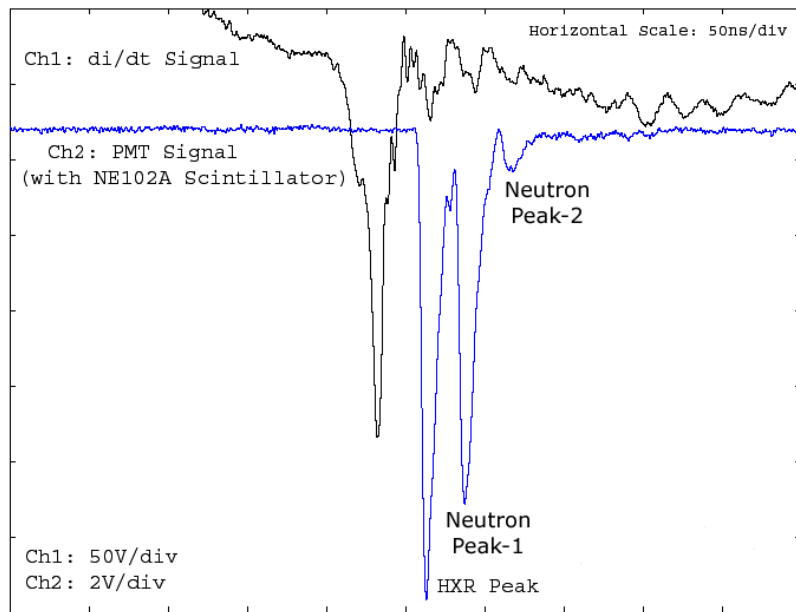


Fig. 5.15 Current derivative signal trace with HXR and neutron signals.

The durations of HXR and neutron pulses, estimated from FWHM of the corresponding peaks are about 8.9 ± 0.8 ns and 6.9 ± 0.8 ns respectively, averaged over 20 shots at 3 mbar gas pressure. Relative measurements (area under the curve) of HXR and neutron emission as recorded by scintillator-photomultiplier detector signal at various pressures are shown in Fig. 5.16. As inferred from the graph, initially with the increase in pressure from 1 to 3 mbar the trend in neutron emission follows the HXR emission trend with the yield for both maximizing at 2 mbar. It may be noted that for all of the shots, (i) the neutron pulse was registered only after the non-thermal hard X-ray pulse (which is mainly

generated due to the accelerated electron beam hitting the anode target), and (ii) the neutron yield predominantly peaked at peak HXR yield in the operation range of 1-3mbar filling gas pressure. These observations point to the dominance of beam-target mechanism for neutron production in the FMPPF-1.

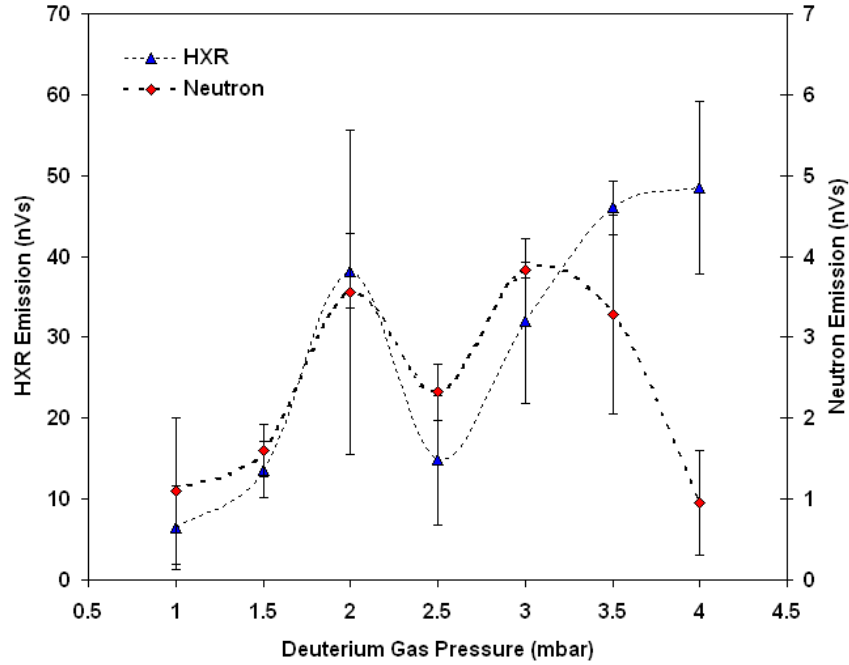


Fig. 5.16 Relative measurements of HXR and neutron emission signals.

At operating pressures >3 mbar, the neutron yield decreased but the HXR emission was still significant. It can be due to difference in acceleration mechanisms of ions (specifically deuterons which produce neutrons) and electrons (which produce HXR).

5.2.4.3 Simulation ‘fitting’ Results with FMPPF-1

The plasma dynamics related to the plasma focus discharge is complicated, as it is inherently 2-D in nature and involves radiation hydrodynamics as well. The ‘Lee code’ [70] provides a useful tool to conduct scoping studies, as it is not purely a theoretical code, but offers a means for the user to conduct phenomenological scaling studies for a given machine, once certain key “fitting” parameters (such as current and mass fractions) and electrical circuit parameters have been adjusted.

Knowing the short circuit discharge current parameters, the approximate anode length, anode radius and inter-electrode separation were chosen for Deuterium operation based on certain universally known range of typical values of axial acceleration and radial compression speeds. Then, the Lee Code can be used to fine-tune the electrode shape, electrode dimensions and operating pressure. The electrode assembly is then constructed and tested. The experimental discharge current traces are then used to fine-tune the ‘fitting’ parameters of the Lee Code for that particular electrode design. The possible pressure and voltage regimes are explored computationally to optimize the pinching efficiency and radiation yield. The experiments are then carried out to verify the simulation results, and possible modifications of the electrode assembly are explored for further enhancement of radiation yield. Hence, the system optimization works on mutual feedback mechanism, from simulation to experiment to simulation and then to experiment again, to continuously evolve the electrode shapes and dimension until the best results are obtained. As an example, in the following paragraph we discuss the results for our fourth version of composite anode geometry (V4/CA/17-3/SS).

The following values of ‘fitting’ parameters were found to have a good fit of the current traces operated for a range of charging voltages and pressures: axial mass factor (f_m) = 0.155; axial current factor (f_c) = 0.6; radial mass factor (f_{mr}) = 0.2 and radial current factor (f_{cr}) = 0.7. The electrical circuit parameters: Capacitance (C_0), Inductance (L_0) and Resistance (R_0) were fixed at 2.4 μ F, 32.9nH and 60m Ω , respectively. The sample current-trace ‘fitting’ result for the ‘FMPF-1’ device operated at 12kV for Deuterium filling at gas pressure of 3mbar with composite anode geometry is presented in Fig. 5.17.

As can be seen, there is good match with the experimentally measured trace, particularly with respect to the rising and roll-off portions (the part of the computed current trace near the end of dip requires further investigation). The maximum current 70kA obtained from simulation matches well

with the experimentally obtained current trace. The Lee Code [70] simulation predicted maximum neutron yield of 4.91×10^4 neutrons per shot at 70kA peak discharge current and 3mbar D₂ filling gas pressure which is of same order of magnitude as obtained experimentally.

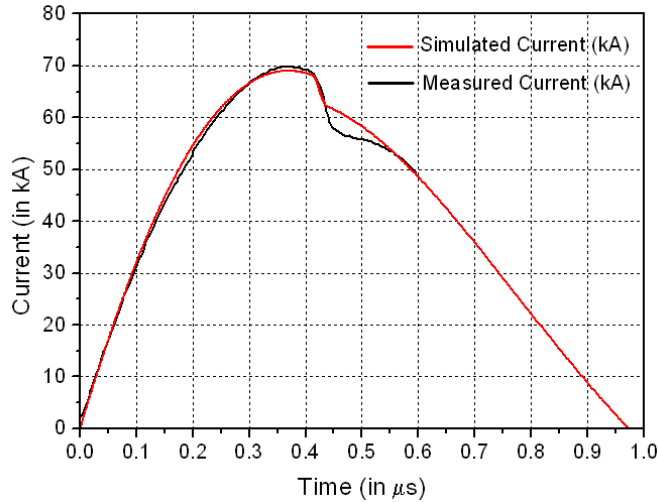


Fig. 5.17 Current-trace ‘fitting’ result for composite anode.

5.2.4.4 Neutron Yield Enhancement with D₂-Kr Admixture Operation

As reported in section 5.2.4.1, while studying focus characteristics of tapered anode geometry V2/TA/20-5/Cu, significant enhancement in focusing efficiency was obtained with D₂-Kr admixture operation, which resulted in enhanced soft X-ray emission [213]. Therefore, after obtaining energetic radiations (i.e. neutrons and hard X-rays) from the improved anode geometry V4/CA/17-3/SS with Deuterium operation, we investigated the effect of D₂-Kr admixture operation on focusing characteristics and radiation yields [214].

To measure time integrated neutron yields, the high sensitivity ³He proportional counter, discussed before, was used. To acquire the time resolved history of emitted radiation, a ‘dual time of flight’ setup was implemented which is shown in Fig. 5.18.

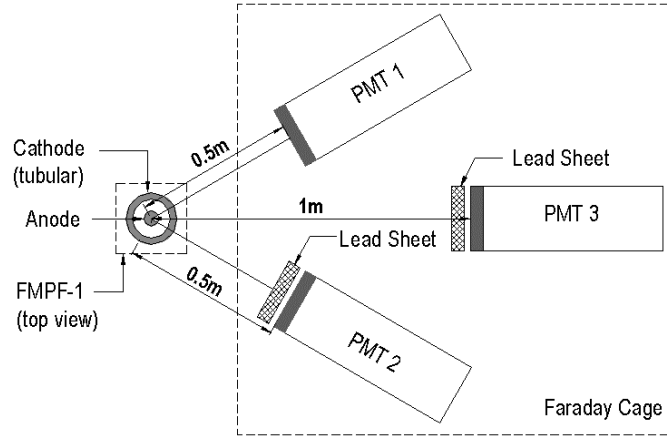


Fig. 5.18 Experimental setup for 'dual time of flight' measurement.

Throughout the experiment, the stored energy of system was kept constant (to deliver about 70kA peak discharge current at 12kV charging voltage) and the effect of varying concentrations of D₂-Kr admixture (with Kr added in the volumetric ratios of 10%, 5% and 2%) on neutron and HXR emissions was investigated for different fill pressures. The results were obtained for averages of 20 shots for every choice of admixture gas pressure. The measured average neutron yields for various volumetric ratios of D₂-Kr admixture are shown in Fig. 5.19.

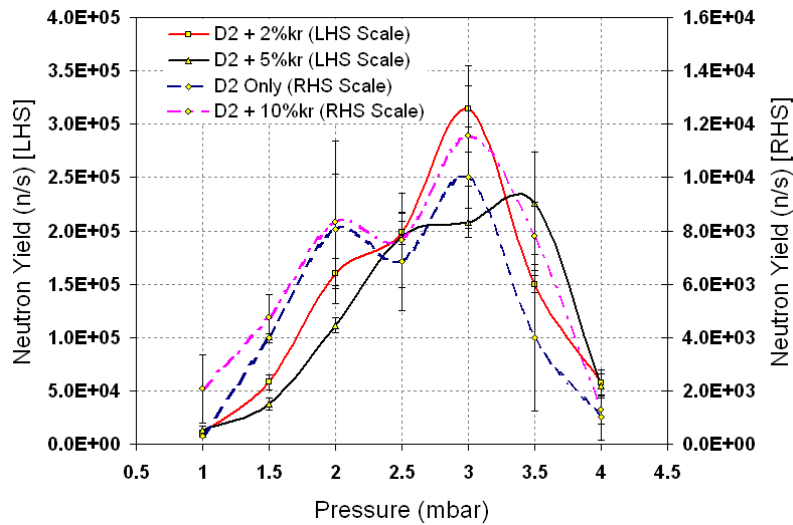


Fig. 5.19 Neutron yields obtained with various admixture concentrations.

A significant enhancement of neutron yield was observed, specifically with $D_2+2\%Kr$ and $D_2+5\%Kr$ admixture ratios [214]. The maximum neutron yield with admixture operation increased by 30, 20 and 1.2 times for $D_2+2\%Kr$, $D_2+5\%Kr$ and $D_2+10\%Kr$ respectively, as compared to pure deuterium operation at 3mbar. Also, more than an order of magnitude enhancement in average neutron yield was observed over the broader pressure range of 1 – 4mbar [214].

The time-resolved information obtained for pure D_2 and $D_2+5\%Krypton$ admixture operation, from the ‘dual time of flight’ measurement setup, are shown in Fig. 5.20 (a) and (b) respectively.

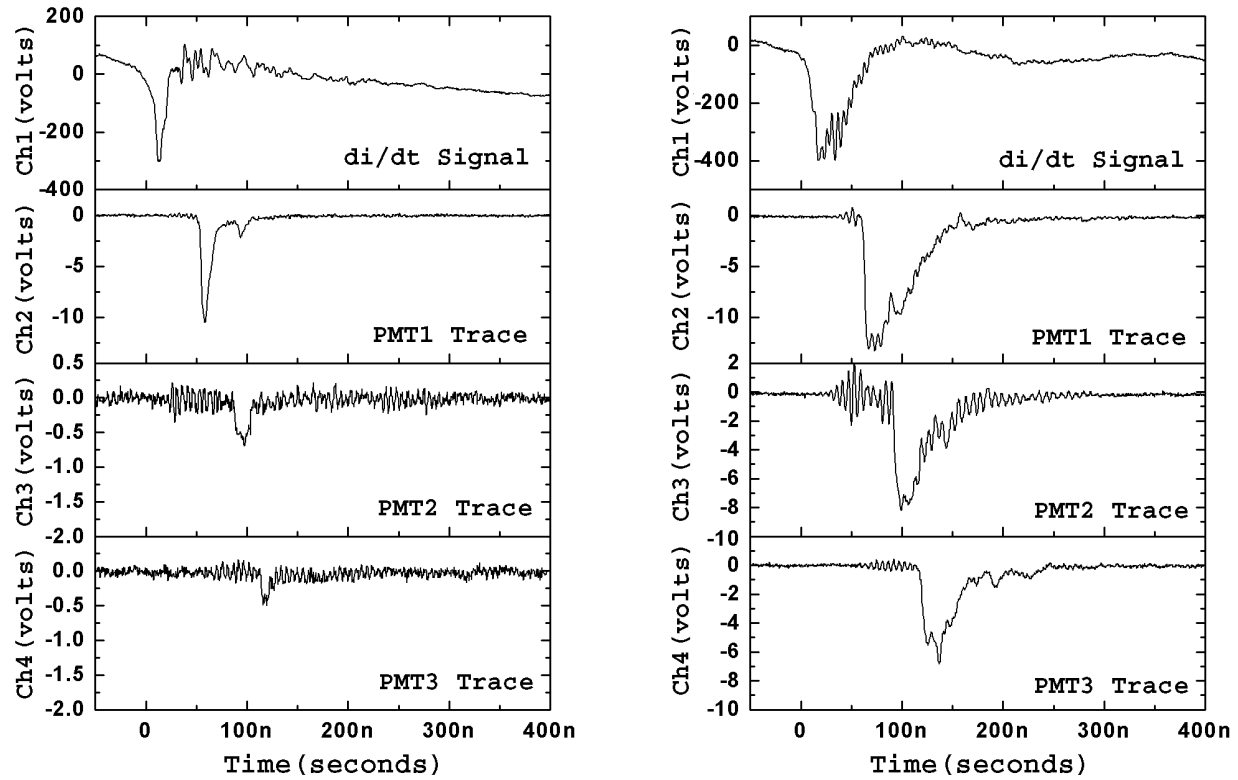
(a) D_2 discharge at 3mbar(b) $D_2+5\%Kr$ admixture discharge at 3mbar

Fig. 5.20 Current derivative (di/dt) trace with HXR and neutron signals.

Time-resolved information obtained using the PMT1 (Ch2 trace) for pure D_2 filling is shown in Fig. 5.20(a). The first peak (PMT1 trace) is of non-thermal HXR whereas the second peak was confirmed to be of neutron on the basis of time of flight consideration. The Ch3 and Ch4 traces,

obtained from lead screened PMT2 and PMT3 placed at a distance of 0.5m and 1m respectively, filter out the HXR pulse and record only the neutron signal. The $\sim 24\text{ns}$ time difference between the neutron pulses recorded by these two channels confirmed the production of 2.45MeV D-D neutrons.

The duration of HXR and neutron pulses, estimated from FWHM of the corresponding peaks, are about $9\pm 1\text{ns}$ and $7\pm 1\text{ns}$ respectively. It is important to note here that the appearance of HXR peak (Ch2 trace) after $\sim 30\text{ns}$ from the peak of the di/dt signal (Ch1 trace) is because of the latency in PMT.

The PMT1 signal (Ch2 trace) with D_2 -Kr admixture operation, as shown in Fig. 5.20(b), depicts burst of radiation consisting of HXR and multiple neutron pulses. The multiple neutron pulses can also be noticed on traces of Ch3 and Ch4. The relative time difference of $\sim 24\text{ns}$ between the neutron pulses of Ch3 and Ch4 confirm the presence of 2.45MeV D-D neutrons. The cumulative pulse widths of HXR and neutron pulses are $\sim 20\text{-}25\text{ns}$ and $\sim 25\text{-}35\text{ns}$ respectively over the entire range of filling gas pressures for various admixture concentrations.

The relative HXR yields, measured using the cumulative area under the identified HXR peaks, at various pressures for different admixture combinations are shown collectively in Fig. 5.21.

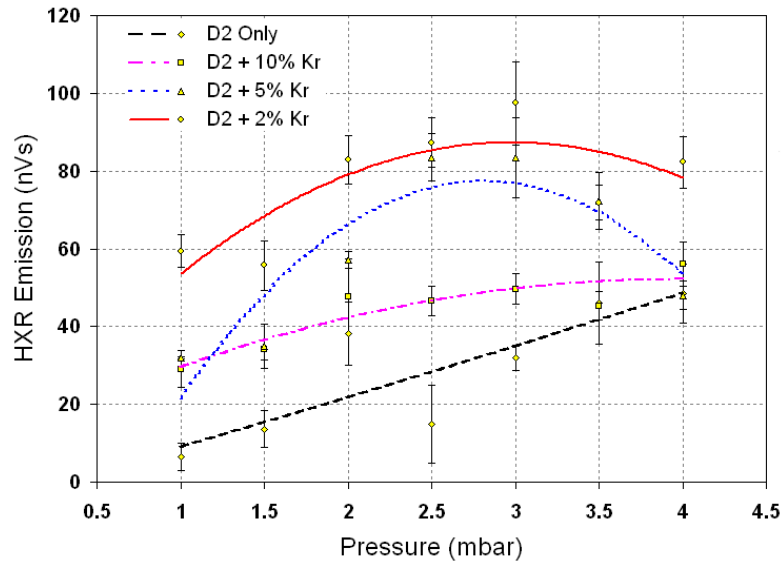


Fig. 5.21 HXR yield for D_2 and D_2 -Kr admixtures at different pressures.

The trends of increasing HXR yield predominantly follows the trend for neutron yield (*i.e.* $Y(D_2+2\%Kr) > Y(D_2+5\%Kr) > Y(D_2+10\%Kr) > Y(D_2)$), indicating that the Kr admixture enhances the contribution of the non-thermal mechanism in neutron production.

The observation of multiple HXR and neutron pulses in Kr seeded operation with enhanced yields can be explained by hypothesizing the formation of several successive micro-pinches initiated by radiative collapse in high-Z admixture operation. It is well known that the use of high-Z impurity increases the effective ion charge (Z_{eff}) of Deuterium plasma. For the fully ionized pure Deuterium plasmas, there are three mechanisms that control the kinetic pressure of the plasma: Ohmic heating rate, radiative cooling rate and the rate of change of current. In general, the Ohmic heating rate is larger than radiative cooling rate until the time at which the pinch current (I_{pinch}) exceeds the Pease-Braginski critical value (I_{PB}) [230]. Beyond the Pease-Braginski current limit, radiative losses exceed the Ohmic heating rate. As the high-Z ions in a <1keV plasma are not fully stripped, line radiation dominates over bremsstrahlung. Thus, for high-Z admixture plasma I_{PB} no longer provides the correct power balance relationship. Instead power balance occurs at a critical current (I_{cr}) which is much less than Pease-Braginski current limit (I_{PB}). Shearer [85] has theoretically calculated Pease-Braginski current limits (I_{PB}) of 650 and 110kA for 1% Carbon and 1% Argon impurity pinches respectively.

According to Koshelev *et al* [231], the critical current limit for radiative collapse to occur is - $I_{cr} = 5 \times 10^{-3} \sqrt{\ln \Lambda} T^{3/4} Z_{eff} / Z_n^2$ (where I_{cr} is in MA, $\ln \Lambda$ is the coulomb logarithm, T is the plasma temperature in eV, Z_{eff} is the effective ion charge and Z_n is the nuclear charge). From this expression it is clear that I_{cr} largely depends on effective ion charge Z_{eff} and nuclear charge Z_n for the plasma of heavy ions, therefore even a small amount of heavy ion impurity drastically modifies the energy balance in the plasma due to intense radiative losses, mainly contributed by line emission. Heavier

atoms, such as Kr, are not fully stripped. For example, 1keV plasma would ionize the Kr to Ne-like stages (Kr^{26+}), therefore many different bound-bound transitions are possible and this makes the high-Z ions radiate copiously. Thus radiation cooling enables the magnetic field to compress the plasma to a higher density and smaller final radius (micro-pinch) [85, 232] while maintaining the plasma in a quasi-equilibrium or Bennett-like state where the average kinetic pressure inside the column is in equilibrium with magnetic pressure at the edge of the plasma [230]. Hence, micro-pinch formations due to the addition of high-Z impurity in pure D_2 in the correct proportion, leads to neutron yield enhancement.

Time-resolved investigation of current derivative (di/dt) signals of D_2 and D_2 -Kr admixtures (refer Ch1 in Fig. 5.20(a) and Fig. 5.20(b) respectively) indicates that radial phase duration, which is approximated to be the FWHM of the dip in the current derivative signals, averages to $30 \pm 5\text{ns}$, $35 \pm 3\text{ns}$, $39 \pm 4\text{ns}$ and $50 \pm 4\text{ns}$ for pure D_2 , $\text{D}_2 + 10\%\text{Kr}$, $\text{D}_2 + 5\%\text{Kr}$ and $\text{D}_2 + 2\%\text{Kr}$ respectively. This implies that the high-Z Kr impurity also plays an effective role in pinch stabilization, depending upon the impurity concentration ratio. The magnitude of the dip of the negative spike in the current derivative signal, which is considered as a measure of focusing efficiency, for the case of D_2 -Kr admixture is on an average ~ 1.5 times higher than with pure deuterium gas discharges. Thus D_2 -Kr admixture, with appropriate volumetric ratio, not only plays an important role in enhancing neutron and HXR yield but also broadens the optimum pressure regime and stabilizes the pinch for a longer duration. Enhancement in pinch stabilization is also supported by the fact that Rayleigh-Taylor instability growth time (which is known to be $t_i = \sqrt{2\pi\lambda_p/g}$, here g is the sheath acceleration and λ_p is the wavelength of perturbation) increases as a consequence of slowing down of current sheath with high-Z admixture operation [90], which was also observed in POSEIDON plasma focus facility for D_2 -Ar admixture operation [233].

In conclusion, our study illustrates that for a judicious choice of concentration of Kr in D₂ (by adding just enough Kr to optimally enhance the radiation by radiative collapse, but not enough to affect the implosion through the higher atomic mass of the Kr atoms) along with adjustment of the filling pressure such that the swept-mass allows sufficient time for first radial collapse to coincide with the peak current, then D₂-Kr admixtures can significantly enhance the radiation yields. The improved pinch stabilization and higher plasma density through improved compression, driven by radiative collapse explains the vigorous enhancement of neutron and HXR yield for D₂-Kr admixture operation.

5.2.5 Effect of Cathode Structure on Neutron Yield in FMPF-1

In all of the experiments reported previously; the wall of the tubular vacuum envelope acted as the cathode. The tubular cathode structure carries the return current and also serves as vacuum envelope, thereby minimizing the size of plasma focus head. In the second phase of the optimization process, our attention was directed towards exploring the effect of cathode structure on the gross dynamics of plasma focus operation. In this section we report the investigation on influence of tubular and squirrel cage cathode structures [211].

The influence of cathode structure on neutron emission is an important issue since in the typical miniature plasma focus devices, the conventional bar/squirrel cage cathodes are replaced with tubular cathodes that also act as vacuum barriers and hence reduce the size of the DPF “head”. We investigated two different types of cathode geometries (whilst maintaining other operating conditions/parameters same and varying only the pressure) on neutron emission from FMPF-1 [211]. The exploration also incorporates numerical experiments that have been performed using ‘Lee Code’, to simulate the modifications in the axial and radial phase dynamics along with gross plasma focus properties, with the use of two different cathode geometries.

5.2.5.1 Experimental Setup and Diagnostics Arrangement

In this experimental setup, the coaxial electrode assembly consisted of a 15mm long stainless steel (SS) anode of composite geometry (tapered over the last 5mm with diameter decreasing from 12mm to 7mm; these dimensions are further refined for the V4/CA/17-3/SS mentioned in previous experiments) with the cathode having a diameter of 30mm. The anode has 15mm deep cavity of 7mm diameter, bored at the center of anode, facing top. Schematic of cathode geometries utilized in the investigation are shown in Fig 5.22 (a) – (b).

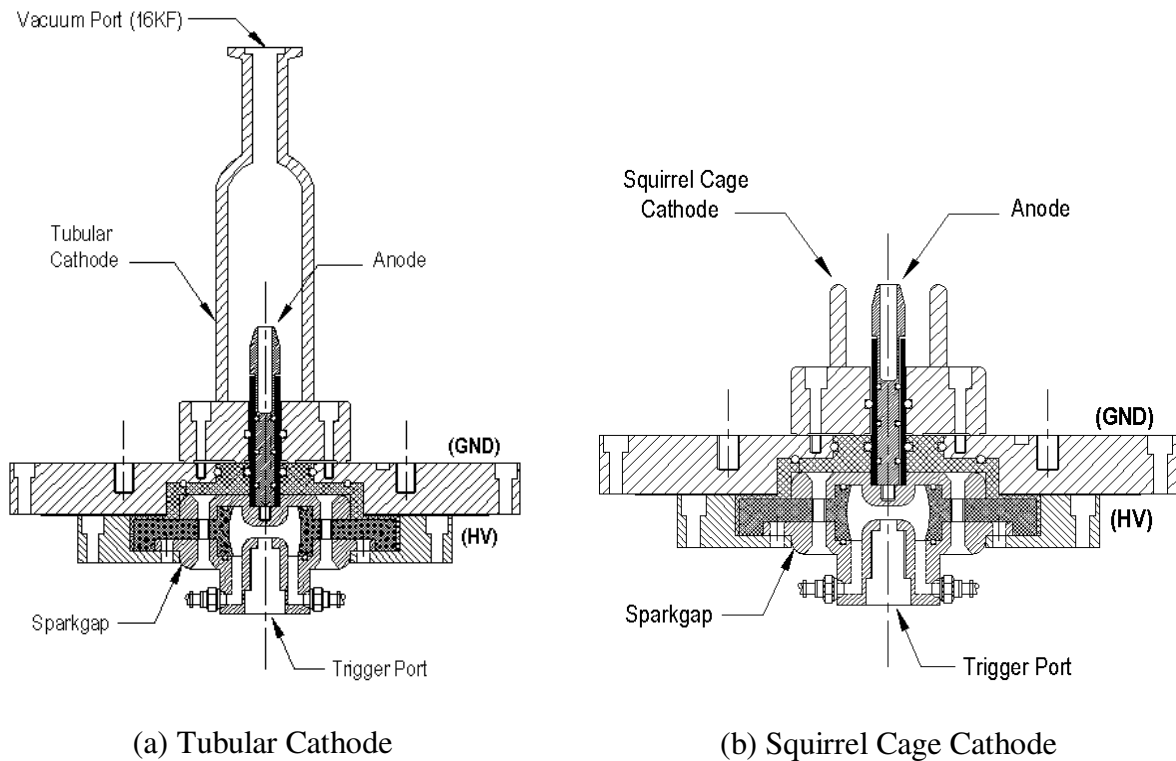


Fig. 5.22 Schematic of utilized cathode structures.

It may be noticed from the schematics shown that the current return path in the case of tubular cathode is through the vacuum envelop where as in squirrel cage cathode, it is through six numbers of Stainless Steel rods (of 6mm in diameter) that are uniformly spaced on a coaxial circumference of 30mm diameter and are concentric with anode axis. The vacuum housing is outer to the squirrel cage cathode and it is excluded from the electrical circuit.

A high bandwidth Rogowski coil was used for electrical diagnostics while time integrated neutron yield was measured by ^3He detector [200]. To acquire the time resolved history of emitted radiation, a ‘dual time of flight’ arrangement consisting of two identical scintillator photomultiplier detectors – PMT1 and PMT2 was used. The PMT1 was placed radially (90°) at a distance of 0.5m from the anode face, whereas PMT2 was placed axially (0°) at a distance of 1.5m from the anode face along the anode axis. A schematic of this arrangement is shown in Fig. 5.23. In the inset, images of the utilized tubular and squirrel cage cathode structures are shown as (a) and (b) respectively.

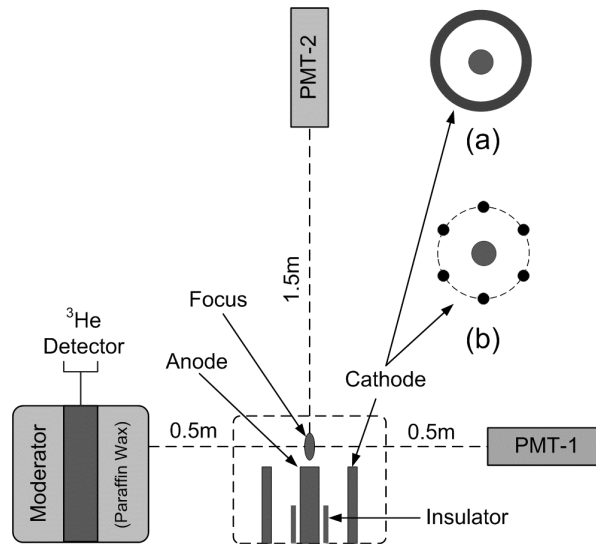


Fig. 5.23 Layout of time resolved and time integrated neutron diagnostic set-up along with schematic of utilized (a) tubular and (b) squirrel cage cathode structure.

The effects of tubular and squirrel cage cathode geometries on neutron and hard X-ray (HXR) emissions for different fill pressures are investigated at the fixed stored energy of 230J (with about 80kA peak discharge current at 13.8kV charging voltage). The results were obtained for averages of 20 shots for each Deuterium gas pressure. To reduce the effect of electrode particulate contamination on neutron output, the gas was refreshed after every five shots. A nominal pressure increase of $\sim 0.05\text{mbar}$ was observed after each set of 5 shots.

5.2.5.2 Average Neutron Yield against Pressure

The measured average neutron outputs for tubular and squirrel cage cathode geometries, for different D₂ filling gas pressures, are shown in Fig. 5.24.

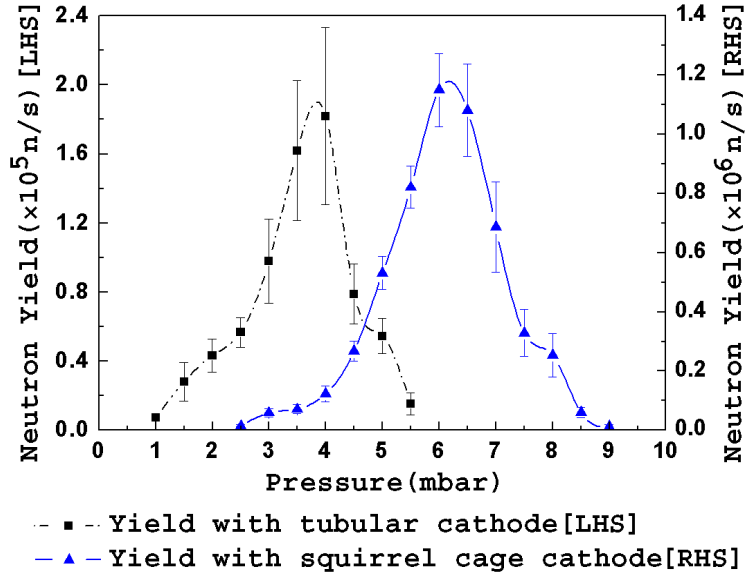


Fig. 5.24 Neutron yield versus D₂ gas pressure for tubular and squirrel cage cathode.

A remarkable enhancement in neutron yield was observed with squirrel cage cathode operation. The maximum average neutron output of $(1.82 \pm 0.52) \times 10^5$ and $(1.15 \pm 0.2) \times 10^6$ neutrons/shot (n/s) were measured for tubular and squirrel cage cathode geometries, respectively. It is also observed that the neutron yield peaked at the higher gas pressure of 6mbar for squirrel cage cathode while it peaked at 4mbar for tubular cathode structure [211]. Our experiments with two different cathode geometries have demonstrated that the final pinch characteristics and in particular, the emission of neutrons is strongly influenced by the cathode structure.

5.2.5.3 Time Resolved Investigation of HXR and Neutron Emission

Time resolved information about the hard X-ray and neutron emission is obtained using the

scintillator photomultiplier detectors PMT1 and PMT2 for tubular and squirrel cage cathodes. Those data are shown in Fig. 5.25 along with corresponding di/dt signals.

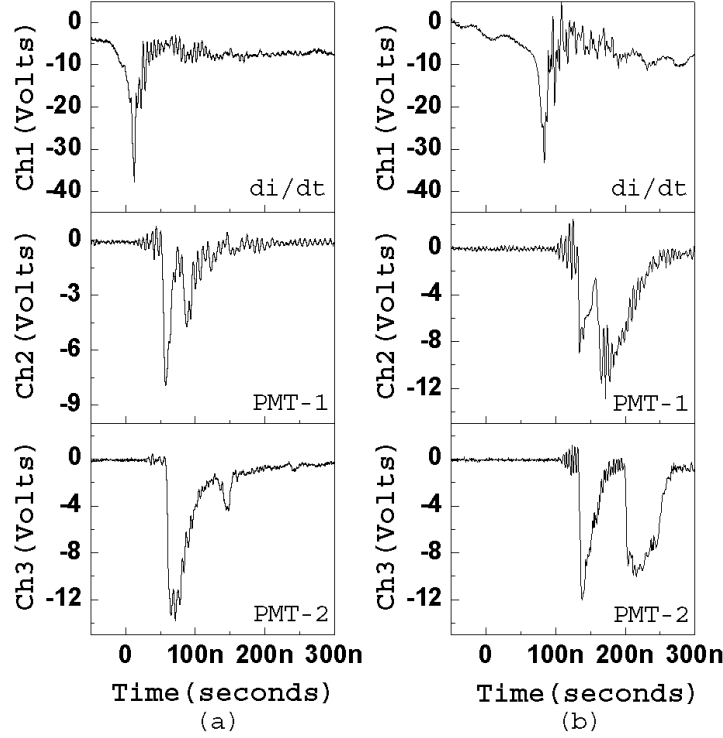


Fig. 5.25 Current derivative signal trace with HXR/ neutron signal recorded with side-on (PMT-1) and end-on (PMT-2) scintillator-photomultiplier detector with
(a) tubular cathode (b) squirrel cage cathode.

The first peak in the PMT signals in Fig. 5.25(a) and (b) (shown as Ch2 and Ch3), is of non-thermal, hard X-rays produced by the instability accelerated electron beam striking the anode. The second peak is confirmed to be that of neutrons, on the basis of time of flight estimates, assuming that the hard X-rays and neutrons are produced at the same time in the pinch. Since PMT1 and PMT2 were placed at distances of 0.5m (radially) and 1.5m (axially), from the anode, subsequent registration of neutron pulses (i.e. second, delayed peak in the respective signals) at about 23 ± 2 ns and 66 ± 3 ns, respectively, after the emission of hard X-ray pulses confirms that the second peak is due to ≈ 2.45 MeV

D-D neutrons. The relative time difference of 45 ± 3 ns, between the neutron pulses, recorded by channels Ch2 (PMT1 trace) and Ch3 9(PMT2 trace) also re-confirms the neutron pulse emission.

The average duration of HXR and neutron pulses, estimated from the FWHM of the corresponding peaks and averaged over 20 shots, in the axial/radial directions are 18 ± 3 ns/ 12 ± 2 ns and 16 ± 3 ns/ 16 ± 2 ns, respectively, for tubular cathode operation (Fig. 5.25(a)), and 18 ± 2 ns/ 15 ± 2 ns and 46 ± 3 ns/ 45 ± 2 ns respectively, for squirrel cage cathode operation (Fig. 5.25(b)). The ~ 30 ns delay in the appearance of the HXR peak (Ch2 and Ch3 trace in the respective figures) from the peak of the current derivative signal (Ch1 trace) is due to the inherent latency in the PMT.

5.2.5.4 Discussion on the Effects of Cathode Geometries

For plasma focus operation, the run-down phase allows the transfer of electrical energy from the capacitor bank to the magnetic energy behind the accelerating current sheath; a fraction of which is rapidly pumped into the pinched plasma column at the end of radial compression phase in a relatively short time, leading to efficient compression and heating [234]. Under stationary initial gas conditions (unlike in a supersonic gas puff) it is impossible to decouple the final radial compression phase and corresponding plasma phenomena from those occurring during the breakdown and axial run down phases [233]. Therefore, while considering losses (loss of pinching efficiency and hence the heating efficiency of the pinched plasma column) in a typical (non gas-puffed) plasma focus device, the thermal flow down the axis during axial acceleration is an important consideration.

The very high temperature achieved in the plasma focus is mainly the result of the axis-symmetric properties of the imploding shock. In the case of the tubular cathode, the outer electrode is an impermeable wall, and there is momentum directed parallel to the shock front. Hence to obtain pressure balance along the shock, the plasma density adjacent to the wall must increase. The associated mass flowing parallel to the shock thus acts as a sink for the incoming plasma and causes a drop in

temperature with increasing radius (as the magnetic pressure falls). With lower temperatures adjacent to the cathode wall and in particular lower electron temperatures, the shock is broadened (i.e. its thickness increases) due to resistive diffusion of the magnetic field into the plasma [235-236]. Aggregation of the plasma mass near the cathode wall also affects the final temperature achieved in the pinch, owing to the fact that compression ratio is dominated by $\rho_{pinch}/\rho_{shock}$ and only a fraction (about 15%) of the plasma mass is collected in the ensuing pinch [235]. Hence, in the case of a tubular cathode, due to its larger contact area (by comparison with a squirrel cage cathode), deleterious effects because of plasma aggregation near the cathode wall are expected to be more pronounced, with a consequent reduction in neutron output.

Another important observation from Fig. 5.24 (neutron output vs. filling gas pressure) is the shift in the optimum filling pressure between tubular and squirrel cage cathode geometries. The difference in the optimum filling pressure range for the two cathode geometries can be explained by the fact that since the geometry of the squirrel cage cathode, allows mass to pass through, during the axial run down phase, the inter-electrode space is not heavily loaded or choked. In the case of tubular cathodes the outward mass flow is blocked, then the channel (i.e. the inter electrode space) tends to get constricted due to growing thickness of 'contact layer'. When this layer gets thick enough the channel tends to get loaded/ choked. In addition, the outward moving particles are reflected back along with impurities from cathode wall with potentially substantial increase in thermal bremsstrahlung radiation loss due to impurity addition. In any case, with the use of tubular cathode geometry the cross sectional area of the channel gets effectively reduced, which accounts for loading of current sheath at comparatively lower operating pressure. This observation is corroborated by the measurements shown in Fig. 5.26, where the variation in time to pinch from the breakdown phase (which includes axial acceleration phase and compression phase, defined as t_p in the current derivative signal shown in the

inset of Fig. 5.26), at different filling gas pressures is plotted, for the tubular and squirrel cage cathode geometries.

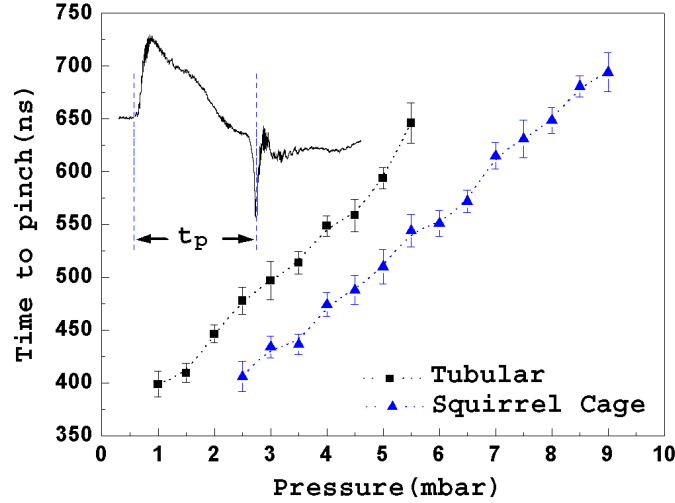


Fig. 5.26 Time to pinch versus D_2 pressure for tubular and squirrel cage cathode operation.

Usually in typical plasma focus device operation, the characteristic time t_p , increases with the increase in operating pressure due to increased load on the current sheath in the axial phase. In the graph shown, larger slope with distinct offset of the operation curve for tubular cathode (by contrast to squirrel cage cathode operation), indicates comparatively higher loading of the current sheath, even at lower pressures. The squirrel cage cathode seems to lessen much of these effects and thus performs with better efficiency resulting in higher yields. It is interesting to note, that for the two cathode geometries, the time-to-pinch is the same (~ 550 ns) for the pressures (4mbar, tubular; 6mbar, squirrel-cage) associated with the maximum neutron yield. This can be seen by considering Fig. 5.24 and 5.26. This observation further corroborates our proposal that main influence of cathode geometry is the current sheath mass loading/choking effect.

Recently, neutron yield scaling has been thoroughly reviewed by S Lee *et al* [128,129,132], using the five-phase Lee model RADPFV5.13 [70], and found to follow $Y_n \sim I_{pinch}^x$ where I_{pinch} is the pinch current that actually participates in the focus pinch phase and x is found to vary in the range of

3 – 5 for different plasma focus devices. To realistically simulate an experiment using the Lee model, firstly the circuit parameters, electrode dimensions, operating voltage and gas pressure are keyed into the code, then the computed current trace is ‘fitted’ to the experimental peak discharge current trace, by adjusting four ‘model parameters’. These four parameters for ‘fitting’ are axial mass swept-up factor f_m and axial current factor f_c (which characterize axial phase electrodynamics), radial mass factor f_{mr} and radial current factor f_{cr} (which characterize radial phase dynamics). These model parameters are then fine-tuned till the computed trace is in close agreement with the experimentally measured discharge current trace.

The main objective of using the Lee code is to investigate the changes in the fitting model parameters and electrodynamic behavior for the two cathode geometries to deduce and understand the corresponding changes in plasma dynamics and pinch current. The fine tuned electrical circuit parameters used for the simulation of respective geometries are: 2.4 μ F, 32.9nH and 60m Ω for tubular cathode operation and 2.4 μ F, 31nH and 67m Ω for squirrel cage cathode operation. A typical ‘fitted’ trace of squirrel cage cathode operation at 6 mbar gas pressure is shown in Fig. 5.27. Values of ‘model parameters’ that have been obtained for ‘best fit’ with tubular and squirrel cage cathodes (at 4 and 6mbar respectively) along with peak/ pinch current estimations are summarized in Table 5.2.

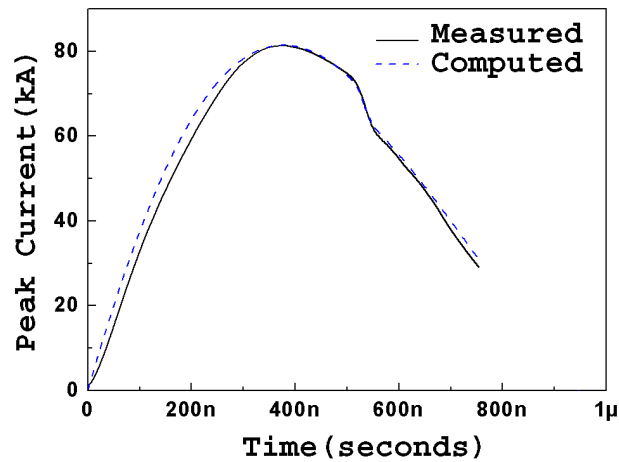


Fig. 5.27 A “fitted” current trace for squirrel cage cathode operation at 6mbar pressure.

Table 5.2 Model parameters and simulation results.

Cathode	f_m	f_{mr}	f_c	f_{cr}	I_{pinch}/I_{peak}	P_o
Tubular	0.31	0.33	0.62	0.84	54kA/81kA	4mbar
Squirrel Cage	0.17	0.15	0.62	0.89	58kA/80kA	6mbar

The two key differences that can be noticed from Table 5.2 are (i) for tubular cathode structure, the values of axial mass swept-up factor f_m and radial mass factor f_{mr} are about ~ 1.82 and ~ 2.2 times higher, respectively, than that obtained for squirrel cage cathode operation which supports the argument and observation of the mass loading of the current sheath for tubular operation and (ii) the estimated pinch current is lower for tubular cathode operation supporting the lower neutron output for this cathode.

In conclusion, our investigation shows a notable enhancement in the neutron output for the squirrel cage geometry as compared to that of a tubular geometry. In a tubular cathode structure, owing to the impermeable wall and relatively larger surface area, the plasma density rises near the cathode wall, causing the shock to be radially broadened through resistive diffusion of the magnetic field. This results in the narrowing of the effective channel cross-section, causing loading of the current sheath along with increased contact layer resistance thereby resulting in lower pinch efficiency. Also in operation with tubular cathodes, addition of impurities (due to back reflected particles from the cathode wall) during the axial flow, may result in a substantial drop in temperature due to increased radiation loss which is also consistent with the reduction in neutron output [211].

5.2.6 Investigation of Anisotropy in Neutron and Hard X-ray Emissions

Since the fluence anisotropy in energetic radiations from plasma foci, due to co-existence of thermal and non-thermal neutron production mechanisms [121,221], is a widely evidenced behavioral

fact, we performed anisotropy characterization of the neutron and hard X-ray bursts, emitted from the FMPF-1 device, to investigate the type of dominant production mechanism in low energy miniature plasma focus devices. The correlation between the neutron and hard X-ray emission characteristics was also explored and is reported in detail under this section.

5.2.6.1 Experimental Arrangement

The coaxial electrode assembly of the plasma focus head consists of a 15mm long, composite tapered anode of stainless steel (SS), with initial diameter of 12mm, tapering down to 7mm at the tip over the last 5mm of the anode and a squirrel cage cathode, consisting of six, 6mm diameter SS rods, uniformly spaced on a coaxial circle of 30mm diameter. A 12mm deep cavity of 6mm diameter was bored at the center of the anode, for higher reproducibility (this is the same anode, improved/modified version of V4/CA/17-3/SS, which was used in experiment reported in section 5.2.5 on the effects of cathode geometries on radiation products). An insulator sleeve of Pyrex glass with a breakdown length of 5mm was placed between the anode and cathode. A schematic drawing of the coaxial electrode assembly is shown in Fig. 5.28(a).

The employed diagnostics include – high bandwidth Rogowski coil, a high sensitivity ^3He proportional counter and two identical scintillator-photomultiplier detectors in the axial (0°) and radial (90°) directions respectively. A schematic drawing of this diagnostics arrangement is shown in Fig. 5.28 (b). The scintillator-photomultiplier detectors designated as PMT1 and PMT2 were placed in the axial and radial directions, respectively, at identical distances of 0.7m from the focus. Both PMT detectors were cross-calibrated before use, and the obtained gain calibration factor has been taken into account for relative flux estimations of hard X-ray and neutron yield [18]. Attenuation of the HXR and neutron flux passing through the vacuum chamber was kept the same for the measurements at 0° and 90° , by using identical thickness of glass windows in the respective directions. In order to match the

timing of PMT signals, similar lengths of double-shielded coaxial cables have been used for signal transport. The synchronization of signals is very important because data analysis requires comparison on the basis of the same time base.

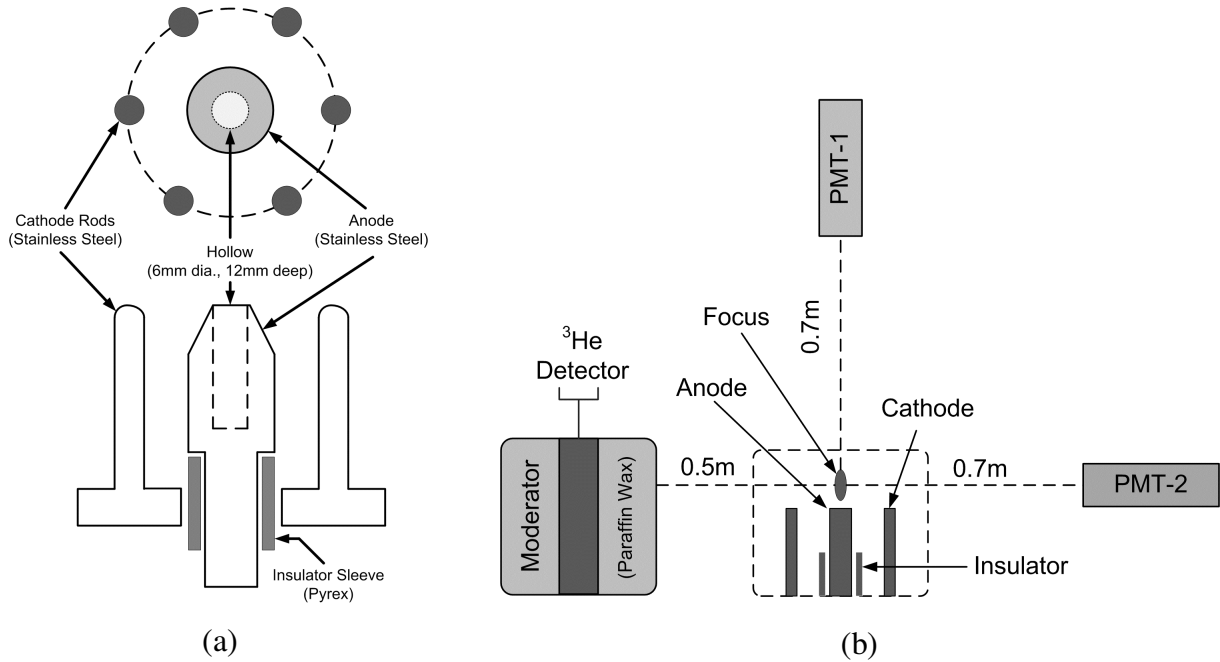


Fig. 5.28 Schematic of the (a) electrode assembly and (b) implemented diagnostics setup.

For the results shown and discussed in this section, to investigate the anisotropy in neutron and HXR emissions for different Deuterium fill pressures in the optimum range; the stored energy of system was maintained constant at 230J (to deliver about 80kA peak discharge current at 13.8kV charging voltage). The reported data are averages over 20 shots for each gas pressure. To reduce the effect of contamination on radiation yields, the gas was refreshed after every five shots.

5.2.6.2 Total Neutron Yield against Pressure

The pressure dependence of the neutron yield per shot, measured using ^3He detector, for pure D_2 discharges in the optimum filling gas pressure range of 4.5 to 6.5mbar is shown in Fig. 5.29. As

may be observed in this graph, the average neutron yield peaked at 5.5mbar, being $(1.2 \pm 0.2) \times 10^6$ neutrons/shot. The highest neutron yield measured for a single shot was 3×10^6 at 5.5mbar.

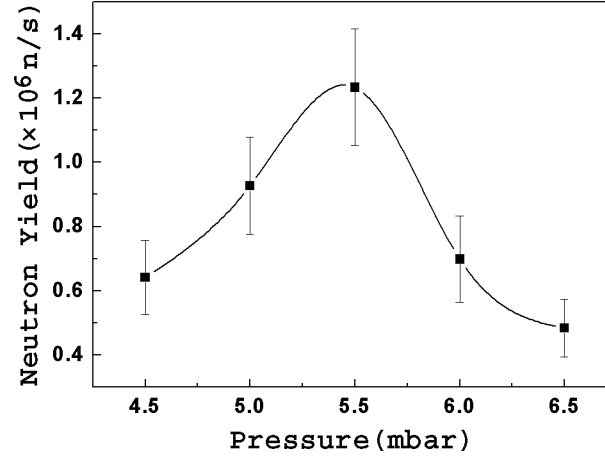


Fig. 5.29 Average neutron yield versus D_2 filling gas pressure.

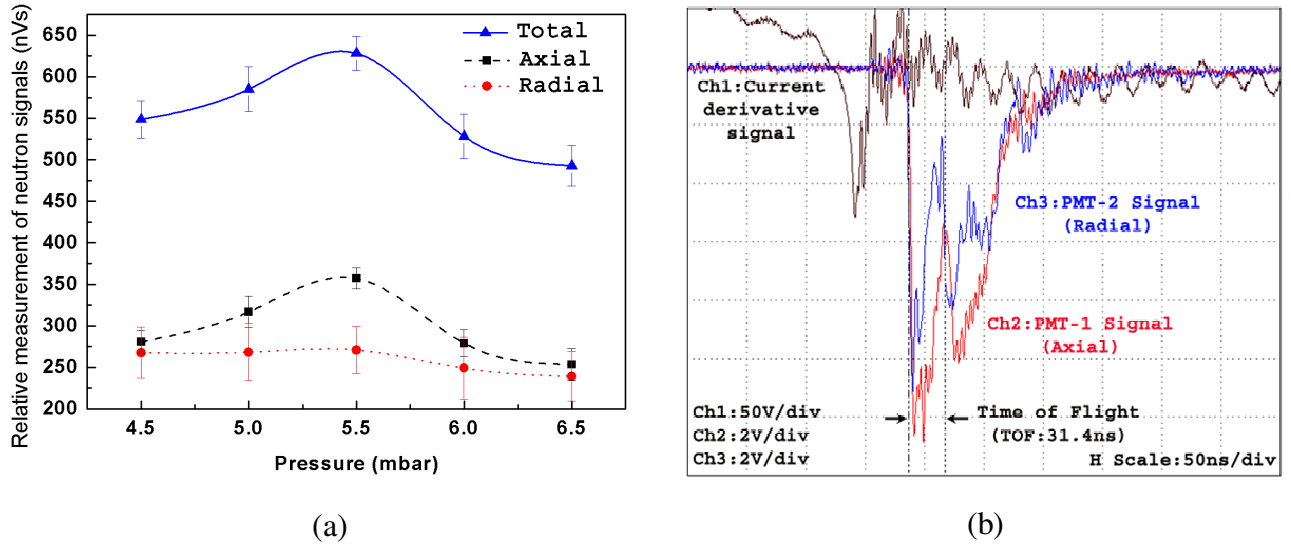


Fig. 5.30 (a) Variation in axial, radial and total relative neutron yields with D_2 filling gas pressure
(b) Current derivative (di/dt) signal trace with HXR and neutron signal recorded with side-on and end-on scintillator-photomultiplier detectors PMT1 and PMT2 respectively.

The results obtained for relative neutron yields estimations using the area under the scintillator-photomultiplier detector signal for neutron emission in the axial and radial directions and total neutron

emission (i.e. sum of areas under the waveforms registered in the axial and radial directions) are shown in Fig. 5.30 (a). The typical signals used for such estimations are shown in Fig. 5.30 (b). It may be noted that the occurrence of maximum average neutron yield measured by ^3He detector (refer to Fig. 5.29) coincides with the measurement of maximum relative neutron yield at 5.5mbar. The relative measurements of neutron flux in the axial and radial directions also imply that the average neutron flux is higher in the axial rather than radial, direction at all pressures, revealing anisotropy in the fluence.

5.2.6.3 Measurement of Neutron and HXR Fluence Anisotropy

Anisotropy in the neutron fluence is an important quantity in neutron emission characteristics of the plasma foci, since it helps to identify the dominant mechanism of neutron production. The typical signals (Ch2 and Ch3) from the two scintillator-photomultiplier detectors (PMT1 and PMT2) and the corresponding current derivative signal (Ch1), for a typical shot at 5mbar Deuterium filling gas pressure are shown in Fig. 5.30 (b). There is an inherent delay of $\sim 30\text{ns}$ in the photomultiplier tube signal due to which the HXR peak is delayed with respect to the steep minimum of the current derivative signal (see Fig. 5.30 (b)). In the PMT signals (Ch2 and Ch3), the first peak is of non-thermal, hard X-rays produced by the interaction of instability generated energetic electrons with the stainless steel anode, whereas the second peak was confirmed to be due to neutrons on the basis of time of flight separation. The observed time delay of $\sim 32\text{ns}$ after registration of the hard X-ray pulse, confirmed the production of fusion neutrons having typical energy $\sim 2.45\text{MeV}$. It may be observed from the sample time-resolved traces (refer to Fig. 5.30 (b)) that as the amplitudes and the area under the signals on Ch2 and Ch3 are different, the fluence of neutrons and hard X-rays in the axial and radial direction is different. The pulse duration and intensity of the hard X-ray and neutron pulse are relatively higher in the end-on (i.e. axial) direction as compared to side-on (i.e. radial) direction. The durations of HXR and neutron pulses in the axial and radial direction, estimated from full width at half

maximum (FWHM) of the corresponding peaks and averaged over 20 shots, in the optimum pressure range of 4.5 to 6.5mbar are shown in Fig. 5.31 (a).

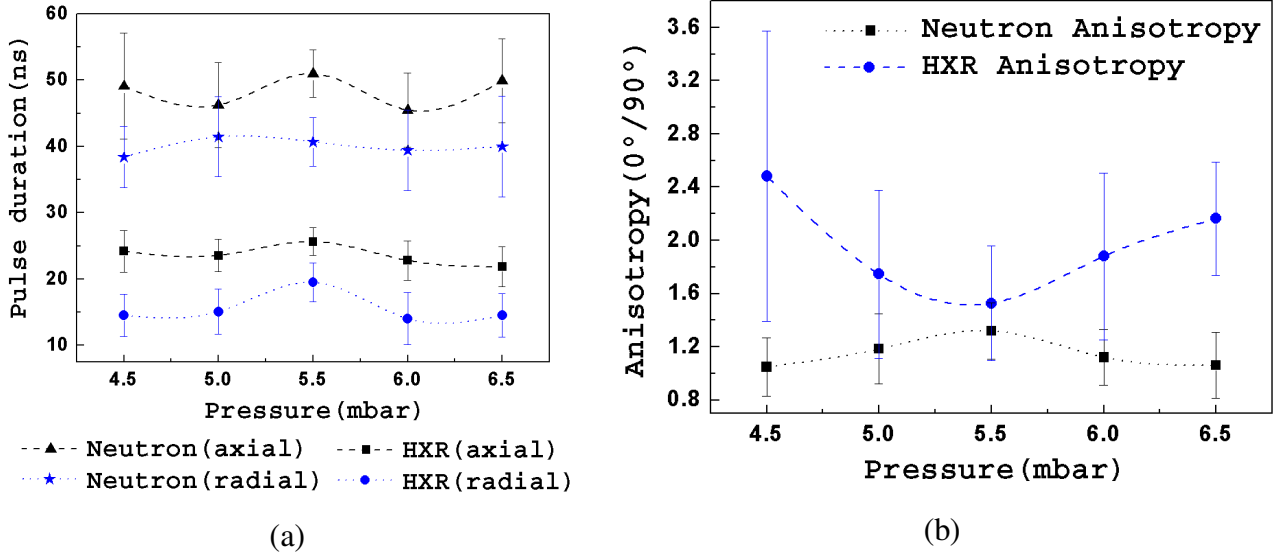


Fig. 5.31 (a) Variation of average pulse duration (FWHM) of HXR and neutron bursts, and (b) Variation of HXR and neutron anisotropy as a function of pressure.

The average pulse durations (FWHM) of HXR/neutron bursts were observed as 24 ± 2 ns / 48 ± 3 ns in axial direction and 16 ± 3 ns / 40 ± 2 ns in the radial direction. In the optimum deuterium filling gas pressure range of 4.5 to 6.5 mbar, the fluence anisotropy Y_{0°/Y_{90° (i.e. hard X-ray and neutron flux ratios in the axial and radial direction) has been estimated by measuring the corresponding HXR/neutron yields by integrating the area under the respective peaks using Yokogawa software, XViewer®. The relative variation of the HXR and neutron anisotropy as a function of pressure is shown in Fig. 5.31 (b).

Upon comparing the obtained neutron anisotropy (see Fig. 5.31(b)) with total neutron yield (Fig. 5.29 and 5.30(a)) at different pressures, it is observed that the neutron anisotropy and total neutron yield trends are similar, both initially increase with the increase in pressure reaching a maximum at optimum pressure of 5.5 mbar and then decrease with further increase in pressure. The maximum average value of 1.3 ± 0.2 for the neutron anisotropy was found to be at the optimum pressure

of 5.5mbar. The higher neutron anisotropy of 1.3 ± 0.2 at optimum pressure of 5.5mbar shows that at this pressure beam-target mechanism plays a dominant role in neutron production whereas at lower and higher deuterium pressures the neutron anisotropy ($Y_{0^\circ} / Y_{90^\circ}$) is lower, indicating a decreased contribution from the beam-target mechanism.

The graph shown in Fig. 5.31(b) also shows that the HXR anisotropy trend is complementary to that of neutron anisotropy (and the neutron yield) as it initially decreases from 2.5 ± 1 at low pressure to a minimum value of 1.5 ± 0.4 at the neutron-optimized pressure of 5.5mbar and then increases again to 2.2 ± 0.4 with the increase in pressure to 6.5mbar.

To understand the decrease of HXR anisotropy for the neutron optimized pressures where neutron anisotropy maximizes, it is important to consider the effect of the anode geometry, Fig. 5.28(a) on measured HXR fluence in the radial direction. As shown in Fig. 5.28(a), since the anode is hollow (12mm deep), therefore the hard X-rays propagating in the radial direction (produced by the energetic electrons impinging on the anode target), are attenuated by the ~3mm thick stainless steel anode wall. The reduction in HXR anisotropy at the very pressures where the neutron anisotropy shows a maximum, suggests that the measured HXR emission in radial direction is relatively higher than at other pressures. Our understanding is that for this observation to be true for the present experimental setup (where HXR emission in radial direction is hindered by 3mm thick Stainless steel wall), the electron beam that produces the higher energy HXR must be more energetic. If the electron beam is more energetic, then so too must be the ion beam propagating in the opposite direction. Thus the observed lowering of the HXR anisotropy is due to increased transparency of high energy X-rays through hollow anode wall in the radial direction at neutron-optimized pressure. This reveals stronger beam target interaction.

Graphs shown in Fig. 5.32(a) and (b) represent the variation of the neutron anisotropy and

HXR anisotropy respectively as a function of the total neutron yield for a set of 20 consecutive shots at the neutron optimized pressure of 5.5mbar.

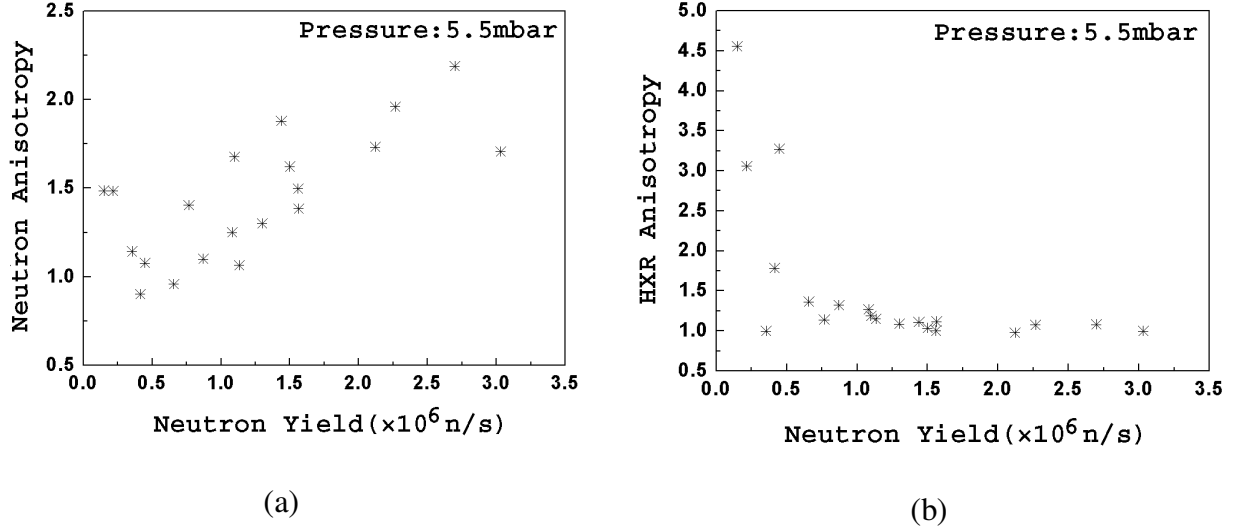


Fig. 5.32 (a) Neutron anisotropy versus total neutron yield, and (b) HXR anisotropy versus total neutron yield for optimum pressure of 5.5mbar.

It may be observed in the respective graphs that higher neutron yields have been obtained with correspondingly higher neutron anisotropies and lower HXR anisotropies. This observation reconfirms dominance of beam target mechanism of neutron production in this miniature plasma focus device.

The anisotropy result obtained at optimum pressure of 5.5mbar corroborates the dominance of neutron production due to axially accelerated deuterons at optimum pressure and supports the beam target postulation. However, the lowering of anisotropy factor in neutron yield with corresponding higher HXR anisotropies, below and beyond the neutron optimized pressure of 5.5mbar indicates that the neutron production mechanism at play depends upon the operating gas pressure, a result which was also described by Zakaullah *et al* [229]. At the lower and higher ends of the operating pressure range the beam target contribution was found to be lower.

According to the beam target hypothesis, the production of accelerated deuterons produced by spatial charge separation in plasma diodes generated by a rapid change in plasma inductance along

with abrupt increase in the plasma resistance (with the onset of $m=0$ instability followed by disruption of the pinched plasma column) plays a crucial role in neutron production [112,113] and hence, the factors affecting instability/stabilization of the pinched plasma column play an important role in the beam target contribution [132]. The growth time (t_i) of the Rayleigh-Taylor (RT) instability in accelerated dynamic pinches is given by $t_i = \sqrt{2\pi\lambda_p/g}$, where g is the sheath acceleration and λ_p is the dominant wavelength of perturbation [90]. In this relation, it may be noted that instability growth time varies inversely with sheath acceleration. As the sheath acceleration mainly depends on the filling gas pressure, therefore probably at low pressures the RT instability develops so rapidly that it prohibits even the formation of a well defined pinched plasma column [229,214] which leads to inefficient beam-target interaction. At higher pressures even though the plasma density is quite high [111], an effect is observed which tends to suppress the $m=0$ sausage instability because a substantial part of the pinch current begins to flow through the rarefied pinch plasma [91] in a corona around the dense core. As a consequence the RT instability formation is damped [91] and produces more weakly accelerated deuteron beams and hence results in less efficient beam target interaction.

The low anisotropy in neutron production at lower and higher ends of the operating pressure range may also be hypothesized from the perspective that since the RT instability is not strong enough, therefore owing to the lower velocity or energy of the accelerated deuterons they get trapped in the local magnetic field [111], gyrate and consequently produce neutrons more or less nearly isotropically (since the ion velocities are randomly oriented because of the curved motion in the magnetic field of the pinch [237]).

5.2.6.4 Energy Spread in the Neutron Spectrum

Since the neutrons of thermonuclear origin are expected to have energy around 2.45MeV, whereas the neutrons produced by beam-target effect are expected have a spread in their energies along

with anisotropic emission (as they are produced by interaction of fast ions with thermal background plasma, when the plasma column expands and $m=0$ instabilities sets in [238]); therefore, by investigating the energy spectrum of neutrons produced by FMPF-1 device, it may be possible to gain further insight of the dominant mechanism responsible for the neutron production [202].

The time of flight (TOF) technique has been used to estimate the average neutron energy. In the present experimental arrangement, the energy spectrum of neutrons is estimated by measuring the time difference between the hard X-ray and neutron peaks and knowing the spatial separation of the scintillator-photomultiplier detectors from the focus anode tip in the respective directions (i.e. axially and radially). It may be noted that the neutron pulse is the second (later) peak in the characteristic signals Ch2 and Ch3 in Fig. 5.30(b). In the estimation, the minimum of the hard X-ray peak has been used as the time reference for the occurrence of the focus and the energy of the fast neutrons has been measured using the difference of time elapsed between the minima of the hard X-ray and neutron peak.

The estimated neutron energy spectrum for the optimum pressure range of 4.5 to 6.5mbar is shown in Fig. 5.33.

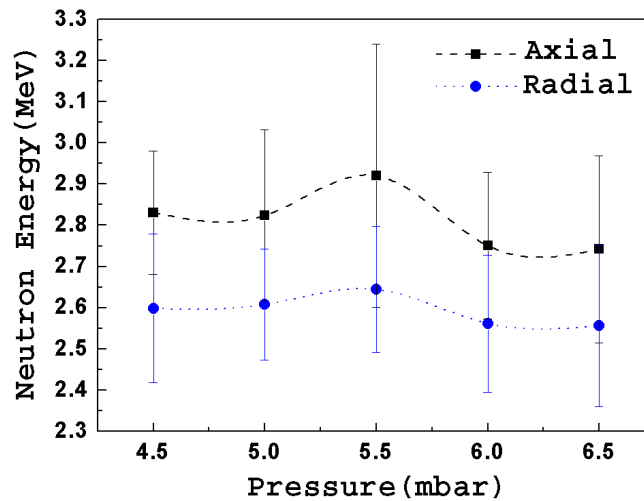


Fig. 5.33 Neutron energy spectrum in the optimized pressure range.

The typical energy of neutrons has an estimated mean value of (2.8 ± 0.2) MeV in the axial direction and (2.6 ± 0.1) MeV in the radial direction in the operating pressure range of 4.5 to 6.5 mbar. The average peak energy of neutrons was observed at neutron-optimized pressure of 5.5 mbar as (2.9 ± 0.3) MeV in the axial direction and (2.6 ± 0.1) MeV in the radial direction. The evidence of spread in the energy of neutrons, with a maximum energy higher than 2.45 MeV (which is the typical energy of thermonuclear neutrons) in the axial and radial direction reveals the presence of non-thermal mechanisms of neutron production.

5.2.6.5 Neutron Production Kinematics and Deuteron Energy Spectra

Considering the beam target hypothesis, the typical energy of deuterons producing neutrons in above mentioned energy ranges in the axial and radial directions could be deduced theoretically [116]. According to the beam target postulate, the D-D reaction is produced by an accelerated deuteron colliding with a thermal deuteron; hence the energy of the neutron is dependent on its direction of travel. The schematic drawing shown in Fig. 5.34(a) illustrates the neutron production kinematics through the D-D fusion reaction via the collision of a fast moving deuterium (D^+) ion. Since in this reaction momentum remains conserved, therefore the momentum triangle deduced from Fig. 5.34(a) is shown in Fig. 5.34(b).

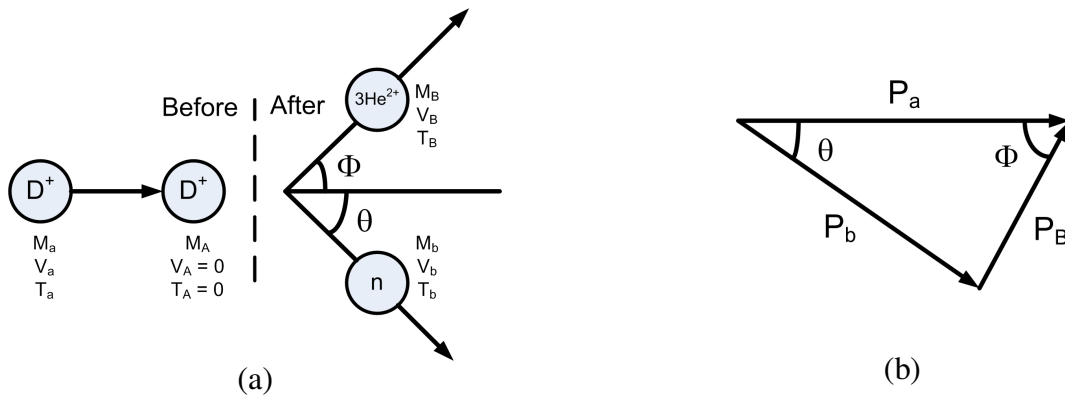


Fig. 5.34 (a) Schematic diagram (a) and momentum triangle (b) of D-D fusion reaction.

In Fig. 5.34(a) and Fig. 5.34(b), notations M , V , T and P denote mass, velocity, kinetic energy (KE) and momentum respectively, associated with corresponding reaction elements. Mathematically, by applying the cosine rule to the momentum triangle the deduced equation is:

$$P_B^2 = P_a^2 + P_b^2 - 2P_a P_b \cos \theta \quad (5.3)$$

Upon substituting the generalized relationship, $P^2 = 2MT$ in Eq. (5.3):

$$2M_B T_B = 2M_a T_a + 2M_b T_b - 4\sqrt{(M_a M_b T_a T_b)} \cos \theta \quad (5.4)$$

Now by rearranging Eq. (5.4) to express T_b (KE of neutrons) in terms of T_a (KE of moving deuterium ions) and θ (angle of detection of neutrons):

$$\sqrt{T_b} = u \pm \sqrt{(u^2 + w)} \quad (5.5)$$

$$\text{where } u = \frac{\sqrt{(M_a M_b T_a)}}{M_B + M_b} \cos \theta \text{ and } w = \frac{M_B Q + T_a (M_B - M_a)}{M_B + M_b}$$

By substituting Q -value for the reaction as $Q = T_B + T_b - T_a$ (i.e. the difference in mass-energy between initial and final states) and the values of respective constants, the plot of T_b was obtained against θ for deuterium ions traveling with different kinetic energies (i.e. $T_a = 1\text{keV}$, 50keV , 100keV , 150keV and 200keV) and the graphs are collectively shown in Fig. 5.35.

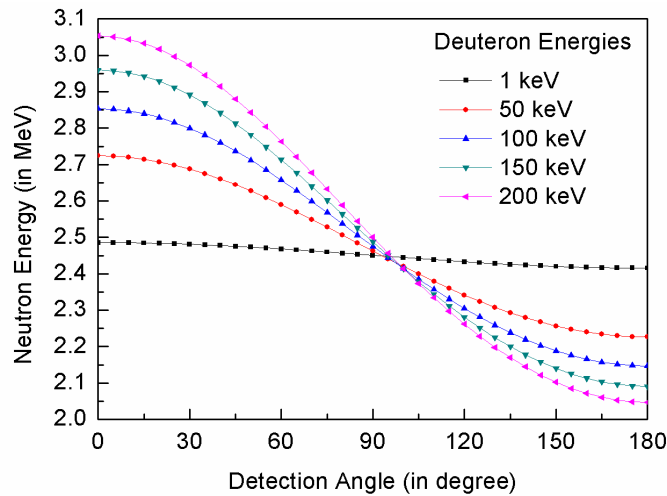


Fig. 5.35 Graph of neutron energy versus angle of neutron detection for various deuteron energies.

The obtained results (as shown in the plot) imply that the experimentally measured mean energy of neutrons of (2.8 ± 0.2) MeV in the axial direction and (2.6 ± 0.1) MeV in the radial direction are mainly contributed by deuterons in the 50 to 150 keV energy range.

For the verification of theoretically deduced results mentioned above; experimentally also the deuteron energy spectra was obtained for optimum D₂ filling gas pressure of 5.5 mbar using the fast response Faraday cup (described under section 4.6). The maxima of dip in the current derivative signal has been taken as time fiducial reference for the time of flight measurement (on the basis of reported observations [93,121] that the maxima of the dip denotes the phase at which the pinched plasma column reduces to minimum radius and it is soon disrupted by the rapid growth of MHD instabilities while the electrons and ions initiate acceleration in the opposite direction). The Faraday cup was located in the end-on direction, at the distance of 15 cm from the pinch. The typical oscilloscope traces of current derivative probe and Faraday cup, obtained for a shot at optimum D₂ filling gas pressure of 5.5 mbar, is shown collectively in Fig. 5.36.

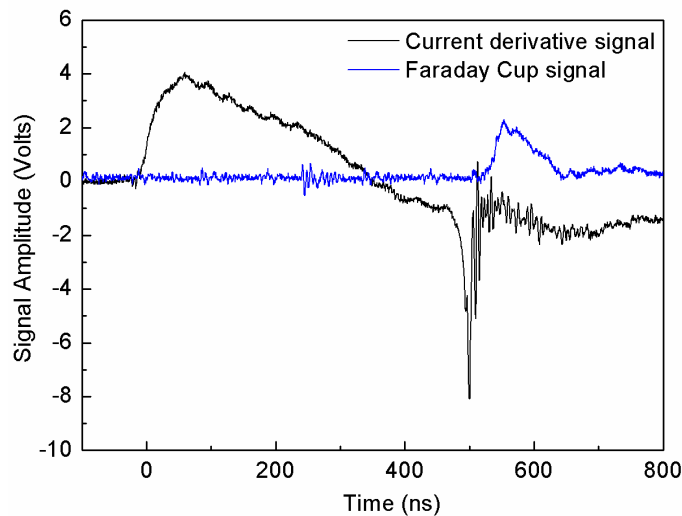


Fig. 5.36 Oscilloscope traces of Faraday cup and current derivative signal.

The ion signal obtained from the Faraday cup was analyzed to deduce the energy spectrum and

density. The energy of deuterons was estimated by their time of flight and the beam evidenced poly-energetic ranging from ~10keV to 600keV. The ion density estimations have been done using the relation [239, 240]:

$$N_i = \frac{V}{RqAv} \quad (5.6)$$

Where N_i is the number density of ions having velocity v and charge q , V is voltage produced corresponding to the ion pulse, R is the termination resistance (of the scope) across which output of the ion pulse is taken and A is the area of the pinhole aperture. The plot for ion density Vs energy spectrum is shown below in Fig. 5.37.

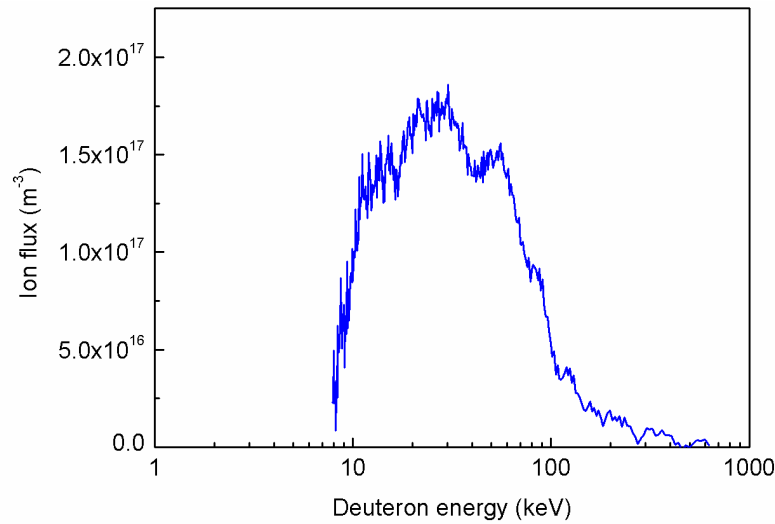


Fig. 5.37 Ion density as a function of deuteron beam energy.

The FWHM (full width at half maximum) of obtained plot depicts that the bulk of the ions have energies in the range of 10 to 70keV. These results are in agreement with theoretically deduced results, where we have illustrated that typical energy of deuteron contributing more significantly in neutron production; typically lie in the range of 50 to 150keV (on the basis of obtained neutron energy spectrum).

5.2.6.6 Findings

In the present experimental investigation, the HXR and neutron emission characteristics were studied in the axial and radial directions and the results confirmed the existence of anisotropy in the HXR and neutrons emissions from our miniature plasma focus device operating in the sub-kilojoule range. The major outcomes of the investigation are:

- (i) The maximum average value of 1.3 ± 0.2 for the neutron anisotropy was obtained at the optimum pressure of 5.5mbar.
- (ii) The presence of stronger neutron anisotropy, indicating the dominance of beam-target mechanism, found to occur under neutron optimized conditions; whereas at the lower and higher end of operating pressure where the neutron yield is low, the thermonuclear mechanism seems to dominate.
- (iii) It has been observed that at the optimum pressure of 5.5mbar, the maximum neutron yield coincides with the peak neutron anisotropy. Also the corresponding HXR anisotropy has a minimum value at the neutron-optimized pressure of 5.5mbar. As the main source of non-thermal HXR radiation is the accelerated electron beam impinging on anode and as the hollow anode then transmits more in the radial direction, we have used this reduction in HXR anisotropy corresponding to peak neutron anisotropy to infer that the electron beam (and hence the complementary ion beam) increases in energy at the optimum pressure.
- (iv) The experimental results show that the average neutron energy is always higher in the forward direction than in the radial direction confirming the beam-target mechanism to be the dominant neutron production mechanism.
- (v) The observed variation in HXR and neutron anisotropy factor at different filling gas pressure suggests that the neutron production mechanism at play has pressure dependence. A plausible explanation is that at low pressures, the instability growth rates are very high and do not allow a

dense pinch to form with well organized dense “spots”. Lack of these spots reduces the local electric fields, hence the accelerated Deuteron energy and consequently the beam-target fusion. At too high a pressure, though such dense spots might form, the current may be easily shunted from the dense core to a lower density through sufficiently conducting corona, effectively shutting off the beam-target mechanism. It was theoretically as well as experimentally deduced that in sub-kilojoule range miniature plasma focus device, mainly low energy deuterons (of typically $<100\text{keV}$) contribute more significantly in the neutron production by D-D interaction.

The fluence anisotropy results obtained from investigation on radiations from FMPF-1 device mainly supports the dominance of beam-target mechanism at neutron optimized pressure. However, the existence of other non-thermal mechanisms cannot be excluded.

5.3 Results of FMPF-2 Device

After gaining experience on optimizing the neutron yield performance of FMPF-1 device, in the next phase of experiments; attention was mainly focused on the characterization of newly devised first repetitive version of miniature plasma focus device i.e. ‘FMPF-2’ [216]. As detailed under chapter 3, the FMPF-2 device was mainly developed as a test bed of high repetition rate miniature plasma focus device with entirely new and upgraded pulsed power system of similar energy. It is still a single module system with 8 Nos. of $0.3\mu\text{F}$, low inductance capacitors connected in parallel which are charged together using a single line and are discharged into the focus load through a single Trigatron sparkgap switch.

In this section, along with the results of basic measurements and device characterization (like electrical parameters and discharge characteristics, determination of neutron optimized pressure, time resolved neutron and hard X-ray measurements) we present (i) the experimental demonstration of enhancement in time averaged neutron fluence using repetitive mode of operation, (ii) time resolved

investigation on stability of neutron yield during a repetitive burst mode, and (iii) time integrated and time resolved investigation on neutron emission characteristics. In the concluding section, the evidenced consequential issues of critical concern affecting the neutron yield performance of the MPF device operation are discussed in detail.

5.3.1 Electrical Characteristics

For investigating the electrical discharge characteristics of FMPF-2 device, a short circuit discharge test was performed under high pressure. Argon gas filling at various charging voltages ranging from 10 – 14kV. The sample oscillogram of short circuit discharge current trace at 14kV operation with 40mbar Argon filling is shown in Fig. 5.38.

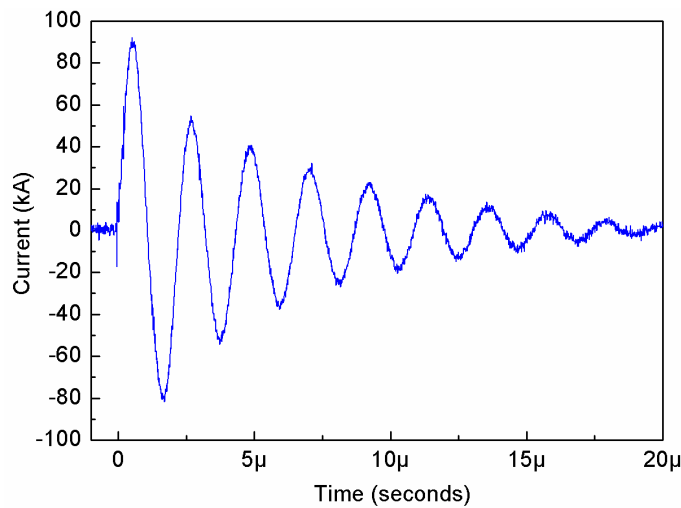


Fig. 5.38 Sample oscillogram of short circuit discharge in FMPF-2 device at 14kV.

The observed higher reversal in the obtained current trace indicates lowering of equivalent circuit resistance in the upgraded pulsed power system of FMPF-2 device (refer Fig. 5.2 for relative comparison). Following the methodology illustrated under section 5.2.1, the obtained main electrical parameters of FMPF-2 device through short circuit tests are summarized in Table 5.3.

Table. 5.3 Main electrical characteristics of FMPF-2 device.	
Energy bank capacitance (C_o)	2.4 μ F
Maximum charging voltage (V_o)	14kV
Maximum stored energy @14kV (E_o)	235J
Maximum current (under short circuit) (I_{sc})	89kA @14kV
Typical operating voltage range	12 – 14 kV
Eq. circuit inductance (L_{eq})	56 $\pm$ 3nH
Eq. circuit resistance (R_{eq})	26 $\pm$ 3m Ω
Voltage Reversal (f)	77%
Quarter time of discharge ($T/4$)	~575ns
Maximum discharge repetition rate (DRR)	10Hz

It may be realized from the obtained results that in FMPF-2 device, although we succeeded in lowering the equivalent resistance R_{eq} in the discharge circuit by replacing the capacitors having low ESR (equivalent series resistance) but owing to the increase in the inductance by (i) use of single discharge switch, and (ii) extended connections of 8Nos. of capacitors to the single switch; there was no significant gain in the peak discharge current level in comparison to FMPF-1 device, while operating at similar energy level. However, the most important upgradation of the FMPF-2 device has been its capability to be operated in repetitive mode up to 10Hz.

5.3.2 Experimental Setup and Diagnostics

In FMPF-2 device, since the peak discharge current and quarter time period of the discharge signal remained close to those of the FMPF-1 device, similar dimensions of anode and cathode have been used. A 15mm long anode of composite geometry along with squirrel cage cathode having diameter of 30mm (as illustrated under section 5.2.6.1, Fig. 5.28(a)) has been used. For separation between electrodes, an insulator sleeve of Pyrex glass with breakdown length of 5mm was used.

The employed diagnostics include [200,216] – (i) high bandwidth Rogowski coil as an electrical diagnostic tool for obtaining current derivative signals, (ii) a high sensitivity ^3He proportional counter for the measurement of neutron yield into 4π sr and (iii) two identical scintillator-

photomultiplier detectors in the axial (0°) and radial (90°) directions respectively with reference to the anode axis), to acquire time resolved neutron and hard X-rays signals. A schematic drawing of the diagnostics arrangement used in FMPF-2 experiments is shown in Fig. 5.39. For measuring the neutron yield of each shot during single/ repetitive mode of operation, a similar ^3He neutron detector setup as described under section 4.4.1 has been used. The ^3He detector tube was placed in the radial direction at distance of 1m from anode axis. During the experimental runs in repetitive mode of operation the analog signals generated by ^3He detector for each shot are captured by the digital oscilloscope and saved in its history mode memory. The signal of ^3He detector was integrated using Yokogawa software, XViewer®, to estimate the area under the curve in μVs and thus the neutron yield [200,216].

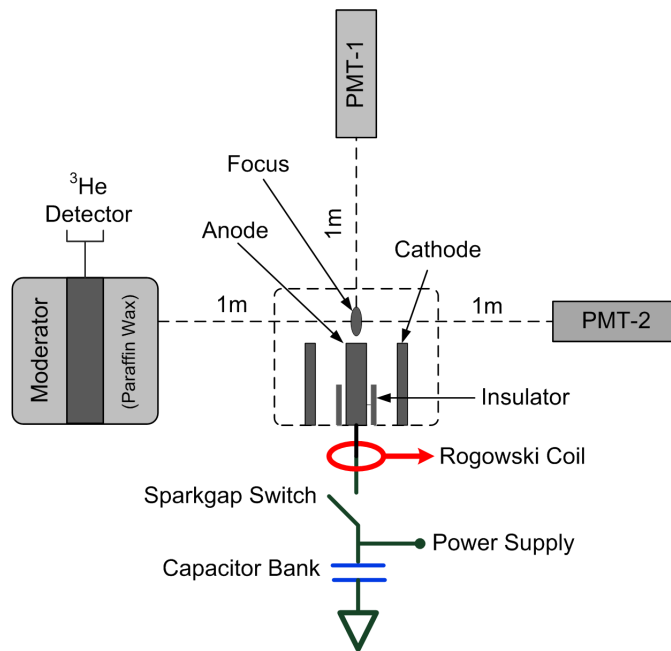


Fig. 5.39 Layout of time resolved and time integrated diagnostic set-up.

The scintillator-photomultiplier detector pair, designated as PMT1 and PMT2 in Fig. 5.39, were placed in the axial and radial directions, respectively, at identical distances of 1m from the focus. Both the detectors have been cross-calibrated before being employed, and the obtained gain calibration factor has been taken into account for relative flux estimations of hard X-ray and neutron yield [18].

Attenuation of the HXR and neutron flux passing through the vacuum chamber was kept the same for the measurements at 0° and 90° , by using identical thickness of glass windows in the respective directions.

For the results shown and discussed in this section, to investigate the optimum performance of the FMPF-2 device in singular and repetitive modes of operation for different Deuterium fill pressures; the stored energy of system was maintained constant at 230J (to deliver about 80kA peak discharge current at 13.8kV charging voltage).

5.3.3 Determination of Optimum D₂ Filling Gas Pressure for FMPF-2

The optimized filling pressure for Deuterium gas was obtained through pressure variation study at ~13.8kV charging voltage. The pressure was scanned in the range of 4 to 11mbar. The most basic and preliminary investigation for optimum operation of plasma focus device is the appearance of distinct and sharp dip/ focusing peak in the current derivative signal (di/dt) of plasma focus discharge. This dip mainly indicates rapid change in plasma impedance [2, 93] on pinching. The oscillograms of typical current derivative trace observed in 'FMPF-2' device for a typical shot in Deuterium at 6mbar pressure is shown in Fig. 5.40.

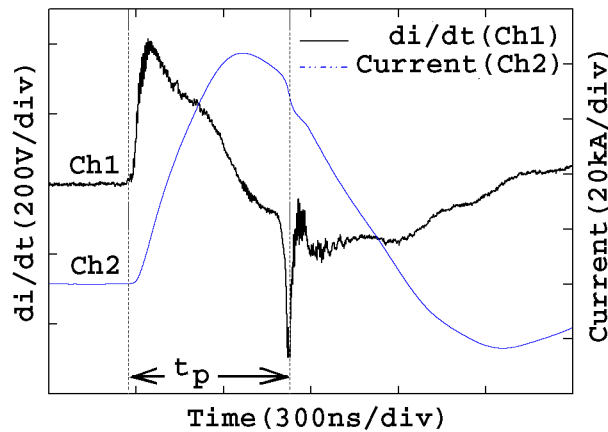


Fig. 5.40 Typical current derivative trace for a shot in Deuterium at 6mbar pressure.

In a typical plasma focus device operation, the variation in time to pinch/ focus (defined as t_p in the current derivative signal shown in the Fig. 5.40) from the breakdown phase (which includes axial acceleration phase and compression phase) increases with the increase in operating pressure due to consequently increased load on the current sheath in the axial phase. This observation is corroborated by the measurements shown in Fig. 5.41, where in for different filling gas pressures, average time to pinch is plotted along with the corresponding average neutron yields. The data were taken as averages of 20 shots, in single shot mode, for every choice of gas pressure. The gas was refreshed after every five shots during the experimental run of single shots, to reduce the effect of contamination on radiation yields.

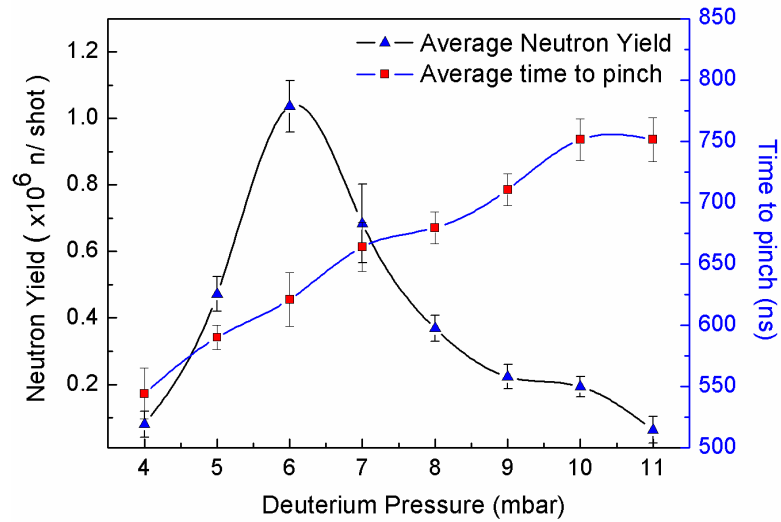


Fig. 5.41 Average neutron yield and time to pinch/ focus at different D_2 pressures.

It may be noted, at 6mbar D_2 filling gas pressure, at the ‘time to pinch’ of ~620ns, the neutron yield maximizes, as the radial collapse coincides with peak discharge current. The measured average neutron yield (Y_{90°) was $(1 \pm 0.3) \times 10^6$ neutrons/shot. Hence, 6mbar was chosen as optimum D_2 filling gas pressure for used electrode dimensions and operating voltage, in the investigation of subsequent studies.

5.3.4 Time Resolved Measurements

For measuring anisotropy (Y_{0°/Y_{90°) in the neutron and hard X-ray fluence from FMPF-2 device, the time resolved investigation on hard X-ray and neutron emission was also performed using two identical high gain scintillator photo-multiplier detectors (described under section 4.3.1), designated as PMT1 and PMT2. The detectors were arranged as shown and illustrated in Fig. 5.39.

The typical signals (Ch2 and Ch3) from the two scintillator-photomultiplier detectors (PMT1 and PMT2 respectively) and the corresponding current derivative signal (Ch1) measured by the Rogowski coil, for a typical shot at 6mbar D₂ filling gas pressure are shown in Fig. 5.42.

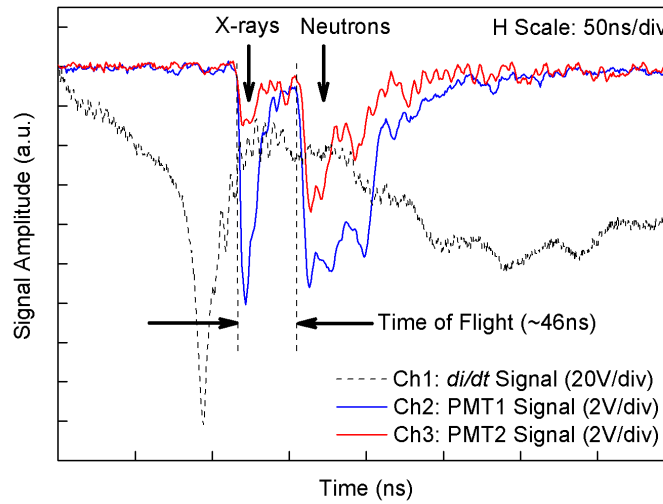


Fig. 5.42 Current derivative (di/dt) and PMT signals at 6mbar D₂ pressure.

In the PMT signals (Ch2 and Ch3), the first peak is of non-thermal hard X-rays (HXR) produced by the interaction of instability generated energetic electrons with the stainless steel anode, whereas the second peak was confirmed to be due to neutrons on the basis of time of flight separation. Since the scintillator photomultiplier detectors were placed at the distance of 1m from the plasma focus, the observed time difference of ~46ns between the two peaks corresponds to the difference in time of flight (TOF) between energetic X-ray photons and 2.45MeV neutrons. As has been mentioned

before, there is an inherent delay of ~30ns in the photomultiplier tube signal due to which the HXR peak is delayed with respect to the steep minimum of the current derivative signal (see Fig. 5.42).

It may be observed from the sample time resolved traces (refer to Fig. 5.42) that the amplitudes and the area under the signals on Ch2 and Ch3 are different hence it may be presumed that the fluence of neutrons and hard X-rays in the axial and radial direction might also be different. The pulse duration and intensity of the hard X-ray and neutron pulse are relatively higher in the end-on (i.e. axial) direction as compared to side-on (i.e. radial) direction. The durations of HXR/ neutron pulses in the axial and radial direction, estimated from full width at half maximum (FWHM) of the corresponding peaks and averaged over 20 shots, at the optimum filling gas pressure of 6mbar were observed as $15\pm 2\text{ns}$ / $45\pm 5\text{ns}$ in axial direction and $12\pm 2\text{ns}$ / $37\pm 6\text{ns}$ in the radial direction.

The fluence anisotropy Y_{0°/Y_{90° (i.e. hard X-ray and neutron flux ratios in the axial and radial direction) has been estimated by measuring the corresponding HXR/neutron yields by integrating the area under the respective peaks using Yokogawa software, XViewer®. The series of measurements at 6mbar optimum D₂ filling gas pressure gave fluence anisotropy of 3.2 ± 0.6 for hard X-rays and 1.6 ± 0.3 for neutrons. The fluence anisotropy results obtained (for HXR and neutrons) in FMPF-2 device are consistent with characterization results obtained for FMPF-1 device and discussions done under section 5.2.6 and support the claim that dominant mechanism of neutron production in sub-kilojoule range plasma focus devices (both FMPF-1 and FMPF-2) at neutron optimized pressures is beam target [215].

5.3.5 Measurement of Time Averaged Neutron Yield in Repetitive Mode

The time averaged neutron fluence Y_{av} (in neutron/ second) at various repetition rates for the burst length of 50 and 30 consecutive shots at different repetition rates have been estimated using the relation [216] –

$$Y_{av} = \frac{\sum Y_n \times DRR}{n} \quad (5.7)$$

Here $\sum Y_n$ is the sum of total neutron yield for the burst duration, DRR is the discharge repetition frequency and n is the total number of shots in the burst.

During repetitive mode of operation, the neutron output corresponding to each discharge has been measured by acquiring ^3He proportional counter signals in a digital storage oscilloscope DL 9140 in the history mode. The sample oscillograms of captured ^3He detector signals from a sequence of shots at 6Hz operation are shown in Fig. 5.43. The correlation between the fixed timings and the incremented shot numbers indicated in the history mode captured real time signals confirms the neutron production at mentioned discharge repetition frequency (*i.e.* 6Hz for the oscillograms shown below) from repetitive operation of FMPF-2 device.

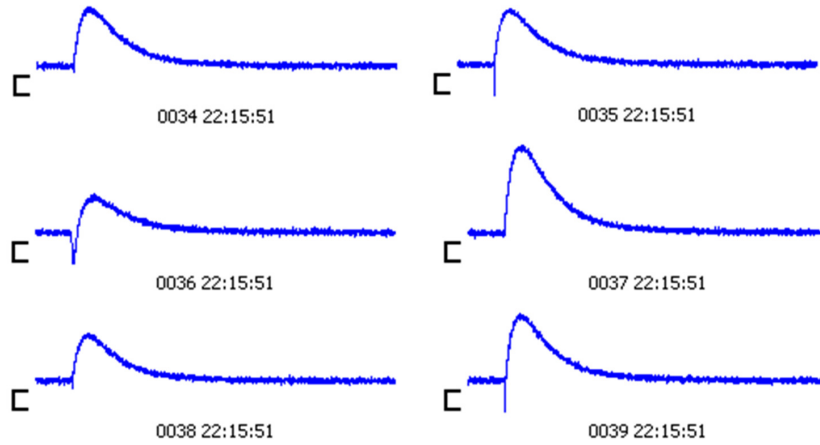


Fig. 5.43 History mode captured ^3He detector signals while operating at 6Hz.

The results obtained for time averaged neutron fluence at various repetition rates for the burst length of 30 and 50 shots are shown in Fig. 5.44. The results shown, imply the scaling in time averaged neutron fluence with discharge repetition frequency. It may be noted that with pure Deuterium as the fueling gas at 6mbar filling gas pressure, the time averaged neutron output of $(6.2 \pm 4) \times 10^5$ neutrons/sec (n/sec) at 1Hz operation was enhanced to $(6.5 \pm 0.6) \times 10^6$ n/sec at 10Hz operation for the burst length of

first 30 consecutive shots. For the burst length of the 50 consecutive shots, the time averaged neutron output of $(4.3 \pm 4) \times 10^5$ n/sec at 1Hz operation was enhanced to $(4.4 \pm 0.6) \times 10^6$ n/sec at 10Hz operation.

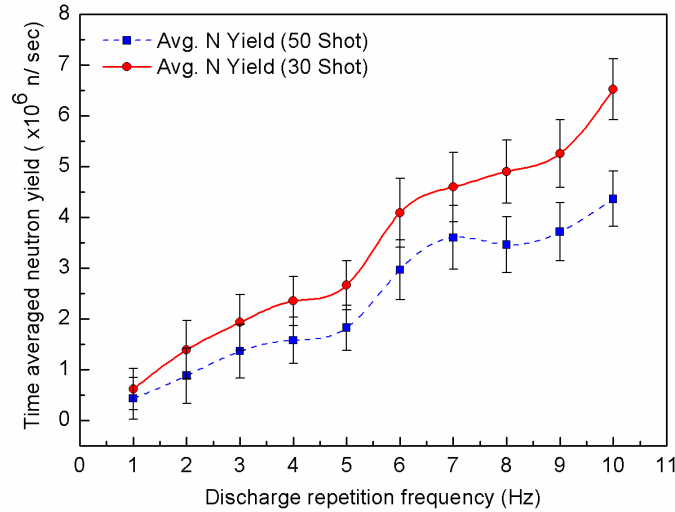


Fig. 5.44 Time averaged neutron output at different discharge repetition rates for 6mbar D₂ filling gas pressure.

The relative lowering of the time averaged output for 50 shots corroborates the results of shot-to-shot variation of neutron output (this has been analyzed in the followed sub-section, refer Fig. 5.45), wherein a descending trend in neutron output was observed typically after ~30 shots at various discharge repetition rates.

5.3.6 Shot-to-Shot Stability of Neutron Emission

The shot-to-shot stability of neutron emission during operation in burst mode, for a range of repetition rates from 1 to 10Hz has been investigated. The shot-to-shot variations in neutron yield for burst length of 50 consecutive shots at different discharge repetition rates, without purging of deuterium gas in between, is collectively shown in Fig. 5.45.

The stability in neutron yield at various discharge repetition rates for the burst length of 50 shots indicate that after few tens of shots, as the shot number increases, the neutron yield decreases

successively. For the present experimental run shown, typically after 30 shots, descending trend in neutron yield at all discharge repetition frequencies was observed to be more significant.

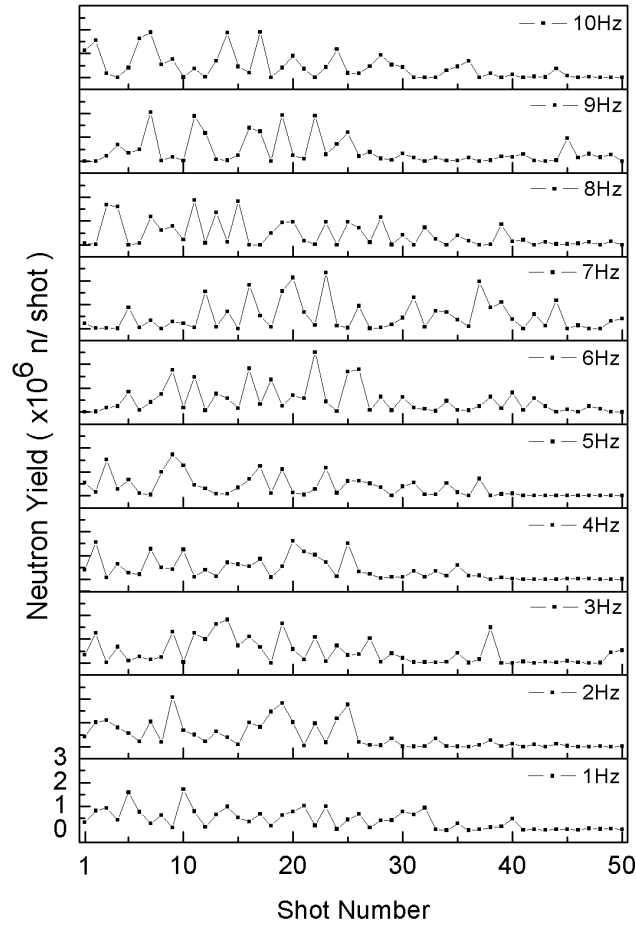


Fig. 5.45 Neutron yield per shot at different repetition rates for 6mbar D_2 pressure.

The probable causes of concerns those might be accounted for successive degradation in neutron yield with increasing number of shots and DRR are explored and analyzed in detail in the next sub-section.

5.3.7 Physics of Repetition Rate Affect on Neutron Yield

The observed degradation in neutron output has been characterized with time to pinch/focus t_p for successive shots during operation in burst mode at various discharge repetition rates and are collectively shown in Fig. 5.46.

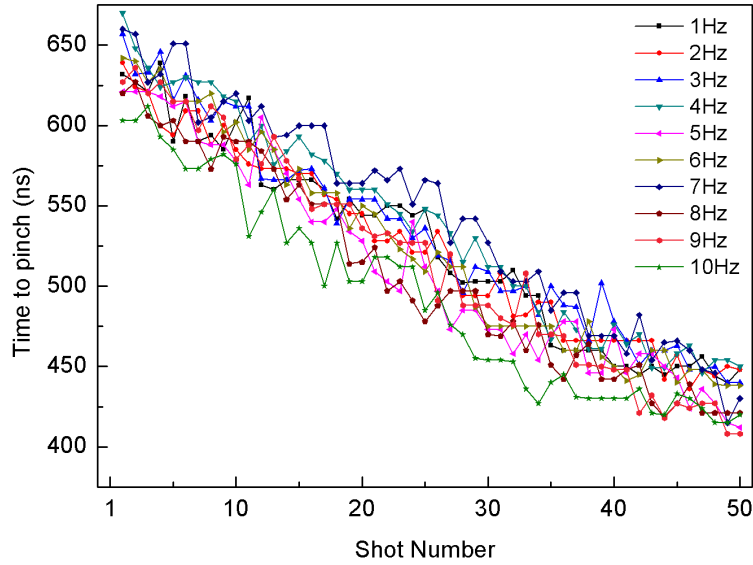


Fig. 5.46 Time to pinch at different repetition rates for 6mbar D₂ filling gas pressure.

The obtained trends reveal that time to pinch/focus is significantly affected by the (i) progressively increasing number of discharges in each of the burst sequence, and (ii) discharge repetition frequency. The measured 50 shot average of time to pinch/focus descended from $\sim 525 \pm 9$ ns at 1Hz to 493 ± 9 ns at 10Hz discharge repetition rate.

It is a well known fact that for optimum neutron output and maximum energy transfer to the pinch, the first radial collapse should occur close to the quarter time period of the signal *i.e.* when the discharge current is near its peak value [121]. Hence, the power transferred to the pinch is significantly reduced when pinching/focusing occurs before the zero-crossing of the dI/dt signal. The evidenced trend of progressively reducing time to pinch/focus during repetitive mode of operation may be held primarily responsible for the corresponding degradation in neutron output from successive shots.

From the consideration of gross dynamics of hot and dense plasma production, the three major distinct phases of typical plasma focus device operation are: initial breakdown phase, axial run down phase and radial collapse phase. The behavioral characteristics of each of these stages strongly influence final compressed plasma and its emissions. While the time durations of the axial run down

and radial compression phase mainly depends on physical dimensions of the anode, mass density of used gas and rate of rise of discharge current dI/dt [241]; the time duration of initial breakdown phase depends on many complex physical processes/factors responsible for the breakdown [63].

In the context of low energy miniature plasma focus device operation, it is important to note that (i) the upper limit for time to pinch/focus is inherently limited in the sub-microsecond regime ($T/4 \sim 0.6\mu s$), therefore there is a relatively short time available for the mentioned distinct phases to occur, and (ii) due to smaller dimensions (length and radius) of anode; even minor alteration in the corresponding durations of the operation phases evidently reflects in the time to pinch/focus, consequently affecting the energy transferred to the pinch. In our experimental runs, while other parameters and operating conditions remain constant, the cause for lowering in time to pinch/focus during repetitive mode of operation may be attributed mainly to significant minimization in the time delay associated with the initial breakdown phase [242].

The initial phase of the discharge mainly comprises of gas breakdown and plasma sheath formation. When the energy stored in the capacitor bank is discharged into the coaxial electrodes, the working gas in the chamber is ionized and the skinning of the current near the insulator leads to the formation of current sheath and its subsequent filamentations [243]. According to the detailed study performed by Kies [61] the deduced expression for sheath formation time t_f is given as – $\sqrt{2\pi r_s l_s d_s w_s L / \eta V_o^2}$. Here r_s and l_s are the radius and length of the insulator sleeve, d_s and w_s are the thickness and energy density of the sheath, V_o and L are the operating voltage and total inductance of plasma focus and η is the efficiency of the sheath formation process. From this expression, fast sheath formation for a given value of energy density (w_s) requires high voltage (V_o), small sheath volume (r_s , l_s and d_s) and high efficiency of the sheath formation process (η). The efficiency of the sheath formation process (η) mainly depends upon ionization by electron avalanche phenomenon that relies on initial availability of free electrons (which are accelerated upon application of high voltage pulse

between coaxial electrodes resulting in avalanche). Therefore, the increased number of free electrons (i.e. pre-ionization) plays a major role in improving the initial breakdown conditions and reducing the sheath formation time [61].

In typical plasma focus device operation, there are always few residual thermal electrons after each discharge and usually these are supportive in the formation of azimuthally uniform current sheath in the following focus discharge.

On the basis of the observation of the reduction in the time to pinch with increasing rep-rate, as seen in Fig. 5.46 we speculate that there might be a dramatic increase in residual electrons as the rep-rate is increased. If true, this would suggest better sheath breakdown and hence a tighter pinch with higher neutron output, as rep-rate is increased. Quantification of this fact requires solution of the conservation equations to track the evolution of free electron density and temperature after a given shot, and to see whether the residual density indeed exhibits a strong dependence upon rep-rate. Such a theoretical model is beyond the scope of this thesis, but reasonable estimates may be made of the relevant physics. Assuming that immediately after a shot, there is a dense hot plasma in front of the anode and some electron sheath along its surface; now since the pinch density is $\sim 10^{19} \text{ cm}^{-3}$, as the pinch disrupts and expands, the electron density will plummet to the local fill gas density, $\sim 10^{17} \text{ cm}^{-3}$, as the plasma radius expands from $\sim 1 \text{ mm}$ to $\sim 1 \text{ cm}$. The temperature will also fall, as the plasma cools by radiative emission and heat transfer to the cold gas envelope. At typical post-pinch conditions of $\sim 10^{17} \text{ cm}^{-3}$ and $< 100 \text{ eV}$, such a plasma remains in coronal equilibrium and the free electron density will stay at $\sim 10^{17} \text{ cm}^{-3}$, until the temperature has fallen to $< 2 \text{ eV}$ or so. Thus the plasma will not recombine rapidly as it might in the case of a denser colder medium where three-body recombination dominates [244]. A second mechanism of electron loss is by diffusion [245]. Assuming that the metal walls of the outer cathode are perfect absorbers of electrons (after the pinch when there is no discharge), for a cylindrical geometry, the residual density after time t is given as [245] –

$$n_{\text{residual}} = n_o \exp(-t/\tau) \times J_o \left[\frac{\gamma}{\sqrt{D_a \tau}} \right] \quad (5.8)$$

Here n_o is the initial electron density, τ the decay time constant, D_a is the ambipolar diffusion coefficient, γ the wall radius. J_o is the zero order Bessel function. The ambipolar diffusion coefficient D_a is given as [245] –

$$D_a = \frac{\mu_i D_e + \mu_e D_i}{\mu_i + \mu_e} \quad (5.9)$$

Here D_i and D_e are diffusion coefficients for positive and negative ions where as μ_i and μ_e are corresponding mobilities. Substituting reasonable values for the various parameters and using a wall radius of 1.5cm, the Eq. 5.8 simplifies to - $n_{\text{residual}} = n_o \exp(-t/3 \times 10^{-2}) \times 0.78$. The plot of estimated residual density versus rep-rate using the above mentioned relation is shown in Fig. 5.47.

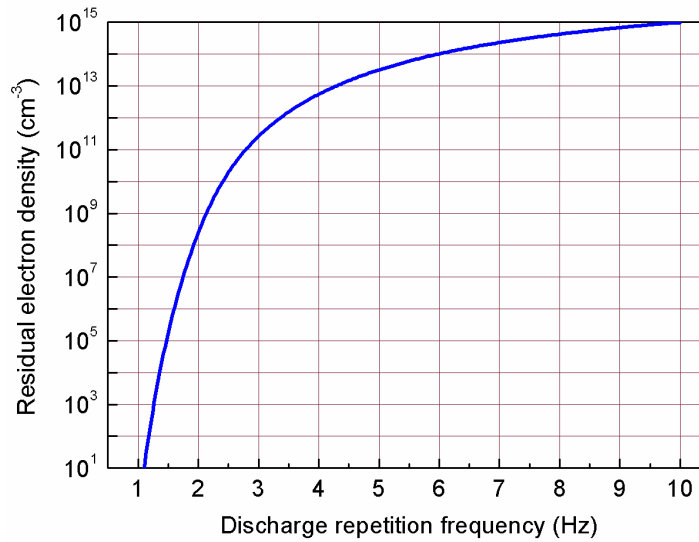


Fig. 5.47 Estimated residual density versus rep-rate (diffusion loss).

In this plot, it may be observed that there is a dramatic change in residual density at rep-rates above 2Hz. Below 1Hz, the density is so low that avalanche breakdown requires some source of seed

ionization such as streamers or cosmic ray induced ionization. Beyond 2Hz, the residual density is so high that avalanche breakdown is no longer as random and should proceed rapidly, with a consequent reduction in the delay for sheath formation. These simple estimates provide a plausible argument for why we have observed such a dramatic effect of rep-rate on neutron output. A more elaborate model would be required to put this on a firmer footing, as well as more detailed measurements.

The probability of contamination in pure Deuterium plasma by anode metal vaporization due to joule heating and electron bombardment and also by the contact of anode surface with hot plasma cannot be ruled out. The condition of the stainless steel (SS) anode after 5000 shots of operation in repetitive mode has been shown in Fig. 5.48.

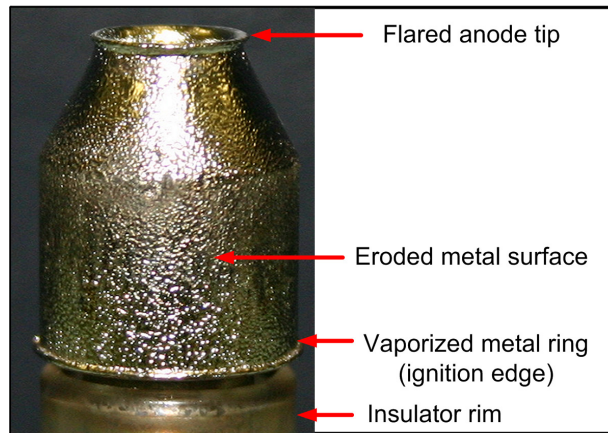


Fig. 5.48 Physical condition of SS anode after 5000 shots of operation in repetition mode.

The visual inspection of the physical condition of anode (post experimental run) shows the formation of a vaporized metal ring at the ignition edge near the top edge of insulator sleeve and flaring of the hollow anode top. It was also noticed that the smooth anode surface turned matt due to vaporization/erosion during successive discharges.

Electrode vaporization at the ignition edge near the insulator sleeve tip occurs during the initial breakdown phase before the well defined current sheath is formed and starts to move axially upwards along the anode (*i.e.* between the sheath formation time and lift off stage). The vaporized metal ring

formation is mainly due to excessive current densities at the ignition edge during the initial breakdown (sheath formation) phase.

The cause for the flaring of the tip may mainly be ascribed to the bombardment of the anode tip by instability accelerated electron beams, physical contact of the tip with hot plasma and joule heating of anode material. An estimate of the current density at which vaporization occurs due to electrode joule heating (J) is given as [246] $-\sqrt{(E_v \times \rho)/(\eta_{res} \times \Delta t)}$. Here E_v , ρ and η_{res} are the heat of vaporization, mass density and resistivity, respectively, of the corresponding anode material and Δt is the pulse duration. The anode materials with higher heat of vaporization E_v (for *e.g.* elkonite i.e. copper-tungsten alloy) and larger volume (so that they have more mass to handle higher thermal load during operation) may mitigate the problem of contamination of the sheath due to metal vaporization up to a certain extent. However, there is a constraint on larger anode dimensions in order to match the sub-microsecond range electrical discharge characteristics in MPF devices.

The contaminants injected by vaporization/erosion of anode, for repetitive focus operation, ultimately results in more rapid radiative cooling by addition of high-Z impurities into pure Deuterium plasma (not only for the same focus discharge but also for the following focus discharge as the residuals of vaporized high-Z impurities may still be around particularly when the focus is fired in the repetitive mode) and hence may lead to the subsequent reduction in neutron yield for repetitive operations, as observed in Fig. 5.45. The radiative cooling might have occurred at both the places i.e. at the ignition edge near the insulator sleeve rim and at the anode tip as material has been seen to vaporize more effectively at these two places. The radiated thermal Bremsstrahlung power density P_B is given as $-5.4 \times 10^{-37} Z^2 n_e^2 T_e^{1/2} \text{ W / m}^3$. Here T_e is in keV, n_e is in m^{-3} , and Z is the atomic number of the fully ionized species. Because $P_B \sim Z^2$, addition of small amounts of high-Z impurities results in (a) substantial drop in temperature due to increased radiation loss [231], and (b) reduction in snowplow efficiency due to thickening of current sheath near the ignition edge by radiative cooling [246].

The mentioned causes are consistent with the reduction in neutron output along with successively increasing frequency of focus shots (as observed in Fig. 5.45) during repetitive mode of operation.

In conclusion, using the repetitive miniature plasma focus device FMPF-2, the enhancement in time averaged neutron output has been successfully demonstrated realizing the long stated concept of ‘enhancement by repetition’ and also the limitations of this approach due to relatively short quarter time period. It was observed during the experiments that under static filling gas conditions, with the progressively increasing number of shots in a burst, there is considerable degradation in the neutron output of successive shots as the time to pinch is continually reduced. This effect worsens at higher discharge repetition rates. However, the optimum neutron output performance of the device relies on evacuation and refueling with pure Deuterium gas. The present device configuration imposes an optimum burst length limitation of 30 shots/burst for discharge repetition rate ranging from 1-10Hz. Hence, in the experiments, it has been conclusively realized that there is an enhancement in overall time averaged neutron output as a function of discharge repetition frequency but the burst lengths (i.e. number of successive discharges in each burst) at various repetition rates is a matter of critical concern in context of (i) enrichment of free electrons (i.e. pre-ionization) which modify the initial breakdown conditions and consequently the time to pinch, and (ii) enhanced addition of high-Z impurities due to contaminants injected by vaporization/erosion of anode material at ignition edge at insulator sleeve rim and at anode top; and their after effects on neutron output.

5.4 Results of FMPF-3 Device

After successfully demonstrating the enhancement in time averaged neutron yield by operating the FMPF-2 device in repetitive mode up to 10Hz [216]; the technical constitution of FMPF-2 device was further refined to enhance the electrical efficiency of the pulsed power system. Hence, 4-module

FMPF-3 device is basically an enhanced version of single-module FMPF-2 device with further upgraded pulsed power system and higher electrical efficiency (i.e. it delivers higher peak discharge current for per joule of stored energy in comparison to two previous versions of FMPF devices). As illustrated under section 3.6, to curtail the inductance contribution due to extended connections of all the eight capacitors of single-module of FMPF-2 device through a common Trigatron sparkgap switch to the PF load; in FMPF-3 device each of four modules with two capacitors each is equipped with a separate Pseudospark switch that closely terminates to a common collector plate assembly.

In this section we report the experiments that have been performed for the complete characterization of FMPF-3 device as the most optimized version of indigenously developed repetitive pulsed neutron source. This mainly include: (i) determination of modified electrical and discharge characteristics, (ii) optimization of anode dimensions and scanning of neutron optimized pressure, (iii) time resolved neutron and hard X-ray measurements, (iv) confirmatory total neutron yield and anisotropy measurements using SSNTD, (v) investigation on neutron yield scaling with peak and pinch current, (vi) measurement of enhancement in time averaged neutron fluence using repetitive mode of operation, and (vii) investigations of shot-to-shot stability of neutron yield during a repetitive burst mode. In the concluding section, experimental results of remedial measures implemented for improvising the optimum neutron burst length of FMPF-3 device have been discussed in detail.

5.4.1 Upgradation in Electrical Characteristics

The foremost investigation performed on FMPF-3 device was to study the modified short circuit discharge characteristics to analyse the improvisation in the electrical efficiency with the implementation of further upgraded pulsed power system. The short circuit discharge was performed under high pressure Argon gas filling at various charging voltages ranging from 10 – 14kV. The sample oscillogram of short circuit discharge current trace at 14kV operation with 40mbar Argon filling is shown in Fig. 5.49.

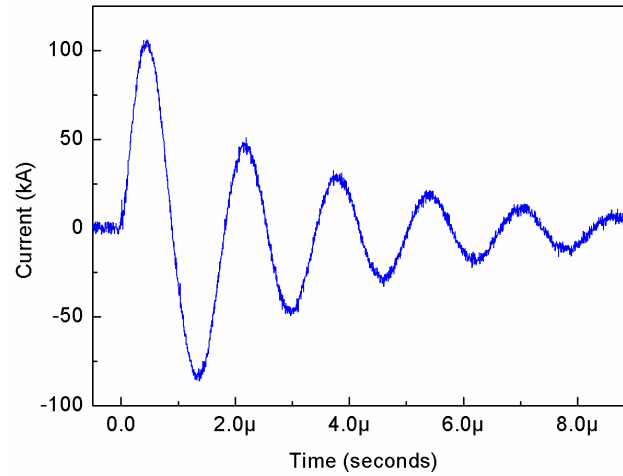


Fig. 5.49 Sample oscillogram of short circuit discharge in FMPF-3 device at 14kV.

The lowering of the system inductance is clearly evident from the observed lowering of time period in obtained short circuit discharge current trace. The quarter time period of the discharge current signal reduced from ~575ns recorded for FMPF-2 device to ~458ns in FMPF-3 device. Following the methodology illustrated under section 5.2.1, the obtained main electrical parameters of FMPF-3 device through short circuit tests are summarized in Table 5.4.

Table. 5.4 Main electrical characteristics of FMPF-3 device.	
Energy bank capacitance (C_o)	2.4 μ F
Maximum charging voltage (V_o)	14kV
Maximum stored energy @14kV (E_o)	235J
Maximum current (under short circuit) (I_{sc})	103kA @14kV
Typical operating voltage range	10 – 14 kV
Equivalent circuit inductance (L_{eq})	34 $\pm$ 2nH
Equivalent circuit resistance (R_{eq})	23 $\pm$ 3m Ω
Voltage Reversal (f)	78%
Quarter time of discharge ($T/4$)	~458ns
Maximum discharge repetition rate (DRR)	10Hz

The results obtained from short circuit discharge tests confirmed the significantly improved electrical efficiency of FMPF-3 device over FMPF-2. For comparison, the investigated peak discharge

currents at different charging voltages/ energies (under short circuit condition by gas filling at high pressures) for all the three different versions of indigenously developed fast miniature plasma focus devices have been collectively summarized and represented in Fig. 5.50.

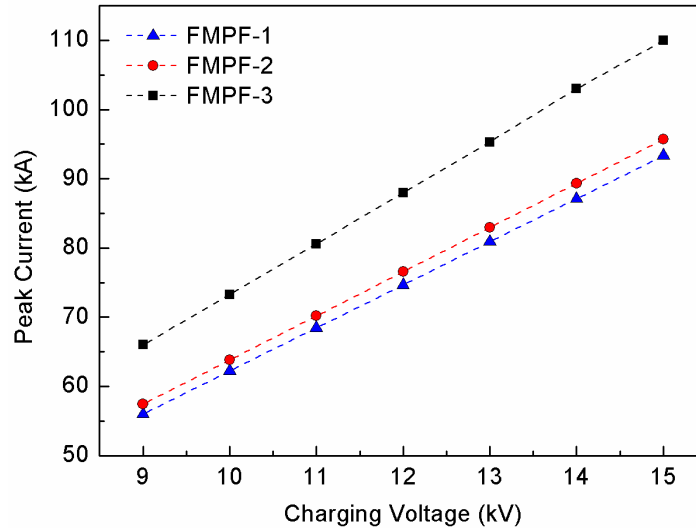


Fig. 5.50 Maximum peak currents (under short circuit test conditions) in FMPF devices at different charging voltages.

Above shown results conclusively indicate that, while operating at similar energies/ charging voltages, the peak discharge current delivered by FMPF-3 device is ~18-20% higher in comparison to the two other versions of FMPF devices.

5.4.2 Experimental Setup and Diagnostics Arrangement

The experimental set-up and diagnostics arrangement for the FMPF-3 device, remains the same as illustrated in section 5.3.2 (refer Fig. 5.39), except for two changes: (i) since in FMPF-3 device higher current is pumped in shorter quarter time period; therefore to match the modified electrical characteristic of the device, experiments were conducted to optimize the new anode design (the details regarding design, dimensions and neutron yield performance of this cylindrical anode are presented

and compared with existing composite anode design in the following subsection), and (ii) as the radiation yields in technologically superior FMPF-3 are expected to be higher so to avoid saturation in the time resolved signals of hard X-rays and neutrons, distances of the scintillator photomultiplier detectors (designated as PMT1 and PMT2 in the diagnostic setup) from anode top were marginally increased from 1m to 1.2m.

5.4.3 Anode Design and Neutron Yield Optimization

For the results shown and discussed in this subsection, to investigate the neutron yield of FMPF-3 device with different anode designs for different Deuterium fill pressures; the stored energy of system was maintained constant at ~200J (to deliver about ~90kA peak discharge current at ~13kV charging voltage). Also, each of the data points shown in the relevant graphs is an average of 20 shots for every choice of D₂ gas pressure.

Since the electrical characteristics of FMPF-3 device were considerably modified in terms of peak discharge current and quarter time period therefore to evaluate the compatibility of existing composite anode design (which was giving optimum neutron yield for FMPF-2 and is shown in subsection 5.2.6.1, Fig. 5.28(a)) with modified electrical discharge parameters, the neutron yield performance was investigated through filling gas pressure variation study. The filling gas pressure of D₂ was scanned in 5 to 9mbar range. The measured average neutron yield and time to pinch/ focus at different D₂ gas pressures are shown in Fig. 5.51 for the composite anode of FMPF-2.

It may be noted, from Fig. 5.51 that the neutron yield maximizes at ~6.5mbar D₂ filling gas pressure with the average maximum neutron yield (Y_{90°) of about $(1.1 \pm 0.1) \times 10^6$ neutrons/shot. Comparing these neutron yields with the one shown in Fig. 5.41 for the same composite anode for FMPF-2 device, we found that, both, neutron yield and its variation trend with filling gas pressure are nearly similar with an exception that optimized pressure shifted to higher value of 6.5mbar for FMPF-3

against 6.0 mbar for FMPF-2. Comparing Fig. 5.51 with 5.41, one can also notice the decrease in time to pinch, for each filling gas pressure, which explains shift in optimum base pressure. It may be noted that as the peak discharge current is increased from ~80kA in FMPF-2 to ~90kA in FMPF-2; so assuming neutron yield to scale as I^4 , the neutron yield was expected to increase from $(1 \pm 0.3) \times 10^6$ neutrons/shot for FMPF-2 to about 1.6×10^6 neutrons/shot for FMPF-3, which was not observed in Fig. 5.51. Another point to note is that the neutron yield using composite anode for both FMPF-2 and FMPF-3 lies in narrow pressure range with the sharp fall in the yield, on both side, about the maximum yield point. We speculate that such response with composite anode design is mainly due to large linear current density (I/a where I is the peak discharge current and a is the anode radius).

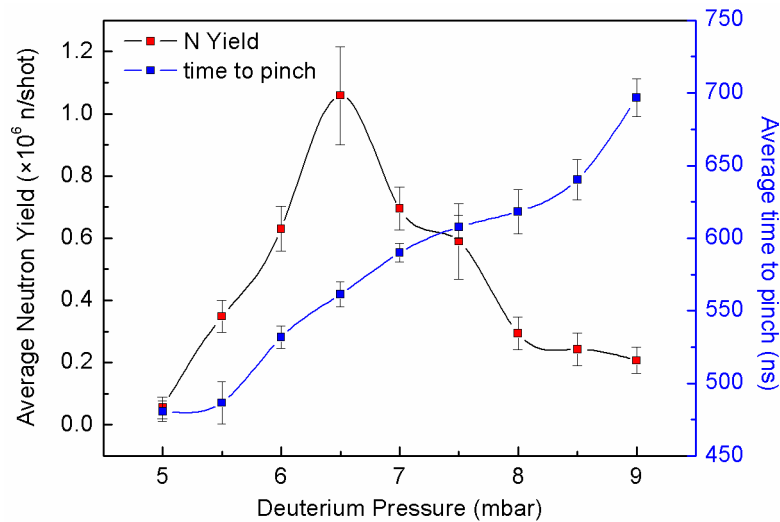


Fig. 5.51 Average neutron yield and time to pinch/ focus for FMPF-3 at different D_2 gas filling pressures for composite anode that provided optimum yield for FMPF-2.

In the modified anode design, two changes were incorporated (i) for broadening the operating pressure range of optimum focusing, the length of the anode was increased from 15mm to 17mm, and (ii) to moderate the large linear current density (I/a); tapering was eliminated from entire anode length by maintaining the anode radius of 6mm throughout. For investigating the neutron yield performance with cylindrical anode design, the D_2 filling gas pressure was scanned in the range of 2 to 7.5mbar.

The measured average neutron yield and time to pinch/ focus at different D₂ filling gas pressures is shown in Fig. 5.52.

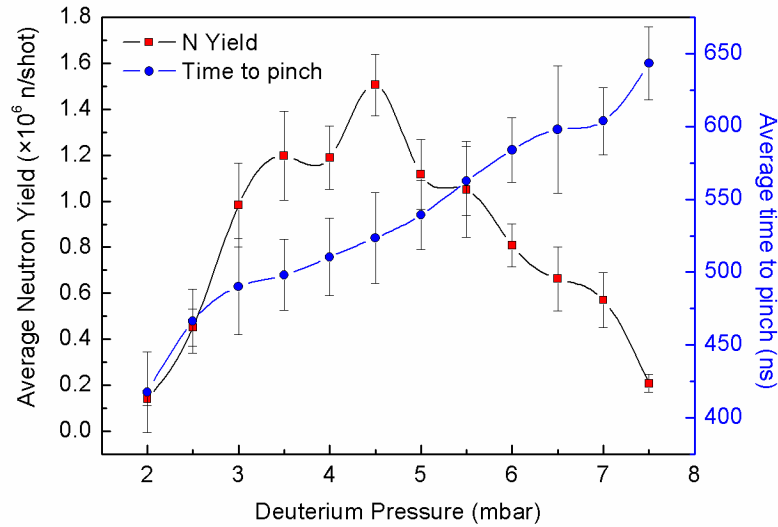


Fig. 5.52 Average neutron yield and time to pinch/ focus at different D₂ gas filling pressures for FMPF-3 device with cylindrical anode.

From the results obtained, the improvisation in the matching of electrical discharge characteristics of FMPF-3 device with cylindrical anode design is clearly evident. The neutron yield maximized at 4.5mbar D₂ filling gas pressure producing average neutron yield of $(1.5 \pm 0.2) \times 10^6$ neutrons/shot. The most important noticeable change is the consistent average neutron yield of the order of $\geq 10^6$ neutrons/shot in the broadened operating pressure range of 3.5 to 5.5mbar as the time to pinch in this range remains close to the quarter time period of capacitor bank that is required for the efficient pinching. In the forthcoming subsections, all results and discussions are done only in the context of cylindrical anode design.

5.4.4 Time Resolved Neutron and HXR Measurements from FMPF-3

The time resolved measurements of neutron and hard X-ray radiation from FMPF-3 device were done in the similar fashion as illustrated under 5.3.4, however the distances of the scintillator

photomultiplier detectors (PMT1 and PMT2) from anode top were kept at 1.2m instead of 1.0m). The typical signals (Ch2 and Ch3) from the two scintillator-photomultiplier detectors (PMT1 and PMT2) and the corresponding current derivative signal (Ch1) measured by the Rogowski coil, for a typical shot at 13kV/ 4.5mbar D₂ filling gas pressure are shown in Fig. 5.53.

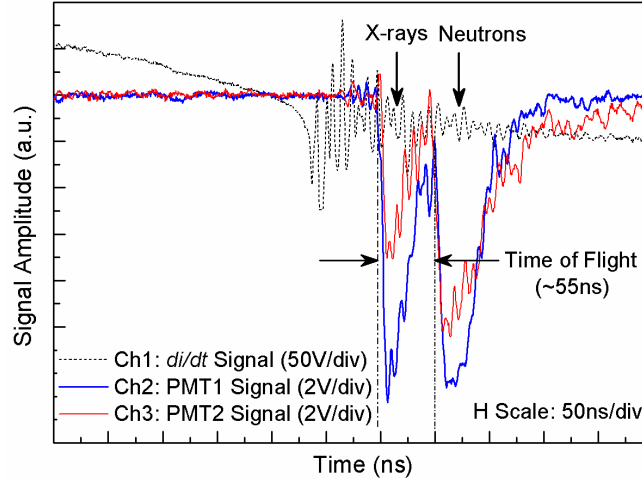


Fig. 5.53 Current derivative (di/dt) and scintillator photomultiplier (PMT) detector signals at 4.5mbar D₂ filling gas pressure.

The two peaks observed in the PMT signals (Ch2 and Ch3) were confirmed to be of hard X-rays and neutrons on the basis of time of flight separation. Since the scintillator photomultiplier detectors were placed at the distance of 1.2m from the plasma focus, the observed time difference of ~55ns between the two peaks is in confirmation with the expected difference in time of flight (TOF) for X-ray and 2.45MeV neutrons.

In the sample time resolved traces (shown in Fig. 5.53) it may be noted that the pulse duration and intensity of the hard X-rays and neutrons pulse are relatively higher in the end-on (i.e. axial) direction as compared to side-on (i.e. radial) direction. The durations of HXR/ neutron pulses in the axial and radial direction, estimated from full width at half maximum (FWHM) of the corresponding

peaks and averaged over 20 shots, at the optimum D₂ filling gas pressure of 4.5mbar were observed as 31 ± 3 ns/ 50 ± 4 ns in axial direction and 16 ± 2 ns/ 33 ± 3 ns in the radial direction.

The average fluence anisotropy Y_{0°/Y_{90° (i.e. hard X-ray and neutron flux ratios in the axial and radial direction); estimated by measuring the corresponding HXR/neutron yields by integrating the area under the respective peaks (using Yokogawa software, XViewer®) is 1.8 ± 0.3 for hard X-rays and 1.4 ± 0.2 for neutrons at the optimum D₂ filling gas pressure of 4.5mbar.

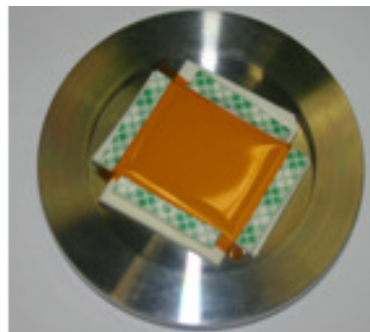
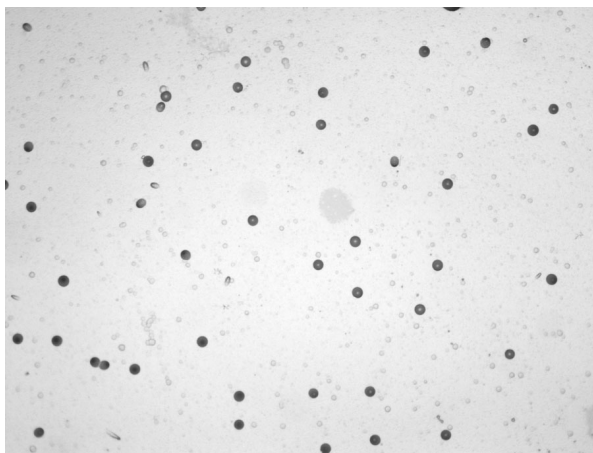
5.4.5 Total Neutron Yield and Anisotropy Measurement using SSNTD

Detailed description of the Solid State Nuclear Track Detector (CR-39/ PM355) has been provided in section 4.4.2. In our experiment, 1mm thick CR-39 plastic track detectors of size 2cm×2cm have been used. The detectors covered with 75μm thick Kapton tape were placed on circular KF40 vacuum flanges in the side-on and end-on directions at the distance of 13cms respectively from the anode top. As summarized in Table 4.2, calculations done using the SRIM code [247] concluded that 75μm thick Kapton window will completely stop all ³He (~0.8MeV) and ³H (~1MeV) ions along with protons and deuterons having energies below 2.3MeV and 3MeV respectively; but will transmit the 3.03MeV fusion Protons with an energy loss of ~1.4MeV.

In the experiment, SSNTD detector samples were exposed to 10 shots at 4.5mbar pure D₂ filling gas pressure. After irradiation, we etched the detector samples in a stirred solution of 6.25M NaOH solution at controlled temperature of 70 ± 1 °C for 9 hours.

The imaging and counting of energetic proton recoil tracks on CR-39 detector sample was done in auto-scan mode by a fully automated optical microscope (Olympus BX51) interfaced with computer. This microscope is also facilitated with auto-focus option. There are four objective lenses with different magnification (×5, ×10, ×20 and ×50) for obtaining different sizes of field of view. In our case magnification of ×20 was used for scanning central area of ~1cm² of the detectors, producing

~500 frames (size of each frame $\approx 528\mu\text{m} \times 398\mu\text{m}$). Fig. 5.54 shows a sample of single frame microphotograph of the etched pits formed on SSNTD detector that was placed in the end-on direction. Snap of CR-39 detector with filter holding arrangement is also shown along the side.



SSNTD CR-39 detector covered with $75\mu\text{m}$ thick Kapton filter

Fig. 5.54 Single frame microphotograph of Proton recoil tracks registered on CR-39 detector (field of view $\approx 528\mu\text{m} \times 398\mu\text{m}$).

Since there is variation in area, roundness and mean density (MD) of etched pits registered on CR-39 detector corresponding to variations in energy of charged particle, angle of incidence, etching time and overlapping; hence for selecting and counting the actual/ real number of proton recoil tracks over scanned area, the frames are analyzed by 'Image-Pro' software along with the implementation of certain boundary conditions (to segregate real proton tracks from other tracks by defining boundaries for parameters like area, brightness and roundness) [122,134,248]. For this purpose Macros are applied to all frames and the relevant data points are exported to excel sheet. The boundary conditions implemented in our analysis follow: $60(\mu\text{m})^2 \leq \text{Area} \leq 90(\mu\text{m})^2$, Mean Density (relative value *w.r.t.* a maximum of 127) ≤ 110 and Roundness ≤ 1.2 . The actual number of proton recoil tracks on CR-39 detector samples, in the end-on and side-on directions, that satisfy the mentioned boundary conditions over the scanned area $\sim 1\text{cm}^2$ are shown in Fig. 5.55 (a) and Fig. 5.55 (b) respectively.

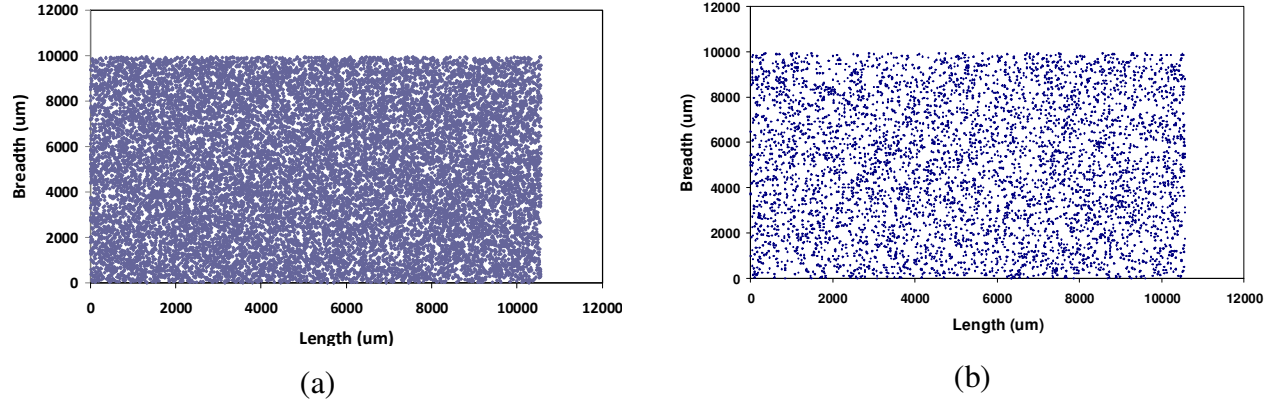


Fig. 5.55 Registered tracks on axially (a) and radially (b) positioned CR-39 detectors satisfying the implemented boundary conditions.

The anisotropy in the fluence may be observed in the shown graphs. The estimated acceptable number of tracks formed by 3.03MeV fusion protons on CR-39 detector samples, placed along the axial and radial direction at a distance of 13cms from anode top are 11275 (out of 17633) and 7760 (out of 10272) respectively for the exposure of 10 focused shots. Hence, the average neutron yield per shot in 4π sr. is estimated to be 2×10^6 neutrons/shot (n/shot) along with anisotropy (Y_{0°/Y_{90°) of ~ 1.45 .

It may be noted that the estimated average neutron yield and anisotropy in the neutron fluence is consistent with the results obtained by other techniques (such as use of ^3He proportional counter for neutron yield measurement and scintillator-photomultiplier tube based relative yield measurement technique for the estimation of anisotropy in neutron flux), as mentioned in the previous subsections 5.4.3 and 5.4.4.

5.4.6 Neutron Yield Scaling in FMPF-3 with Peak and Pinch Current

Recent analysis of experimental data by Lee and Saw [128] for plasma focus devices with a wide range of stored bank energies from 400J to 1 MJ showed the observed neutron yield Y_n scaling law with peak (I) and pinch current (I_p) as $Y_n \sim I^{3.9}$ and $Y_n \sim I_p^{4.7}$, respectively. Since our miniature plasma focus device lies down further in the low energy range of $< 240\text{J}$, hence it becomes relevant to

investigate experimentally whether neutron yield scaling in FMPF-3 device is consistent with the recently updated scaling laws [128,132].

It may be noted that the experimental measurement of pinch current is complicated; particularly in miniature devices, because of space constraint and much smaller size of the pinch (also there is a probability that the probe may inevitably interfere with imploding current sheath and affect the dynamics); therefore the pinch current estimations, mentioned in Table 5.5, corresponding to device operation at various charging voltages/ energies were done using the five-phase Lee model RADPFV5.13 [70]. Since Lee Model couples the electrical circuit with plasma focus dynamics, thermodynamics and radiation enabling realistic simulation of all gross focus properties [129]; the experiments performed were realistically simulated by ‘fitting’ the computed current trace with experimentally obtained peak discharge current trace, by adjusting the four ‘model parameters’ (axial mass swept-up factor f_m , axial current factor f_c , radial mass factor f_{mr} and radial current factor f_{cr}), as mentioned under previous sub-sections 5.2.4.3 and 5.2.5.4. Lee Model defines pinch current I_p as the instantaneous value of current that flows at the start of the quiescent (or pinch) phase of the plasma focus radial dynamics [132].

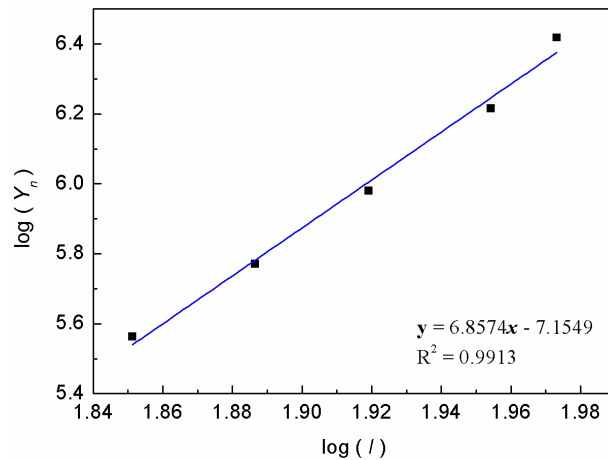
The neutron yield scaling study was performed by varying the charging voltage of FMPF-3 device in the range of 10-14kV and operating the device at their corresponding optimum pressures. The obtained average maximum neutron yield of 10 shots and the average peak discharge current at respective charging voltages/ pressures with pure D₂ filling have been summarized in Table 5.5.

The obtained neutron yield scaling results against variations in peak and pinch current are shown in Fig. 5.56 (a) and Fig. 5.56 (b). The gradients obtained for logarithmic plots in respective graphs deduce neutron yield scaling with peak (I) and pinch current (I_p) as $Y_n \sim I^{6.87}$ and $Y_n \sim I_p^{5.47}$ for the operation of FMPF-3 device in the energy range of 120J – 235J. The index obtained for neutron yield scaling against peak (I) and pinch current (I_p) as 6.85 and 5.47 respectively, are on the higher side

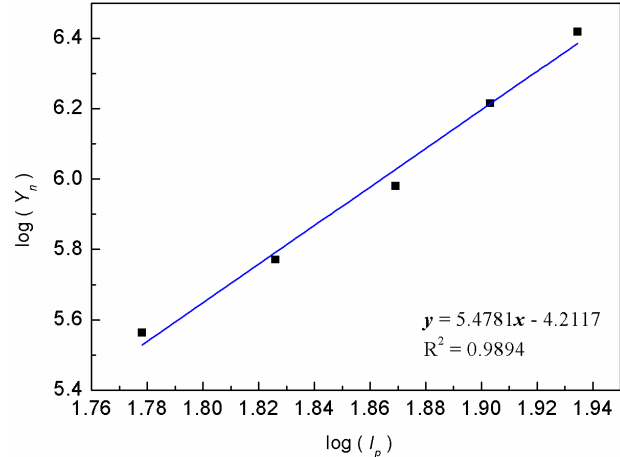
than the typically observed range [128]. This finding indicates that neutron yield scaling has higher dependence on peak/ pinch currents for the sub-kilojoule range devices typically operating with currents $<100\text{kA}$.

Table 5.5 Results obtained from neutron yield scaling experiments.

Charging Voltage (kV) / Energy (J)	Optimum Pressure	Peak Current (I in kA)	Pinch Current (I_p in kA)	Average Neutron Yield (n/shot)	S Factor kA/cm $\sqrt{\text{torr}}$
10kV/ 120J	3.5mbar	71kA	60kA	3.66×10^5	75
11kV/ 145J	3.5mbar	77kA	67kA	5.90×10^5	83
12kV/ 172J	4.0mbar	83kA	74kA	9.56×10^5	85
13kV/ 203J	4.5mbar	90kA	80kA	1.64×10^6	86
14kV/ 235J	4.5mbar	94kA	86kA	2.62×10^6	93



(a) Neutron Yield Vs. Peak Current



(b) Neutron Yield Vs. Pinch Current

Fig. 5.56 Neutron yield versus peak and pinch current in FMPF-3 device in log-log scale to determine the scaling laws for miniature plasma focus devices.

It is remarkable that in deciding the optimum device performance, role of anode dimension is specially more critical in low energy plasma focus devices (than medium/ large energy range plasma focus devices) due to the lower peak/ pinch currents and inherently short quarter discharge time

periods typically lying in the sub-microsecond regime. Hence, we also speculate that observed higher dependence, to a certain extent, may also be due to the use of common anode design and dimension at all charging voltages/ energies which had specifically been optimized for operation at ~90kA/13kV/ 200J. This is infact also corroborated by the variation in speed factor estimations (as shown in Table 5.5) obtained for operation at various charging voltage/ energies [33].

5.4.7 Measurements of Time Averaged Neutron Yield in Repetitive Mode

From the experience gained in repetitive mode of operation with FMPF-2 device, the repetitive mode operation results with FMPF-3 device were obtained for optimum and higher than optimum D₂ filling gas pressures i.e. at 4.5mbar and 5.5mbar. As discussed under section 5.3.7, we expect the mitigation of problem of successive degradation in time to pinch/focus (and consequently the neutron yield) with increasing number of shot and discharge repetition rate (DRR), during repetitive mode of operation, to a larger extent, by operating the FMPF-3 device at higher than the optimum pressure that was obtained for operation in singular mode during the pressure scan study. We speculate that by using filling gas pressure higher than singular mode optimum, as with the increasing shot number, the time to pinch/ focus will start to decrease, as observed before, then the pinch/ radial collapse will start to occur at the right instant i.e. when the discharge current is at its peak; thereby resulting in an increase in neutron output.

The measurements of time averaged neutron yield in repetitive mode in FMPF-3 device, at corresponding pressures were performed in the similar fashion as mentioned under subsection 5.3.5. The sample oscillograms of captured ³He detector signals from a sequence of 10 successive shots at 13kV/ 4.5mbar/ 10Hz operation are shown in Fig. 5.57 to confirm the shot-to-shot stability of the yield. The correlation between the fixed timings and the incremented shot numbers indicated in the history mode captured real time signals (note the incremental shot number from 29 to 38 and at the

same time of 10:47 in Fig. 5.57), confirms the neutron production in each of the 10 successive discharges at 10Hz repetition rate.

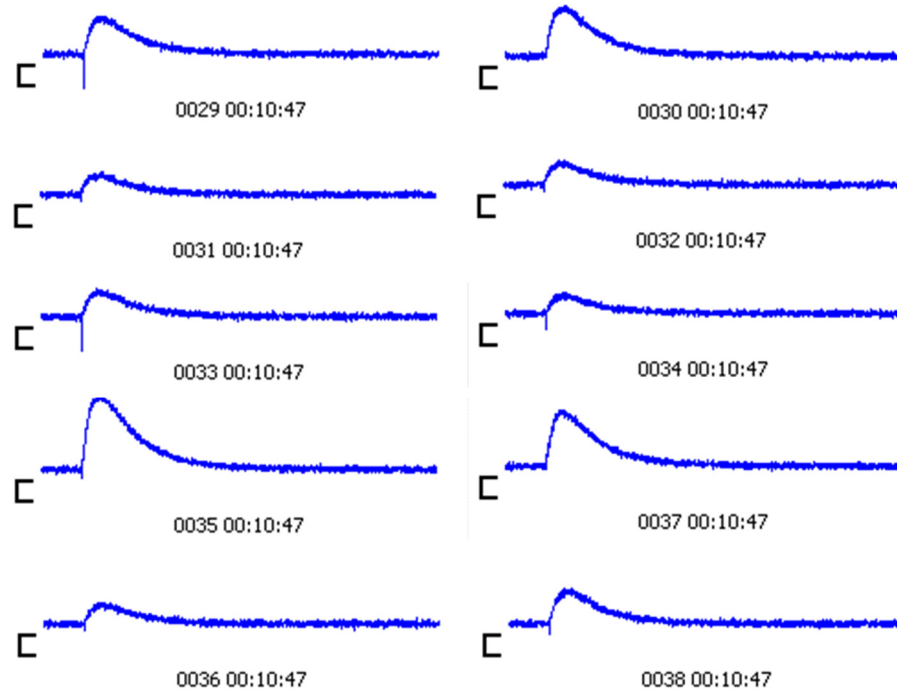
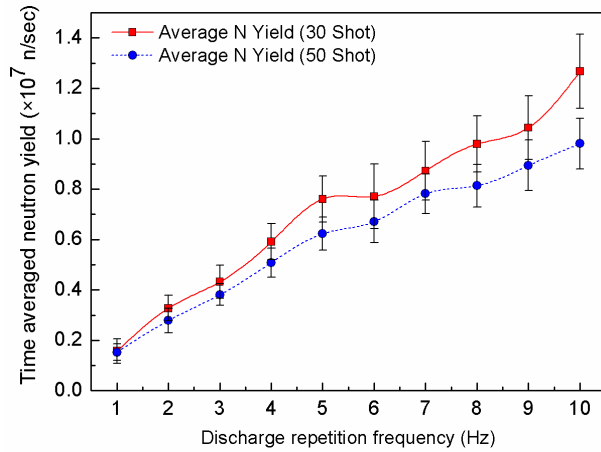
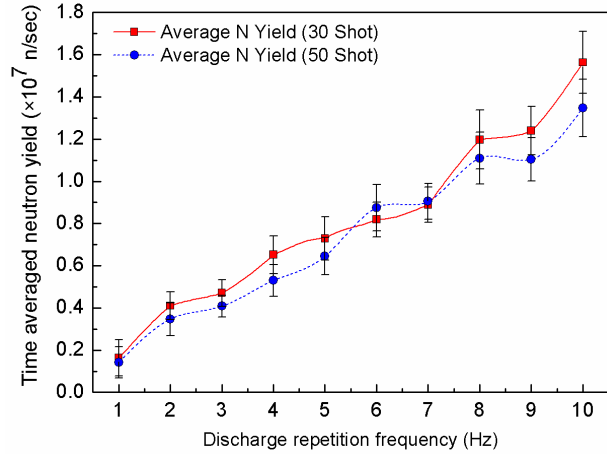


Fig. 5.57 History mode captured ^3He detector signals while operation at 10Hz.

As mentioned under subsections 5.3.2 and 5.3.5, for the estimation of neutron yield in each discharge during the experimental runs in repetitive mode of operation, such real time analog signals from ^3He detector (as shown in Fig. 5.57) are individually captured by the digital oscilloscope and saved in the memory of History mode for the entire burst duration (of 50 shots). Later on the data is extracted from the memory and the registered signals of ^3He detector are integrated using Yokogawa software, XViewer®, to estimate area under the curve in μVs and then by using the calibration factor [200], the neutron yield of individual shots is calculated.

The results obtained for enhancement in time averaged neutron fluence at various repetition rates for the burst length of 30 and 50 shots, at 4.5mbar and 5.5mbar D_2 filling gas pressures, are shown in Fig. 5.58 (a) and Fig. 5.58 (b) respectively.

(a) at 4.5mbar D₂ filling gas pressure(b) at 5.5mbar D₂ filling gas pressure**Fig. 5.58** Time averaged neutron output at different discharge repetition rates.

The results shown in Fig. 5.58 infer the scaling in time averaged neutron fluence with discharge repetition frequency. The enhancements observed in time averaged neutron outputs for 30 and 50 number of consecutive shots at 1Hz and 10Hz operation, for the D₂ filling gas pressures of 4.5mbar and 5.5mbar are collectively summarized for comparison in Table 5.6.

Table 5.6 Time averaged neutron yields for experimental run in repetitive mode at different D₂ filling gas pressures/ discharge repetition rates (DRR).

D ₂ filling pressure →	4.5mbar		5.5mbar	
Repetition rate →	DRR: 1Hz	DRR: 10Hz	DRR: 1Hz	DRR: 10Hz
Time averaged neutron yield for 30 shots →	$(1.6 \pm 0.5) \times 10^6$ n/ sec	$(1.3 \pm 0.2) \times 10^7$ n/ sec	$(1.6 \pm 0.8) \times 10^6$ n/ sec	$(1.6 \pm 0.2) \times 10^7$ n/ sec
Time averaged neutron yield for 50 shots →	$(1.5 \pm 0.3) \times 10^6$ n/ sec	$(0.9 \pm 0.1) \times 10^7$ n/ sec	$(1.4 \pm 0.6) \times 10^6$ n/ sec	$(1.4 \pm 0.2) \times 10^7$ n/ sec

It may be noticed from the results summarized above that FMPF-3 operation at higher than the optimum pressure of 5.5mbar resulted in the enhancement in time averaged neutron yield for 10Hz operation. For 5.5mbar operation, the time averaged neutron output for 10Hz operation was enhanced

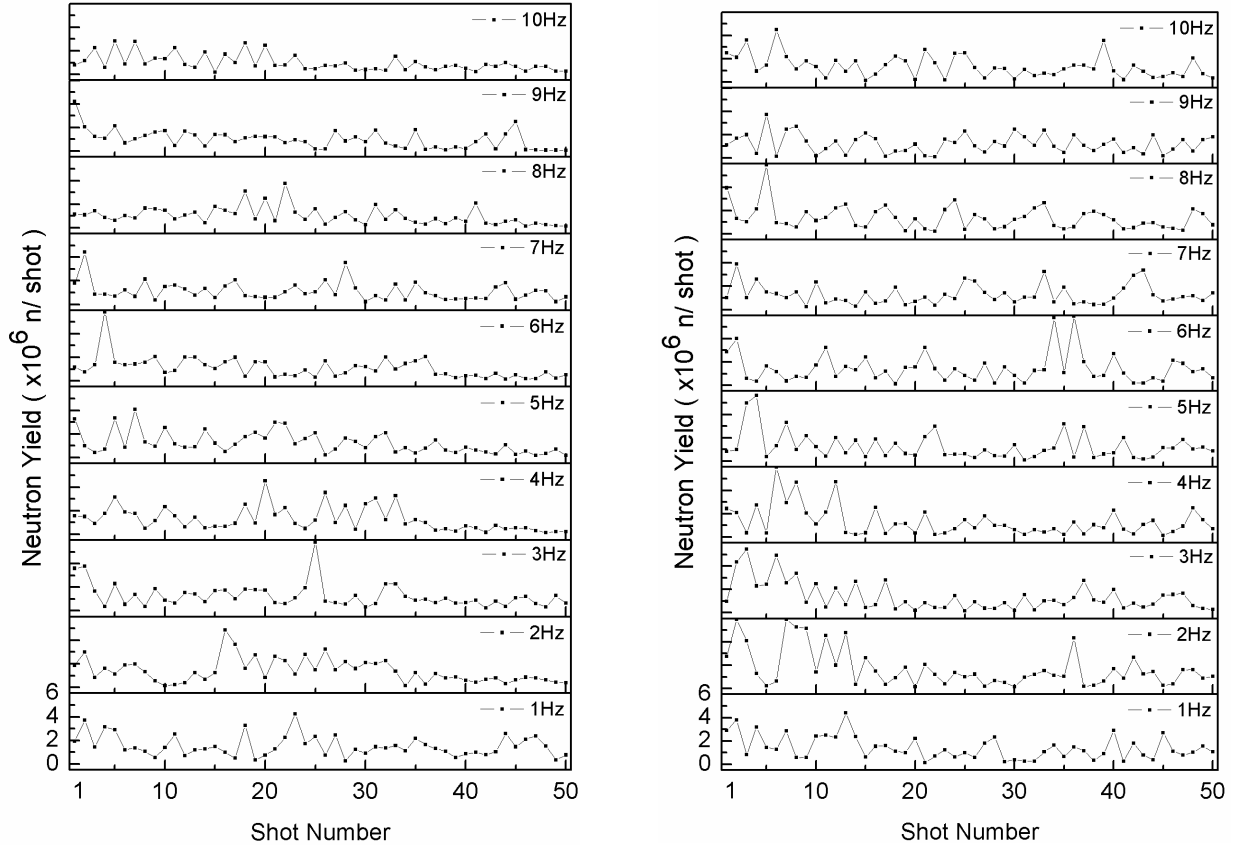
by about an order of magnitude for 30 as well as 50 shot bursts as compared to 1 Hz operation. However, for 4.5mbar operation, the relative lowering of the time averaged neutron output for 50 shots is consistent with the similar trend as observed in FMPF-2 device [216].

For gaining further insight of the obtained trends/ results evidenced in the measurements of time averaged neutron fluence, for 4.5 and 5.5mbar operation; in the next subsection, the investigation on stability in neutron emission was performed by analyzing the shot-to-shot variation in the neutron yield during the entire burst duration while the device was operated at different pressures and various discharge repetition rates.

5.4.8 Investigation on Shot-to-Shot Stability in Neutron Emission

The investigation of stability in neutron emission in successive shots during operation in burst mode, for a range of repetition rates from 1 to 10Hz has been performed. The shot-to-shot variations in neutron yield for burst length of 50 consecutive shots at different discharge repetition rates, without purging of D₂ gas in between, for the D₂ filling gas pressures of 4.5mbar and 5.5mbar are collectively shown in Fig. 5.59(a) and Fig. 5.59(b).

From the obtained results on shot-to-shot stability in neutron yield at various discharge repetition rates (DRR) for the burst length of 50 consecutive shots, improved constancy in neutron output for entire burst duration is clearly evident for repetitive mode of operation at 5.5mbar D₂ filling gas pressure (refer Fig, 5.59 (b)). For repetitive mode of operation at 4.5mbar D₂ filling gas pressure, the relative lowering of the time averaged output for 50 shots corroborates the results of shot-to-shot variation in neutron output (refer Fig, 5.59 (a)) where typically after 30 shots, descending trend in neutron yield at all discharge repetition frequencies was observed. This observation is consistent with the results/ trends obtained for FMPF-2 device [216].

(a) at 4.5mbar D₂ filling gas pressure(b) at 5.5mbar D₂ filling gas pressure**Fig. 5.59** Neutron yield per shot at different repetition rates for(a) 4.5mbar and (b) 5.5mbar D₂ filling gas pressures.

Further characterization of neutron output with time to pinch/ focus for successive shots during operation in burst mode at various discharge repetition rates, provided in the next subsection, unfolds the underlying cause of (a) deterioration in neutron output after few tens of shots and its further worsening at higher discharge repetition rates, for operation at 4.5mbar, and (b) constancy in neutron yield for entire burst length of 50 shots, for operation at 5.5mbar D₂ filling gas pressure.

5.4.9 Effect of Repetition Rate on Time to Pinch at Different Pressures

The measured average time-to-pinch/focus (for entire sequence of 50 shots) in repetitive mode

of operations at various discharge repetition rates, for operation at 4.5mbar and 5.5mbar D₂ filling gas pressure is collectively shown in Fig. 5.60.

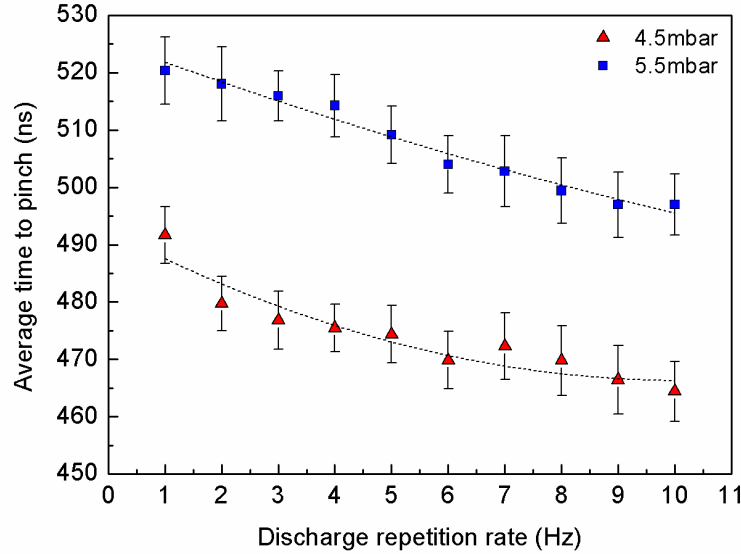


Fig. 5.60 Time-to-pinch at different repetition rates for 4.5 and 5.5mbar D₂ filling gas pressures.

For repetitive mode of operation at 4.5mbar, the measured 50 shot average of time to pinch/focus descended from $\sim 492 \pm 5$ ns at 1Hz to 465 ± 6 ns at 10Hz discharge repetition rate where as for device operation at 5.5mbar, the measured descend (50 shot average) in time to pinch/focus is from $\sim 520 \pm 6$ ns at 1Hz to 497 ± 6 ns at 10Hz discharge repetition rate. It may be noted from the neutron yield curves for single shot operation, in Figure 5.52, that the time to pinch corresponding to 4.5mbar pressure is about 520ns which provides the maximum neutron yield. The Fig. 5.60 shows that for the operation at 5.5mbar, the range of time to pinch values at different repetition rate are closer to the 520ns time to pinch observed for single shot operation, where the range of the time to pinch values for 4.5mbar operation are far below this optimum time to pinch value. This explains the better stability in neutron yield at 5.5mbar as compared to that at 4.5mbar for repetitive operations.

The obtained key results shown in Fig. 5.58, Fig. 5.59 and Fig. 5.60, for operation in repetitive mode at 4.5mbar and 5.5mbar D₂ filling gas pressure, conclusively indicate that the descend in time to

pinch with successively increasing discharge repetition frequency (and with the increasing number of shots within the burst), as a consequence of successive enhancement in residual electron density with increasing discharge repetition rate (detailed under subsection 5.3.7) can be minimized effectively by using judiciously chosen higher filling gas pressure for repetitive mode of operation. Under such condition the time to pinch under repetitive operation is much closer to time to pinch of single shot operation resulting in increase in neutron output while improvising the burst length response at the same time.

Although we succeeded in mitigating the problem of fast descending trend in time to pinch/focus observed during repetitive mode of operation, which affects the shot-to-shot neutron yield stability, by pressure regulation but as discussed before under section 5.3.7, another major cause of concern for successive degradation in neutron yield with increase in the discharge repetition rate is the enhancement in erosion/vaporization of anode. The physical condition of the cylindrical stainless steel anode after 5000 shots of operation in FMPPF-3 device has been shown in Fig. 5.61.

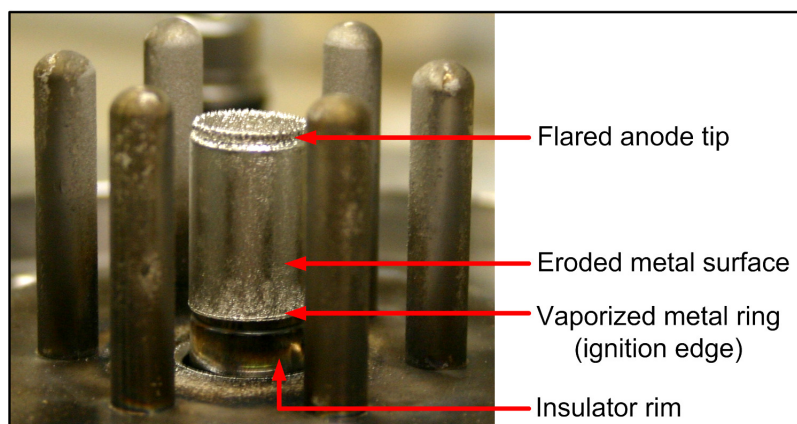


Fig. 5.61 Physical condition of cylindrical anode after 5000 shots of operation in repetitive mode in FMPPF-3 device.

The wearing of cylindrical anode (post experimental run) is clearly evident in the photograph shown in Fig. 5.61. The anode damage exhibits flared tip with eroded metal on top (due to contact with

hot plasma and instability accelerated electrons) and the formation of vaporized metal ring at the ignition edge. Considerable erosion in cathode plate has also been evidenced (post experimental runs) in repetitive mode of operation.

In the photograph shown in Fig. 5.62 (a), vaporization/ erosion of cathode plate ignition edge may be observed clearly. In the initial phase of experiments single piece assembly of cathode plate was used. Later on, when it was experienced that significant erosion of cathode plate ignition edge results in loss of focusing reproducibility (which might be due to failure/ delay in sheath lift off); the surrounding portion of cathode plate edge adjacent to insulator rim was modified and plugged with a replaceable brass bush, as shown in Fig. 5.62(b). Brass was chosen as material for bush due to its self-healing characteristics [185] and to eliminate the possibility of welding with the main body of cathode plate, which is of stainless steel.

The damages shown and discussed for anode and cathode affirm deleterious effects of thermal loading during repetitive mode of operation.

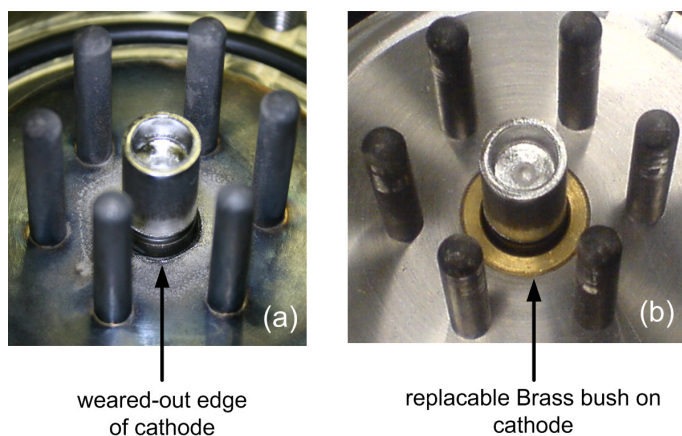


Fig. 5.62 Photographs of (a) eroded cathode edge, and (b) cathode plate with replaceable brass bush in the modified assembly.

The deterioration in the life of Pyrex glass insulator sleeve has also been found to be significant under experimental runs in repetitive mode. The typical life of glass sleeve insulator used under

repetitive mode of operation in FMPF-3 device is ~20000 shots. It may be noted that the breakdown characteristics in plasma focus device operation largely depends on insulator surface conditions [61]. Upon doing the microscopical study of damaged insulator under scanning electron microscope (SEM), substantial deposition of anode material was observed on the surface. Fig. 5.63 shows the obtained typical surface patterns (with different magnifications) of a damaged glass insulator, under different magnifications.

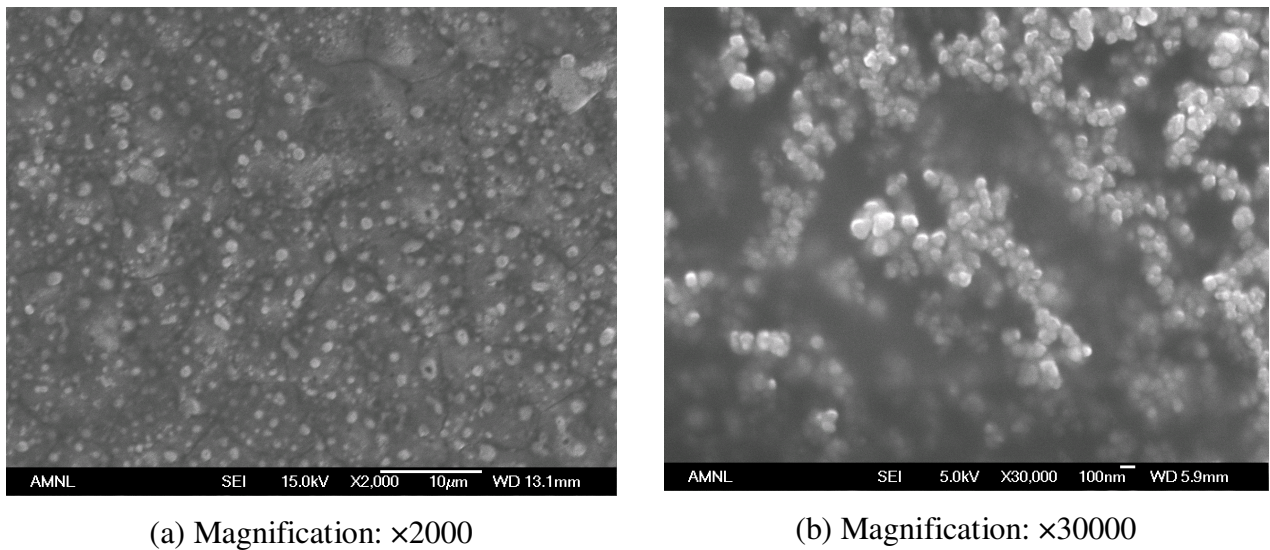


Fig. 5.63 Microscopical images of damaged insulator surface.

At magnification of $\times 2000$ (i.e. Fig 5.63 (a)), deposition of micro-size ablated anode material droplets were seen all over insulator surface, where as at higher magnification of $\times 30000$ (i.e. Fig 5.63 (b)), formation of nano-particles were clearly observed. The obtained EDX results confirmed the elemental presence of Iron (Fe) as 48% (by weight) over the insulator surface. Thus, microscopical observation affirms that deposition of anode metal vapors and ablated particles from anode is also a prominent cause of modification in the breakdown characteristics and contributes deleterious effects to the neutron yield performance with successively increasing number of shots.

In summary, experiments performed on FMPF-3 device at 13kV/ 200J not only demonstrated successful enhancement in time averaged neutron yield by operating the device in repetitive mode but it unambiguously led to an important conclusion that “optimum pressures for the plasma focus device operation in singular and repetitive mode are different”. Obtained trends of results (at higher filling gas pressure of 5.5mbar) also indicate that discrete choices of D₂ filling gas pressures for specific discharge repetition rate may lead to further optimization in time averaged neutron yield performance for repetitive mode of operation.

In general, from the experience gained on repetitive operation of fast miniature plasma focus devices (FMPF-2 and FMPF-3), it has been realized that the major technological challenges along the path to higher repetition rate/larger burst length, miniature plasma focus devices at present are as follows:

- (i) The enhanced pre-ionization as a consequence of cumulative effect of subsequent series of discharges (*i.e.* aggregation of left over thermal electrons after each discharge) that worsens at higher discharge repetition rates as the recovery time goes successively smaller at higher repetition rates [216]. This problem has been resolved to a certain extent by using higher filling gas pressures for repetitive operation so that even if for initial shots the pinch may be pushed into a non optimized regime with lower output due to higher filling gas pressure, with increasing number of shots, due to shortening of time to pinch, the pinch starts to occur at the right instant, resulting in increase in neutron output. An alternative approach is also to maintain the optimum pressure by purging the Deuterium continuously at an optimal flow rate. The second option, however, removes the possibility of developing MPF device as a repetitive and portable sealed tube neutron source.
- (ii) The enhanced erosion/vaporization of anode, particularly in repetitive mode, due to bombardment by instability accelerated electrons, softening of anode material due to enhanced

joule heating and contact with hot plasma is another major technological problem, which as explained earlier not only reduces the neutron output, but also significantly reduces the useful life of a particular anode. Anodes of materials with higher heat of vaporization (like Tungsten, Molybdenum, and Zirconium) and larger volume may mitigate this problem to a certain extent. Use of such high impulsivity anode materials will also be helpful in minimizing the fast deterioration in shot life of insulators, which is also an important concern.

The next chapter summarizes key research results and corresponding conclusions of the investigations performed on indigenously developed three different versions of fast miniature plasma focus devices (FMPF-1, FMPF-2 and FMPF-3) that are included in this thesis. The chapter also suggests future research directions in which the further studies can be carried out.

Chapter 6

Conclusions and Suggested Future Work

During the course of this PhD Programme several aspects of low energy sub-kilojoule range repetitive plasma focus devices have been investigated. This thesis presented fundamentals, detailed design and construction of three different versions of indigenously developed fast miniature plasma focus devices, diagnostics setup and results and discussions of key investigations on the three miniature plasma focus devices. In this chapter, the key results and findings are summarized and the possible future work is suggested.

6.1 Summary and Conclusions

Three different versions of fast miniature plasma focus devices – FMPF-1, FMPF-2 and FMPF-3 have been indigenously designed, constructed and optimized as compact and repetitive pulsed neutron source. The first version FMPF-1 device (with stored energy of <240J) was mainly developed and utilized as a test module for investigating and optimizing the neutron yield performance of the device, by fine tuning focus tube parameters and other operating parameters like filling gas pressure, charging voltage, admixture compositions *etc.* The second and third versions, FMPF-2 and FMPF-3 devices were developed and customized as high repetition rate pulsed neutron sources with consecutively higher electrical efficiencies, while operating in the similar energy range. The preliminary version FMPF-1 device can be operated at maximum repetition rate of 0.5Hz where as the advanced versions FMPF-2 and FMPF-3 devices can be operated at maximum repetition rate of 10Hz in quasi-continuous mode. Many technological challenges were confronted and resolved for making the successful operation of the FMPF-2 and FMPF-3 devices in repetitive mode up to 10Hz. Constructionally, FMPF-1 (comprising of four 0.6 μ F/30kV capacitors and one Trigatron switch) and

FMPF-2 (comprising of eight 0.3 μ F/30kV capacitors with single Trigatron) devices are single module systems whereas the FMPF-3 device is a four module system comprising of eight 0.3 μ F/30kV capacitors in four modules of two capacitors each with each module having a separate Pseudospark switch.

The main results and conclusions obtained from investigations on successive modifications in pulsed power system and other geometrical and operating parameters of FMPF devices are as following:

- 1) After passing several stages of design and experimental iterations of coaxial electrode assembly and other operating parameters, the 'FMPF-1' device was successfully demonstrated as a portable neutron source producing maximum average neutron yield of $(1.15 \pm 0.2) \times 10^6$ neutrons/ shot at about 230J/80kA peak discharge current [211,212,215]. The conceived model of 'FMPF-1' is extremely compact and aids the feature of portability/ table top model. Unlike conventional plasma focus machines, in this device, the connections between sub-systems like capacitor bank, spark gap and plasma focus tube has been minimized by embedding the sparkgap within transmission line assembly of capacitor bank and the plasma focus head was directly interfaced to the discharge end of the sparkgap. The overall dimensions of the apparatus, which includes capacitor bank, sparkgap switch and the focus chamber is 0.2m $\times$ 0.2m $\times$ 0.5m and the total weight of the system is ~25kgs [200].
- 2) The FMPF-2 device was developed as the first prototype of high repetition rate miniature plasma focus device, powered by novel high wattage pulsed power system. To the best of our knowledge for the first time, using the low energy miniature plasma focus device in repetitive mode, the enhancement in time averaged neutron yield has been successfully demonstrated by an order of magnitude [216], realizing the long stated concept of 'enhancement by repetition'. This device has been successively fired and tested up to 10Hz repetition rate under different

operating conditions. Using pure Deuterium as the filling gas, at ~230J operation (delivering ~80kA peak discharge current) the time averaged neutron yield of $(6.2 \pm 4) \times 10^5$ neutrons/second at 1Hz discharge repetition rate was enhanced to $(6.5 \pm 0.6) \times 10^6$ neutrons/second at 10Hz discharge repetition rate for the burst length of 30 consecutive shots [216].

- 3) The FMPF-3 device is demonstrated as the most optimized version of the repetitive miniature plasma focus device with stored energy of about 200J. It is basically an upgraded version of FMPF-2 device having superior pulsed power system efficiency along with high repetition rate capability. The electrical efficiency of FMPF-3 device is the highest; it delivers higher peak discharge current (~18-20%) per joule of stored energy in comparison to two previous versions of FMPF devices and thus produces maximum neutron yield, while operating at the similar charging voltage and energy. Using pure Deuterium as the filling gas, at ~200J operation (delivering ~90kA peak discharge current) the time averaged neutron yield of $(1.4 \pm 0.6) \times 10^6$ neutrons/second at 1Hz operation was enhanced to $(1.4 \pm 0.2) \times 10^7$ neutrons/second at 10Hz operation for the improved burst length of 50 consecutive shots. From the experiments performed on FMPF-3 device, it was also conclusively deduced that optimum filling gas pressures are different for singular and repetitive mode of operation [217].
- 4) The potential of sub-kilojoule range repetitive miniature plasma focus devices has been successfully demonstrated by enhancing the time averaged neutron fluence but it has been conclusively realized that there are critical consequences and limitations of repetitive mode of operation, responsible for subsequent degradation in neutron yield during the repetitive mode of MPF device operation [216]: (i) modification in the initial breakdown conditions and consequently the time to pinch because of enrichment of free electrons by cumulative effect of subsequent series of discharges (*i.e.* aggregation of left over thermal electrons after each discharge) that worsens at higher discharge repetition rates as the recovery time becomes

successively smaller [216], (ii) the enhanced erosion/vaporization of anode due to excessive joule heating, successive bombardment by instability accelerated electron beam and contact with hot plasma (anodes of materials with higher heat of vaporization like Tungsten, Molybdenum, Zirconium and larger volume may mitigate this problem up to a certain extent), and (iii) the enhanced addition of high-Z impurities in the pure D₂ plasma due to contaminants injected by vaporization/erosion of anode material at ignition edge at insulator sleeve rim and at anode top [216].

- 5) The discharge/ focusing characteristics of FMPPF devices with various anode geometries have been studied in detail [200]. In our investigations, the critical role of the speed factor has been observed on focusing characteristics. It has also been found that, even in sub-kilojoule miniature plasma focus devices (for Deuterium operation) the optimum neutron production efficiency range lies within speed factor range of $89 \pm 7 \text{ kA/cm/(Torr)}^{1/2}$ (which is the typical range for Deuterium operated plasma focus devices having energies in the range of kJ to MJ). Our interpretation of this fact is that – as with conventional kilojoule range plasma focus devices, in sub-kilojoule range plasma focus devices too, similar orders of plasma density and temperature are created. Another important conclusion obtained, while performing neutron yield optimization study of various anode geometries having speed factors/drive parameters in various range, from our investigations is that speed enhancement does work and plays a very important role in deciding the axial and radial velocities (and thus focusing efficiency). But if the drive parameter is increased beyond certain limit, the performance of the device as neutron source decreases. Very high speed factors may cause effective separation of the magnetic piston from the plasma layer thereby resulting in inefficient collapse of the plasma layer in the radial phase as the piston continues to drive axially [222]. However, it was interesting to note that significant enhancement in the soft X-ray yields was observed (though there was no or

very low yield of neutrons) for anodes that were operated at relatively high speed factor ($>200\text{kA/cm}/(\text{Torr})^{1/2}$), using long tapered anode geometry (V2/TA/20-5/Cu) [213]. This result supports the study proposed by Serban and Lee [222] that soft X-ray emission (Y_{SXR}) has correlation with peak discharge current (I) and axial sheath velocity (v_{axial}) as: $Y_{\text{SXR}} \approx I^2 v_{\text{axial}}^4$.

- 6) In the experiments performed on FMPF-1 device to study the effect of tubular and squirrel cage cathode structure on neutron yield performance of the device while maintaining similar operating conditions it was found that with D_2 operation, the maximum average neutron yield of $(1.82 \pm 0.52) \times 10^5$ neutrons/shot for tubular cathode at 4mbar was enhanced to $(1.15 \pm 0.2) \times 10^6$ neutrons/shot with squirrel cage cathode at 6mbar operation at $\sim 230\text{J}/80\text{kA}$ peak discharge current [211,212]. In this investigation it was been found that in operation with tubular cathode structure, owing to the impermeable wall and relatively larger surface area, the plasma density rises near the cathode wall, causing the shock to be radially broadened through resistive diffusion of the magnetic field. This results in the narrowing of the effective channel cross-section, causing loading of the current sheath along with increased contact layer resistance thereby resulting in lower pinch efficiency. Also in operation with tubular cathodes, addition of impurities (due to back reflected particles from the cathode wall) during the axial flow, may result in a substantial drop in temperature due to increased radiation loss which is also consistent with the reduction in neutron output. These conclusions appear to be confirmed by the experimentally measured yields of neutrons, as a function of filling gas pressure range, for both cathode geometries. The outcome of this investigation reveals penalty for using a non-transparent cathode, which negates the advantage of the smaller size of the DPF head [211].
- 7) The experiments performed on FMPF-1 device with varied concentrations of D_2 -Kr admixtures have been found to yield some important results [213,214]. It was observed that (irrespective of anode geometry/ design), a few percent of admixture gas aids in the transfer of directed kinetic

energy in the implosion into thermal energy in the pinch plasma and helps to stabilize the dense pinch for longer duration. While operating FMPF-1 device at $\sim 200\text{J}$, the maximum average neutron yield of $(1 \pm 0.27) \times 10^4$ neutrons/shot for pure D_2 filling at 3mbar was enhanced to $(3.14 \pm 0.4) \times 10^5$ neutrons/shot with $\text{D}_2 + 2\% \text{Kr}$ admixture operation, which represents a >30 fold increase. More than an order of magnitude enhancement in the average neutron yield was observed over the broader operating range of 1– 4mbar for $\text{D}_2 + 2\% \text{Kr}$ and $\text{D}_2 + 5\% \text{Kr}$ admixtures [214]. Similarly about 17 and 10 fold increase in X-ray yield in spectral ranges of 0.9 to 1.6keV and 3.2 to 7.7keV, respectively, have been obtained with $\text{D}_2 + 10\% \text{Kr}$ admixture at operating pressures ≤ 0.4 mbar [213]. It is remarkable to note that characteristic of the admixture gas influences the dynamics in two aspects: (i) consume extra ionization/ radiation energies and lower down the plasma temperature, and (ii) produce extra particles by ionization and affect the plasma dynamics by thermodynamic parameters. In conclusion, our study on use of D_2 + high-Z gas admixtures illustrates that if judicious concentration of high-Z gas impurity in pure D_2 is used (by just seeding enough to optimally enhance the radiation by radiative collapse, but not enough to affect the implosion though the higher atomic mass of the impurity atoms) along with adjustment of the base pressure such that the swept mass allows the time for first radial collapse to coincide with the peak current, then D_2 + high-Z gas admixtures can significantly enhance the radiation yields. The improved pinch stabilization and higher plasma density through improved compression, actuated by radiative collapse explains vigorous enhancement in neutron and X-ray yield for D_2 + high-Z admixture operations.

- 8) The investigation on neutron emission characteristics of sub-kilojoule range miniature plasma focus devices and their correlation with hard X-ray emission revealed dominance of neutron production due to axially accelerated deuterons and supports the beam target postulation [215]. It has been commonly observed in all the three different versions of FMPF devices that at their

respective optimum pressures, the maximum neutron yield coincides with the peak neutron anisotropy and dip in the HXR anisotropy. As the main source of non-thermal HXR radiation is the accelerated electron beam impinging on anode and as the hollow anode then transmits more in the radial direction, we have used this reduction in anisotropy to infer that the electron beam (and hence the complementary ion beam) increases in energy at the optimum pressure. The presence of anisotropy, under neutron optimized conditions, indicates deuteron velocity distribution along the anode axis and rules out the possibility of purely thermal emission. The average forward to radial neutron yield anisotropy in FMPF-1, FMPF-2 and FMPF-3 devices is found to be 1.3 ± 0.2 , 1.6 ± 0.3 and 1.4 ± 0.2 , respectively. The experimental results also showed that the average neutron energy is always higher in the forward direction than in the radial direction, confirming the beam target mechanism to be the dominant neutron production mechanism. It was theoretically as well as experimentally deduced (by neutron production kinematics and Faraday Cup measurements) that low energy deuterons, of typically $<100\text{keV}$, contribute more significantly in the neutron production in sub-kilojoule low energy miniature plasma focus devices [215].

- 9) The obtained neutron yield scaling results with peak current (I) and pinch current (I_p) show the scaling laws as $Y_n \sim I^{6.85 \pm 0.3}$ and $Y_n \sim I_p^{5.47 \pm 0.2}$ for the operation of FMPF-3 device in the energy range of 120J – 235J. The indexes obtained as 6.85 ± 0.3 and 5.47 ± 0.2 for peak (I) and pinch current (I_p) respectively, are relatively on the higher side than the typically observed range [128]. This finding indicates that neutron yield scaling has higher dependence on peak/ pinch currents for the sub-kilojoule range devices typically operating with currents $<100\text{kA}$.
- 10) Numerical experiments were also performed using ‘Lee Code’ [70] to simulate the modifications in the axial and radial phase dynamics along with gross plasma focus properties for various focus tube parameters that have been experimented. The designing and optimization

of device performance was a complex procedure based on multi-step of refinement to obtain best neutron yield. In our investigations, Lee Code simulation was used as rapid tool to tune the geometrical and electrical parameters of the fast miniature plasma focus devices during the various stages of optimization [200] [211-217].

Apart from demonstrating the FMPF-1, FMPF-2 and FMPF-3 devices as compact and efficient neutron generators; on the application side, the versatility of its usage has also been demonstrated in the other areas of basic and applied research such as fast radiography (hard X-ray imaging of sub-millimeter range spatial resolution with nanosecond exposure time) [249] and in material modification studies like deposition of FeCo thin films with soft magnetic properties [250] and synthesis of diamond like carbon coatings [251].

6.2 Suggested Future Work

Based on the work done during the PhD Programme following investigations may be done for future research work.

- 1) An important and the major outcome of the project done during this PhD Programme is not only the successful demonstration of low energy miniature plasma focus device as repetitive pulsed neutron source but also acquaintance with consequential affects of repetitive modes of operation on radiation products are of paramount importance. The two such major issues of concern in which the future research work can be done are:
 - a. The issue of progressively descending trend of time to pinch with the successively increasing number of shots and discharge repetition rate during repetitive mode of operation requires further in-depth understanding and experimental investigations of the breakdown phase. Despite the fact that the dynamics of acceleration and compression phases of plasma focus device operation are well examined theoretically as well as experimentally; the study

of breakdown phase is far from being complete. In our investigations, while searching for the possible reasons, it has been realized that, till date, there is a very limited availability of theoretical and quantitative information on physical processes involved in the initial breakdown phase. This fact necessitates detailed study to theoretically simulate the entire breakdown process up to lift off stage and elucidate the involved physical processes. Experimentally time resolved spectroscopy/ laser shadowgraphy investigations, performed in sync with repetitive operation of device may also lead to some conclusive observations that might be helpful in finding remedial measures of this cause. Further more investigations may be conducted by adjusting the gas pumping/ filling speeds (i.e. continuous gas flow at fixed chamber pressure) at high repetition rates. The geometry of focus tube may also be redesigned to study the effect of modification in gas refilling/ pumping path under continuous gas flow conditions. For static fillings, the discrete choice of gas pressures for specific rep-rates can be further iterated to obtain optimum results.

b. The second major issue that requires to be handled tactfully during the repetitive mode of operation is ‘thermal loading’. Issue of ‘thermal loading’ is not only problematic from a macroscopic point of view, resulting in the fast erosion of electrodes, but also from the microscopic point of view of plasma purity: metal particles ablated by the high temperature plasma from the electrodes are carried by the sheath into the pinch region, resulting in cooling of the plasma (by enhanced line radiation losses) and consequent degradation in the neutron yield. Explorations of proper cooling systems/ alternative techniques are therefore necessarily required, to overcome ‘thermal loading’ in the repetitive plasma focus devices. One of the major reasons for excessive joule heating during the repetitive mode of operation is because at each pulse/ shot, due to high voltage reversal in circuit, the discharge oscillates even after termination of the pinch and a significant amount of input

electrical power is dissipated in the anode and cathode. The possible solution that may be incorporated to eliminate the voltage reversal is to use crowbar switch that will short circuit/ bypass current just after the pinch. Implementation of this measure may not only eliminate the requirement of cooling but will also result in enhancing the shot life of the pulsed power switches and capacitors, used in the repetitive system. However, this technique may add further complexity in the pulsed power system.

- 2) Fluence anisotropy measurements in our experiments have only been performed in axial and radial directions, in future; detailed measurement of angular neutron yield distribution can be done to obtain the absolute isotropic and anisotropic contributions, to gain insight of the prevailing mechanism of neutron production in sub-kilojoule range MPF devices.
- 3) Experiments targeting enhancement in the neutron yield with higher reproducibility can be explored using Deuterium-Tritium (DT) as gas fill. Since the cross section for DT reaction is ~ 100 times higher than the DD reaction, typical gain in the neutron yield by two orders of magnitude is expected in operations with DT gas mixtures. Using DT as gas fill in FMPF-3 device at 200J, 10Hz operation, it will be possible to achieve 10^8 - 10^9 n/s (of 14MeV energy). An added advantage with use of DT as gas fill is also that the rate of rise of reaction rate with respect to temperature for DT is comparatively lower than that for DD, for temperature $>10\text{keV}$, hence smaller rate of change of cross section with energy for DT reaction may also result in lower shot to shot variation in neutron yield.
- 4) The use of FMPF-3 device as repetitive pulsed neutron source can be explored for application, as interrogation tool for the detection of hidden explosives and narcotic drugs (by using Pulsed Fast/Thermal Neutron Analysis (PFTNA) technique).
- 5) In different FMPF devices the investigation of change in surface resistivity of the insulator with the ageing/use may be of interest to understand the problem of neutron yield variation.

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